GMN | The Global MTCC Network
A global network for energy-efficient shipping

MTCC PACIFIC
Maritime Technology Cooperation Centre

MARITIME TECHNOLOGY COOPERATION CENTRE IN THE PACIFIC (MTCC-PACIFIC)

CAPACITY BUILDING FOR CLIMATE MITIGATION IN THE MARITIME SHIPPING INDUSTRY

THE GLOBAL MTCC NETWORK (GMN) PROJECT

MTCC-PACIFIC REGIONAL CONFERENCE AND OFFICIAL LAUNCH

Suva, Fiji

12-15 December, 2017

EUROPEAN UNION

IMO INTERNATIONAL MARITIME ORGANIZATION

Pacific Community
Communauté du Pacifique

HOST INSTITUTIONS OF MTCC-PACIFIC

SPREP
Secretariat of the Pacific Regional Environment Programme

The Global MTCC Network (GMN) project is funded by the European Union and implemented by the IMO.
This publication was produced with the financial assistance of the European Union. The contents of this publication are the sole responsibility of MTCC-Pacific and can in no way be taken to reflect the views of the European Union.

CONTENTS

EXECUTIVE SUMMARY 1

INTRODUCTION 2

WORKSHOP FOR HEADS OF MARITIME 3

- INTRODUCTION 3
- INTRODUCTION TO GMN PROJECT AND MTCC-PACIFIC 3
- NATIONAL TECHNOLOGY NEEDS, BARRIERS 4
- ENERGY MANAGEMENT IN PORTS: METHODS AND TOOLS 4

SECOND MTCC-PACIFIC STEERING COMMITTEE MEETING 6

- INTRODUCTION 6
- ACTION POINTS 6
- ADOPTION OF THE 2ND STEERING COMMITTEE MEETING AGENDA 6
- REVIEW OF THE LAST MEETING MINUTES 6
- IMO WORK TO ADDRESS ENERGY EFFICIENCY AND GHGE 7
- UPDATES FROM THE COMMITTEE ON COP23 AND OTHER GLOBAL EVENTS 7
- MTCC-PACIFIC PROGRESS REPORT 8
- OTHER BUSINESS 9
- CLOSING REMARKS 9

MTCC-PACIFIC REGIONAL CONFERENCE AND OFFICIAL LAUNCH 10

RELEVANT ACTIONS TO IMPLEMENT THE TRANSPORT MINISTERS' RESOLUTION AND COMMUNIQUE 10

GMN INTRODUCTION AND STATUS UPDATE 10

WELCOME CEREMONY AND OFFICIAL LAUNCH OF MTCC-PACIFIC 11

- Welcome Address 11
- Opening Address 12
- Keynote Address 13
- Handover ceremony from IMO to MTCC-Pacific of its official plaque 14

SESSION 1: PACIFIC ISLAND COUNTRIES AND TERRITORIES PERSPECTIVES 15

- Micronesian Centre for Sustainable Transport: A Framework for Transition 15

Country Presentations and Updates 16
Pacific Islands Development Forum 21
A Comprehensive Imo Strategy on Reduction of GHGE from Ships 22
Low Carbon Sea Transport Project 22
New-Caledonia toward Sustainable Blue Growth 23
Norway Webinar On New Ship Technologies 23
Reducing GHGE from Domestic Ships: Ship Owners Perspective 23
A Concept for Inter-Island Sustainable Sea Transport in the Pacific 24
Reviving our Sustainable Sea Transportation Culture in Fiji 24
SESSION 3: MTCC-PACIFIC WORK 25
MTCC-Pacific Progress Report 25
Energy Management in Ports – Activities and Results 25
MTCC-Asia Summary 25
Pilot Project and Data Collection 26
Country Priority Questionnaire 26
Mainstreaming Gender in Maritime Transport Sector 26
REVIEW AND ADOPTION OF THE REGIONAL CONFERENCE OUTCOME 27
CLOSING REMARKS 27
IMO, Aubrey Botford 27
SPC/MTCC-Pacific, Thierry Nervale 28
Government of Fiji, Lui Naisara 28
COMMUNICATION AND VISIBILITY 29
LIST OF ANNEXES 32

1 List of Participants
2 Agenda
3 Outcomes Document
4 Presentations
5 Evaluation Analysis
6 Photos
7 Communications and Visibility Activities

EXECUTIVE SUMMARY

The First Regional Conference of the Maritime Technology Cooperation Centre in the Pacific (MTCC-Pacific) was held in Suva, Fiji from 12 to 15 December 2017 at the Novotel Conference Centre in Lami. The meeting was organised and hosted by the MTCC-Pacific with the support of the Pacific Community (SPC), its Host Institution, and the International Maritime Organization (IMO). The Conference was co-hosted by Fiji and attended by Permanent Secretaries, Secretaries, Deputy Secretaries, Directors and Officers from Cook Islands, Kiribati, Marshall Islands, Nauru, New Caledonia, Papua New Guinea, Samoa, Solomon Islands, Tokelau, Tonga, Tuvalu and Vanuatu. International and regional partners also attended such as IMO, the European Union, Secretariat of the Pacific Regional Environment Programme (SPREP), Pacific Islands Development Forum (PIDF), Australia Maritime Safety Agency (AMSA), MTCC-Asia, and representatives from Pacific Ports, the Pacific Islands Maritime Conference (PIMC), representatives from Fiji shipowners, Okeanos Foundation and the Uto Ni Yalo Trust. The list of participants is attached in Annex 1.

The objective of the Regional Conference was to officially launch MTCC-Pacific as a centre of excellence for the region with the view to build the capacity of Pacific Small Islands Developing States (SIDS) and Least Developed Countries (LDCs) for climate mitigation in the maritime industry. The Conference was also the opportunity to showcase the Global MTCCs Network (GMN), a project implemented by IMO and funded by the European Union that aims to establish five MTCCs in Africa, Asia, Caribbean, Latin America and the Pacific regions. The Conference gathered representatives from Pacific Islands Countries and Territories (PICTs) and partners to inform them of the activities of the MTCC-Pacific and to agree on actions to give effect to the resolution of the 2017 Pacific Regional Energy and Transport Ministers' Meeting to transition to low-carbon maritime transport in the Pacific. The week-long agenda is attached as Annex 2.

On 12 December 2018, a half-day workshop for Heads of Maritime from the Pacific Islands region was also organised to review the information collected from previous consultation about the drivers, needs, barriers and relevant actions to progress towards low-carbon maritime transport in the Pacific. This information was provided to participants in a tabulated form grouped by areas such as technology, capacity-building, etc. with recommendations for short-, medium- and long-term actions. It was proposed to use this matrix of needs and actions as an output of the Regional Conference and were reviewed throughout the conference following specific presentations. At the end of the workshop, Heads of Maritime proposed to prioritise the actions in four categories (Capacity Building; Collection of data and information; Pilot-projects; Policy/Legal Frameworks) to be endorsed in the Regional Conference outcome.

In the afternoon of 12 December, the Second MTCC-Pacific Steering Committee meeting was chaired by the the Fiji Ministry of Infrastructure and Transport and attended by the Steering Committee members from the Pacific Community (SPC), the Secretariat of the Pacific Regional Environment Programme (SPREP), the Fiji Delegation of the European Union, Marshall Islands, Samoa, Solomon Islands, the Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH, the Pacific Islands Development Forum (PIDF), the Pacific Islands Forum Secretariat (PIFS) and the University of the South Pacific (USP). The Steering Committee went through the usual proceedings to report on members actions and be updated on MTCC-Pacific progress. The IMO was represented and they updated on the work at IMO related to greenhouse gas (GHG) emissions from shipping.

The MTCC-Pacific Regional Conference was officially opened on 13 December with the Official Launch ceremony in the morning in the presence of the Honourable Ratu Osea Naiqama, Fiji Minister for Forestry (and Acting Minister of Fisheries), the Ambassador of the European Union Delegation for the Pacific, Julian Wilson, Dr. Audrey Aumua, Deputy Director General of the Pacific Community (Suva) and Mr. Aubrey Botsford, Deputy Director/Chief Officer of General Services of the IMO. To mark this particular event and better integrate MTCC-Pacific in the region, a video was displayed showing a Fijian traditional outrigger canoe (Drua) bringing the MTCC-Pacific plaque that was handed over from IMO to MTCC-Pacific in the presence of traditional Fijian warriors. The video will be used as part of the MTCC-Pacific promotional video to be developed in late 2018. Supporting images of conference activities are attached as Annex 6.

On the last day 15 December, a space was given to representatives of Pacific Women in Maritime (PacWiMA) to raise the challenges and opportunities for women's contribution to the maritime sector and particularly the opportunities presented in the area of maritime energy efficiency. Before closing the Regional Conference, the participants adopted an '*Outcome*' document with commitments, requests for support and immediate actions to be commenced. The outcomes document is attached as Annex 3. All participants expressed their satisfaction to SPC and MTCC-Pacific for the organisation of the conference and confirmed their commitment to MTCC-Pacific pilot-projects and capacity-building activities in 2018-2019. All workshop presentations are attached as Annex 4 in consecutive order as outlined in the Agenda.

INTRODUCTION

The MTCC-Pacific Regional Conference and Official Launch was held at the Novotel Convention Centre, Suva, Fiji, from the 12th to 15th December 2018 (refer to Annex 1 Agenda). The Conference included a preparatory workshop for Heads of Maritime and the Second MTCC-Pacific Steering Committee.

The objectives of the Conference were to agree with representatives from Pacific Islands Countries and Territories (PICTs) on the needs, barriers and relevant actions for transitioning to low-carbon maritime transport in the Pacific and the way forward in the short-medium term. It was also the opportunity to officially launch MTCC-Pacific and present its project and how it will assist PICTs to respond to the issues raised.

The following report is in three parts for each of the events: the workshop for Heads of Maritime, the Second Steering Committee and the Conference with the MTCC-Pacific Official Launch. In each parts, the sections have been broken down according to the Agenda items including the discussion points. The Conference has produced an outcome document which will guide the region in the medium term.

A media release was sent out at the end of the workshop noting the important role of the MTCC-Pacific in ensuring the representation of the Pacific in the global network given the central role maritime plays in the region; and in assisting Pacific countries in accelerating the uptake of technical and operational possibilities to improve the energy efficiency of shipping and promote clean energy shipping; which will have immediate impact on global GHG emissions. Several tweets/retweets and posts/re-posts were sent out as well on social media. Attached as Annex 7 are examples of communication and visibility activities that took place during this event.

WORKSHOP FOR HEADS OF MARITIME

INTRODUCTION

The Workshop for the Heads of Maritime (HoM) in the Pacific was held on the morning of 12 December 2017. The workshop was attended by the MTCC-Pacific member countries naming Cook Islands, Fiji, Kiribati, Nauru, New Caledonia, Papua New Guinea, Republic of the Marshall Islands, Samoa, Solomon Islands, Tonga, Tuvalu, Vanuatu; MTCC-Pacific partners such as the International Maritime Organization (IMO), Fiji Ministry of Infrastructure and Transport (MoIT), Pacific Community (SPC), Pacific Islands Development Forum (PIDF), Secretariat of the Pacific Regional Environment Programme (SPREP), Vanuatu Maritime College (VMC); and private sector shipping companies including Samoa Shipping and Swire Shipping.

INTRODUCTION TO GMN PROJECT AND MTCC-PACIFIC

(Anton Rhodes, IMO)

In 2013, Regulation 23 of MARPOL Annex VI was adopted by IMO member states. This focuses specifically on providing capacity building and technology transfer to support member states. The emphasis is on LDCs and SIDSs. Since 2013, IMO has focused on raising capacity in developing regions, through its Integrated Technical Cooperation Program (ITCP) i.e. supporting and organising workshops and activities, providing guidance and training materials to countries. This is all part of IMO's strategy to empower countries to move forward with better energy efficiency. Another key part of the IMO strategy since 2013, to support energy efficiency, has been to establish major projects. For example, the Global Maritime Energy Efficiency Project (GloMEEP), which focuses on helping legislators and implementation of MARPOL Annex VI i.e. providing support to particular countries and helping them move through the legal process of implementation. The GMN project, which was launched in 2016, fits within the major projects of the IMO. It focuses on creating a network of centres to promote technology cooperation and transfer and works with regional partners.

The five MTCCs of the Global MTCC Network (GMN) are established in Latin America, Caribbean, Africa, Asia and now the Pacific. The network was formally launched at the IMO in London in December 2017 and all the MTCCs are now established and operational. The initiative is funded by the EU and implemented by IMO. The role of the IMO is to provide coordination at a global level. The total funding of the project is EUD10 million provided by the EU and channelled through IMO. The MTCC-Pacific signed its contract in May 2017 and, similar to the other MTCCs, is currently delivering on the agreed work plans. MTCC-Pacific was acknowledged as having progressed in a very short amount of time.

The GMN project will end in 2019; however, it is intended that the network will continue with a new financial model that can be applied beyond 2019. The overall objectives of the GMN is to: build capacity, improve energy efficiency, reduce GHG emissions is their respective regions, and to contribute to the fight against climate change at the highest level.

The specific objectives are as follows:

- To establish the five MTCCs;
- To improve human and institutional capacities in maritime administrations and relevant stakeholders;
- To promote the uptake of low carbon technologies and operations;
- To establish pilot projects on data collection on fuel consumption and reporting systems;
- To focus on communication and dissemination of information.

NATIONAL TECHNOLOGY NEEDS, BARRIERS AND REQUIRED RELEVANT ACTIONS FOR PACIFIC LOW-CARBON MARITIME TRANSPORT AND DISCUSSION OF RECOMMENDATIONS

(Mark Davis, MTCC-Pacific)

The Matrix of *Recommendations for Low Carbon Maritime Transport in the Pacific* was consolidated following three consultation workshops in Fiji, New Zealand, and Vanuatu together with a submission by the Solomon Islands Maritime Safety Administration. The session introduced participants to the matrix of recommendations for further discussion.

Comments on the Matrix

- Pilot projects – it was identified that Dry Docks were not included under the actions/recommendations of the pilot projects and need to be captured.
- Referring to R21 and R22, Solomon Islands expressed concern that countries are not receiving consistent resourcing to fund capacity or systems development. Samoa agreed with Solomon Islands pointing out that there is a need to identify the specific needs of each member state i.e. from developing legislation to the collection of data. How do countries obtain financing without having proper systems in place? Countries have different approaches to adopting new tools and have different infrastructure standards. Therefore, the implementation of various IMO tools will be a challenge and will not be applicable to all member states. SPC responded highlighting that the project is looking at ensuring that there is some level of consistency across countries.
- Referring to R48 – the sentence is to be amended and to insert *'Adopt economically transparent franchise schemes on non-economical routes …'*
- The difficulty for a small island atoll like Tokelau is the availability of technology. There is a need to reduce the use of technology and to look at other ways to reduce GHGe.
- Shipping companies/industries and businesses play an important role and need to be involved through this consultation process. It needs to be communicated and promoted by MTCC-Pacific that shipping companies have to operate now more sustainably. MTCC-Pacific acknowledged the feedback and comments and recalled its focus on capacity building towards shipping companies and the private sector.
- HoM will provide long-term direction to the MTCC-Pacific to operate in the region under the GMN. The solutions around energy efficiency is based on cooperation, collaboration and partnerships.
- Shipping industries are mostly private sector therefore they need to be involved and to contribute to the process. Future workshops must include private sector needs.

The session chair concluded the discussions and highlighted that the discussions around the table revolved around - capacity development, money, cost, business, private sector inclusion, and making the case i.e. the value of budgets and speaking the same language. All comments were noted and MTCC-Pacific indicated that the matrix will be again reviewed to finally be part of the Regional Conference outcome.

ENERGY MANAGEMENT IN PORTS: METHODS AND TOOLS

(Bruce Rowse, Consultant, MTCC-Pacific)

The session introduced participants to the Port Energy Management concept emphasising that it is an active ongoing process to reduce energy use and GHGes and with the aim to improve productivity. The following tools have been developed by MTCC-Pacific that will help member countries through the process i.e.:

- Energy Management Policy template;
- Port Energy Measurement and tracking spreadsheet;
- Energy Management Plan template; and
- Guides accompanying spreadsheets and templates.

The presentation also highlighted the benefits of port energy management stating that it reduces operating costs, reduces emissions (both CO2-e and local air pollution), lowers maintenance expenses, and reduces downtime due to equipment failure, improves comfort and safety, improves company reputation and results in a greater employee retention.

In addition, there are many challenges to effective energy management in ports. It needs a cultural change and is effective when enforced at all levels of the organisation. Purchasing equipment's that have high-energy rating also provides an opportunity to improve energy efficiency.

Discussion

- People need to be convinced that saving costs is revenue, in particular for Ports as State Owned Enterprises (SoEs) so that will facilitate the implementation of energy efficiency measures.
- It was suggested that trials should be carried out to show impact on energy management when it involves change in behaviour such lighting or air conditioning management or eco-driving – the Head of MTCC-Pacific confirmed that MTCC-Pacific is targeting measures that require change in behaviour rather than purely technology measures even though it is more complicated.
- The example of a shipping company in Australia was shared on energy management practices i.e. shifting to low energy lighting, installing solar panels on warehouse roofs and the installation of light sensors in every room that is not occupied, which reduced their bill by 50%. Port energy management should be practiced by all levels of the organisation and for all operations so that there is a common understanding of the importance and benefits of improving energy efficiency.

SECOND MTCC-PACIFIC STEERING COMMITTEE MEETING

INTRODUCTION

The Second MTCC-Pacific Steering Committee (SC) was held on the afternoon of 12 December 2017 and attended by the following members: Fiji Ministry of Infrastructure and Transport (MoIT), Pacific Community, Secretariat of the Pacific Regional Environment Programme (SPREP), Fiji Delegation of the European Union, Samoa, Solomon Islands, Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH, Pacific Islands Development Forum (PIDF), Pacific Islands Forum Secretariat (PIFS) and the University of the South Pacific. Also present were representatives from New Caledonia and the International Maritime Organization as observers.

ACTION POINTS

- The Maritime Technology Cooperation Centre for the Pacific (MTCC-Pacific) Secretariat will share the TOR with the members of the Steering Committee.
- Proper minutes of the meeting will be produced to highlight main points of discussions and action points that the steering committee can follow-up and to report back on matters discussed at the last meeting.
- All relevant documents relating to the meeting will be uploaded on the MTCC-Pacific website.

ADOPTION OF THE 2ND STEERING COMMITTEE MEETING AGENDA

The meeting adopted the meeting agenda.

REVIEW OF THE LAST MEETING MINUTES

(Thierry Nervale, MTCC-Pacific)

SPC presented and guided the meeting through the minutes and discussions of the last meeting held on the 16th -17th of August 2017. The Chair then requested the MTCC-Pacific Secretariat that the minutes are prepared apart from other updates, to allow members to follow-up on discussions of the previous meeting.

Discussion

- SPC and MCST clarified the working relationship between MTCC and (Micronesian Centre for Sustainable Transport) MCST-USP (University of the South Pacific). MTCC work is focused on two pilot projects with no research component. The role of the MSTC in relation to MTCC is complimentary given that the MCST has a long standing ambition in relation to an extensive research agenda that informs policies and developments of practice, in addition to supporting member countries to deliver on their own objectives. MCST therefore anticipates a strong commitment by the two centres to collaborate on a way forward.
- In addition to the discussion, MSTC stated that it responds to the development needs of the Pacific and is not member driven, thus operating beyond the requirements of IMO. Therefore, the needs identified by the MTCC must reflect the research work of the MCST, while the data gathered by the MTCC is expected to complement the research activities of the MCST.
- Members of the Steering Committee echoed its support for the MTCC and MCST and to share current work to check complementary. In closing the session, the Chair requested that the TOR is circulated to members of the Steering Committee.

IMO WORK TO ADDRESS ENERGY EFFICIENCY AND GREENHOUSE GAS EMISSIONS

(Leigh Mazany, GMN Consultant)

The presentation focused on (i) the Marine Environment Protection Committee (MEPC) work (ii) the data collection system and (iii) the Roadmap for an IMO Strategy. To begin with, it was highlighted that the Pacific small island states were instrumental in the development of the IMO Energy Efficiency Regulations.

In July 2009, the MEPC on behalf of the IMO developed the first set of regulations to address energy efficiency. This was followed by the development of the resolution on technical cooperation and capacity building.

The IMO Data Collection System (DCS) for Fuel Oil Consumption (FOC) of ships is the product of 3 years of discussion resulting in the adoption of Resolution MEPC.278(70) amending MARPOL Annex VI in 2016 to add a new regulation 22A on 'Collection and reporting of ship fuel oil consumption data'. The regulation applies only to ships that are 5000GT and above which captures roughly 80% of all CO2 emissions. Regulation 22A will enter into force in March 2018 and where the Data Collection System confirmation of compliance will be issued as well as the Ship Energy Efficiency Management Plan (SEEMP) implementation. Based on compliance standards, ships will start collecting data at the beginning of 2019 and will report to administration by April 1st, 2020. The Administration will report to IMO by August 2020.

In October 2016, MEPC approved a Roadmap for developing a comprehensive IMO strategy on reduction of GHG emissions from ships, which foresees a Reduction Strategy to be adopted in 2018. The Roadmap contains a list of activities, including further IMO GHG studies and significant intersessional work, with relevant timelines and provides for alignment of those new activities with the ongoing work by the MEPC on the three-step approach to ship efficiency improvements.

Discussion

- The IMO is a member state driven organisation therefore the will to progress and accomplish according to the timelines depends on the member states.
- The Marshall Islands pointed out that there is a need for IMO to expedite the gradual progress to meeting the 1.5 degrees limit of temperature increase from ships GHG emissions. Before 2015, there has been little effort from IMO placed on prioritising the discussion and debate within the last decade, on reducing ships emissions. Every sector needs to be ambitious and genuine about addressing their emissions. According to recent studies, it is a win-win situation for the industry. IMO responded and reiterated that it is a member state driven organisation and the Pacific now have the Paris Agreement as an enabling mechanism that will support and encourage the debate over energy efficiency. As member states of IMO, the Pacific is reassured to continue to build coalitions and to keep submitting papers that will push discussions forward.

UPDATES FROM THE COMMITTEE ON COP23 AND OTHER GLOBAL EVENTS

Members and partners of the MTCC-Pacific Steering Committee participated at the Global MTCC Network (GMN) Side Event of the 2017 COP23 Summit held in Bonn, Germany. The Side Event was organised by GMN in collaboration with the IMO High Commission in London. Committee members and partners that attended the Summit shared the following updates.

- SPC as the Host Institution of MTCC-Pacific participated at the Side Event and presented on the progress and achievements of MTCC-Pacific. In addition, SPC participated in Side Events related to international shipping, GHG emissions and sustainable sea transport organised by the Republic of the Marshall Islands, Palau and other countries but also by Okeanos. Overall, the Pacific was well represented at various Side Events of the COP 23 summit.

- PIDF also participated and presented on the Framework for Resilient Development in the Pacific (FRDP) referring to *Goal 2 on Low Carbon Development*. Goal 2 is directed towards development sectors, which include the Shipping sector.
- USP through their Pacific Centre for Environment and Sustainable Development (PACE-SD) worked with a number of government delegations at the summit. SPREP participated at the *3rd Meeting of the UN Environment Assembly (UNEA 3)*. The theme of the conference *'Towards a Pollution-Free Planet'* aligns with the objectives of MTCC-Pacific to mitigate and address climate change issues relating to maritime. The Steering Committee was than informed of two main discussions resulting from UNEA 3 and its relevance to the MTCC i.e.
  i. The development of an International Framework to address plastic pollution is being developed by an open-ended technical working group.
  ii. The need for the Pacific to implement and take actions in addressing pollution.

EU delegation for the Pacific and Fiji reaffirmed its support for the MTCC's initiatives and the need to address from all angles some of the Pacific's complex issues such as those relating to climate change, pollution and energy management.

Marshall Islands is working in partnership with the German government through the Low Carbon Project to address low carbon transition for the maritime sector. Based on the discussions around the table, Marshall Islands recalled the Pacific leaders call to IMO to reduce emissions from shipping and reiterated that Pacific Island countries have continued to advance discussions with IMO to address emissions from shipping and other sectors.

Marshall Islands also recalled the *2017 Third Pacific Regional Energy and Transport Ministers Meeting* in Tonga, where discussions revolved around the existing need for collaboration to tackle shipping emissions. Recognising that it is an area already being addressed by USP's MCST, it is recommended that the MTCC-Pacific and the MCST foster alliances and to look at joint efforts to address shipping emissions.

MTCC-PACIFIC PROGRESS REPORT

(Thierry Nervale, MTCC-Pacific)

SPC presented and provided an update of the MTCC-Pacific highlighting that the office has been established and is operational with the availability of an MTCC-Pacific website, which is online and accessible.

The overview highlighted the following achievements since the signing of the SPC-IMO contract on 12 May 2017:

Result 1 – Establishment and operations
Result 2 – Capacity building activities
Result 3 – Pilot projects on energy efficiency
Result 4 – Pilot projects on data collection
Result 5 – Communication and visibility

In implementing the project work plan, the following activities have been completed and/or carried out:

- 7 Project Monthly Meetings have been completed: 18th May, 28th June, 19th July, 24th August, 28th September, 12th October and 1st November
- First Quarterly Progress Reports submitted and approved by IMO
- Second Quarterly Report submitted and approved by IMO
- Participated at COP23 GMN side-event
- Participated at the MTCCs Technical Workshop at IMO in December – MTCCs MoUs signed and MTCCs Coordinating Committee established

SPC also guided the committee through the MTCC-Pacific Fundraising Strategy, which is expected to directly benefit the Pacific domestic ships and international and domestic port operators. The strategy will also be of benefit to the Pacific Islands countries governments that will provide progress towards their NDC targets.

Discussion

- In the next two or three years the MTCC-Pacific project will look at ship energy efficiency management as a pilot project to assist ship owners. This is in response to the sentiments raised on the existing need to strengthen the capacity of ship owners. The activity will require the development of bigger proposals that will encompass climate financing with the support of partners and members.
- IMO supported the comments raised by MTCC-Pacific and emphasised that most of the work undertaken by the project is replicated across the MTCC regions, which have the potential to be expanded. The MTCC-Pacific existing pilot projects have and will continue to establish and strengthen a network and platform that can be utilised to prompt future projects and activities. The intention is for the Pacific to be able to take ownership of the work and progress MTCC-Pacific into the future.

OTHER BUSINESS

- The Project Manager for the Marshall Islands German government project briefed the committee on the 5-year project on Low Carbon Sea Transport. It is aimed at identifying options on how to reduce fuel consumption and emissions of Marshall Islands government owned cargo ships. The project is financed by the German Corporation for International Cooperation (GIZ) and is anticipated that the project will be expanded to other pacific island countries in the future.
- The MTCC-Pacific Head advised the committee that there would be two Steering Committee Meetings per year following 2017 meeting – preferably in August and December. Specific dates will be confirmed.

CLOSING REMARKS

The Chair closed the meeting and acknowledged the MTCC-Pacific Steering Committee for a productive and informative meeting and discussion.

DISCUSSION ON RELEVANT ACTIONS TO IMPLEMENT THE TRANSPORT MINISTERS' RESOLUTION AND COMMUNIQUE

(Thierry Nervale, MTCC-Pacific)

The MTCC-Pacific Head provided a recap on discussions and comments raised at the Workshop for the Heads of Maritime (HoM) and then proposed that the meeting further prioritise and discuss the comprehensive list of recommendations according to international and regional priorities that countries identify with. Priorities identified and preferred by members are in the order of:

a. **Capacity Building** – preferred by Solomon Islands and Kiribati specifically for ship owners and awareness raising.
b. **Collection of data and information -** there is a lack of awareness about information and data that is available or existing frameworks that countries and industries are bound by. The focus on data collection as a first step must be a priority of the MTCC-Pacific, followed by the development of policies, frameworks, and capacity building, all of which rely on the information gathered.
c. **Policy/Legal Frameworks** – there are existing frameworks and policies that countries are not aware of and should be implemented.

Discussion

- The gathering of information and data is a tedious exercise for shipping owners because it is a process that is carried out on a one-on-one basis (per vessel) as opposed to filling in a simple form (previously fuel oil consumption was only requirement). In addition, shipping companies are part of the private sector whose priority is profitability rather than collecting data with no direct impact on business. Data collection standards need to be institutionalised by the government to ensure that it becomes a priority. MTCC-Pacific responded to the sentiments raised and stated that a generic draft regulation on data collection is in preparation with SPREP and will work with the private sector on an agreement that will ensure the collection of data.
- Establishing the MTCC-Pacific is important i.e. gathering the right data, having the right technology, and training the right people to use the technology. Shipping is a challenging business and unless we have good data and a national policy there will continue to be issues with contracts and shipping lines.
- It was recommended that the MTCC-Pacific project recruit local staff to carry out data collection if the data collection process is to become successful, in particular for small island states like Tuvalu.
- The MTCC-Pacific project and members have established a reputation in the region and implementing activities on the ground; however, communication and the sharing of experiences and stories need to be strengthened. For instance, there is a need to share ideas and experiences on how ISO standards can be achieved.
- It was suggested that the MTCC-Pacific adopt a different approach to ensuring that data is captured i.e. by presenting some of the benefits of data collection and its impact on the industry. Furthermore, shipping operators need to understand that there is available expertise and data that can be utilised to help the industry transition to a more energy, cost and greenhouse gas (GHG) efficient business.

GMN INTRODUCTION AND STATUS UPDATE

(Anton Rhodes, IMO)

The International Maritime Organization (IMO) representative presented and introduced the status of the Global MTCC Network (GMN) stating that the GMN is focused on regions with Least Developed Countries (LDC) and Small Islands Developing States (SIDS). The GMN is an initiative funded by the European Union

and implemented by IMO with the overall objective to build capacity, improve energy efficiency, reduce GHG emissions (GHGE) and contribute to the fight against climate change (mitigation). The programme has a total funding of EUR 10,000,000 for a period of 4 years. The European Union is also actively involved in the governance of the project.

The five MTCC centres of excellence are in Latin America, Caribbean, Africa, Asia and now the Pacific. The network was formally launched on the 4th of December 2017 in London (IMO HQ), and all the MTCCs are now established and operational. The role of the IMO is to provide coordination at a Global level. The MTCC-Pacific signed its contract in May 2017 and, similar to the other MTCC regions, is currently delivering on the agreed work plans (WP). MTCC-Pacific was acknowledged as having progressed in a very short amount of time.

The MTCC-Pacific project will end in December 2019; however, the MTCCs plan to become financially self-sustained.

Discussion

- In the past, the United Nations Development Program (UNDP) previously supported two people from each developing country to study at the World Maritime University (WMU), an opportunity which has been lost to the Pacific. It was suggested that similar opportunities are made available to the Pacific again, through the MTCC-Pacific, the UNDP or other organisations willing to provide that level of support, for the opportunity to study at WMU and acquire a higher level of knowledge and understanding of the maritime transport sector (MTS), and its issues.

WELCOME CEREMONY AND OFFICIAL LAUNCH OF MTCC-PACIFIC

The MTCC-Pacific Regional Conference was officially opened on 13 December with the Official Launch ceremony in the morning in the presence of the Honourable Ratu Osea Naiqama, Fiji Minister for Forestry (and Acting Minister of Fisheries), the Ambassador of the European Union Delegation for the Pacific, Julian Wilson, Audrey Aumua, Deputy Director General of the Pacific Community (Suva) and Aubrey Botsford, Deputy Director/Chief Officer of General Services of the International Maritime Organisation.

Welcome Address

(Audrey Aumua, Deputy Director General, SPC)

The Deputy Director General for the Pacific Community (SPC), Suva (DDG-Suva), addressed and extended a warm welcome to the Fiji Minister for Forests (and Acting Minister of Fisheries), the Ambassador to the Delegation of the European Union (EU), Permanent Secretaries, Secretaries to Government, representatives of the International Maritime Organisation (IMO), representatives of women's regional organisations, the President and representatives of the Pacific Women in Maritime (PacWIMA) Network, partners of the MTCC-Pacific and representatives of the MTCC-Asia Centre.

In welcoming everyone to the launch of the MTCC-Pacific, the SPC DDG Suva highlighted that the MTCC-Pacific, hosted by SPC in collaboration with SPREP, is one of the five centres in the world established to build the capacities of developing and least developed countries for climate intervention and sustainable development in the maritime industry. In addition, the Global MTCC Network (GMN) is a project implemented by the IMO and funded by the EU. SPC further acknowledged the funding and presence of the EU delegation, its collaborative effort, foresight and support toward the establishment of the centre. SPC also acknowledged the IMO for the leadership support and technical guidance in the movement of the programme. Fiji is the host country of the MTCC-Pacific and SPC values the commitment of the Fiji

Government in supporting the initiative. The launching of the MTCC-Pacific Centre shows the Pacific's commitment to COP23 and will assist the maritime industry to play their part in reducing GHG emissions.
The SPC's DDG further highlighted the significance of the Framework for Resilient Development in the Pacific (FRDP), which provide a strong direction and guide toward addressing climate change, disasters and resilience building. The establishment of the MTCC-Pacific Centre will play a critical role in implementing the FRDP framework to the region as the Pacific delivers on the leader's aspirations. She further acknowledged the PaCWIMA Side Event where discussions revolve around improving the participation of women in the maritime sector and reducing the barriers of engagement for women.

Opening Address

(Ratu Osea Naiqama, Fiji Minister for Forestry and Acting Minister of Fisheries)

The Minister acknowledged and addressed the EU Delegation representative in the Pacific, the Deputy Director IMO, the Pacific Community (SPC) DDG-Suva, development partners, international and regional organisations, Permanent Secretaries, Deputy Secretaries, Directors and Delegates from Pacific Island Countries and Territories (PICTs).

The Minister also welcomed everyone to the 1st Regional Conference and Official Launch of the MTCC-Pacific of which Fiji is proud to provide support to the Pacific Community (SPC) and the Secretariat of the Pacific Regional Environment Programme (SPREP) as the host institutions. The platform is an opportunity to discuss and agree on credible actions, in meeting the Pacific's commitments to climate change.

The Minister recalled that since the 2017 3rd Energy and Transport Ministers Meeting in Tonga, several meetings have taken place and have identified actions to be implemented. The Energy and Transport Ministers acknowledged the commitment of the region to demonstrate leadership, in the fight against climate change and the pivotal role of affordable, reliable and sustainable energy and transport services for the Pacific.

Two relevant outcomes that are now guiding the Pacific in the medium term was highlighted by the Minister i.e.:

a. The vision of 100% renewable energy generation for the region.
b. The obligation for the maritime sector to contribute to global efforts to reduce GHG emissions and the same for all stakeholders to take appropriate actions to progress low carbon maritime transport in the Pacific (PLCMT) islands region.

It was then mentioned that Fiji's Presidency for the COP23 will be significant for the Pacific to drive plans for a Pacific low carbon maritime transport sector. Fiji, thanked the European Union for the financial support and the International Maritime Organization (IMO) as the implementing agency for the global MTCC network.

The conference is a call for the Pacific to take action in transitioning towards green shipping, which requires a strong partnership with the private sector, the maritime industry, the support from regional partners and international organisations. The Minister acknowledged other Pacific Island Countries in taking up the initiative to promote the reduction of GHGEs, which is an inclusive approach for the transition towards low carbon maritime transport.

Keynote Address

(Aubrey Botsford, Deputy Director/Chief Officer of General Services of the International Maritime Organisation)

The Deputy Director General Services of IMO acknowledged the Fiji Minister for Forestry and the Deputy Director General SPC and extended good wishes towards the conference from the Secretary General of the IMO and also towards the launch of MTCC-Pacific. The Deputy Director expressed appreciation in being able to address the Pacific at the launch of the MTCC-Pacific Centre, which is part of an ambitious project to establish a global network that will contribute to the worldwide effort to address climate change, one of the greatest challenges of the century.

IMO congratulated the Pacific Community (SPC) and the Secretariat of the Pacific Regional Environment Programme (SPREP) for their successful bid to host the MTCC in the Pacific region. It has been recognised by IMO that Fiji and the Pacific Islands play an important role in the fight against climate change and in pursuing efforts towards a low carbon future. The challenge is particularly complex when considering how dependent the Pacific island states are on maritime transport. The Pacific's efforts to mitigate climate change and promote energy efficiency in the maritime sector, through actions undertaken by governments and the maritime industry, has shown the dynamic approach taken by the region. IMO is therefore supportive of the approach and of the various actions taken within the Pacific.

Global efforts to reduce GHGEs from shipping rely heavily on improvements in energy efficiency and the increased uptake of low carbon technologies. IMO is continuously engaged in protecting the environment, both marine and atmospheric. The IMO's efforts to mitigate the impacts of shipping on the atmosphere stem from the genuine concern for the environment and for the health of citizens, expressed by IMO's member states as well as by the industrial organisations that help in pursuing the organisation's objectives. To that end, IMO is now working towards the finalisation of a robust regime that will regulate shipping at the global level and contribute to efforts to address climate change.

IMO further recognises that crucial to effective implementation, is the provision of technical assistance, to support capacity building in member states and the shipping sector. As part of the wider initiative the EU and IMO entered into partnership in December 2015 to address climate change issues in the shipping industry and establish a Global MTCC Network (GMN). In 2017, with financial support from the EU and technical assistance from IMO, five MTCC's have been established in Africa, Asia, Caribbean, Latin America and the Pacific. Delegates at the launch were then informed that alongside the MTCC-Pacific's event, MTCC-Africa was formally launched in Kenya on that same week.

IMO acknowledged SPC's commitment and historical emphasis on the importance of sustainable development and efficient use of resources. SPREP for its part will develop partnerships for implementing adaptation and mitigation policies in the region. Both organisations are guided by a deep understanding of the needs of their country and territory members. The global network of MTCC's will promote understanding and knowledge of technologies and operations and will help to steer shipping toward a low carbon future. The network will act as regional centres of excellence aimed at improving compliance with existing and future international energy efficiency regulations and promoting the uptake of low carbon technologies and operations in maritime transport.

On behalf the IMO Secretary General, special thanks was extended toward the Government of Fiji for supporting the project, as well as to SPC and SPREP for hosting the MTCC for the Pacific region. The Director General also extended deep appreciation to the EU for its financial support and the European Commission for its cooperation and implementation of the project from the initial stages. The IMO's Project Coordination Unit (PCU) was also acknowledged for the hard work and contribution in establishing the MTCC-Pacific.

Keynote Address

(Ambassador of the European Union Delegation for the Pacific, Julian Wilson)

The Ambassador addressed the Fiji Minister for Forestry, the SPC Deputy Director General Suva, the IMO Deputy Director and Officer of General Services and representatives of the Council of Regional Organisations in the Pacific (CROP). The launch of the MTCC-Pacific was recognised by the EU as a timely event considering the tragic events that are being witnessed every day, and increasingly around the world and in particular for the Pacific, as a result of climate change. The EU is particularly concerned about the position of the least developed countries (LDC) and the Small Island Developing States (SIDS).

It is therefore important to gravitate into areas that have been neglected, like the maritime sector, and the GMN recognises that addressing the sector is an important solution to reducing GHG emissions. The MTCC-Pacific also enables the opportunity to further the EU-Pacific alliance in the fight against climate change considering that the EU and the Pacific are two key-players that have kept the actions against climate change alive. At the 20th Anniversary of the Kyoto Protocol, the EU-Pacific alliance was as crucial as it has ever been.

In its strive to address the maritime problem, the IMO has developed and adopted the Energy Efficiency Measures and also established a data collection system. Furthermore, the 2016 agreement in the IMO to adopt by spring 2018 an initial strategy on the reduction of GHGE from ships was an important step forward. The GMN will therefore play an important part in this global drive with the financial support of the EU.
MTCC-Pacific along with the other MTCC regions has an important role to play in capacity building for climate mitigation in the maritime shipping industry. The establishment of the MTCC-Pacific in Fiji offers a great opportunity to support the regional authorities to develop domestic policies to boost implementation and make changes on the ground.

The Ambassador further congratulated Fiji as the host country of the MTCC-Pacific and congratulated the Fiji Ministry of Infrastructure and Transport (MoIT) for its commitment to making MTCC-Pacific the regional centre of excellence. The CROP agencies and regional organisations are encouraged to strengthen their collaboration and data collection, the implementation of energy efficiency pilot projects and capacity building. The Ambassador then concluded by raising the point that there is a need to ensure the empowerment of Ministers in the region who represent the people, in order to translate the vision of energy efficiency and sustainable transport into actual projects on the ground.

Handover ceremony from IMO to MTCC-Pacific of its official plaque

To mark this particular event for the region and better integrate MTCC-Pacific in its region, a video was displayed showing a Fijian traditional canoe bringing the MTCC-Pacific plaque that was physically handover from IMO to MTCC-Pacific in presence of traditional Fijian warriors. The video clip depicting the journey of the plaque across the ocean by two MTCC-Pacific staff representing the necessary transfer of technology and assistance from overseas but keeping in mind the particularities of the navigation in the Pacific Islands region and traditional knowledge that recall the need for sustainable use of ocean. The plaque handover was followed by a benediction of MTCC-Pacific and a Vote of Thanks from the Government of Fiji. A group photo was made following the ceremony and photos of guest of honour with MTCC-Pacific plaque. All material will be used for the creation of a promotional video of MTCC-Pacific and its Official Launch.

SESSION 1: PACIFIC ISLAND COUNTRIES AND TERRITORIES PERSPECTIVES

(Facilitated by Anthony Talouli, SPREP)

The Fiji Deputy Secretary for Policy and Planning, Mr. Lui Naisara chaired the MTCC-Pacific Regional Conference and acknowledged the MTCC-Pacific for the opportunity given to Fiji to host and chair the 2017 Conference. The chair also acknowledged the Solomon Island Ports Authority for its contribution towards the conference.

Micronesian Centre for Sustainable Transport: A Framework for Transition

The presentation introduced the Micronesian Centre for Sustainable Transport (MCST) is a joint program between the Republic of the Marshall Islands (RMI) and the University of the South Pacific (USP), highlighting its history and the MCST Framework Priorities for Action.

In 2013, at the Inaugural Pacific Islands Development Forum (PIDF) Leaders' Summit, RMI committed to reducing fossil fuelled electricity and to setting a transport target. In 2014 at the 2nd Sustainable Sea Transport Talanoa, the RMI called for a National Program for Low Carbon Transport and sort USP'S support to design a whole of country transition program. Concurrently, the United Nations (UN) recognized sustainable transport as a regional priority. In 2015, at the endorsement of the MCST, Tony De Brum led the Pacific delegation to call for a shipping emissions target that would meet the 1.5°C Paris Agreement goal.

In 2016, USP endorsed the MCST framework as a catalyst for research and identified ways in which the initiative could involve other Pacific Island Countries and Territories (PICT). The Centre, with the support of USP, looked at new technologies to address the challenges of maritime transportation in the region.

RMI added that time is of the essence in particular for the Pacific. The transition to low carbon transportation is paramount, with the need to have the support and contributions of all parties. Six months into the opening of the MCST centre, there is potential to achieve greater things. Sustainable development in maritime cannot be addressed without addressing sustainable maritime transportation in the Pacific.

Country Presentations and Updates

Country	What are the drivers in your country for reducing GHGE from your maritime transport sector (MTS) including ports and shipping?	What are the needs to achieve the country objectives (if any) in reducing GHGE from MTS and/or to take action to address GHGE from MT?	What are the identified barriers that prevent from implementing country approach to reduce GHGE from MT?	What are the possible relevant actions to address GHGE from MT?
Cook Islands	This is achieved through major investments in renewable energy, with a target of 100% reliance on solar power by 2020. In addition, through private sector initiatives where wind/diesel hybrid systems have been trialled on board the inter-island domestic shipping vessels, primarily to lower operational costs.	▪ Promoting compensatory mechanisms for SIDS ▪ Subsidisation to offset economic impacts, ▪ Provision of targeted technical assistance, and/or ▪ Direct financial assistance with acquisition of new technology and/or new vessels to replace older less efficient technology	▪ Slow onset impacts of human induced climate change and natural disasters ▪ Sensitive to any measures that would significantly increase the costs of shipping in the region. ▪ Cook Islands heavily rely on imported goods and foodstuffs and there is a risk of increase in cost down to the consumer, which may increase the cost of living on the islands	▪ Since costs of shipping in the region is already very high and already impact on our ability to meet key Sustainable Development goals (SDG) related to the *2030 Development Agenda*, we need to explore ways to bring these costs down, not increase them. This is why the Cook Islands often links these challenges with the GHG issue. Indeed, it's important to acknowledge that in meeting the relevant SDGs, we also increase the ability for our countries to become more resilient to the impacts of human-induced climate change. ▪ So whatever actions decided must always be considered within the broader context of meeting the development aspirations of the countries.
Kiribati	▪ National Action Plan on Marine Pollution (NATPLAN) ▪ Energy Management in Port ▪ Kiribati Nationally Determined Contributions (NDCs) on GHGE reduction targets ▪ Strategies in reducing imported fossil fuel dependency	▪ Strengthening GHGE Policies, Strategies and Legislation ▪ The need to deliver appropriate education, training and awareness programmes ▪ The need to improve knowledge and information generation, management and sharing ▪ The need to promote the use of sustainable renewable sources of energy and energy efficiency	▪ Lack of national GHGE legislation in place in addressing GHGE ▪ No Technical Expertise in GHGE within the Country ▪ Lack of detailed information and understanding in anthropogenic CO_2 emissions and its impact from the Maritime, Port and Shipping Sectors ▪ No study undertaken yet in addressing the impacts of GHG by maritime transport	▪ Seeking support by Government - full enactment and implementation of GHG legislation, Policies and Strategies ▪ Assistance by foreign technical experts to assist in coordination of awareness program on GHG from Ports ▪ Shipping Communities and relevant key stakeholders ▪ Entailing benefits to the shipping market, in terms of emission reductions (cost vs revenue)

		▪ The need to strengthen capacity to access finance, monitor expenditure and maintain strong partnerships		▪ Provision of an energy-efficiency culture in both port and shipping sectors
Nauru		• Nauru lacks the capacity to actively progress with work on the ground and needs the support and expertise of the MTCC-Pacific network. Nauru had also signed up to the Paris Agreement, and there is much work that needs to done on the port, therefore the conference platform will be a starting point for further collaboration and support.		
Fiji	▪ National Framework ▪ Majuro Declaration ▪ Fiji Development Plan - National Framework ▪ Fiji's Strategic Plan ▪ Annual Corporate Plan ▪ High level executives/champions ▪ Fiji Ports Corp Ltd (FPCL) ensures that MTCC-Pacific objectives are integrated into their Corporate Plans	▪ Lack of human resources and technical expertise ▪ Infrastructure and funding ▪ Coordination to develop initiatives		▪ The Fiji government is working on capturing data of GHGEs from the Suva harbour to be analysed so that it gives an indicator of the level of GHGEs. ▪ The operations of the FPCL, the maritime companies, various partners and organisations that work alongside the maritime sector, are familiarised with objectives of the MTCC-Pacific and the industry's Corporate Plans. The intention is to involve all levels of operations in meeting the objectives of the MTCC-Pacific. This includes the private sector. It is everyone's business. ▪ The participation of the public sector, all levels of operations, down to the shipping crew is important to ensure that every member of the organisation and relevant stakeholders are made aware of the need and benefits of energy efficiency. It must be an all-inclusive approach to ensure that people take ownership in tackling the goal of energy management and reducing GHGEs.

New Caledonia	▪ French state: MARPOL VI, IEEC, MRV (Europe area) and PSC. ▪ New-Caledonia government Energy Transition Plan for next 20 years. ▪ Sulphur Emission Control Area (SECA) to be implemented in the Coral Sea National Park ▪ In the provinces: SECA will be implemented in the harbour area (to limit GHGEs from cruise liners).	▪ Need to improve maritime traffic monitoring and to check the compliance with regulations ▪ Need to coordinate the different measures to reduce GHGE ▪ Need to support innovating technical projects for vessels propulsion		
Papua New Guinea	▪ Regulations in place which put a cap on sulphur content ▪ Strategy as a directive from the government to phase out the use of fossil fuel	▪ Need to develop and enforce national legislation ▪ The need for capacity building and resources ▪ The need for data collection and monitoring	▪ Lack of legislation ▪ Lack of collection, collation and analysis of data to make informed decisions ▪ Lack of equipment	
Samoa	▪ International Conventions – comply with requirements that call for protection of the marine environment from climate change ▪ National Framework - initiatives involving relevant maritime stakeholders that align with international maritime instruments to ensure enabling laws, action plans and procedures to reduce GHGE are developed and implemented	▪ MARPOL – comply with applicable provisions of the Convention (e.g. Annex VI) ▪ Implement Marine Pollution Prevention Act 2008 (Regulations near completion) ▪ Enforce National Disaster Management Plan ▪ Implement GHG Abatement Strategy 2008-2018.	▪ Scarcity of resources – finance, technology and lack of expertise (human resources). ▪ Cost of new technology on existing ships - to comply with Annex VI of MARPOL and reduce SOx and NOx on vessels >500gt.	▪ Enforcement of Port State and Flag State inspections ▪ Public Awareness programmes ▪ Application of International Rules and Regulations ▪ Develop Energy National Strategy to reduce GHGE in the MTS in Samoa
Solomon Islands Maritime Safety Administration (SIMSA)	▪ Alternative fuels ▪ Improve ship design ▪ Operations	▪ Consciousness raising ▪ The need for improved planning ▪ Need to have better statistical information ▪ A better understanding of “cost of ship’s time” is needed ▪ Need to remove socio-political barriers	▪ Socio-political barriers i.e. Provincial rivalry and exclusivity, and “shipping grant ships” vs “private sector ships.” ▪ Socio-economic barriers i.e. high cost of trading licenses to more than one province and rationalisation of shipping services	▪ Solomon Islands is very active at IMO Marine Environment Pollution Committee (MEPC) in the fight to reduce GHGE internationally ▪ The Solomon Islands Government need to strengthen its focus on concerns relating to shipping matters: safety, environment and commercial ▪ Increase powers of SIMSA, the administration, to take on responsibility for marine environmental protection (as mandated, but not resourced) and also become involved in research

				into commercial shipping matters as an "Authority" (this is in the pipeline) ▪ Rationalisation of port operations to increase productivity ▪ Strengthen the capacities of shipping and ports personnel through professional education and training programmes (this is in the pipeline)
Solomon Islands Port Authority (SIPA) Perspective)	▪ Rising sea level – climate change ▪ High electricity cost	▪ Government policies ▪ Ports initiative to become 'GREEN' ▪ Commitment to the environment	▪ Lack of awareness and importance of GHGE reduction ▪ High cost of imported material to the country (solar panels) ▪ Regulatory framework review ▪ High cost of power generation (diesel power – as a result of providing shore power for vessels will be very expensive)	▪ Use of renewable energy for yard lighting ▪ Use of hybrid vehicles ▪ Reduce unnecessary electricity usage ▪ Maintain vehicles for better performance
Port of Tonga (PAT)	▪ Political Support - strong support from the government/Cabinet (shareholder) will help PAT Board of Directors (BoD) and Management successfully achieve the output objectives set in the PAT Business Plan to reduce GHGE in the Port of Nuku'alofa ▪ BoDs Support - the BoDs must have the values of good governance and strong leadership in order to make concise and informative policy decisions concerning the reduction of GHGE within the ports operations ▪ CEO and Managements full commitment through PAT internal policies, board and management- Strategic Objectives and KPIs, procurement strategies, and through cargo handling processing and wharf layout	▪ The Marine & Ports Division (MPD) as a regulatory body needs to be more proactive in enforcing existing maritime environmental legislation and regulations ▪ Maritime Division to set and monitor the level of GHGEs in the MTS in Tonga ▪ It needs to be a consensus approach at the national and regional levels	▪ Budget constraint - funding ▪ Poor enforcement from regulatory body - MPD ▪ Lack of clear legal framework to address GHGE in the MTS in Tonga ▪ Lack of coordination between relevant stakeholders i.e. MPD, PAT & Local boat owners	▪ Implement effective environmental policies and procedures to minimise the risks of environmental pollution ▪ Work closely with MPD in enforcing maritime environmental legislation and regulations ▪ More and clearer strategies and KPIs to address GHGE ▪ Report and analyse annual fuel consumption to determine GHGE to assist setting KPIs
Vanuatu	▪ Vanuatu commitment towards implementing its National Energy Roadmap 2016-2030 in reducing GHGE in the MTS (land and sea). ▪ Vanuatu commitment to the SDGs	▪ Need capacity building for maritime administrators and shipping stakeholders ▪ Need awareness on the importance of reducing GHGEs for shipping	▪ Lack of awareness by ship owners ▪ Lack of understanding of economic benefits for promoting energy efficiency in shipping and ports	▪ Established proper policies and regulations to address GHGEs. ▪ Engaged concessionaires in early discussions and negotiations to promote energy efficiency

	▪ The National Energy Road Map (NERM) 2016-2030, found that by 2030, Vanuatu consumption of imported petroleum products (petroleum, diesel, etc.) is expected to grow by 50% ▪ All vessels operated in Vanuatu rely on diesel fuel ▪ Demand for shipping in Vanuatu increased in the past 5 years ▪ Majority of the population relies on shipping for transportation ▪ More vessels imported into Vanuatu ▪ International port of entry upgraded and expanded with state of the art facilities ▪ high demand for electricity	▪ Private sector engagement ▪ Proper policy and regulations needed to legalize the requirement for minimising GHGEs ▪ Access to Green Funds to support Vanuatu implement GHGE programs	▪ Lack of partnership between stevedoring operators and port administrators (two different entities operate the ports and stevedoring) in terms of reducing GHGEs ▪ Lack of policies and regulations to regulate GHGEs in ports and vessels	▪ Create a national Working Group to advocate and implement GHGE programs ▪ Provide capacity for maritime administrations and ship owners on the benefit of GHGEs (economic and technical) ▪ Access Green Funds where possible ▪ Government must take ownership and support GHGE programs (tax incentive, exemptions and subsidies, etc.)
Tuvalu	▪ National Strategy for Sustainable Development 2016 -2020 ▪ SDGs - goal 13 and 14	▪ National legislation needs to be updated so that it includes requirements on the reduction of GHGE ▪ Expand workforce at small administrations ▪ All MTS agree that shipping should reduce its GHGEs in the long term ▪ Trading routes and level playing field should not suddenly change as a result of CO_2 reduction measures	▪ There is no national legal framework in place ▪ There is minimal public knowledge on GHGE ▪ Stimulate/support cooperation between Governments – Research – Shipping sector (golden triangle) ▪ As in other sectors, development of appropriate technologies, fuels and infrastructure for shipping must be incentivized to reduce GHGEs ▪ Business relationships between charter and ship owner will need improvement ▪ Measures of impacts on all States must be considered before implementation, especially impacts on SIDS and LDCs ▪ Negative impacts on vulnerable states, ship registries and the shipping sector will be greater and more difficult to adapt to later in the century when measures are finally implemented ▪ All SIDs and LDC's must maintain solidarity in global debate to reduce GHGEs	▪ All ships should target to have almost a zero Co_2 footprint in the near future by green (environmental friendly) ship, instead of black ships. ▪ Energy efficiency measures alone will not achieve zero emission from the sector (2035, 2050, 2060 and 2100) ▪ A clear roadmap on CO_2 reduction to guide the shipping sector to undertake proper planning

Discussions

- Nauru is a small island nation in Micronesia and has been using open sea moorings for the last 20 years. Damage to the moorings has been a costly exercise for the island and earlier in 2017, the government has managed to repair the mooring system allowing Nauru to transport cargo from ship to shore. Construction of the new Nauru wharf will commence in June 2018 through the Green Climate Fund programme.
- The Automatic Identification System (AIS) used by New Caledonia is a very useful technology for navigation safety. SPC is interested in collaborating further with New Caledonia on the use of AIS. Furthermore, the concept of Emission Control Areas (ECA), which will be implemented in the New Caledonia in the harbour, is an interesting concept and initiative.
- PNG highlighted that there is a need to collect GHGE data, analyse and interpret the information to make informed decisions in order to address climate change. Furthermore, the younger generations need to understand the importance of managing GHGEs.
- Tonga highlighted that the construction of separate wharfs for different purposes compliments the development of an Energy Management Plan. Movements of ships in and out of ports causes congestion, which consumes a lot of fuel. Therefore, the design of the new Tonga wharf will contribute to reducing the use of energy and fuel on ships. In addition, the Tonga terminal was built with a solar panel roof, which reduces the use of electricity.

SESSION 2: PARTNERS PERSPECTIVES

(Facilitated by Mark Borg, PIDF)

The Chair welcomed the meeting to Session 2 of the MTCC-Pacific Conference and acknowledged the International Union for the Conservation of Nature (IUCN) and SPC for hosting the opening cocktail. The session was an opportunity for partners and maritime operators to share their experiences and perspectives towards energy efficiency of local maritime transport.

Pacific Islands Development Forum

(Mark Borg, PIDF)

The Pacific Islands Development Forum (PIDF) was established in 2013 with 12 member countries and two NGOs (i.e. Pacific Island Association of Non-Government Organisation [PIANGO) and Pacific Island Private Sector Organisation [PIPSO]). It officially became an organisation through the adoption of their Charter in 2015 and is a multi-stakeholder organisation, which provides a platform for governments and the private sector in PICTs to collaborate without the influence of bigger countries. PIDF recognises the need to continue to address sustainable transport in order to meet the 1.5 degrees increase, as indicated in the Suva Declaration.

PIDF in partnership with IUCN, USP and World Wide Fund for Nature (WWF) will soon launch the Oceania Centre for Sustainable Transport (OCST) website. The centre is more of an idea than an organisation, which will bring under one umbrella the three transport sectors (Sea, Land and Air) to enable collaboration. The PIDF also provides support to countries in addressing issues related to the different transport sectors.

A Comprehensive Imo Strategy on Reduction of GHGE from Ships

(Leigh Mazany, IMO Consultant)

The presentation provided an overview of the IMO Strategy to reduce GHGEs from international shipping. The overview highlighted IMO initiatives, the IMO data collection system (DCS), their Strategy, capacity building and technical cooperation, and potential impacts, costs and benefits of the project.

In setting the context of the IMO work, Dr. Mazany provided a brief background on the studies and findings that have been conducted by IMO i.e.

- The 2nd IMO GHG Study 2009, provided the estimated cost of implementing energy efficiency technologies i.e. payback period versus how much it costs. The report provides graphs that show the payback period is very quick for some, whereas for others it might take longer.
- The 3rd IMO GHG Study 2014, did not update the marginal abatement costs, however it updated the estimates and produced new projections. Therefore the projected increase of 50% to 250% by 2030 for the growth in international shipping emissions, was a result of the 3rd GHG Study.
- Dr. Mazany also addressed the issue of port emissions, which countries referred to as the major contributing factor to air pollution and informed the meeting that the 3rd GHG Study also looked at emissions of sulphur oxides, nitrogen oxides and particulate matter.
- The IMO Road Map will look at conducting the 4th GHG Study that will update the data from the 3rd Study.

In introducing the IMO Strategy, it was highlighted that the document contains a list of activities, including further IMO GHG Studies and significant intersessional work, with relevant timelines and provides for alignment of those new activities with the ongoing work by the MEPC on the three-step approach to ship energy efficiency improvements.

Low Carbon Sea Transport Project

(Alexander Beetz, Project Manager GIZ)

The presentation introduced the GIZ Sustainable Sea Transport Project based in the Marshall Islands with a EUR 9.5 million funded by the German Federal Ministry for the Environment, Nature Conservation, Building and Nuclear Safety (BMUB). The project is being implemented through the German Embassy and the Marshall Islands Ministry of Transport and Communication.

The 5-year project (2017 – 2022) is looking at achieving the following outcomes i.e.:

- Contribution to reduction of emissions from shipping
- Contribution to achieving RMI's NDC
- Lower shipping costs
- Saving on fossil fuel imports
- Proposal for upscaling
- Inputs of RMI to the UN Framework Convention on Climate Change (UNFCCC), IMO and regional meetings

The University of Emden in Germany will oversee the refitting of cargo vessels. A local Marshallese organisation *Waan Aelon in Majel-Wa Kuk Wa* (WAM) specialised in building traditional canoes, will work with the project to develop a carbon free intra-lagoon vessel for transport within the atoll. The USP will be engaged to oversee the economic and social aspects of sea transport in the Marshall Islands and the MCST will provide support in terms of training and research.

New-Caledonia toward Sustainable Blue Growth – Programme and Projects of the New-Caledonia Maritime Cluster

(Lionel Loubersac, New Caledonia)

The presentation provided a brief overview of the Cluster Maritime de Nouvelle-Calédonie (CMNC) highlighting that it is a non-profit organisation (NPO), which was established on August 5th, 2014. The organisation currently represents more than 3000 employees with a budget of EUR 160,000 i.e. members' fees and grants from local authorities.

Priorities of the CMNC are: Blue Growth promotion, Wealth creation, Diversification of economic activities, Federation of actors, French speaking axis, and recognising that the ocean is 98% of the New Caledonia territory. Lionel provided a few examples of projects carried out by CMNC, amongst a number of other projects, Marine Technology platforms i.e.

- Equipping ships with continuous marine environment monitoring systems at a lower cost than physically recording.
- Using AIS to estimate near real time current fields for better routing and economical use of fuel…

It was then suggested through the overview that the CMNC concept could be expanded for the wider Pacific region, to provide assistance and to set up public/private partnerships in order to achieve sustainable Blue Growth excellence.

Norway Webinar on New Ship Technologies

(Iver Iversen, IMO)

The IMO consultant who presented on the approach in Norway towards designing new vessels and installing new technologies on vessels of less than 500. He highlighted the opportunities, benefits and limitations of such technologies from the pilot-projects conducted. The presentation focused on demonstrated various options for new technology vessels from new hull design and type of propulsion to use of alternative fuels.

The participants were interested to see experimentation done on the ground and noted that while the size of vessels was similar to those operated in the Pacific, the type of vessels and operations were significantly different. Indeed most of maritime routes in the Pacific, despite being within countries, are similar to deep sea route. Also the capacity and resources of Pacific shipowners are barriers to purchase new vessels and new technologies. Unfortunately most of the vessels operated in the Pacific are second to fifth hand. Lastly a question raised on the reliability and durability of those new designed vessels and technologies and whether these vessels will still be operated in 20 years.

Reducing GHGE from Domestic Ships: Ship Owners Perspective

(Josateki Tagi, Fiji Local Shipowners Association)

The presentation introduced the perspectives of the Fiji Local Ship Owners Association (LSOA) approach to reducing GHGEs. LSOA was established in the 1970's to: Protect ship operators, to act as an advisory role to the Government on marine transportation, to advocate for fare trade and, to observe compliance with IMO Conventions and the Maritime Safety Authority of Fiji (MSAF).

The Fiji shipping industry operates on second hand ships across 17 shipping companies. The LSOA confirms its support for the MTCC-Pacific project in:

- Supplying the necessary information that is required for analysis
- Championing the initiative (Placards and MTCC-Pacific stickers on board vessels)
- Proving regular updates of progress in the implementation

It is anticipated that the Fiji shipping industry will have improved and modernised domestic fleets that comply with GHGE requirements in the future.

A Concept for Inter-Island Sustainable Sea Transport in the Pacific

(Peia Patai, Okeanos Vaka Moana Fleet)

In 2009, Okeanos built seven regional ocean going canoes or 'Vakas' for the purpose of reviving canoe culture, traditional navigation and vessels to carry an environmental message of ocean awareness and climate change. The idea of using traditional canoes as a sustainable sea transport to help the pacific people was very strong.

In 2011, that idea and dream became a reality and the first Vaka Motu was launched. The Vaka Motu was built maintaining traditional knowledge and measures, traditional sail design, traditional steering method called the 'Ue' and a modern navigation aid. It has a very shallow draft, which gives it the ability to dock on the beach. The Vaka uses solar to generate electricity and a coconut oil engine as a secondary propulsion. It is considered a robust and safe vessel to sail.

Okeanos Foundation has active sustainable sea transport vessels based in the Pacific i.e:

- Okeanos Vanuatu based in Port Vila
- Okeanos Marshall Islands based in Majuro
- Okeanos Marianas based in Saipan
- Okeanos Aotearoa based in the Bay of Islands, New Zealand, and
- The intent of reactivating eco-whale-watching in Tonga.

There are a variety of ways in which the vessel can be utilised i.e. on business initiatives, disaster relief work, servicing remote countries in particular for islands with irregular or no shipping services.

Reviving our Sustainable Sea Transportation Culture in Fiji

(Monifa Fiu, Uto Ni Yalo Trust)

The Uto ni Yalo Trust shared some experiences of sustainable sea transportation culture in Fiji and acknowledged the presence of the Okeanos Foundation of which the Uto ni Yalo vessel was a part of the original fleet.

The Vaka and the Uto ni Yalo Trust promotes communities and sustainability, and inspires the Pacific to take action and relive moments of history in traditional navigation. In addition to sustainable sea transportation, the Vaka also offers multiple cultural, economic, environmental, and social benefits for people living out in the islands. In the future, it is anticipated that there will be an increase in smaller fleets, which will ease the burden of logistics. Therefore, the agenda for sustainable sea transport need to revitalise all aspects of traditional sea transport and navigation i.e. from the construction to operation.

The Uto ni Yalo Trust hopes that the vision will go beyond the shores of the Pacific and to be able to re-tell the stories and aspects of the Pacific's sailing history.

SESSION 3: MTCC-PACIFIC WORK

(Facilitated by the John Tunidau, MSAF)

MTCC-Pacific Progress Report

(Thierry Nervale, MTCC-Pacific)

SPC presented and provided an update of the MTCC-Pacific highlighting that the office has been established and is operational with the availability of an MTCC-Pacific website, which is online and accessible. The overview highlighted the following achievements since the signing of the SPC-IMO contract on 12 May 2017:

Result 1 – Establishment and operations
Result 2 – Capacity building activities
Result 3 – Pilot projects on energy efficiency
Result 4 – Pilot project on data collection
Result 5 – Communication and visibility

As a way forward for 2018, the following activities are planned by MTCC-Pacific. Targeted countries are Fiji, Kiribati, Marshall Islands, Samoa, Solomon Islands, Tuvalu and Vanuatu.

- National workshop on Ship Energy Efficient Operations (SEEO) and Index (EEOI)
- Data Collection
- Development and implementation of Ship Energy Efficiency Management Plan (SEEMP)
- Port Energy Management

Energy Management in Ports – Activities and Results

(Bruce Rowse, MTCC-Pacific Consultant)

The presentation is an overview of the activities and results of energy management in ports highlighting that the activities involved:

- The Energy Management Workshop, which was held in Auckland, New Zealand, August 2017.
- The development of an Energy Management Guide with templates to assist in the development of energy management plans and policies.
- Energy audits have been carried out at the Fiji, Solomon Islands and Samoa Ports.
- Energy management trials– Eco driving trial is being carried out for the Port of Suva looking at how behaviour can save energy. The project is also upgrading the tower lighting for the port.

The overview then briefly covered the results of the above activities. Drivers and barriers to energy management were identified at the Energy Management Workshop while at the same time a number of solutions to barriers have also been listed.

A number of post workshop events had been carried out in Cook Islands, Fiji, Kiribati, Nauru, Solomon Islands, Tonga, and Vanuatu.

MTCC-Asia Summary

(Xie Jieying, MTCC-Asia)

The session introduced the scope of MTCC-Asia's activities highlighting that the vision of the Centre is *'Uptake, Share, Sustainability for Greener Shipping'*. Stakeholders of MTCC-Asia are Cambodia, Thailand,

Det Norske Veritas (DNV), private shipping companies including COSCO and Shanghai Governmental Commissions, etc. There are about 25 partners and stakeholders.

MTCC-Asia's 2017 activities included:

- Capacity building/training for staff
- MTCC-Asia launch event in May
- Development and launching of the Online Clean Shipping Air Forum
- Regional Workshop in China in September
- Enrolment of post-graduate candidate in ship energy efficiency program
- National Workshop in Cambodia etc.

The presentation also provided an overview on the action plans of 2017, pilot projects, upcoming events and activities planned for 2018.

Pilot Project and Data Collection

(Zullah Mohammed, MTCC-Pacific)

The session provided a brief overview of the MTCC-Pacific pilot projects to reduce fuel oil consumption and GHGEs from the MTS. There are 2 pilot projects which have been approved:

- Pilot-project on "uptake of ship energy efficient technologies and operations"
  - Focus mainly on domestic shipping and ports.
  - Provide tools to develop/improve energy management.
- Pilot-project on "fuel oil consumption data collection and reporting"
  - Data collection on fuel oil consumption in domestic shipping.

The projects will be implemented by the MTCC-Pacific through the Pacific Island Domestic Ship Safety (PIDSS) Programme of SPC. PIDSS have established contacts in PICTs and have facilitated baseline data collection that is required to initiate the project.

A data collection template has been developed and distributed in Fiji and Kiribati. In addition, an EEOI calculation template has also been developed that will calculate the carbon emission of a vessel carrying cargo. It will be used as a performance indicator to judge the efficiency of a vessel on a particular voyage. The presentation than introduced the Ship Energy Efficiency Management Plan (SEEMP) and distribution materials, innovative technologies and energy saving devices (ESD) for the Pacific.

Country Priority Questionnaire

IMO distributed questionnaires and requested member countries to respond to the questions and to state their country priorities. This was to enable MTCC-Pacific to identify how best to assist and respond to the needs of PICs.

Mainstreaming Gender in Maritime Transport Sector

(Mary Narfi-Navaika, Office of the Maritime Regulator, Vanuatu)

The presentation introduced the Pacific Women in Maritime (PacWIMA) network, which was re-launched in April 2016. At the 3rd Pacific Regional Energy and Transport Minister's Meeting in April 2017, Nukualofa, the meeting noted that it was important to strengthen PacWIMA as the key tool for addressing women in

maritime issues. It also noted the improvements needed in coordination and communication at local and international levels. The meeting then agreed that PacWIMA develop a Regional Strategy for the recognition, visibility, and training of Pacific women in the MTS.

PacWIMA, in its presentation then called on participants of the MTCC-Pacific Regional Conference to:

- support the recommendations made in the Outcome document of the meeting of December 17th, 2017 Pacific Women in Maritime, which was held in parallel with the MTCC-Pacific Conference
- adopt measures to implement those recommendations in their country and workplace
- share the outcomes widely in their country and through various networks
- support the development of Women in Maritime networks in their respective countries.

The MTCC-Pacific conference expressed its support for the PacWIMA for gender equality in the MTS. However SPC's Deputy Director Transport questioned participants and particularly Heads of Maritime whether enough was done to give space for gender equality and if maritime administrations were in position to give this space physically and administratively to Women in Maritime networks, associations and focal points. The Conference was informed of the Regional Conference for Women in Maritime to be organised in 2018.

Discussion

- MTCC-Asia's participation at the Pacific conference reinforces the scope of the GMN. The MTCC regions differ in their priorities and approaches; however, there is an incredible resource in MTCC-Asia in terms of their expertise and knowledge. In moving forward to and beyond 2019, there is potential for the two regions to collaborate and provide training opportunities at their MTCC-Asia facilities. There is also opportunities to look at synergies between the MTCC regions recognising that there is potential for MTCC-Asia to help support the other MTCC regions.
- The Uto ni Yalo Trust expressed their interest to be part of the PacWIMA network noting that the Vaka has a 45% participation of women as crew.

REVIEW AND ADOPTION OF THE REGIONAL CONFERENCE OUTCOME

(Thierry Nervale, MTCC-Pacific)

The conference was guided through the listed priority actions under *'Appendix 1 and 2 – Matrix of drivers, needs, barriers and recommendations for relevant actions'* (Annex 3).

Participants were than invited to comment on and to propose changes to the MTCC-Pacific Conference Outcome statement (Annex 3).

Amendments to the conference Outcome statement was then captured, consolidated and adopted by the MTCC-Pacific member countries and partners.

CLOSING REMARKS IMO, SPC and Government of Fiji

IMO, Aubrey Botsford

The IMO acknowledged and praised the MTCC-Pacific member countries and participants expressing that the Conference was a very informative two days. The objective of the Conference was to:

- Launch the MTCC-Pacific which was carried out in an extremely effective and moving opening ceremony.

- Brought together members and partners of the MTCC-Pacific project to build ties and to show a common purpose, which was very well received.

Furthermore, the Conference has been a very educational platform, which displayed incredibly different societies, heritages and environments. A powerful message coming out of the Conference was that a forum like the MTCC-Pacific allows people with common interests to speak with one voice, an approach that always has the potential to be extremely effective.

IMO thanked the MTCC-Pacific Secretariat team for organising the event and for the efficient support during the Conference. IMO also extended its acknowledgement to SPC and SPREP for hosting MTCC-Pacific. Further acknowledgements and thanks were extended to the Government of Fiji for supporting the MTCC-Pacific project and most importantly to the EU for funding the project.

SPC/MTCC-Pacific, Thierry Nervale

MTCC-Pacific acknowledged the remarks by IMO and supported the sentiments raised by IMO that the Conference has indeed been an educational platform, in particular for the MTCC-Pacific member countries, partners, and the Secretariat. The conference has resulted in the establishment of stronger partnerships and fostered the exchange of experiences and ideas. MTCC-Pacific acknowledged participants for the productive discussions over the two days and for adopting the Outcome statement.

SPC is proud to be the host of the MTCC-Pacific and acknowledged the support of the EU and IMO. Special thanks and acknowledgements was also extended to SPREP for the partnership and support of the project. SPC also thanked the Government Fiji for chairing the conference and for the continuous support since the establishment of the Centre in Fiji.

Government of Fiji, Lui Naisara

The Closing Remarks on behalf of the Government of Fiji was delivered by the Deputy Secretary Policy and Planning, Mr. Lui Naisara. Special thanks was extended to the EU and the IMO for the funding and technical support enabling Fiji to host MTCC-Pacific. The Fiji Government also praised SPC and SPREP for organising the conference and the launch of the MTCC-Pacific project.

COMMUNICATION AND VISIBILITY

All MTCC-Pacific communication and visibility materials were displayed and disseminated at the Conference. This includes the pull-up banner of GMN and presentations with all logos. A media alert was sent 10 days before the event and a media release at the Official Launch highlighting the establishment of MTCC-Pacific as a centre of excellence for the Pacific. The table below shows the communication actions related to the Regional Conference and the media coverage. This was relayed on Social Media through SPC network and in news from IMO website.

Intermediate Results (IR)	Expected result	Time	Status	Communication and visibility
IR5.2 Commemorative plaque, posters and brochures produced for the launch of the Pacific MTCC	Commemorative plaque, poster, brochures (e- and hard copies), banner, stationary/USB sticks, presentation background, flyers available for all activities	12-15 December 2017	Commemorative plaque offered by IMO to MTCC-Pacific at the MTCC-Pacific Regional conference and official launch Banners and pull up banners displayed throughout this event Brochures distributed during this event Previously developed dedicated logo displayed on all MTCC-Pacific communication and visibility materials on display at this event.	• Commemorative plaque presented at the conference • Poster displayed and brochures, flyers, presentations and USB sticks distributed at the event
IR5.5 Progress on regional conference activities is systematically reported	Media release issued major the regional conference and official launch	12-15 December 2017	MTCC-Pacific Regional conference and official launch Media release on 13 December 2017 http://www.spc.int/blog/launch-of-maritime-technology-cooperation-centre-in-the-pacific-establishes-centre-of-excellence-for-the-region/ MTCC-Pacific launch on 13 December 2017 http://www.imo.org/en/MediaCentre/WhatsNew/Pages/default.aspx Articles in Fiji Times on the MTCC-Pacific Launch http://www.fijitimes.com/story.aspx?id=427343 http://www.fijitimes.com/story.aspx?ref=archive&id=427020 http://www.fijitimes.com/story.aspx?id=427324 Fiji One News Video link. News on MTCC-Pacific @ 12.48 timeline https://youtu.be/5xCM9X8XX9E Fiji Government video on YouTube about MTCC-Pacific Launch. https://www.youtube.com/watch?v=N0a_kjTogF8	• Media release issued and published on various media platforms

			Fiji Sun article – Major Maritime Meeting Launched At Novotel Lami http://fijisun.com.fj/2017/12/12/major-maritime-meeting-launched-at-novotel-lami/ http://fijisun.com.fj/2017/12/20/regions-maritime-centre-of-excellence-based-in-fiji/ Green 4 sea article – Maritime cooperation centres launched in Africa and Pacific https://www.green4sea.com/maritime-cooperation-centres-launched-in-africa-and-pacific/ Scoop article – Launch of Maritime Technology Cooperation Centre http://www.scoop.co.nz/stories/WO1712/S00161/launch-of-maritime-technology-cooperation-centre.htm Ship and Bunker article – Center of excellence for low carbon shipping technology launched for the Pacific https://shipandbunker.com/news/apac/924294-centre-of-excellence-for-low-carbon-shipping-technology-launched-for-pacific	
	Social media is used at the regional conference and official launch	12-15 December 2017	Social media (Twitter on 12 Dec 2017 Heads of Maritime meeting to prepare for the regional conference https://twitter.com/DrZullah/status/940669426365370368 Social media (Twitter on 12 Dec 2017 Steering Committee meeting) https://twitter.com/DrZullah/status/940670745486639104 https://twitter.com/DrZullah/status/940683321209581568 https://twitter.com/BBernedine/status/940685247569526784 Social media (Twitter on 13-15 Dec 2017 MTCC-Pacific Regional Conference and Official Launch) https://twitter.com/BBernedine/status/940707385772933120 https://twitter.com/spc_live/status/940708673550086144 https://twitter.com/spc_live/status/940709309234626560 https://twitter.com/andrewj_spc/status/940754537706676225 https://twitter.com/Lore_Croker/status/941076638485245954 https://twitter.com/spc_cps/status/941852910677381121 Video of official handing over of the MTCC-Pacific plaque	• Social media was fully utilised

			https://twitter.com/spc_live/status/940711111329923072	
	MTCC-Pacific is an agenda item of the meeting of the Heads of Maritime	12 Dec 2018	**Quarter 13 November 2017-12 February 2018** GMN Project and MTCC-Pacific introduction at the Heads of Maritime meeting on Dec 12 2017	• MTCC-Pacific an agenda item at the annual Heads of Maritime meeting

ANNEXES

ANNEX 1: Signed List of Participants

	MTCC-PACIFIC REGIONAL CONFERENCE AND OFFICIAL LAUNCH, WEDNESDAY																
	Name	Job Title	Organisation	Gender	Line 1 address	Line 2 Address	Building number	Post code	City	Country where based	Telephone	Mobile Number	Email Address 1	Email Address 2	Source of Funding	Role	SIGNATURE
1	Ambassador Julian wilson	Ambassador EU Pacific	EU Pacific	M					Suva	Fiji					In-country	Participant	Attended
2	Honorable Osea Naiqamu	Minister for Fisheries and Forests	Ministry of Fisheries and Forests	M					Suva	Fiji					In-country	Participant	Attended
3	David Kolitagane	Permanent Secretary	Ministry of Infrastructure and Transport	M					Suva	Fiji			psit@moit.gov.fj		In-country	Participant	Attended
4	Makereta Konrote	Permanent Secretary	Ministry of Economy	F					Suva	Fiji			makereta.konrote@economy.gov.f		In-country	Participant	Attended
5	Mikaele Belena	Director of Energy	Ministry of Infrastructure and Transport	M					Suva	Fiji			mikaele.belena@moit.gov.fj		In-country	Participant	Attended
5	John Tunidau	CEO	MSAF	M					Suva	Fiji			jtunidau@msaf.com.fj		In-country	Participant	Attended
6	Biu Tawake	Commander Navy		M					Suva	Fiji			humphrey.tawake@rfmf.gov.gj		In-country	Participant	Attended
7	Jonathan Moore	CEO	Fiji Road Authorities	M					Suva	Fiji			jonathan.moore@fijiroads.org		In-country	Participant	Attended
8	Lui Tuisiga Naisara	Deputy Secretary Policy & Planning	Ministry of Infrastructure & Transport	M	84 Ratu Mara Road				Suva	Fiji	(679) 3389547	9904282	lui.naisara@moit.gov.fj		In-country	Participant	Attended
8	Stephen Lee Simpson	Director of Marine	Ministry of Transport	M	PO Box 61				Rarotonga	Cook Islands	682 28810	682 55605	stephen.simpson@cookislands.gov.ck	stevels11@hotmail.com	IMO	Participant	Attended
9	Nooroa Tou	General Manager	Cook Islands Ports Authority	M	PO Box 84				Rarotonga	Cook Islands	682 21921	682 55750	Nooroa.tou@cookislands.gov.ck	twins@oyster.net.ck	IMO	Participant	
10	Ngatokorua Ngatokorua	Compliance Manager	Ministry of Transport	M	PO Box 61				Rarotonga	Cook Islands	682 28810	682 563392	junior.ngatokorua@cookislands.gov.vk	jkorua@gmail.com	IMO	Participant	
11	Maryanne Mikaere Namakin	Deputy Secretary	Ministry of Information, Communication, Transport and Tourism Development	F	PO Box 487				Tarawa	Kiribati	686 26003	686 73062233	ds@mcttd.gov.ki	barrykatoy@gmail.com	IMO	Participant	
12	Ruoikabuti Tioon	Director of Marine	Ministry of Information, Communication, Transport and Tourism Development	M	PO Box 487				Tarawa	Kiribati	686 26003	686 73055552	dom@mcttd.gov.ki	jrruoi@gmail.com	IMO	Participant	
13	Ruatu Titaake	CEO	Kiribati Port Authority	M	KPA				Tarawa	Kiribati	686 26972	686 73046436	ceo@kpa.com.ki	ruatit@gmail.com	IMO	Participant	
14	Joshua Lakabung	Acting Director, Maritime and Safety	Ministry of Transportation, Communication and Information Technology	M	PO Box 1079				Majuro	RMI	692 6258869	692 4567400	jlakabung3@gmail.com		IMO	Participant	

15	Kemp Wade Detenamo	Maritime & Ports Operations Manager	Maritime and Ports Authority	M					Aiwo District	Nauru	674 5573089	674 5573089	Kemp.detenamo@gmail.com		IMO	Participant	
16	Mathew Wowoni	First Assistant Secretary	Department of Transport	M	PO Box 1489				Port Moresby	PNG	675 323 6187	675 76050001	mwowoni@transport.gov.pg	mjwowoni@gmail.com	IMO	Participant	
17	Pawa Limu	Manager	Marine Environment Protection, PNG National Maritime Saftey Authority	M	PO Box 668				Port Moresby	PNG	675 3211244	675 72335599	plimu@nmsa.gov.pg	pawalims401@hotmail.com	IMO	Participant	
18	Su'a Afamasaga Pou Onesemo	CEO	Ministry of Works, Transport & Infrastructure	M	Private Mail Bag				Apia	Samoa	685 21858	685 7770528	pou.onesemo@mwti.gov.ws	pou.onesemo.mwti@gmail.com	IMO	Participant	
19	Fepuleai Faleniu Alesana	Assistant CEO Maritime Division	Ministry of Works, Transport & Infrastructure	M	Private Mail Bag				Apia	Samoa	685 21611	685 771 9996	faleniu.alesana@mwti.gov.ws	faleniu.alesana15@gmail.com	IMO	Participant	
20	Timothy Whiteford Harris	Acting Director of Marine	Solomon Islands Maritime Safety Administration	M	PO Box G62				Honiara	Solomon Islands	677 28157	677 7791562	Tharris@mid.gov.sb	timharris@gmail.com	IMO	Participant	
21	Jimmy Nuake	Acting Permanent Secretary	Ministry of Infrastructure Development	M	PO Box G8				Honiara	Solomon Islands	677 2865	677 7521819	Jnuake@mid.gov.sb	jnuake@gmail.com	IMO	Participant	
22	James Alfred Gerea	Training Manager	Solomon Islands Ports Authority	M	PO Box 307				Honiara	Solomon Islands	677 22646	677 7245370	j.gerea@sipa.com.sb		IMO	Participant	
23	Mosese Lavemai	Chief Executive Officer	Ports Authority Tonga	M	PO Box 144				Nuku'alofa	Tonga	676 23168	676 8447005	mlavemai@portsauthoritytonga.com		IMO	Participant	
24	Falasese Tupau	Assistant Secretary	Ministry of Communication and Transport	M	Government Building, Vaiaku				Funafuti	Tuvalu	688 20055	688 7002926	ftupau@gov.tv	falasese@gmail.com	IMO	Participant	
25	Salolo Tepoga	Assistant Manager	Ministry of Communication and Transport	M					Funafuti	Tuvalu	688 20055	688 7005419	siaositepoga@gmail.com	siaositepoga@gmail.com	IMO	Participant	
26	Lloyd Maiki Fikiasi	Office of Maritime Regulator	Legal Officer	M	PO Box 6499				Port Vila	Vanuatu	678 35455	678 7753473	lfikiasi@omr.vu		IMO	Participant	
27	Henry Worek	Director	Ports Marine and Department	M	PMB 9046				Port Vila	Vanuatu	678 33600	678 7743972	hworek@vanuatu.gov.vu	henzyb@gmail.com	IMO	Participant	
28	Adrian Nicolae	Team Leader Climate Chang	EU Pacific	M	Private Mail Bag, GPO				Suva	Fiji	3313633	9484486	Adrian.nicolae@eeas.europa.eu		EU Pacific	Participant	Attended
29	Sylvain Rabeau	Deputy Director	New Caledonia Maritime Affairs	M					Noumea	New Caledonia	687270692	687787440	sylvain.rabeau@gouv.nc	sylvain.rabeau@hotmail.com	SPC	Participant	
30	Mapu Jnr Tapuai	Transport Operations Manager	Government of Tokelau Transport Support Services	M	PO Box 3298				Apia	~~Samoa~~ Tokelau	685 20822	685 7774971	mapu.tapuai@tokelau.org.nz	jtapuai@yahoo.con.nz	SPC	Participant	

31	Annalisse Sly	Acting Manager, Environment Standards	Australian Maritime Safety Authority	F	GPO Box 2181				Canberra	Australia	61 2 62795040	61 410755481	annalisse.sly@amsa.gov.au	maggie_sly@hot	AMSA	Participant	
32	Nanise Kabakoro	Trainee Port Engineer	Fiji Ports Authority	F	GPO Box 780				Suva	Fiji	679 3279203	679 9908598	nanise@fijiports.co	nkabakoro@gmail.com	In-country	Participant	
33	Lopeti Radravu	Operations Manager	Fiji Ships and Heavy Industries Limited	M	Lot 1 Eliza St WBay Suva, Fiji				Suva	Fiji	679 3306229	679 9907176	lopeti@fijiports.com.fj		In-country	Participant	
34	Scott Hook	Resilience Team Leader	Pacifc Islands Forum Secretariat	M	PMB				Suva	Fiji Islands	679 3312600		scotth@forumsec.org		PIFS	Participant	
35	Teea Tira	Climate Change Coordinator	Pacifc Islands Forum Secretariat	F	PMB				Suva	Fiji Islands	679 3312600		teeat@forumsec.org		PIFS	Participant	
36	Johan Fred Amoa	President	Pacific International Maritime Law Association	M	PO Box 2814				Apia	Samoa	685 20014	685 7634102	fred.amoa302@gmail.com		PIMLA	Participant	
37	Tafaigata Toilolo	Port Operations Manager	Samoa Port Authority	M	PO Box 2279				Apia	Samoa	685 770691				Ports Samoa	Participant	
38	Eranda Kotelawala	CEO	Solomon Islands Ports Authority	M					Honiara	Solomon Islands			ekotelawala@sipa.com.sb		Ports Solomon Islands	Participant	
39	Anthony Talouli	Pollution adviser	Secretariat of the Pacific Regional Environment Programme	M	SPREP, Vailima				Apia	Samoa	679 7806000		anthonyt@sprep.org		SPREP	Participant	
40	Philip Colin	President	Uto ni Yalo	M	5 Maafu Street				Suva	Fiji		679 9992340			In-country	Participant	
41	Monifa Fiu	Vice President	Uto ni Yalo	F	6 Maafu Street				Suva	Fiji		679 9965663	monifafiu@gmail.com		In-country	Participant	
42	Richard Coleman	CEO	Vanuatu Maritime College	M	PO Box 201				Luganville	Vanuatu		678 36547	marico@vanuatu.com.vu		SPC	Participant	
43	Alo ki Hakau Maileseni	Chief Financial Officer	Ports Authority Tonga	M	PO Box 144				Nuku'alofa	Tonga		676 23168	amaileseni@portsauthoritytonga.com		Ports Tonga	Participant	
44	Tevita Kaufusi	Manager Infrastructure & Technical	Ports Authority Tonga	M	PO Box 144				Nuku'alofa	Tonga		676 23168	portsauthority.tonga.com		Ports Tonga	Participant	
45	Sione Akauola	CEO	Ministry of Public Enterprises	M	PO Box 19				Nuku'alofa	Tonga		676 28144	sakauola@mpe.go	folaosi@gmail.com	Ports Tonga	Participant	
46	Lionel Loubersac	Deputy Director	New Caledonia Maritime Affairs	M					Noumea	New Caledonia	687 763208		lionel.loubersac@outlook.fr		SPC	Participant	
47	Mark Borg	Team Leader Programme Management	Pacific Islands Development Forum	M	56 Domain Road				Suva	Fiji		9907633	markborg@pacificidf.org		PIDF	Participant	

48	Derrick Armstrong	Deputy Vice-Chancellor	USP	M									derrick.armstrong@usp.ac.fj		USP	Participant	
49	Peter Nuttall	Research Fellow	USP	M									peter.nuttall@usp.ac.fj		USP	Participant	
50	Josateki Tagi	Fleet Superintendent/Capta	Billett Wright & Associates	M	P O Box 3092, Lami							3300711	7776919/ 7776918/ 7776920	billett@unwired.c josatagi@yahoo	In-country	Participant	
51	Alexander Beetz	Project Manager	[redacted]GIZ	M											[redacted]	Participant	
52	Peia Tatai	Project Manage															
53	Joseph Arc	Corporate Affair	PNG Ports	M	Box 671 Pom, NCD.												
54	Avnita Goundar		USP	F													
55	LEAFUAGA JAKARIA	SHIPSURVEYOR	SSC	M	SAMOA												
56	Jieying Xie	Technical officer	MTCC-Asia	F													
57	AVIOUR	"	IMO	M													
58	A. BOTSFORD	Chief, OGS	IMO	M													
59	A. FREEMAN	DIRECTOR	WFP	M													
60	Mavis J. Logavatu	LRAD	MSAF	F													
61	CATHY INDU KALIOVA	DIREGISTRAR	SIMSA	F													
62	KADONAVATU	MC		M													
63	Lesi V	APTP	MOIT	F													
64	Ruamau	MLO	✓	M													
65	Rusi R	UNDP	UNDP	M													
66	JTagi	Billett Wright	Director														
67	Jerusi V	Billett Wright	Consultant														
68	T.A. Isoroda																
69																	
70																	
71																	
72																	
73																	
74																	
75																	
76																	
77																	

	MTCC-PACIFIC REGIONAL CONFERENCE AND OFFICIAL LAUNCH (13/12/2017)																
	Name	Job Title	Organisation	Gender	Line 1 address	Line 2 Address	Building number	Post code	City	Country where based	Telephone	Mobile Number	Email Address 1	Email Address 2	Source of Funding	Role	SIGNATURE
1	Anton Rhodes	Project Officer	IMO	M	4 Albert Embarkment				London, SE17SR	UK			AntonR@imo.org			Resource	
2	Aubrey Botsford	Deputy Director/Chief	IMO	M	4 Albert Embarkment				London, SE17SR	UK			AubreyB@imo.org			Resource	
3	Jieying ~~Cie~~ Xie	MTCC-Asia	Shanghai Maritime Univeristy	F	Room 401, No. 110, Lane 366				Shanghai	China						Resource	
4	Robin Leigh Mazany	Consultant	IMO	F					Ottawa	Canada			leigh.mazany@yahoo.ca			Resource	
5	Bruce Rowse	Consutant	SPC	M									bruce@8020green.com			Resource	
6	Audrey Aumua	Deputy Director General	SPC	F		Nabua			Suva	Fiji			audreya@spc.int			Resource	
7	Andrew Jones	Director GEM Division	SPC	M	Mead Road	Nabua			Suva	Fiji			andrewj@spc.int			Resource	
8	Thierry Nervale	Deputy Director Transport	SPC	M	Lotus Building	Nabua			Suva	Fiji			thierryn@spc.int			Resource	Attended
9	Mark Davies	Transport Greenhouse Gas Adviser	SPC	M	Lotus Building	Nabua			Suva	Fiji			markd@spc.int			Resource	
10	Mohammed Zullah	Maritime Industry Energy Efficiency Officer	SPC	M	Lotus Building	Nabua			Suva	Fiji			zullahm@spc.int			Resource	
11	Ore Toua	Maritime Training Adviser	SPC	F	Lotus Building	Nabua			Suva	Fiji			oret@spc.int			Resource	Attended
12	Francesca Pradelli	Policy and Legal Officer	SPC	F	Lotus Building	Nabua			Suva	Fiji			francescap@spc.int			Resource	
13	Yolisaguyau Tomtavala	Policy and Legal Adviser	SPC	M	Lotus Building	Nabua			Suva	Fiji			yolisaguyaut@spc.int			Resource	
14	Lore Coker	Admin & Comms. Assistant	SPC	F	Lotus Building	Nabua			Suva	Fiji			lorec@spc.int			Resource	
15	Bernedine Managreve	Admin. Assistant	SPC	F	Lotus Building	Nabua			Suva	Fiji			bernedinem@spc.int			Resource	Attended
	P. Foster	Dir Comms	SPC	M	Nabua								PeterF@spc.int			I T	Attended

MTCC-PACIFIC REGIONAL CONFERENCE AND OFFICIAL LAUNCH, 14-15 December 2017

	Name	Job Title	Organisation	Gender	Line 1 address	Line 2 Address	Building number	Post code	City	Country where based	Telephone	Mobile Number	Email Address 1	Email Address 2	Source of Funding	Role	SIGNATURE (Thursday 14/12/2017)
1	John Tunidau	CEO	MSAF	M					Suva	Fiji			jtunidau@msaf.com.fj		In-country	Participant	
2	Lui Tuisiga Naisara	Deputy Secretary Policy & Planning	Ministry of Infrastructure & Transport	M	84 Ratu Mara Road				Suva	Fiji	(679) 3389547	9904282	lui.naisara@moit.gov.fj		In-country	Participant	
3	Stephen Lee Simpson	Director of Marine	Ministry of Transport	M	PO Box 61				Rarotonga	Cook Islands	682 28810	682 55605	stephen.simpson@cookislands.gov.ck	stevels11@hotmail.com	IMO	Participant	
4	Nooroa Tou	General Manager	Cook Islands Ports Authority	M	PO Box 84				Rarotonga	Cook Islands	682 21921	682 55750	Nooroa.tou@cookislands.gov.ck	twins@oyster.net.ck	IMO	Participant	
5	Ngatokorua Ngatokorua	Compliance Manager	Ministry of Transport	M	PO Box 61				Rarotonga	Cook Islands	682 28810	682 563392	junior.ngatokorua@cookislands.gov.vk	jkorua@gmail.com	IMO	Participant	
6	Maryanne Mikaere Namakin	Deputy Secretary	Ministry of Information, Communication, Transport and Tourism Development	F	PO Box 487				Tarawa	Kiribati	686 26003	686 73062233	ds@mcttd.gov.ki	barrykatoy@gmail.com	IMO	Participant	
7	Ruoikabuti Tioon	Director of Marine	Ministry of Information, Communication, Transport and Tourism Development	M	PO Box 487				Tarawa	Kiribati	686 26003	686 73055552	dom@mcttd.gov.ki	jrruoi@gmail.com	IMO	Participant	
8	Ruatu Titaake	CEO	Kiribati Port Authority	M	KPA				Tarawa	Kiribati	686 26972	686 73046436	ceo@kpa.com.ki	ruatit@gmail.com	IMO	Participant	
9	Joshua Lakabung	Acting Director, Maritime and Safety	Ministry of Transportation, Communication and Information Technology	M	PO Box 1079				Majuro	RMI	692 6258869	692 4567400	jlakabung3@gmail.com		IMO	Participant	
10	Kemp Wade Detenamo	Maritime & Ports Operations Manager	Maritime and Ports Authority	M					Aiwo District	Nauru	674 5573089	674 5573089	Kemp.detenamo@gmail.com		IMO	Participant	
11	Mathew Wowoni	First Assistant Secretary	Department of Transport	M	PO Box 1489				Port Moresby	PNG	675 323 6187	675 76050001	mwowoni@transport.gov.pg	mjwowoni@gmail.com	IMO	Participant	
12	Pawa Limu	Manager	Marine Environment Protection, PNG National Maritime Saftey Authority	M	PO Box 668				Port Moresby	PNG	675 3211244	675 72335599	plimu@nmsa.gov.pg	pawalims401@hotmail.com	IMO	Participant	
13	Su'a Afamasaga Pou Onesemo	CEO	Ministry of Works, Transport & Infrastructure	M	Private Mail Bag				Apia	Samoa	685 21858	685 7770528	pou.onesemo@mwti.gov.ws	pou.onesemo.mwti@gmail.com	IMO	Participant	
14	Fepuleai Faleniu Alesana	Assistant CEO Maritime Division	Ministry of Works, Transport & Infrastructure	M	Private Mail Bag				Apia	Samoa	685 21611	685 771 9996	faleniu.alesana@mwti.gov.ws	faleniu.alesana15@gmail.com	IMO	Participant	
15	Timothy Whiteford Harris	Acting Director of Marine	Solomon Islands Maritime Safety Administration	M	PO Box G32				Honiara	Solomon Islands	677 28157	677 7791562	Tharris@mid.gov.sb	timharris@gmail.com	IMO	Participant	
16	Jimmy Nuake	Acting Permanent Secretary	Ministry of Infrastructure Development	M	PO Box G8				Honiara	Solomon Islands	677 2865	677 7521819	Jnuake@mid.gov.sb	jnuake@gmail.com	IMO	Participant	
17	James Alfred Gerea	Training Manager	Solomon Islands Ports Authority	M	PO Box 307				Honiara	Solomon Islands	677 22646	677 7245370	j.gerea@sipa.com.sb		IMO	Participant	
18	Mosese Lavemai	Chief Executive Officer	Ports Authority Tonga	M	PO Box 144				Nuku'alofa	Tonga	676 23168	676 8447005	mlavemai@portsauthoritytonga.com		IMO	Participant	
19	Falasese Tupau	Assistant Secretary	Ministry of Communication and Transport	M	Government Building, Vaiaku				Funafuti	Tuvalu	688 20055	688 7002926	ftupau@gov.tv	falasese@gmail.com	IMO	Participant	
20	Salolo Tepoga	Assistant Manager	Ministry of Communication and Transport	M					Funafuti	Tuvalu	688 20055	688 7005419	siaositepoga@gmail.com	siaositepoga@gmail.com	IMO	Participant	
21	Lloyd Maiki Fikiasi	Office of Maritime Regulator	Legal Officer	M	PO Box 6499				Port Vila	Vanuatu	678 35455	678 7753473	lfikiasi@omr.vu		IMO	Participant	
22	Henry Worek	Director	Ports Marine and Department	M	PMB 9046				Port Vila	Vanuatu	678 33600	678 7743972	hworek@vanuatu.gov.vu	henzyb@gmail.com	IMO	Participant	
23	Adrian Nicolae	Team Leader Climate Chan	EU Pacific	M	Private Mail Bag, GPO				Suva	Fiji	3313633	9484486	Adrian.nicolae@eeas.europa.eu		EU Pacific	Participant	
24	Sylvain Rabeau	Deputy Director	New Caledonia Maritime Affairs	M					Noumea	New Caledonia	687270692	687787440	sylvain.rabeau@gouv.nc	sylvain.rabeau@hotmail.com	SPC	Participant	
25	Mapu Jnr Tapuai	Transport Operations Manager	Government of Tokelau Transport Support Services	M	PO Box 3298				~~Apia~~	~~Samoa~~ Tokelau	685 20822	685 7774971	mapu.tapuai@tokelau.org.nz	jtapuai@yahoo.con.nz	SPC	Participant	

26	Annalisse Sly	Acting Manager, Environment Standards	Australian Maritime Safety Authority	F	GPO Box 2181				Canberra	Australia	61 2 62795040	61 410755481	annalisse.sly@amsa.gov.au	maggie_sly@hotmail	AMSA	Participant		
27	Nanise Kabakoro	Trainee Port Engineer	Fiji Ports Authority	F	GPO Box 780				Suva	Fiji	679 3279203	679 9908598	nanise@fijiports.com	nkabakoro@gmail.com	In-country	Participant		
28	Lopeti Radravu	Operations Manager	Fiji Ships and Heavy Industries Limited	M	Lot 1 Eliza St WBay Suva, Fiji				Suva	Fiji	679 3306229	679 9907176	lopeti@fijiports.com.fj		In-country	Participant		
29	Scott Hook	Resilience Team Leader	Pacifc Islands Forum Secretariat	M	PMB				Suva	Fiji Islands	679 3312600		scotth@forumsec.org		PIFS	Participant		
30	Teea Tira	Climate Change Coordinator	Pacifc Islands Forum Secretariat	F	PMB				Suva	Fiji Islands	679 3312600		teeat@forumsec.org		PIFS	Participant		
31	Johan Fred Amoa	President	Pacific International Maritime Law Association	M	PO Box 2814				Apia	Samoa	685 20014	685 7634102	fred.amoa302@gmail.com		PIMLA	Participant		
32	Tafaigata Toilolo	Port Operations Manager	Samoa Port Authority	M	PO Box 2279				Apia	Samoa	685 770691				Ports Samoa	Participant		
33	Eranda Kotelawala	CEO	Solomon Islands Ports Authority	M					Honiara	Solomon Islands			ekotelawala@sipa.com.sb		Ports Solomon Islands	Participant		
34	Anthony Talouli	Pollution adviser	Secretariat of the Pacific Regional Environment Programme	M	SPREP, Vailima				Apia	Samoa	679 7806000		anthonyt@sprep.org		SPREP	Participant		
35	Philip Colin	President	Uto ni Yalo	M	5 Maafu Street				Suva	Fiji		679 9992340			In-country	Participant		
36	Monifa Fiu	Vice President	Uto ni Yalo	F	6 Maafu Street				Suva	Fiji		679 9965663	monifafiu@gmail.com		In-country	Participant		
37	Richard Coleman	CEO	Vanuatu Maritime College	M	PO Box 201				Luganville	Vanuatu		678 36547	marico@vanuatu.com.vu		SPC	Participant		
38	Alo ki Hakau Maileseni	Chief Financial Officer	Ports Authority Tonga	M	PO Box 144				Nuku'alofa	Tonga		676 23168	amaileseni@portsauthoritytonga.com		Ports Tonga	Participant		
39	Tevita Kaufusi	Manager Infrastructure & Technical	Ports Authority Tonga	M	PO Box 144				Nuku'alofa	Tonga		676 23168	portsauthority.tonga.com		Ports Tonga	Participant		
40	Sione Akauola	CEO	Ministry of Public Enterprises	M	PO Box 19				Nuku'alofa	Tonga		676 28144	sakauola@mpe.gov.t	folaosi@gmail.com	Ports Tonga	Participant		
41	Lionel Loubersac	Deputy Director	New Caledonia Maritime Affairs	M					Noumea	New Caledonia	687 763208		lionel.loubersac@outlook.fr		SPC	Participant		
42	Mark Borg	Team Leader Programme Management	Pacific Islands Development Forum	M	56 Domain Road				Suva	Fiji		9907633	markborg@pacificidf.org		PIDF	Participant		
43	Derrick Armstrong	Deputy Vice-Chancellor	USP	M									derrick.armstrong@usp.ac.fj		USP	Participant		
44	Peter Nuttall	Research Fellow	USP	M									peter.nuttall@usp.ac.fj		USP	Participant		
45	Josateki Tagi	Fleet Superintendent/Capt	Billett Wright & Associates	M	P O Box 3092, Lami						3300711	7776919/ 7776918/ 7776920	billett@unwired.co	josatagi@yahoo.com	In-country	Participant		
46	Alexander Beetz	Project Manager	GIZ	M											GIZ	Participant		
47	Peia Patai	Project Manager	Vaka	M									peia.patai@okeanos-foundaion		Okeanos	Participant		
48	Leatuasa Sakaraia	Ship Surveyor	SSC	M					Apia	Samoa								
	Simon Bonnett	General Manager - Sustainability	Swire Shipping	M					Suva/ Singapore	Fiji/ Singapore	+65 6632 0310	+65 9643 3492	simon.bennett@swire.com.sg		Self	observer		
	Mary Nawicka	Finance Manager	OMC	F														
	CATHY IVEY	SIMSA																
	Shirleen Swapna	Programme Manager	EU	F					Suva	Fiji	9062538	3313633 ext 147	shirleen.swapna@eeas.europa.eu		EU	Participant		

	Avnita Goundar	PhD in Marine Affairs candidate	USP	F								avnitag@siyaahi.com.fj				
	Alesi Waqanivalu	Policy Analyst	MoIT	F								alesi.waqanivalu@govnet.gov.fj				
	RESOURCE															
	Anton Rhodes	Project Officer	IMO	M	4 Albert Embarkment				London, SE17SR	UK			AntonR@imo.org			
	Aubrey Botsford	Deputy Director/Chief	IMO	M	4 Albert Embarkment				London, SE17SR	UK			AubreyB@imo.org			
	Jieying Xie	MTCC-Asia	Shanghai Maritime Univeristy	F	Room 401, No. 110, Lane 366				Shanghai	China						
	~~Robin~~ Leigh Mazany	Consultant	IMO	F					Ottawa	Canada			leigh.mazany@yahoo.ca			
	Bruce Rowse	Consutant	SPC	M									bruce@8020green.com			
	Thierry Nervale	Deputy Director Transport	SPC	M	Lotus Building	Nabua			Suva	Fiji			thierryn@spc.int			
	Mark Davies	Transport Greenhouse Gas	SPC	M	Lotus Building	Nabua			Suva	Fiji			markd@spc.int			
	Mohammed Zullah	Maritime Industry Energy Efficiency Officer	SPC	M	Lotus Building	Nabua			Suva	Fiji			zullahm@spc.int			
	Francesca Pradelli	Policy and Legal Officer	SPC	F	Lotus Building	Nabua			Suva	Fiji			francescap@spc.int			
	Lore Coker	Admin & Comms. Assistant	SPC	F	Lotus Building	Nabua			Suva	Fiji			lorec@spc.int			
	Bernedine Managreve	Admin. Assistant	SPC	F	Lotus Building	Nabua			Suva	Fiji			bernedinem@spc.int			

Maria Elder — Regional Environmental Specialist — US Embassy — F — ElderMB@state.gov

Mark Mineo — Regional Environmental Officer — USEmbassy — M. — MineoMR@state.gov

MTCC-PACIFIC REGIONAL CONFERENCE AND OFFICIAL LAUNCH, 15 December 2017

	Name	Job Title	Organisation	Gender	Line 1 address	Line 2 Address	Building number	Post code	City	Country where based	Telephone	Mobile Number	Email Address 1	Email Address 2	Source of Funding	Role	SIGNATURE (Friday 14/12/2017)
1	John Tunidau	CEO	MSAF	M					Suva	Fiji			jtunidau@msaf.com.fj		In-country	Participant	
2	Lui Tuisiga Naisara	Deputy Secretary Policy & Planning	Ministry of Infrastructure & Transport	M	84 Ratu Mara Road				Suva	Fiji	(679) 3389547	9904282	lui.naisara@moit.gov.fj		In-country	Participant	
3	Stephen Lee Simpson	Director of Marine	Ministry of Transport	M	PO Box 61				Rarotonga	Cook Islands	682 28810	682 55605	stephen.simpson@cookislands.gov.ck	stevels11@hotmail.com	IMO	Participant	
4	Nooroa Tou	General Manager	Cook Islands Ports Authority	M	PO Box 84				Rarotonga	Cook Islands	682 21921	682 55750	Nooroa.tou@cookislands.gov.ck	twins@oyster.net.ck	IMO	Participant	
5	Ngatokorua Ngatokorua	Compliance Manager	Ministry of Transport	M	PO Box 61				Rarotonga	Cook Islands	682 28810	682 563392	junior.ngatokorua@cookislands.gov.vk	jkorua@gmail.com	IMO	Participant	
6	Maryanne Mikaere Namakin	Deputy Secretary	Ministry of Information, Communication, Transport and Tourism Development	F	PO Box 487				Tarawa	Kiribati	686 26003	686 73062233	ds@mcttd.gov.ki	barrykatoy@gmail.com	IMO	Participant	
7	Ruoikabuti Tioon	Director of Marine	Ministry of Information, Communication, Transport and Tourism Development	M	PO Box 487				Tarawa	Kiribati	686 26003	686 73055552	dom@mcttd.gov.ki	jrruoi@gmail.com	IMO	Participant	
8	Ruatu Titaake	CEO	Kiribati Port Authority	M	KPA				Tarawa	Kiribati	686 26972	686 73046436	ceo@kpa.com.ki	ruatit@gmail.com	IMO	Participant	
9	Joshua Lakabung	Acting Director, Maritime and Safety	Ministry of Transportation, Communication and Information Technology	M	PO Box 1079				Majuro	RMI	692 6258869	692 4567400	jlakabung3@gmail.com		IMO	Participant	
10	Kemp Wade Detenamo	Maritime & Ports Operations Manager	Maritime and Ports Authority	M					Aiwo District	Nauru	674 5573089	674 5573089	Kemp.detenamo@gmail.com		IMO	Participant	
11	Pawa Limu	Manager	Marine Environment Protection, PNG National Maritime Saftey Authority	M	PO Box 668				Port Moresby	PNG	675 3211244	675 72335599	plimu@nmsa.gov.pg	pawalims401@hotmail.com	IMO	Participant	
12	Su'a Afamasaga Pou Onesemo	CEO	Ministry of Works, Transport & Infrastructure	M	Private Mail Bag				Apia	Samoa	685 21858	685 7770528	pou.onesemo@mwti.gov.ws	pou.onesemo.mwti@gmail.com	IMO	Participant	
13	Fepuleai Faleniu Alesana	Assistant CEO Maritime Division	Ministry of Works, Transport & Infrastructure	M	Private Mail Bag				Apia	Samoa	685 21611	685 771 9996	faleniu.alesana@mwti.gov.ws	faleniu.alesana15@gmail.com	IMO	Participant	
14	Timothy Whiteford Harris	Acting Director of Marine	Solomon Islands Maritime Safety Administration	M	PO Box G32				Honiara	Solomon Islands	677 28157	677 7791562	Tharris@mid.gov.sb	timharris@gmail.com	IMO	Participant	
15	Jimmy Nuake	Acting Permanent Secretary	Ministry of Infrastructure Development	M	PO Box G8				Honiara	Solomon Islands	677 2865	677 7521819	Jnuake@mid.gov.sb	jnuake@gmail.com	IMO	Participant	
16	James Alfred Gerea	Training Manager	Solomon Islands Ports Authority	M	PO Box 307				Honiara	Solomon Islands	677 22646	677 7245370	j.gerea@sipa.com.sb		IMO	Participant	
17	Mosese Lavemai	Chief Executive Officer	Ports Authority Tonga	M	PO Box 144				Nuku'alofa	Tonga	676 23168	676 8447005	mlavemai@portsauthoritytonga.com		IMO	Participant	
18	Falasese Tupau	Assistant Secretary	Ministry of Communication and Transport	M	Government Building, Vaiaku				Funafuti	Tuvalu	688 20055	688 7002926	ftupau@gov.tv	falasese@gmail.com	IMO	Participant	
19	Salolo Tepoga	Assistant Manager	Ministry of Communication and Transport	M					Funafuti	Tuvalu	688 20055	688 7005419	siaositepoga@gmail.com	siaositepoga@gmail.com	IMO	Participant	
20	Lloyd Maiki Fikiasi	Office of Maritime Regulator	Legal Officer	M	PO Box 6499				Port Vila	Vanuatu	678 35455	678 7753473	lfikiasi@omr.vu		IMO	Participant	
21	Henry Worek	Director	Ports Marine and Department	M	PMB 9046				Port Vila	Vanuatu	678 33600	678 7743972	hworek@vanuatu.gov.vu	henzyb@gmail.com	IMO	Participant	
22	Adrian Nicolae	Team Leader Climate Chan	EU Pacific	M	Private Mail Bag, GPO				Suva	Fiji	3313633	9484486	Adrian.nicolae@eeas.europa.eu		EU Pacific	Participant	
23	Sylvain Rabeau	Deputy Director	New Caledonia Maritime Affairs	M					Noumea	New Caledonia	687270692	687787440	sylvain.rabeau@gouv.nc	sylvain.rabeau@hotmail.com	SPC	Participant	
24	Mapu Jnr Tapuai	Transport Operations Manager	Government of Tokelau Transport Support Services	M	PO Box 3298				Apia	Samoa	685 20822	685 7774971	mapu.tapuai@tokelau.org.nz	jtapuai@yahoo.con.nz	SPC	Participant	
25	Annalisse Sly	Acting Manager, Environment Standards	Australian Maritime Safety Authority	F	GPO Box 2181				Canberra	Australia	61 2 62795040	61 410755481	annalisse.sly@amsa.gov.au	maggie_sly@hotmail	AMSA	Participant	
26	Nanise Kabakoro	Trainee Port Engineer	Fiji Ports Authority	F	GPO Box 780				Suva	Fiji	679 3279203	679 9908598	nanise@fijiports.com	nkabakoro@gmail.com	In-country	Participant	

27	Lopeti Radravu	Operations Manager	Fiji Ships and Heavy Industries Limited	M	Lot 1 Eliza St WBay Suva, Fiji				Suva	Fiji	679 3306229	679 9907176	lopeti@fijiports.com.fj		In-country	Participant	
28	Scott Hook	Resilience Team Leader	Pacifc Islands Forum Secretariat	M	PMB				Suva	Fiji Islands	679 3312600		scotth@forumsec.org		PIFS	Participant	
29	Teea Tira	Climate Change Coordinator	Pacifc Islands Forum Secretariat	F	PMB				Suva	Fiji Islands	679 3312600		teeat@forumsec.org		PIFS	Participant	
30	Johan Fred Amoa	President	Pacific International Maritime Law Association	M	PO Box 2814				Apia	Samoa	685 20014	685 7634102	fred.amoa302@gmail.com		PIMLA	Participant	
31	Tafaigata Toilolo	Port Operations Manager	Samoa Port Authority	M	PO Box 2279				Apia	Samoa	685 770691				Ports Samoa	Participant	
32	Eranda Kotelawala	CEO	Solomon Islands Ports Authority	M					Honiara	Solomon Islands			ekotelawala@sipa.com.sb		Ports Solomon Islands	Participant	
33	Anthony Talouli	Pollution adviser	Secretariat of the Pacific Regional Environment Programme	M	SPREP, Vailima				Apia	Samoa	679 7806000		anthonyt@sprep.org		SPREP	Participant	
34	Philip Colin	President	Uto ni Yalo	M	5 Maafu Street				Suva	Fiji		679 9992340			In-country	Participant	
35	Monifa Fiu	Vice President	Uto ni Yalo	F	6 Maafu Street				Suva	Fiji		679 9965663	monifafiu@gmail.com		In-country	Participant	
36	Richard Coleman	CEO	Vanuatu Maritime College	M	PO Box 201				Luganville	Vanuatu		678 36547	marico@vanuatu.com.vu		SPC	Participant	
37	Alo ki Hakau Maileseni	Chief Financial Officer	Ports Authority Tonga	M	PO Box 144				Nuku'alofa	Tonga		676 23168	amaileseni@portsauthoritytonga.com		Ports Tonga	Participant	
38	Tevita Kaufusi	Manager Infrastructure & Technical	Ports Authority Tonga	M	PO Box 144				Nuku'alofa	Tonga		676 23168	portsauthority.tonga.com		Ports Tonga	Participant	
39	Sione Akauola	CEO	Ministry of Public Enterprises	M	PO Box 19				Nuku'alofa	Tonga		676 28144	sakauola@mpe.gov.to	folaosi@gmail.com	Ports Tonga	Participant	
40	Lionel Loubersac	Deputy Director	New Caledonia Maritime Affairs	M					Noumea	New Caledonia	687 763208		lionel.loubersac@outlook.fr		SPC	Participant	
41	Mark Borg	Team Leader Programme Management	Pacific Islands Development Forum	M	56 Domain Road				Suva	Fiji		9907633	markborg@pacificidf.org		PIDF	Participant	
42	Derrick Armstrong	Deputy Vice-Chancellor	USP	M									derrick.armstrong@usp.ac.fj		USP	Participant	
43	Josateki Tagi	Fleet Superintendent/Capt	Billett Wright & Associates	M	P O Box 3092, Lami						3300711	7776919/ 7776918/ 7776920	billett@unwired.com.fj	josatagi@yahoo.com	In-country	Participant	
44	Alexander Beetz	Project Manager	GIZ	M											GIZ	Participant	
45	Peia Patai	Project Manager	Vaka	M									peia.patai@okeanos-foundaion		Okeanos	Participant	
46	Leatuasa Sakaraia	Ship Surveyor	SSC	M					Apia	Samoa							
47	Simon Bonnett	General Manager - Sustain	Swire Shipping	M					Suva/Singapoer	Suva/Singapore	65 6632 0310	65 9643 3482	simon.bennett@swire.com.sg		Self-funded	Participant	
48	Mary Narfi CAPTAIN/USP																
49																	

									RESOURCE									
	Anton Rhodes	Project Officer	IMO	M	4 Albert Embarkment				London, SE17SR	UK			AntonR@imo.org					
	Aubrey Botsford	Deputy Director/Chief	IMO	M	4 Albert Embarkment				London, SE17SR	UK			AubreyB@imo.org					
	Jieying Xie	MTCC-Asia	Shanghai Maritime Univeristy	F	Room 401, No. 110, Lane 366				Shanghai	China								
	Robin Leigh Mazany	Consultant	IMO	F					Ottawa	Canada			leigh.mazany@yahoo.ca					
	Bruce Rowse	Consutant	SPC	M									bruce@8020green.com					
	Thierry Nervale	Deputy Director Transport	SPC	M	Lotus Building	Nabua			Suva	Fiji			thierryn@spc.int					
	Mark Davies	Transport Greenhouse Gas	SPC	M	Lotus Building	Nabua			Suva	Fiji			markd@spc.int					
	Mohammed Zullah	Maritime Industry Energy Efficiency Officer	SPC	M	Lotus Building	Nabua			Suva	Fiji			zullahm@spc.int					
	Francesca Pradelli	Policy and Legal Officer	SPC	F	Lotus Building	Nabua			Suva	Fiji			francescap@spc.int					
	Lore Coker	Admin & Comms. Assistant	SPC	F	Lotus Building	Nabua			Suva	Fiji			lorec@spc.int					
	Bernedine Managreve	Admin. Assistant	SPC	F	Lotus Building	Nabua			Suva	Fiji			bernedinem@spc.int					

ANNEX 2: Agenda

Maritime Technology Cooperation Centre - Pacific
Pacific Community, Suva Regional Office
Private Mail Bag, Suva, Fiji
Tel: (679) 337 0733 | Fax: (679) 337 01 46
Email: mtcc-pacific@spc.int | Web: www.spc.int

WORKSHOP FOR HEADS OF MARITIME
Suva, Fiji, 12 December 2017

PROGRAMME

Workshop for Heads of Maritime
All partners can attend as observers

WORKSHOP FOR HEADS OF MARITIME			
0830 – 0900	**Registration**		
0900 – 0910		Administrative remarks	MTCC-Pacific
		Prayer	Country
0910 – 0920		Welcome address	Fiji
0920 – 0925		Opening address	MTCC-Pacific
0925 – 0940	**1**	Introduction to GMN project and MTCC-Pacific	IMO MTCC-Pacific
0940 – 1000	**2**	National technology needs, barriers and required relevant action for a Pacific low-carbon maritime transport	MTCC-Pacific
1000 – 1030	**Photo Session and Morning Tea**		
	Facilitated by: SPREP		
1030 – 1100	**3**	New technologies and operations to meet the requirements of improved efficiency and reduced GHG emissions on small vessels	IMO
1100 – 1130	**4**	Energy Management in ports: methods and tools	8020Green
1130 – 1230	**Discussion** *A regional action plan to implement the 2017 Ministers' Communique*		All
1230 – 1330	**Lunch**		

Project funded by the EUROPEAN UNION
Host institutions of MTCC-Pacific

This document was produced for approval by IMO. It was prepared by MTCC-Pacific for the GMN Project funded by the European Union and implemented by IMO

SECOND MTCC-PACIFIC STEERING COMMITTEE MEETING
Suva, Fiji, 12 December 2017

PROGRAMME

Only partners and countries members of the MTCC-Pacific Steering Committee to attend as member:

- *Fiji Ministry of Infrastructure and Transport (Chair);*
- *Pacific Community (SPC);*
- *Secretariat of the Pacific Regional Environment Programme (SPREP);*
- *Fiji Delegation of the European Union*
- *Maritime Safety Authority of Fiji (MSAF)*
- *Marshall Islands;*
- *Samoa;*
- *Solomon Islands;*
- *Fiji Maritime Academy (FMA);*
- *Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH;*
- *Pacific Islands Development Forum (PIDF);*
- *Pacific Islands Forum Secretariat (PIFS);*
- *University of the South Pacific (USP);*
- *Private sector shipping representative.*

Other countries can attend as observers

SECOND MTCC-PACIFIC STEERING COMMITTEE MEETING			
1300 – 1330	**Registration**		
1330 – 1400		Administrative remarks	MTCC-Pacific
		Prayer	Country
		Welcome address	Chair
	1	Adoption of the 2nd Steering Committee Meeting agenda	Chair
1400 – 1430	**2**	IMO MEPC work progress	IMO
1430 – 1500	**3**	Update on side-events at COP23 related to shipping	Partners
1500 – 1530	**Tea/Coffee Break – Group Photo**		
1530 – 1600	**4**	MTCC-Pacific Progress Report	MTCC-Pacific
1600 – 1630	**5**	Other business	All

MTCC-PACIFIC REGIONAL CONFERENCE AND OFFICIAL LAUNCH
Suva, Fiji, 13-15 December 2017

PROGRAMME

Day 1: Wednesday, 13 December 2017			
0830 – 0900	**Registration**		
0900 – 0930		Discussion on relevant actions to implement the Transport Ministers' Resolution and Communiqué	All
0930 – 1000		Global MTCCs Network (GMN)	IMO
1000 – 1030	**Morning Tea**		
1030 – 1130	**1**	Official Launch of MTCC-Pacific *See separate programme for the Welcome Ceremony and Official Launch of MTCC-Pacific*	
1130	**Group Photo**		
1200 – 1330	**Lunch**		
	Session 1: Pacific Islands Countries and Territories perspectives *Facilitated by:* USP		
1330 – 1400	**2**	Micronesia Center for Sustainable Transport: Framework and approach	RMI/USP
1400 – 1500	**3**	Countries presentations (max 10 min each) *Cook Islands; Fiji; French Polynesia; Kiribati; Nauru; New Caledonia*	Countries
1500 – 1530	**Afternoon Tea**		
1530 – 1630	**4**	Countries presentations (max 10 min each) *Papua New Guinea; Samoa; Solomon Islands; Tonga; Tuvalu; Vanuatu*	Countries
1800 – 2100	Welcome Cocktail (Novotel Lami)		
Day 2: Thursday, 14 December 2017			
	Session 2: Partners perspectives *Facilitated by:* PIDF		
0900 – 0940	**5**	A comprehensive IMO strategy on reduction of GHG emissions from ships	IMO
0940 – 1000	**6**	Project for Transitioning to Low Carbon Sea Transport in Marshall Islands	GIZ
1000 – 1030	**Morning Tea**		
1030 – 1045	**7**	Reducing GHG form domestic shipping: shipowners perspective	Fiji Shipowners Representative
1045 – 1100	**8**	Reducing GHG from port operations: port operators perspective	Ports representative
1100 – 1130		**Discussion**	
1130 – 1145	**9**	New Caledonia Towards Sustainable Blue Growth	Cluster Maritime Nouvelle-Caledonie
1145 – 1200	**10**	A concept for inter-island sustainable sea transport in the Pacific	Okeanos

1200 – 1215	**11**	Reviving our Sustainable Sea Transportation culture in Fiji	Uto ni Yalo Trust
1215 – 1230		**Discussion**	
1230 – 1330	**Lunch**		
	Session 3: MTCC-Pacific work *Facilitated by:* Fiji		
1330 – 1400	**12**	MTCC-Pacific progress and way forward	MTCC-Pacific
1400 – 1430	**13**	Energy management in ports: activities and results	8020Green
1430 – 1500	**14**	Presentation of MTCC-Asia work	MTCC-Asia
1500 – 1530	**Afternoon Tea**		
1530 – 1545	**15**	MTCC-Pacific Pilot-Projects	MTCC-Pacific
1545 – 1630	**Discussion** *Countries engagement with MTCC-Pacific: data collection & energy management measures*		
Day 3: Friday 15 December 2017			
0900 – 1000	**16**	Adoption of a Regional Action Plan to implement the Transport Ministers' Resolution and Communiqué	All
1000 – 1030	**Morning Tea**		
1030 – 1115	**17**	Review of the Regional Conference Outcome	All
1115 – 1130	**Break – Evaluation form**		
1130 – 1145		Adoption of the Regional Conference outcome	All
1145 – 1200		Certificate of attendance	All
1200 – 1210		Closing remarks	IMO
1210 – 1220		Closing remarks	MTCC-Pacific
1220 – 1230		Closing remarks	Fiji
1230 – 1330	**Lunch**		

AGENDA

12th December 2017, Suva, Fiji

SIDE EVENT FOR WOMEN IN MARITIME			
0830 – 0900	Registration		
0900 – 0910		Administrative remarks	SPC GEM Division - Maritime Programme, Ore Toua
		Prayer	Country
0910 – 0920		Welcome & Opening address	SPC DDG Suva, Dr. Audrey Aumua
0920 – 0940	1	Social Development and Gender Lens into sectoral fields, maritime programmes and projects	SPC Social Development Programme - Brigitte Leduc, Social Development Adviser (Gender Equality)
0940-1000	2	The role of maritime transport from the perspective of energy and gender: The case of the Pacific Islands	SPC GEM Division - Maritime Programme, Ore Toua
1000 – 1030	Photo Session and Morning Tea		
1030 – 1045	3	Women in Ocean and Coastal Geoscience	SPC GEM Division – Maritime & Ocean Programme – Judith Giblin, Senior Technical Officer, Oceanography
1045-1100	4	Project Proposal – Women in Maritime Programme	SPC GEM Division - Maritime Programme, Ore Toua
1100-1115	5	Experiences of Women in Maritime (National Level)	Representative - Fiji Women in Maritime Association
1115-1130	6	Role of Pacific States in decarbonisation of the international shipping sector	Avnita Goundar, PhD in Marine Affairs (Candidate) School of Marine Studies, USP
1130-1145	7	Regional Strategy for Pacific Women in Maritime	Mavis Joseph Logavatu PacWIMA Chairperson
1145-1215	**Open Discussion**		All
1215-1230	**Summary & Conclusion**		SPC GEM Division - Maritime Programme, Ore Toua
1230 – 1330	Lunch		

ANNEX 3: Outcomes Document

FIRST REGIONAL CONFERENCE AND OFFICIAL LAUNCH

OF THE MARITIME TECHNOLOGY COOPERATION CENTRE IN THE PACIFIC

Suva, Fiji, 13-15 December 2017

OUTCOME

1. The First Regional Conference of the Maritime Technology Cooperation Centre in the Pacific (MTCC-Pacific) was held in Suva, Fiji from 12 to 15 December 2017 at the Novotel Conference Centre in Lami. The meeting was organised and hosted by the MTCC-Pacific with the support of the Pacific Community (SPC), its Host Institution, and the International Maritime Organization (IMO). The Conference was co-hosted by Fiji and attended by Permanent Secretaries, Secretaries, Deputy Secretaries, Directors and Officers from Cook Islands, Kiribati, Marshall Islands, Nauru, New Caledonia, Papua New Guinea, Samoa, Solomon Islands, Tokelau, Tonga, Tuvalu and Vanuatu. International and regional partners also attended such as IMO, the European Union, the Secretariat of the Pacific Regional Environment Programme (SPREP), the Pacific Islands Development Forum (PIDF), the Australia Maritime Safety Agency (AMSA), MTCC-Asia, representatives from Pacific Ports, the Pacific Islands Maritime Conference (PIMC), representative from Fiji shipowners, Okeanos Foundation and the Uto Ni Yalo Trust.

2. The objective of the Regional Conference was to officially launch MTCC-Pacific as a centre of excellence for the region with the view to build the capacity of Pacific Small Islands Developing States (SIDS) and Least Developed Countries (LDCs) for climate mitigation in the maritime industry. The Conference was also the opportunity to showcase the Global MTCCs Network (GMN), a project implemented by IMO and funded by the European Union that aims to establish five MTCCs in Africa, Asia, Caribbean, Latin America and the Pacific regions. The Conference gathered representatives from Pacific Islands Countries and Territories (PICTs) and partners to inform them of the activities of the MTCC-Pacific and to agree on actions to give effect to the resolution of the 2017 Pacific Regional Energy and Transport Ministers' Meeting to transition to low-carbon maritime transport in the Pacific.

3. The conference expressed its deepest appreciation to the Government and the people of Fiji, SPC, SPREP and IMO for the excellent arrangements made in funding and organising the conference including the preceding meetings for Heads of Maritime, Pacific Women in Maritime, the Pacific Islands Maritime Conference and the MTCC-Pacific Steering Committee. It also acknowledged the contribution and support from the International Union for Conservation of Nature (IUCN), the Port Authority of Tonga (PAT), the Solomon Islands Port Authority (SIPA) and Fiji Ports Corporation Limited (FPCL). The Conference further expressed its appreciation to the Fiji Deputy Secretary Policy and Planning, Lui Naisara, for his leadership in chairing the Conference and to the facilitators from PIDF, SPREP and the Maritime Safety Authority of Fiji (MSAF).

The Conference:

4. **Applauded** the launch of the MTCC-Pacific in the presence of the Minister for Forests and Acting Minister for Fisheries, Honourable Osea Naiqamu; the Ambassador of the European Union to Fiji and the Pacific, His Excellency Julian Wilson; the SPC's Deputy Director-General, Dr. Audrey Aumua; and the IMO Deputy Director, Office of General Services, Mr. Aubrey Botsford.

5. **Acknowledged** the 2017 Energy and Transport Communiqué and Resolutions that set out the direction to transition towards low-carbon maritime transport and that MTCC-Pacific provides an effective platform for the implementation of a regional climate mitigation strategy for Pacific maritime transport.

6. **Recognised** that the development and implementation of a comprehensive IMO strategy with ambitious targets to reduce greenhouse gas emissions from shipping will create an enabling environment for capacity development, technical cooperation and the uptake of new technologies and operations in maritime transport in the future.

7. **Noted** existing initiatives, programmes and projects related to low-carbon maritime transport in response to the call of Energy and Transport Ministers who urged all stakeholders to take appropriate action to progress low-carbon maritime transport in the Pacific Islands region.

8. **Supported** MTCC-Pacific and GMN approach and urged PICTs to become involved in MTCC-Pacific pilot-projects or similar projects on energy efficiency and data collection.

9. **Acknowledged** the important role of industry and business in the maritime sector and their challenges to invest in new technology and operations to transition towards low-carbon maritime transport in the Pacific. Accordingly, it urged governments to create an enabling environment for business investment.

10. **Agreed** that solutions to transition to low-carbon futures in maritime transport will require infrastructure development and a combination of technical and operational measures, including options such as traditional navigation for inter-island mobility of our people and goods in harmony with culture and heritage.

11. **Emphasised** the need for collaboration, cooperation and partnerships between all stakeholders from the international to regional, national and community levels to ensure adapted solutions and services are provided to countries.

12. **Supported** the outcome of the meeting for Pacific Women in Maritime held on 12 December and agreed to address gender issues through mainstreaming gender in maritime policies, programmes and project activities, including maritime energy, that provide new opportunities to advance and empower women in the maritime sector.

13. **Adopted** the immediate priority actions shown in Appendix 1 to ensure critical issues related to capacity building, collection of data and information, implementation of pilot-projects and policy and legal frameworks are addressed as first priorities.

14. **Recommended** the implementation of the relevant actions shown in Appendix 2 in response to the needs and barriers identified in various fora. These actions constitute the basis for a regional action plan with the view to develop a future Regional Strategy for the Pacific region.

Appendix 1 – Priority areas and actions

Priority areas	Priority actions[1]
Capacity Building – there is a need to build the capacity of governments and maritime administrations related to policy, legal and regulatory framework and also the private sector to raise awareness and train on new technologies and operations and the need for quality, relevant data.	a. Provide short training courses and vocational trainings on energy efficient operation of ships and new technologies for the maritime industry and crews (R23, R45, R57 and R58); b. Train maritime personnel for the implementation of energy efficient measures in the maritime industry (R24); c. Government support for training, improved infrastructure including dry-docks, technology transfer and competiveness (R46).
Collection of data and information – there is a need to develop awareness and capacity to collect relevant information and data and to make data accessible to measure progress.	d. Enact laws to support data collection and sharing domestically and with the IMO; e. Adopt data collection methods and systems to establish the country GHG emissions status and identify gaps (R10); f. Collect, analyse and communicate data related to GHG emissions from the maritime sector to raise awareness on targets for domestic shipping and ports (R37).
Pilot-projects – "proof-of-concept" pilot-projects are essential to show results and replicate across the region.	g. Conduct annual energy audits and implement short-term energy savings projects in maritime transport (ships & ports) (R30); h. Develop and implement pilot-projects and adapted tools/methods on the uptake of new technologies and operations in maritime transport (R25 and R26); i. Assess feasibility of Onshore Power Supply (OPS) in Pacific ports (R34); j. Conduct cost-benefit analysis for new technologies on board small vessels (less than 50 meters) (R44); k. Provide media updates on results from success stories (R51); l. Implement vessel replacement programmes and major green infrastructure projects in ports (Long-term >10 years) (R29).
Policy/Legal Frameworks –policy and legal frameworks need to be reviewed to integrate energy efficiency related targets and standards and to be effectively implemented.	m. Review/draft generic laws to address energy efficiency in maritime transport aligned with international mandatory instruments, recommendations and guidelines (R7) n. Ratify international maritime instruments related to energy efficiency and emissions (R8) o. Draft instructions to maritime administrations for the implementation of energy efficiency laws (R9) p. Adopt laws including target for the reduction of GHG emissions (R41).

[1] In 2018-2019, MTCC-Pacific can assist with actions a, b, f, g and h; especially in MTCC-Pacific priority countries.

Appendix 2 – Matrix of drivers, needs, barriers and recommended relevant actions

Theme	Drivers	Needs	Barriers	Relevant actions
PICTs perspectives				
International Framework	• International framework and negotiations on climate change • Participation/contribution to international negotiations to defend SIDS/LDCs special circumstances	• Coordinate efforts for a continued and strong engagement of the PICTs with a unified voice in the international negotiations and to raise the special circumstances of the Pacific SIDS and LDCs	• Lack of resources to attend international meetings • Lack of expertise and coordination to prepare, submit and support submissions/side-vents to international meetings	• Raise funding support to ensure PICTs representation at international meetings • Put in place coordination mechanisms with PICTs and relevant regional partners to prepare submissions
Recommendations	**Short-term (<5 years)** R1. **Implement the 2017 Transport Ministers' Resolution with mechanism to facilitate communication between all stakeholders to coordinate the drafting and submission of cosponsored regional documents to IMO** R2. **Coordinate the regular submission of cosponsored regional documents to IMO** R3. **Secure external funds to attend international meetings regularly**		**Medium-term (5-10 years)** R4. **Secure internal funds to attend international meetings regularly** R5. **Identify and train suitable maritime personnel to attend relevant international meetings related to maritime issues**	**Long-term (>10 years)** R6. **Secure long-term representation (permanent or occasional) at relevant international maritime organisations**
National Framework	• National initiatives/plans e.g. Green Growth frameworks, GHG emissions targets aligned to international obligations and supported by regional frameworks and declarations including 2013 Majuro Declaration, FRDP Goal 2 on low carbon development, FATS Theme 5	• Have specific objectives in policies and develop National Strategies on energy efficiency in maritime transport • Analysis of PICTs' Nationally Determined Contributions (NDCs) for maritime transport • Implement relevant international maritime instruments related to the reduction of GHG emissions	• Absence or lack of reliable data regarding GHG emissions by sectors including air, land and maritime transport • Absence of specific targets for reducing GHG emissions from maritime transport to guide efforts and initiatives • Absence of specific strategy or action plan to address GHG emissions from maritime transport aligned with the sectors' contribution to the country's GHG emissions	• Collect and analyse relevant data related to GHG emissions by sectors for informed decision-making and adoption of relevant indicators and targets • Develop specific national strategies and/or action plans aligned with the sectors' contribution to the country's GHG emissions • Development regional and national initiatives that integrate the special circumstances related to maritime transport in each country and align with national priorities and regional frameworks

		• Collect and share reliable, accurate and quality maritime transport oriented data • Develop and implement initiatives in line with regional frameworks to ensure consistent and coordinated approach for better impact • Develop enabling policies, laws, action plans and procedures for reduced GHG emissions and energy efficiency in the Pacific maritime industry	• International maritime instruments related to GHG emissions not ratified • Lack of policies, laws, action plans and procedures creating an enabling environment for reduced GHG emissions and energy management in the maritime industry	• Ratify international maritime instruments related to GHG emissions • Development of policy and legal frameworks implemented by adapted procedures for reduced GHG emissions and energy management in the Pacific maritime industry
Recommendations	**Short-term (<5 years)** R7. **Review/draft generic laws to address energy efficiency in maritime transport aligned with international mandatory instruments, recommendations and guidelines** R8. **Ratify international maritime instruments related to energy efficiency and emissions** R9. **Draft instructions to maritime administrations for the implementation of energy efficiency laws** R10. **Adopt data collection methods and systems to establish the country GHG emissions status and identify gaps** R11. **Review and adopt country NDCs targets desegregated by sector including maritime transport based on relevant data related to GHG emissions from the Pacific maritime transport** R12. **Incorporate in National Maritime Policies objectives on energy efficiency and GHG emissions** R13. **Develop/review National Strategies on energy efficiency and GHG emissions** R14. **Develop and adopt a Regional Strategy for low-carbon maritime transport in the Pacific**		**Medium-term (5-10 years)** R15. **Measure progress and review country NDCs targets for maritime transport** R16. **Assess implementation of laws and amend laws to incorporate new obligations** R17. **Assess implementation of National Strategies and review targets**	**Long-term (>10 years)** R18. **Assess progress in reduction of GHG emissions over the period of policy/strategy implementation and identify best impact projects**

Climate Financing	• Climate financing and available expertise with international and regional partners	• Make use of climate financing opportunities and available expertise among international and regional partners • Coordination among countries and regional partners	• Absence or lack of a formally established network gathering international and regional partners and PICTs • Lack of funding to upscale existing projects and initiatives • Lack of coordination to develop initiatives that can access climate financing mechanisms	• Establish a platform for networking, coordination and information sharing
Recommendations	**Short-term (<5 years)** R19. **Develop proposals to submit to climate funds and frameworks – proposals to be on new technologies but also capacity development and development of policy and legal frameworks** R20. **Map out climate financing mechanisms and donors** R21. **Develop the capacity of country to access climate financing and country financial processes and systems**		**Medium-term (5-10 years)** R22. **Enhance country financial systems for the implementation of major climate projects**	**Long-term (>10 years)**
Pilot-projects	• Lead by example to reduce GHG emissions from maritime transport in the Pacific under regional frameworks	• Reduced fuel oil consumption from domestic ships and energy consumption in ports • Proof-of-concept initiatives • Adapted tools and small-scale projects • Enhance PICTs capacity to implement and verify compliance with international regulations	• Very old domestic fleet with some vessels engines designed before the fuel crisis of 1973 • Small port with limited financial resources • Poor understanding and management of energy efficient operations on-board domestic vessels and in ports – change in personnel behaviour • Lack of expertise and capacity to use new technologies • Lack of expertise and adapted tools and technical solutions readily available to improve energy management in the Pacific maritime industry • Lack of capacity and expertise in maritime administrations to implement effective	• Reduction in GHG as a long term exercise: ○ Raise awareness and develop capacity of ship and port operators on new technologies and operations ○ Cost-benefit analysis for replacing/retrofitting domestic vessels ○ Finance (loan facility, incentives, etc.) ○ Awareness and capacity building activities on Ship Energy Efficiency Management Plan (SEEMP) ○ Implement projects that includes capacity development activities and proof-of-concept initiatives ○ Develop adapted tools and small-scale projects to introduce energy management, improve quality management and implement energy audit in the maritime industry

			Flag State Implementation (FSI) and Port State Control (PSC) regimes	○ Implement research and economic, policy, safety, technical and legal analysis on new technologies adapted to the Pacific maritime industry • Develop FSI and PSC regimes: ○ Train marine surveyors and inspectors to carry out FSI and PSC on board all type of vessels including requirements for energy efficiency ○ Establish a platform for experience sharing
Recommendations	**Short-term (<5 years)** R23. **Develop vocational trainings and qualifications for energy efficient transport and new technologies** R24. **Train maritime personnel for the implementation of energy efficient measures in the maritime industry** R25. **Implement pilot-projects on the uptake of new technologies and operations in maritime transport** R26. **Develop adapted tools and small-scale projects for energy management in maritime** R27. **Facilitate research programmes in the Pacific**		**Medium-term (5-10 years)** R28. **Implement green port and shipping programmes**	**Long-term (>10 years)** R29. **Implement vessel replacement programmes and major green infrastructure projects in ports**
Pacific Maritime Industry perspectives				
Costs	• Costs of energy	• Reduce energy consumption in ports • Reduced fuel oil consumption from domestic ships • Cold ironing/Onshore Power Supply (OPS) to save energy/costs • Implement Energy Management Systems in shipping companies to improve practices	• Costs of implementation of energy consumption reduction projects • Lack of government support (most ports are SoEs) • Transition costs for additional or change of equipment	• Conduct energy audit to determine energy consumption reduction project technical feasibility, return on investment, benefit for reducing GHG emissions • Infrastructure development in ports to provide onshore power supply including dry-docks – support from Government • Implement SMS/MMS including energy management with the support of SPC (PIDSS and MTCC-Pacific)

Recommendations	**Short-term (<5 years)** R30. **Conduct annual energy audits and implement short-term energy savings projects in maritime transport** R31. **Develop integrated solutions for green port/green shipping/green supply chain** R32. **Implement existing tools such as Ship Energy Management Plan to improve energy efficiency and start plans for long-term alternatives through vessels replacement programme** R33. **Assess feasibility of Onshore Power Supply (OPS) in Pacific ports**		**Medium-term (5-10 years)** R34. **Assess costs-benefit of long-term energy savings project in maritime transport involving hard infrastructure development such as alternative fuels, marine energy**	**Long-term (>10 years)** R35. **Invest in green technologies for the maritime industry** R36. **Implement vessels replacement programmes**
National Framework	• National initiatives/plans e.g. Green Growth frameworks, GHG emissions targets for environmental protection	• Integrate national targets in port management • Collect data for baseline to show reduction in fuel consumption • Consistent approach for environment protection in domestic shipping and ports	• Resistance to change • Political priority changes • No transparency in the use of data collected to improve systems – lack of open and timely process	• Awareness on national targets • Awareness and capacity building for the implementation of innovative technical solutions • Consistent energy management with indicators and targets for ports • Collect data and provide accessibility to data to use in fuel savings measures through a Public Website with dedicated independent data • Consistent approach in domestic shipping and port for environment protection that integrate pollution prevention, reducing GHG, water quality • Implementation of Green Maritime Industry initiatives and incentives including Green Ports and Green Shipping concepts
Recommendations	**Short-term (<5 years)** R37. **Collect, analyse and communicate data related to GHG emissions from the maritime sector to raise awareness on targets for domestic shipping and ports** R38. **Review/develop policies and laws including short-, medium- and long-term target for domestic shipping and ports.**		**Medium-term (5-10 years)** R40. **Assess achievement of target and adopt/adjust target** R41. **Adopt laws including target for the reduction of GHG emissions**	**Long-term (>10 years)** R42. **Assess effective implementation and adaptation of policies and laws by the maritime industry**

	R39. **Improve awareness on national target and contribution of the maritime sector.**			
Profitability	• Improve profitability of ships and reliability and efficiency of domestic shipping • Competitive advantage of ports	• Maintain competitive advantage and increase port profitability • Keep costs of energy at the lowest possible • Domestic shipping arrangements such as franchise scheme to support energy efficiency of domestic shipping • Reduced competition that can compromise safety, efficiency and reliability of shipping services provided • Appropriate/relevant legal, regulatory and technical measures adapted to the size of the vessels and the capacity and resources of shipowners • Training on safety, efficiency including energy efficiency, etc. • Reduce lost time for berthing	• Available budget against dividend claimed by government and shareholders • Costs of implementation of energy consumption reduction projects • Political interference and priority changes • Population demand for direct service instead of shared service with another Province • Political interference to serve provinces with individual ships creating over-tonnaging, freight wars and excessive demand on domestic wharves • Overall economic in-efficiency • Over-regulation and not locally adapted measures that are too prescriptive • Lack of crew qualification • Measures are often adapted to vessels of more than 50 meters while most of domestic vessels are under this size and old • Lack of infrastructure in outer islands to accommodate domestic vessels	• Cost benefit analysis associated to energy audits to show long-term benefits to all stakeholders including economic benefits of government and shareholders and well-being of people living around ports • Review Franchise Shipping Schemes efficiency • Organise liner service for a weekly regular round trip to service a group of main ports in outer islands • Government financial through subsidies and incentives • Waiver depending the size and age of vessels • Implementation of measures adapted to the Pacific domestic fleet • Control domestic fleet tonnaging and pre-inspection/limitation for vessels purchase overseas • Capacity building of ship operators and crews on energy efficiency measures and practises • Infrastructure development in outer islands to facilitate domestic shipping
Recommendations	**Short-term (< 5 years)** R43. **Review franchise schemes to support energy efficient and profitable shipping services to reduce costs, fuel consumption and GHG emissions and increase profitability** R44. **Conduct cost-benefit analysis for new technologies on board small vessels (less than 50 meters)** R45. **Train specifically crews on energy efficient operations of ships**		**Medium-term (5-10 years)** R48. **Adopt economic & transparent franchise scheme on non-economical routes conducive to energy efficiency and reduction of GHG emissions while increasing profitability** R49. **Create an enabling environment for affordable new technologies**	**Long-term (>10 years)** R50. **Green maritime infrastructure in outer islands in place**

	R46. **Government support for training, improved infrastructure including dry-docks, technology transfer and competiveness.** R47. **Maritime infrastructure study made available for public submissions (open access).**		**and safety equipment for small ships**	
Reputation	• Good reputation	• Good reputation of ports leading to commercial attractiveness • Improve the reputation of shipping with regards to the protection of the environment and the emissions of GHG	• Lack of communication on efforts to reduce energy consumption and GHG in shipping and ports • Lack of cooperation between all shipowners due to high competition	• Awareness, storytelling on existing initiatives in ports to reduce energy consumption and GHG • ISO certification for energy management, quality management and environmental protection • Communicate efforts to implement energy efficient measure in Fiji domestic shipping • Revive the Fiji Ship Owners Association to support Green Shipping in Fiji
Recommendations	**Short-term (<5 years)** R51. **Provide media updates on results from success stories** R52. **Support Ship Owners Associations to promote Green Shipping programmes** R53. **Promote and advocate Green port approaches**		**Medium-term (5-10 years)** R54. **Awareness campaigns on Green Port and Shipping** R55. **Review/develop green awards and similar incentives in Port and for domestic shipping**	**Long-term (>10 years)** R56. **Promote green port and shipping governance**
Standards	• Improved standards of the maritime industry including safety, training, pollution prevention and energy efficiency	• Reduced competition that compromise safety, efficiency and reliability of shipping services • Financial support from governments to support new measures • Revive or develop capabilities in the PICTs to build and maintain new concept equipment and vessels • Compliance of ports with international requirements and best practises • Improve shore services and supply for shipboard electronic equipment and systems (radars, gyro-	• Lack of support to provide information and technical tools on energy efficiency • Lack of government support/incentives to revive/develop ship building and maintenance • Lack of expertise and capacity to implement international requirements • Lack of shore service and repair contractors • Lack of available electronic equipment and systems • Awareness and practices in communities that must travel between islands and use small outboard powered vessels	• Support from government through subsidies and tax incentives for safety, energy efficiency equipment, ship building and maintenance • Incentives for the establishment of competitive shore services for ships • Technology transfer • Reduce costs of electronic systems for ships through reduced taxes and duties • Availability of new equipment affordable to communities • Technical support and capacity building provided by MTCC-Pacific to implement adapted measures

		compasses, GPS, Radios, GMDSS, AIS, computerised systems, etc.) • Consider the inclusion of outboard small vessels in measures to reduce fuel consumption and GHG emissions taking into account the basic needs of communities mobility • Regular hull cleaning with adapted hull coating systems compliant with best standards and independent inspection	• Lack of dry-dock facility • Lack of infrastructure in outer islands to accommodate domestic vessels	• Infrastructure development in outer islands to facilitate domestic shipping and Pacific ports to provide onshore power supply • Capacity-building in ports on international requirements and best practises
Recommendations	**Short-term (<% years)** R57. **Develop vocational training for the maritime industry on sustainable shipping and green maritime infrastructure** R58. **Develop and implement short training courses for the maritime industry on energy efficiency standards and technologies** R59. **Assess feasibility of energy efficient technologies for small vessels** R60. **Review/adopt government incentives for safety, energy efficiency equipment, ship building and maintenance.** R61. **Review/adopt incentives to install minimum safety electronic equipment and energy efficient technologies.** R62. **Review/develop operational, energy and environmental systems and processes in ports aligned to quality standards and IAPH tools** R63. **Develop/review adapted quality systems on board vessels and in shipping companies including energy efficiency processes**		**Medium-term (5-10 years)** R64. **Revive local shipyards for new concept Implement certified quality standards for operational, energy and environmental systems and processes in ports** R65. **Review/develop green awards and similar incentives in Port** R66. **Implement quality standards on board ships for safety and energy efficiency that are regularly audited**	**Long-term (>10 years)** R67. **Implement sustainable sea transport and green maritime infrastructure education** R68. **Operate shipyards equipped for new concept vessels** R69. **Implement consistent green port governance based on quality standards and incentive to green shipping** R70. **Implement consistent green shipping management based on quality standards**

MEETING OF PACIFIC WOMEN IN MARITIME

Suva, Fiji 12th December 2017

Meeting Outcome

1. The meeting was held in Suva, Fiji on 12 December 2017. The meeting was organized by the Pacific Community (SPC) and attended by representatives from the Pacific Women in Maritime Association (PacWIMA) Executive Committee, representatives from the Fiji Women in Maritime Association, Vanuatu Women in Maritime Association, Solomon Islands Women in Maritime Association, University of South Pacific, Government representative from Ministry of Women, Children and Poverty Alleviation, Ministry of Infrastructure and Transport in Fiji, colleagues from SPC dealing with development programmes such as Energy, Ocean and Social Development, and observers from the Swedish Government. The meeting participants list is attached in Annex B.

2. The Deputy Director General, Suva, Dr. Audrey Aumua delivered the welcome and opening address, reiterating SPC's position in integrating SDG5s targets in its various business area programmes and projects. Further highlighting the campaign of 16 days of activism with a strong message "sexual harassment is not joke" encouraging representatives to reflect and bring back the message to their countries.

3. The meeting expressed its appreciation of the meeting to pull resources within the region to discuss enabling vectors to raise the profile of women in the various development sectors and sharing networks for opportunities to mainstream gender and social inclusiveness.

The meeting:

.1 **RECOGNISED** the role of women in the maritime sector, their barriers, their aspirations, the opportunities, the perceptions and proposed mainstreaming gender in policies, programmes and project activities as a mechanism to advance women in the maritime sector.

.2 **ACKNOWLEDGED** the joint Pacific Campaign of the 16 days of activism with the strong message of "Sexual Harassment is no Joke" and the importance to eliminate sexual harassment thereby a creating a safe working environment in the maritime sector.

.3 **RECOMMENDED the adoption of** measures to promote and guarantee that all places of learning and work related to maritime sector are safe places for women and young women of all diversities.

.4 **ACKNOWLEDGED** the progress made in the development of ***the Regional Strategy for the recognition, visibility and training of Pacific women in the maritime sector***, as a strategic means to increasing visibility, training and participation of pacific women in the maritime sector

.5 **ACKNOWLEDGED** the Outcomes of the 13th Triennial Conference of the Pacific Women and the 6th Meeting of the Ministers for Women held in Fiji from 2 to 5 October 2017, and supported the relevant recommendations on women's economic empowerment to advance women in maritime: -

a. Promote lifelong learning for all women, of all diversities, and encouraging and supporting young women to develop skills in science, technology, engineering, mathematics and economics, and gain qualifications that enhance their employment prospects.

b. Adopt measures to promote and guarantee that all places of learning and work are safe places for women, young women and girls of all diversities.

c. Remove biases and stereotypes in human resource policies, adopting practices to reduce gender gaps in hiring, promotion and pay, and establishing accountability mechanisms including appropriate protocols for pay equity processes, with objective criteria for initial pay and promotion, and regular reviews of pay equity.

d. Encourage women's participation and leadership in trades unions, organizations for informal workers, employers' and business organizations and professional associations.

e. Developing and implement legislative measures, policies and programmes to eliminate gender-based discrimination and violence and harmful gender norms in the workplace.

f. Establish high-level corporate leadership of gender equality to advance policies for addressing gender-based violence, discrimination and sexual harassment, and promote the appointment of women to leadership positions

g. Support the initiatives of CSOs to challenge harmful gender norms, gender-based violence and discriminatory practices, and to promote equality and the human rights of women and girls of all diversities

6. **AGREED** to inform the MTCC Meeting held in Suva, Fiji on 15th December 2017 about the outcome of today's side meeting of pacific women in maritime.

12 December 2017

ANNEX 4: Workshop Presentations

HEADS OF MARITIME

GMN | The Global MTCC Network
A global network for energy-efficient shipping

GMN introduction and status Update

Anton Rhodes| Suva, Fiji, Dec 2017

The Global MTCC Network (GMN) project
is funded by the European Union and implemented by IMO
The views expressed in this presentation
can in no way be taken to reflect the views of the European Union

GMN | The Global MTCC Network
A global network for energy-efficient shipping

IMO's path to promote tech. transfer and capacity building

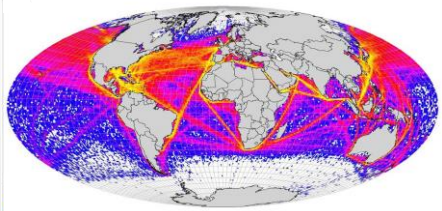
GMN Project

Res. 23 -MARPOL Annex VI (2013), MEPC Resolution on TT and CB,	ITCP: Awareness raising and capacity building tools	Major Projects: Capacity building & private sector partnerships	Global network to promote technology cooperation and transfer	Institutionalize capacity building and tech. transfer support at regional level for sustainable impact

IMO INTERNATIONAL MARITIME ORGANIZATION

Project funded by the EUROPEAN UNION

Initiation of a Network through GMN Project



- Focus on regions with large number of LDCs and SIDS (Africa, Asia, Caribbean, Latin America and Pacific)

GMN | The Global MTCC Network

5

GMN overview

- An initiative funded by EU and implemented by IMO
- Total Funding Euro 10,000,000
- Jan, 2016 – Dec, 2019 (4 Years)
- During the project, 5 MTCCs to become fully operational
- Beyond the project, the MTCCs will become financially self-sustained

Overall Objective:

- Build capacity – Improve energy efficiency – Reduce GHG emissions – Contribute to fight against climate change

6

GMN - Specific objectives

1. Establish 5 MTCCs
2. Improve human and instiutional capacities in maritime administrations and relevant stakeholders
3. Promote the uptake of low carbon technologies and operations
4. Establish pilot voluntary data collection and reporting systems
5. Communicate and disseminate information

GMN | The Global MTCC Network

Objective 1 - All MTCCs are established and operational

MTCC CARIBBEAN
Maritime Technology Cooperation Centre

University of Trinidad & Tobago

MTCC AFRICA
Maritime Technology Cooperation Centre

Jomo Kenyatta University of Agriculture & Technology, Kenya

MTCC ASIA
Maritime Technology Cooperation Centre

Shanghai Maritime university, China

MTCC PACIFIC
Maritime Technology Cooperation Centre

Pacific Community (SPC), Fiji

MTCC LATIN AMERICA
Maritime Technology Cooperation Centre

International Maritime University of Panama, Panama

GMN | The Global MTCC Network

11

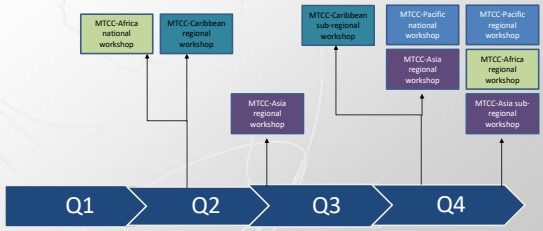
Launch of Global MTCCs Network

IMO HQ, London, 4 December

7

Objective 2 - Capacity building

MTCC capacity building activities in 2017



8

Objective 3 - Low Carbon technologies

- MTCC Africa - Establish on-shore power supply at Port of Mombasa (Cold Ironing)
- MTCC Asia - Update the ship trim optimization management system
- MTCC Caribbean - Capture the current baseline on the use of ship energy efficient technology within the region
- MTCC Pacific - Conduct energy audits in maritime ports
- MTCC Latin America - TBC

9

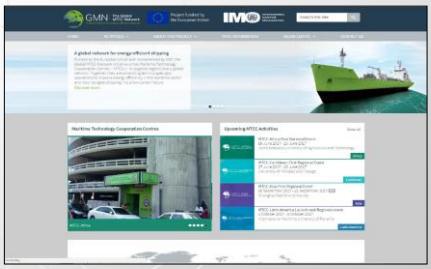
Objective 4 - Data collection

- **Caribbean** - collection of data
- **Pacific** – data templates developed
- **Africa** – Staff training on-board target vessels
- **Asia** – Data collection seminar – Role of key stakeholders confirmed

10

Objective 5 - Communication and dissemination

- Launch of GMN and MTCC websites



11

Objective 5 - Communication and dissemination

- GMN and MTCC newsletters, articles and social media

12

Objective 5 - Communication and dissemination

- International and regional events

13

Overview of GMN deliverables

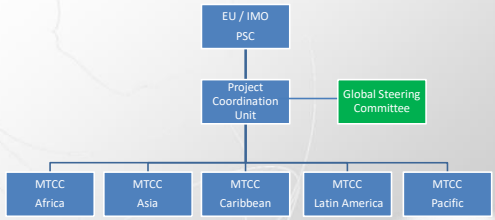
2016
- Established PCU
- Completed international tender process

2017
- MTCCs established and operational
- MTCC Launch Events and Regional Workshops
- Established MTCC network
- Established GSC

2018
- Develop pilot projects
- Deliver national and sub-regional workshops
- Develop sustainability plans

2019
- Complete Pilot Projects
- Deliver regional conferences
- Finalize sustainability plans

GMN MANAGEMENT STRUCTURE



GMN | The Global MTCC Network

15

Thank you for your attention!
http://gmn.imo.org/

GMN | The Global MTCC Network

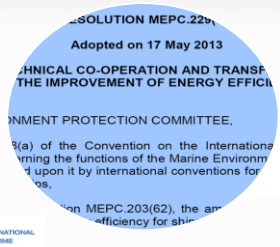
16

Roadmap for developing a comprehensive IMO Strategy on reduction Of GHG emissions from ships

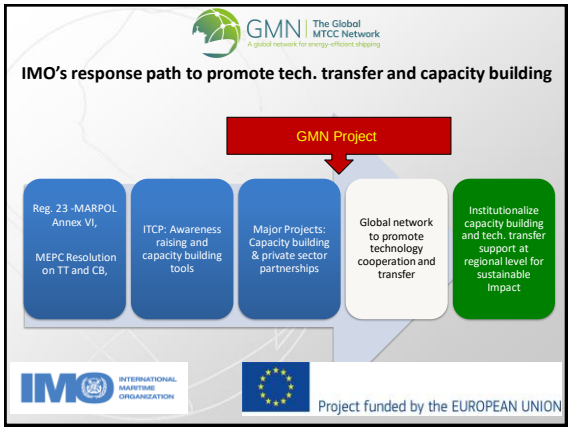
January 2019	- Start of Phase 1: Data collection (Ships to collect data)
Spring 2019 (MEPC 74)	- Discussion continues - Initiation of Fourth IMO GHG Study using data from 2012-2018
Summer 2020	- Data for 2019 to be reported to IMO
Autumn 2020 (MEPC 76)	- Start of Phase 2: data analysis (no later than autumn 2020) - Discussion continues - Publication of Fourth IMO GHG Study for consideration by MEPC 76[3]
Spring 2021 (MEPC 77)	- Initiation of work for adjustments on Initial IMO Strategy, based on DCS data - Secretariat report summarizing the 2019 data pursuant to regulation 22A.10
Summer 2021	- Data for 2020 to be reported to IMO
Spring 2022 (MEPC 78)	- Phase 3: Decision step - Discussion continues - Secretariat report summarizing the 2020 data pursuant to regulation 22A.10
Summer 2022	- Data for 2021 to be reported to IMO
Spring 2023 (MEPC 80)	- Adoption of Revised IMO Strategy, including short-, mid- and long-term further measure(s), as required, with implementation schedules - Secretariat report summarizing the 2021 data pursuant to regulation 22A.10

MARPOL Annex VI & Technical Cooperation

Regulation 23 of MARPOL Annex VI, Chapter 4: "Promotion of technical co-operation and transfer of technology relating to the improvement of energy efficiency of ships"



IMO INTERNATIONAL MARITIME ORGANIZATION


GMN
The Global MTCC Network
A global network for energy-efficient shipping
IMO's response path to promote tech. transfer and capacity building
GMN Project
Reg. 23 -MARPOL Annex VI,
MEPC Resolution on TT and CB,
ITCP: Awareness raising and capacity building tools
Major Projects: Capacity building & private sector partnerships
Global network to promote technology cooperation and transfer
Institutionalize capacity building and tech. transfer support at regional level for sustainable impact
IMO INTERNATIONAL MARITIME ORGANIZATION
Project funded by the EUROPEAN UNION

GloMEEP Project Components
Legal, Policy and Institutional Reforms (LPIR)
Capacity building and knowledge sharing
Public-private partnerships
IMO INTERNATIONAL MARITIME ORGANIZATION
gef
UNDP
GloMEEP

MARITIME TECHNOLOGY COOPERATION CENTRE – PACIFIC (MTCC-PACIFIC)

CAPACITY BUILDING FOR CLIMATE MITIGATION IN THE MARITIME SHIPPING INDUSTRY
THE GLOBAL MTCC NETWORK (GMN) PROJECT

Recommendations for Low Carbon Maritime Transport in the Pacific

Workshop for Heads of Maritime
Suva Fiji, 12th December 2017

The Global MTCC Network (GMN) project is funded by the European Union and implemented by the IMO.

PLCMT Consultation

Three workshops and a submission:

- Regional Workshop on Energy Efficiency in Maritime Transport (RWMTEE)- Vanuatu December 2016;
- Port Energy Management (PEM)- Auckland New Zealand August 2017;
- National Workshop on Ship Energy Efficient Operations (SEEO)- Suva Fiji October 2017;
- Expert submission from Solomon Islands Maritime Safety Administration (SIMSA)- Acting Director Capt. Tim Harris.

D-N-B-A Matrix Drivers-Needs-Barriers-Action

- MTCC-Pacific Report- Recommendations for Low Carbon Maritime Transport in the Pacific (PLCMT) to be Submitted to IMO by Dec 2017;
- Aim- to implement Transport Ministers' Resolution;
- Headings- International and National Frameworks, Climate Financing, Costs and Profitability, Pilot-Projects, Reputation and Standards.
- Actions- Short (<5yr), Medium (5-10yr) and Long-term (>10yr)

PLCMT Strategy PICTs Recommendations

International Framework:

Drivers:

International framework and negotiations on climate change

Participation/contribution to international negotiations to defend SIDS/LDCs special circumstances

Recommendations:

Facilitate communication between all stakeholders to coordinate the drafting and submission of regular cosponsored regional documents to IMO and secure funding to attend meetings.

Identify and train suitable maritime personnel to attend and become long term representatives to international meetings.

PLCMT Strategy PICTs Recommendations

National Framework:

Drivers:

National initiatives/plans e.g. Green Growth frameworks, GHG emissions targets supported by regional frameworks and declarations including 2013 Majuro Declaration, FRDP Goal 2 on low carbon development, FATS Theme 5

Recommendations:

Draft model laws and instructions to MSAs, ratify laws and National MT Policies to address SEE in MTI aligned with international obligations.

Adopt data collection systems to determine GHGEs and meet MT NDCs.

Develop Regional Strategy for low carbon MT in Pacific.

PLCMT Strategy PICTs Recommendations

Climate Financing:

Drivers:

Climate financing and available expertise with international and regional partners

Recommendations:

Seek Green Funds and Donors, Develop Capacity and Proposals to access.

Enhance country financial systems to utilise green funds.

The Global MTCC Network (GMN) project is funded by the European Union and implemented by the IMO

PLCMT Strategy PICTs Recommendations

Pilot-Projects:

Drivers:

Lead by example to reduce GHG emissions from maritime transport in the Pacific under regional frameworks

Recommendations:

Develop training, tools, research capacity and qualifications, and implement small-scale pilot-projects for SEE operations and technologies.

Implement Green Ports and Shipping Programs.

Implement vessel replacement and green infrastructure in ports.

The Global MTCC Network (GMN) project is funded by the European Union and is implemented by the IMO

PLCMT Strategy PICTs Recommendations

Costs:

Drivers:

Cost of Energy

Recommendations:

Conduct annual audits and implement short-term energy savings projects.

Develop integrated solutions for Green Port-Shipping-Supply chain.

Implement SEEMP and start plans for vessel replacement.

CBA of long-term savings for port infrastructure (OPS) and bio-fuels.

The Global MTCC Network (GMN) project is funded by the European Union and is implemented by the IMO

PLCMT Strategy PICTs Recommendations

Profitability:

Drivers:

Improve profitability of ships and reliability and efficiency of domestic shipping and competitive advantage of ports

Recommendations:

Review government franchise schemes (GFS) to support SEE and profitable services to reduce costs, fuel and GHGEs.

CBA of new technologies on board smaller vessels (<50m).

Government support for training crews on SEE Operations, improved infrastructure, technology transfer and competitive operations.

Assist GFS operators on uneconomic routes with SEE measures and GHGE reduction.

Assist with new technologies and safety equipment for smaller vessels.

The Global MTCC Network (GMN) project is funded by the European Union and is implemented by the IMO

PLCMT Strategy PICTs Recommendations

Reputation:

Drivers:

Good Reputation

Recommendations:

Regular success stories published.

Support Ship Owners Associations to promote Green Ports & Ships.

Develop Green Awards for Ports & Ships.

The Global MTCC Network (GMN) project is funded by the European Union and is implemented by the IMO

PLCMT Strategy PICTs Recommendations

Standards:

Drivers:

Improved standards of MTI including safety, training, pollution prevention and energy efficiency

Recommendations:

Develop short training courses for SEE, standards, technologies and green port infrastructure.

Review government incentives for SEE technologies, ship building and maintenance, and safety equipment.

Review incentives for installing minimum electronic safety & SEE equipment.

Review Port & SEE, Operational & Environmental Systems to International standards.

The Global MTCC Network (GMN) project is funded by the European Union and is implemented by the IMO

The Global MTCC Network (GMN) project is funded by the European Union and implemented by the IMO

MARITIME TECHNOLOGY COOPERATION CENTRE – PACIFIC (MTCC-PACIFIC)

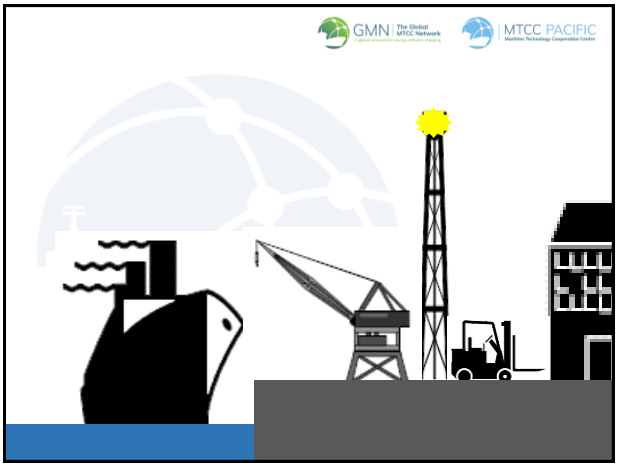
CAPACITY BUILDING FOR CLIMATE MITIGATION IN THE MARITIME SHIPPING INDUSTRY
THE GLOBAL MTCC NETWORK (GMN) PROJECT

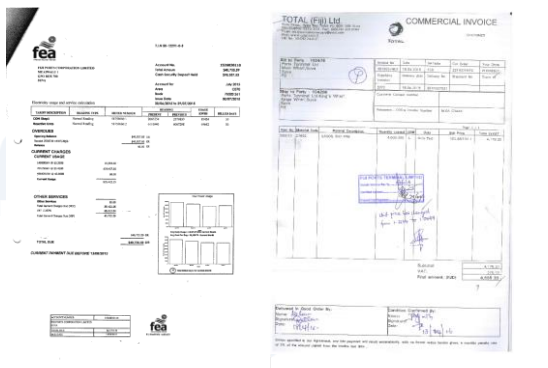
Energy Management in Ports

Methods and Tools

Suva, 12 December 2017

The Global MTCC Network (GMN) project is funded by the European Union and implemented by the IMO.





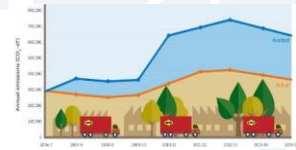
Energy management

Benefits

- ❑Reduced operating costs.
- ❑Reduced emissions (both CO2-e and local air pollution).
- ❑Lowered maintenance expenses, and reduced downtime due to equipment failure.
- ❑Improved comfort and safety.
- ❑Improved company reputation.
- ❑Greater employee retention.
- ❑An edge when it comes to recruitment.

Examples

- Logistics company, Linfox, operating trucks and warehouses, reduced its carbon intensity by 47% over a 8 year period. https://www.youtube.com/watch?v=DSO3dV-HIQU



http://www.linfox.com/about-us/sustainability.htm

Methods

2 general approaches

Energy Management
- Sets targets
- Has a policy
- Allocates responsibility, sets KPIs
- Has an annual plan
- Ongoing measurement of energy use and carbon emissions
- Allocates Resources

Ad-hoc, opportunistic
- Undertakes an energy audit (maybe)
- Implements one or two measures to save energy
- No ongoing measurement
- No ongoing resourcing

Benefits of the EM approach

$

- "Organizations that set a reduction **goal**, **analyze** energy data frequently, add internal or external **resources**, and use external financing were found to implement four times as many improvement measures as those who did none" Fifth Annual Global Energy Efficiency Survey, June 2011

Tools

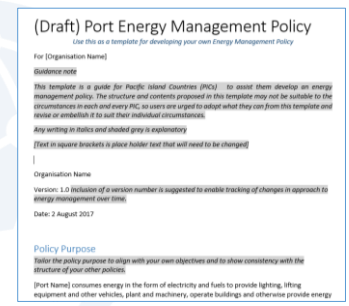
Developed by MTCC Pacific to help Pacific Ports save money and reduce emissions

Tools overview

- EM Policy Template
- EM Measurement and tracking spreadsheet
- EM Plan Template
- Port Energy Management Guide and accompanying spreadsheet

Policy

- Commitment
- Targets
- Links to other policies
- Communication



(Draft) Port Energy Management Policy

Use this as a template for developing your own Energy Management Policy

For [Organisation Name]

Guidance note

This template is a guide for Pacific Island Countries (PICs) to assist them develop an energy management policy. The structure and contents proposed in this template may not be suitable to the circumstances in each and every PIC, so users are urged to adopt what they can from this template and revise or embellish it to suit their individual circumstances.

Any writing in italics and shaded grey is explanatory

[Text in square brackets is place holder text that will need to be changed]

Organisation Name

Version: 1.0 *Inclusion of a version number is suggested to enable tracking of changes in approach to energy management over time.*

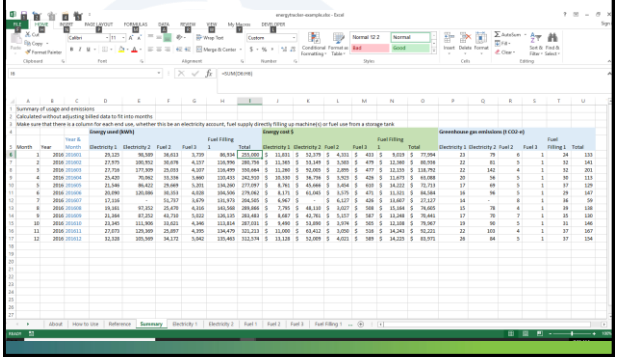
Date: 2 August 2017

Policy Purpose

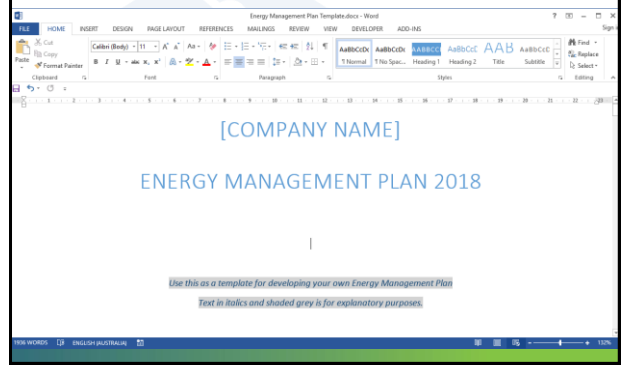
Tailor the policy purpose to align with your own objectives and to show consistency with the structure of your other policies.

[Port Name] consumes energy in the form of electricity and fuels to provide lighting, lifting equipment and other vehicles, plant and machinery, operate buildings and otherwise provide energy

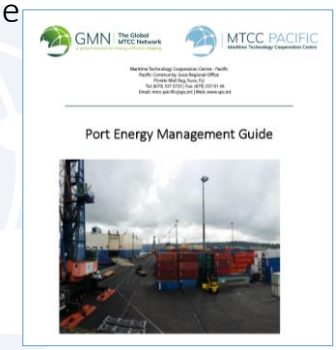
Measurement



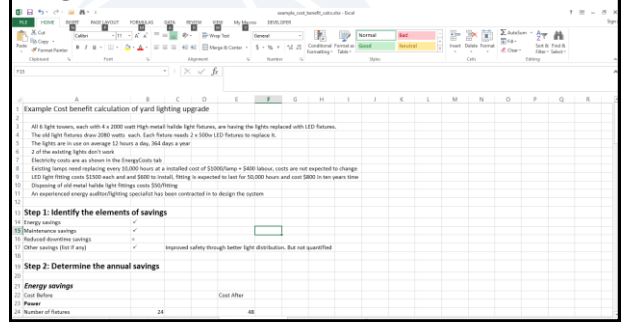
Annual Action Plan



Port EM Guide



Spreadsheet accompanying the guide



Summary

- Energy management: what's involved, benefits
- Method: Energy Management achieves much more than an ad-hoc approach:
  - Sets targets
  - Has a policy
  - Allocates responsibility, sets KPIs
  - Has an annual plan
  - Ongoing measurement of energy use and carbon emissions
  - Allocates Resources
- MTCC Pacific tools developed to assist ports:
  - Energy Policy Template
  - Measurement spreadsheet
  - Energy Management Plan
  - EM Guide and opportunity spreadsheet

On Thursday…

- Presenting Activities and Results in Pacific Ports

Thank you

Bruce Rowse, 8020Green Pty Ltd
Port Energy Management Consultant to MTCC-Pacific
bruce@8020green.com

2nd STEERING COMMITTEE MEETING

IMO Work to Address Energy Efficiency and Greenhouse Gas Emissions

Dr. Leigh Mazany| GMN Consultant
MTCC-Pacific 1st Regional Workshop | 12-15 December 2017 | Suva, Fiji

The Global MTCC Network (GMN) project
is funded by the European Union and implemented by IMO
The views expressed in this presentation
can in no way be taken to reflect the views of the European Union

Content

1. Estimating GHG Emissions from International Shipping
2. Marine Environment Protection Committee Work
3. Data Collection System
4. Roadmap for an IMO Strategy

Estimating Greenhouse Gas (GHG) Emissions from International Shipping

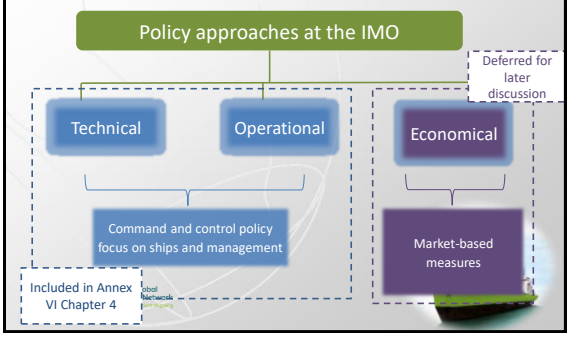
IMO GHG studies

- Three major studies have been undertaken over the last 15 years
  - First IMO GHG Study 2000
  - Second IMO GHG Study 2009
  - Third IMO GHG Study 2014
- Studies looked at both GHG emissions and air pollutant emissions
- IMO Roadmap calls for a fourth study in 2019

2nd IMO GHG Study 2009

- Introduction to shipping and its legislative framework
- Emissions from shipping 1990–2007
- Technological and operational potential for reduction of emissions
- Policy options for reductions of GHG emissions scenarios for forecast of future shipping emissions
- Climate impact
- Comparison of emissions of CO_2 from ships with emissions from other modes of transport

2nd GHG Study: Policy approaches to reduce GHG emissions



2nd GHG Study: Main Findings

- Shipping was estimated to have emitted 3.3% of the global emissions during 2007
- International shipping was estimated to have emitted 870 million tonnes, or about 2.7% of the global emissions in 2007
- Carbon dioxide is the most important GHG emitted by ships
- A significant potential for reduction of GHG emissions through technical and operational measures was identified
- Energy efficiency of ships could potentially be 25% to 75% below the then current levels
- A number of policies to reduce GHG emissions from ships were conceivable
- Shipping was shown, in general, to be an energy-efficient means of transportation compared to other modes

3rd IMO GHG Study 2014

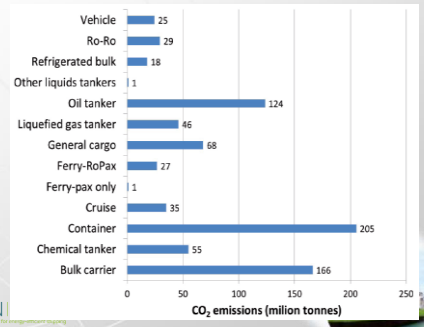
- The Third IMO GHG Study 2014 aimed to update the 2nd IMO GHG Study 2009
- The main objective was to focus on the following topics:
  - Development of the inventories of CO_2 emissions from international shipping for 2007–2012
  - Development of the inventories of other air emissions from international shipping for 2007–2012
  - Development of future shipping scenarios and projection of shipping emissions for 2012–2050

3rd IMO GHG study findings: Shipping share of global CO_2 emissions

		IMO GHG Study 2014 CO_2			
Year	Global CO_2[1]	Total shipping	Percent of global	International shipping	Percent of global
2007	31,409	1,100	3.5%	885	2.8%
2008	32,204	1,135	3.5%	921	2.9%
2009	32,047	978	3.1%	855	2.7%
2010	33,612	915	2.7%	771	2.3%
2011	34,723	1,022	2.9%	850	2.4%
2012	35,640	938	2.6%	796	2.2%
Average	33,273	1,015	3.1%	846	2.6%

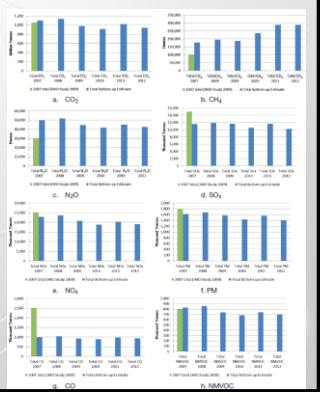
International shipping share of total global GHG emissions was 2.2% in 2012

3rd GHG Study findings: GHG emissions per ship type for 2012



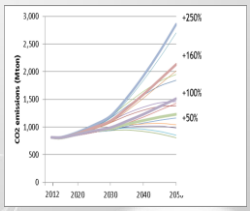
3rd GHG Study findings: Emissions estimates

- Emissions estimates for all shipping for period 2007 to 2012
- Green bar represents the 2nd IMO GHG Study 2009 estimate



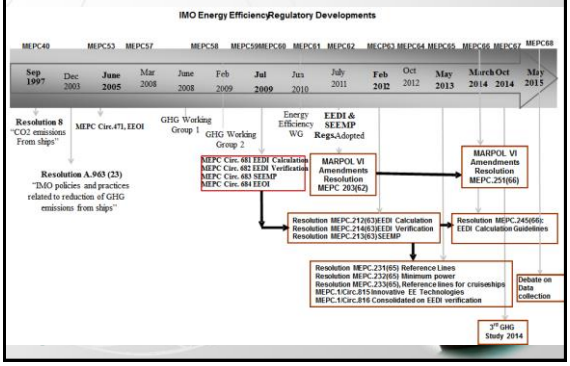
3rd IMO GHG Study: CO_2 emissions projections

- Various scenarios modelled
- An estimated average increase of 50% to up to 250% in shipping emissions by 2050
- This study, plus a number of other studies, all point to an increase in future shipping GHG emissions
- Regulators aim to stabilise or reduce future emissions



Marine Environment Protection Committee Work

Work leading to Chapter 4 of MARPOL Annex VI



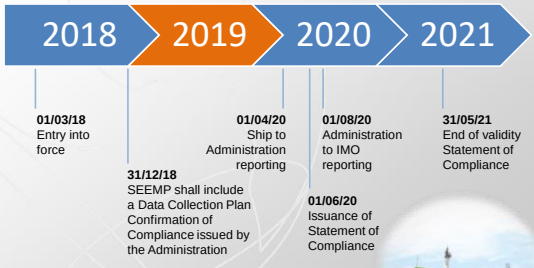
Data Collection System

IMO Data Collection System for Fuel Oil Consumption of Ships

- **Resolution MEPC.278(70)** adopted on 28 October 2016 amending MARPOL Annex VI
- New regulation 22A on «Collection and reporting of ship fuel oil consumption data»
  - SEEMP part II (Data Collection Plan)
  - Applies to ships 5,000 GT and above
  - Ship to report data to its Administration on an annual basis
  - Data to be transferred to the IMO database
  - Parties to MARPOL Annex VI shall have access to anonymized data for their analysis and consideration
  - Based on reported data, the Secretary-General shall produce an annual report to the MEPC

16

IMO Data Collection System for Fuel Oil Consumption of Ships in MARPOL Annex VI



17

IMO Data Collection System for Fuel Oil Consumption of Ships

- Information to be submitted to the IMO Ship Fuel Oil Consumption Database
  - IMO number of the ship
  - Start date - End date
  - Ship type
  - GT - NT
  - DWT (if applicable)
  - Power output of main and auxiliary engines over 130 kW
  - EEDI (if applicable)
  - Ice class (if applicable)
  - Fuel oil consumption, by fuel oil type, in metric tonnes
  - Methods used for collecting fuel oil consumption data
  - Distance travelled
  - Hours underway

18

IMO Data Collection System Guidelines and Database

- Guidelines on development of a ship energy efficiency management plan (SEEMP) – Resolution MEPC.282(70) adopted 28 October 2016
- Guidelines for Administration verification of ship fuel oil consumption data – Resolution MEPC.292(71) adopted 7 July 2017
- Guidelines on development and management of IMO ship fuel oil consumption data base – Resolution MEPC.293(71) adopted on 7 July 2017
- Development of the IMO Ship Fuel Oil Consumption Database
  - GISIS model user interface
  - Extensible Markup Language (XML) format
  - IMO Web Accounts system of permissioning

GMN | The Global MTCC Network

19

Submission of data from a State non-Party to MARPOL Annex VI

- **Circular MEPC.1/Circ.871** approved by MEPC 71 and issued on 4 September 2017
- Provides measures for voluntary submission of data from non-Party States
- Provisions for recognized organizations
- As of 8 November 2017, MARPOL Annex VI had 88 contracting States representing 96.16% of the world fleet

GMN | The Global MTCC Network

20

Roadmap for an IMO Strategy

GMN | The Global MTCC Network

Comprehensive IMO strategy on reduction of GHG emissions

MEPC 70 (October 2016)

- MEPC approved a Roadmap for developing a comprehensive IMO strategy on reduction of GHG emissions from ships, which foresees an initial GHG reduction strategy to be adopted in 2018.
- It contains a list of activities, including further IMO GHG studies and significant intersessional work, with relevant timelines and provides for alignment of those new activities with the ongoing work by the MEPC on the three-step approach to ship energy efficiency improvements.
  - This alignment provides a way forward to the adoption of a revised strategy in 2023 to include short-, mid-, and long-term further measures, as required, including implementation schedules.
- The Committee also agreed to hold intersessional working group meetings on reduction of GHG emissions from ships.

GMN | The Global MTCC Network

Development of an IMO Strategy on reduction of GHG emissions from ships

Outcome of MEPC 71 (July 2017) & ISWG-GHG 2 (October 2017)

- Draft outline of the structure of the initial IMO GHG Strategy (following the Roadmap for developing a comprehensive IMO Strategy agreed at MEPC 70)
  1. Preamble/introduction/context/objectives including emission scenarios
  2. Vision
  3. Levels of ambition
     Guiding principles
  4. List of candidate short-, mid- and long-term further measures with possible timelines and their impacts on States
  5. Barriers and supportive measures; capacity building and technical cooperation; R&D
  6. Follow-up actions towards the development of the revised Strategy
  7. Periodic review of the Strategy
- ISWG-GHG 3 planned for April 2018

GMN | The Global MTCC Network

Latest progress made in developing GHG Strategy for international shipping

Outcome of ISWG-GHG 2 (23-27 October 2017)

- ISWG-GHG 2 has made progress in starting to shape a draft initial IMO GHG Strategy including refining the **vision** for IMO, which will express IMO's further commitment to reducing GHG emissions from international shipping.
- While the structure of the Strategy has been largely agreed, the detailed text to be included is still under discussion.
- The group agreed that candidate short-term measures could be finalized and agreed by the Marine Environment Protection Committee (MEPC) between 2018 and 2023; candidate mid-term measures could be measures finalized and agreed by the MEPC between 2023 and 2030; and candidate long-term measures could be measures finalized and agreed by the MEPC beyond 2030.
- Dates of entry into force and when the measure can effectively start to reduce GHG emissions would be defined for each measure individually.
- The group supported the need for early action.

GMN | The Global MTCC Network

Thank you for your attention!
http://gmn.imo.org/

GMN | The Global MTCC Network
A global network for energy-efficient shipping

25

MARITIME TECHNOLOGY COOPERATION CENTRE – PACIFIC (MTCC-PACIFIC)

CAPACITY BUILDING FOR CLIMATE MITIGATION IN THE MARITIME SHIPPING INDUSTRY
THE GLOBAL MTCC NETWORK (GMN) PROJECT

MTCC-Pacific Progress Report

Second MTCC-Pacific Steering Committee Meeting
Suva, Fiji, 12 December 2017

The Global MTCC Network (GMN) project is funded by the European Union and implemented by the IMO.

Result 1: Establishment & Operations

IR1.1 MTCC established in the office of SPC

- SPC-IMO contract signed on 12 May 2017 - Project ending on 31 December 2019
- A closed office is allocated by SPC at the third floor of Lotus Building at its Regional Office in Suva, Fiji
- Website published at www.mtccpacific.spc.int

IR1.2 Appoint qualified staff to MTCC

- 2 October 2017: all MTCC-Pacific staff appointed
  - ✓ Head of MTCC-Pacific
  - ✓ Transport GHG Adviser
  - ✓ Administration and Information Assistant
  - ✓ Maritime Industry Energy Efficiency Officer
- SPC Project Kick-off Team de-established

Result 1: Establishment & Operations

IR1.4 Project workplan implementation

- 7 Project Monthly Meetings have been completed: 18 May, 28 June, 19 July, 24 august, 28 September, 12 October and 1 November
- First Quarterly Progress Report submitted and approved by IMO
- Second Quaterly Report submitted and approved by IMO
- Participated at COP23 GMN side-event
- Participated at the MTCCs Technical Workshop at IMO in December – MTCCs MoU signed and MTCCs Coordinating Committee established.

IR1.5 Partnerships implementation

- MTCC-Pacific Partners' Meeting on 27 June attended by EU delegation, FNU, FMA, PIDF, PIFS, USP
- First Steering Committee Meeting 16-17 August – ToR adopted
- Second Steering Committee Meeting 12 December

Result 2: capacity-building activities

Fiji National Workshop on Energy Efficient Operations of Ships

- Date: 24-27 October in Suva, Fiji
- Representatives from
  - Fiji Ministry of Infrastructure and Transport (MoIT)
  - Maritime Safety Authority of Fiji (MSAF)
  - Fiji Port Corporation Limited (FCPL)
  - Fiji Ships and Heavy Industry Limited (FSHIL)
  - Pacific Islands Development Forum (PIDF)
  - Patterson Shipping/Searoad Shipping
  - Seaquest Fiji Ltd/Sealand Processors
  - Inter Link Shipping Line Ltd
  - Government Shipping Services
  - Tokalau Shipping, All Barging and Marine (Fiji) Limited
  - Billett Wright & Associates (Fiji) Limited
  - Solander Pacific Limited.
- The objective of the National Workshop was to inform the participants on new energy efficient technologies, tools and methods
- This workshop elicited great interest in improved ship propulsion efficiency, especially in Energy Saving Devices (ESD).

Result 3: pilot-projects on energy efficiency

- Pilot-project on Energy Efficiency approved on 19 July by IMO
- Guide for Energy Management in Port and tools developed
- Port Energy audits completed:
  - In August in Samoa and Solomon Islands
  - In Fiji in September for trials, 2nd week of December and scheduled last visit in March
- Ship Energy Management visit in Kiribati in November

Result 4: Pilot-project on Data Collection

- The pilot-project on Energy Efficiency was approved by IMO on 19 July
- SPC visit in Vanuatu in July under the Pacific Islands Domestic Ship Safety (PIDSS) programme
- MTCC-Pacific visit in Kiribati in November in conjunction with PIDSS
- All shipowners provided with data collection templates – awaiting for on-board training

The Global MTCC Network (GMN) project is funded by the European Union and implemented by the IMO

Result 5: communication and visibility

IR5.1 Communication and visibility plan designed and approved
- MTCC-Pacific C&V Plan approved by IMO on 26 June

IR5.2 Communication and Visibility materials
- All communication and visibility materials delivered on 31 July

IR5.3 Dedicated website for the MTCC-Pacific designed and launched
- Website published

IR5.5 Progress communicated
- GMN Project and MTCC-Pacific presented at SPC CRGA (with booth) in July and at September SPREP meeting

IR5.6 Relevant publications
- Article in August *Pacific Maritime Watch* and in next December issue

This Global MTCC Network (GMN) project is funded by the European Union and is implemented by the IMO

2017 Worplan implementation Progress

- As of 12 November, 72% of the Result 1 activities for 2017 completed - on track
- 50% of the Capacity-building activities, rest Regional Conference – on track
- 36 % of the pilot-project on Energy Efficiency activities have been completed – behind schedule for Fiji Port work and legal review that is postponed early 2018
- 0% Result 4 shows 0% completion as the indicators for 2017 activities are related to the dissemination of templates which will happen at the Regional Conference in December
- 50% of the communication and visibility activities – rest is website and dissemination of all template developed (completed this week) – on track

The Global MTCC Network (GMN) project is funded by the European Union and is implemented by the IMO

MARITIME TECHNOLOGY COOPERATION CENTRE – PACIFIC (MTCC-PACIFIC)

CAPACITY BUILDING FOR CLIMATE MITIGATION IN THE MARITIME SHIPPING INDUSTRY

THE GLOBAL MTCC NETWORK (GMN) PROJECT

Fundraising Strategy

Second MTCC-Pacific Steering Committee Meeting
Suva, Fiji, 12 December 2017

The Global MTCC Network (GMN) project is funded by the European Union and implemented by the IMO.

Vision, Mission and Values

Vision: *A low-carbon Pacific maritime transport that contribute to global climate mitigation efforts and reduction of greenhouse gas emissions*

Mission: *To be the Pacific regional centre of excellence in the Global MTCC Network to build capacity, transfer technology and knowledge for climate mitigation in the Pacific maritime industry through the uptake of low-carbon technologies and operations*

SPC's Values which are :

- *We treasure the rich and diverse cultures, traditions and environments of the people of the Pacific, and our sustainable solutions are based on a sound understanding of their strengths and needs, and of the challenges they face*
- *We are committed to adding value to the scientific and technical capacities of our members, in pursuing transformational development*
- *We are committed to the principles of good governance, to the defence and promotion of gender equality and human rights, and to ensuring that the needs of the most vulnerable in our societies are at the forefront of our work*
- *We work collaboratively with existing partners and build new relationships so that the collective strength of all can be harnessed for the interests of our members.*
- *We are committed to developing an organisational culture that enables our staff to continually improve their effectiveness in working for the people of the Pacific.*

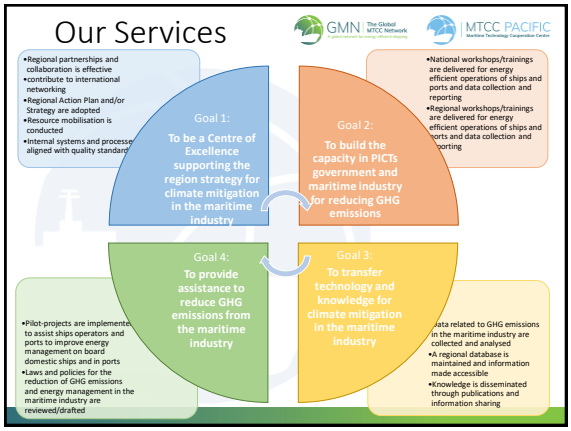
Beneficiaries

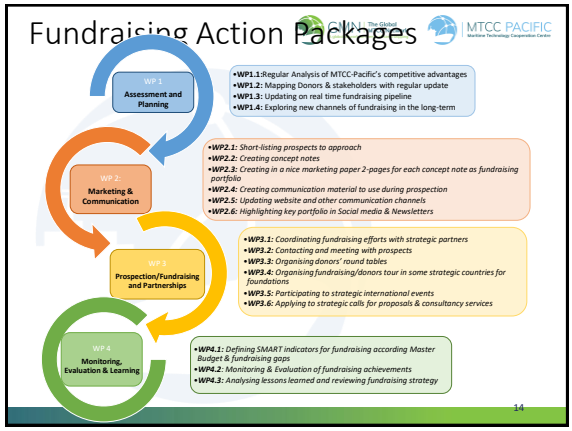
- Direct beneficiaries:
  - Pacific domestic ships and international and domestic ports operators – MTCC-Pacific assist them in reducing the energy consumption and identifying and implementing short- and long-term technical solutions for low-carbon futures
  - Pacific Islands countries governments that will progress towards their NDCs' targets and show their leadership in the global efforts
- Indirect beneficiaries:
  - Pacific people that will enjoy maritime transport services that reduce their carbon footprint while providing the necessary safe, reliable and affordable services to remote communities

SWOT analysis

	Helpful	*Harmful*
Internal	**Strength** • Integrated to SPC organisational, management and financial structure • Overall and specific objectives, results and activities guided and supported by a project funded in 2017-2019 • Technical expertise provided by the International Maritime Organization (IMO) • Part of an international network • Additional technical expertise on climate change issues and support is provided by SPREP • Expert staff dedicated full time and embedded within SPC maritime and oceans • Lump-sum service contract with IMO based on deliverables (encouraging results-based approach)	**Weakness** • End of funding at the end of the project date (31 December 2019) • Dedicated staff contracts aligned with the end of project date (31 December 2019) • Deliverables and reporting process strictly determined by contract between IMO and SPC (less flexibility) • No physical representation in other countries than Fiji in the Pacific • Lack of awareness from international donors and partners on MTCC-Pacific role
External	**Opportunities** • Pacific Leaders are strong advocates to reducing GHG emissions • Shipping GHG emissions identified as one of the main issues and priorities in climate change negotiations • IMO is working on a long-term Strategy to address GHG emissions from shipping that should further support technical cooperation in the future • Several donors are in support to ambitious GHG reduction targets • Climate financing opportunities could benefit MTCC as well	**Threats** • Lack of capacity and awareness in Pacific Islands countries that prevent from prioritising energy efficiency in maritime transport • Technology not readily available in the Pacific • Small Pacific ports and shipping companies operating old infrastructure and equipment • In conflict with other priorities at the national level • Climate financing requires long and complicated process • Technical solutions requiring combination of innovative systems that would not be affordable

The Global MTCC Network (GMN) project is funded by the European Union and implemented by the IMO





Budget Master Plan

GMN The Global MTCC Network
MTCC PACIFIC

MTCC-Pacific Master Business Plan Budget January 2018 to December 2021 (4 years)	Jan - Dec 2018 (Y1)	Jan - Dec 2019 (Y2)	Jan - Dec 2020 (Y3)	Jan - Dec 2021 (Y4)	TOTAL 2018-2021 (4 years)	Funding Plan 2018 (EUR)	Funding Plan 2019 (EUR)	Funding Plan 2020 (EUR)	Funding Plan 2021 (EUR)
	EUR	EUR	EUR	EUR	EUR	Unfunded / To Fundraise	Unfunded / To Fundraise	Unfunded / To Fundraise	Unfunded / To Fundraise
HUMAN RESOURCES	486,287	510,601	536,131	562,937	2,095,956	129,044	135,523	536,131	562,937
TECHNICAL SPECIALISTS	429,293	450,664	473,196	496,856	1,849,918	105,688	110,967	473,196	496,856
Maritime Technical Specialists	422,092	443,197	465,357	488,625	1,819,271	98,572	103,501	465,357	488,625
Cross-Cutting Technical Specialists	7,111	7,466	7,840	8,232	30,648	7,111	7,466	7,840	8,232
SUPPORT STAFF	44,180	46,384	48,714	51,156	190,443	23,285	24,554	48,714	51,156
OPERATIONAL COSTS	89,508	101,566	101,709	101,855	394,638	29,214	24,499	101,709	101,855
PROGRAMME ACTIVITIES COSTS	361,626	297,714	186,626	71,268	917,234	163,144	141,459	186,626	71,268
Goal 1: Be a Centre of Excellence	26,944	153,289	26,944	26,944	234,122	14,144	126,993	26,944	26,944
R1.1: Regional partnerships and collaboration is effective	15,500	15,500	15,500	15,500	61,000	12,856	12,856	15,500	15,500
R1.2: Contribute to international networking	5,988	39,888	5,988	5,988	57,660	1,288	28,488	5,988	5,988
R1.3: Regional Action Plan and/or Strategy are adopted	0	98,401	0	0	98,401	0	85,649	0	0
R1.4: Resource mobilisation is conducted	5,456	5,500	5,456	5,456	21,868	5,456	5,456	5,456	5,456
R1.5: Internal systems and processes aligned with quality standards	0	0	0	0	0	0	0	0	0
Goal 2: Build capacity	125,018		125,018		250,037	71,348		125,018	
R2.1: National Workshops/Trainings are delivered for energy efficient operations of ships and ports and data collection and reporting	93,670	0	93,670	0	187,340	0	0	93,670	0
R2.2: Regional Workshops/trainings are delivered for energy efficient operations of ships and ports and data collection and reporting	71,348	0	71,348	0	142,696	71,348	0	71,348	0
Goal 3: Transfer technology and knowledge	34,756	9,756	9,756	9,756	64,024	34,756	9,756	9,756	9,756
R3.1: Data related to GHG emissions in the maritime industry are collected and analysed	25,000	0	0	0	25,000	25,000	0	0	0
R3.2: A regional database is maintained and information made accessible	4,756	4,756	4,756	4,756	19,024	4,756	4,756	4,756	4,756
R3.3: Knowledge is disseminated through publications and information sharing	5,000	5,000	5,000	5,000	20,000	5,000	5,000	5,000	5,000
Goal 4: Provide assistance to reduce GHG emissions	174,914	34,374	24,914	34,574	268,976	24,914	4,754	24,914	34,574
R4.1: Pilot-projects are implemented to assist ships operators and ports to improve energy management on board domestic ships and in ports	170,160	29,618	20,160	29,820	249,757	20,160		20,160	29,820
R4.2: Laws and policies for the reduction of GHG emissions and energy management in the maritime industry are reviewed/drafted	4,754	4,756	4,754	4,754	19,019	4,754	4,756	4,754	4,754
TOTAL Direct Costs	937,421	909,881	824,465	736,060	3,407,828	311,836	301,494	824,465	736,060
Monitoring, Evaluation and Learning (M&E)	37,497	6,389	32,974	29,445	136,213	37,497	16,389	32,974	29,445
Programme Management Fees (PMF)	146,238	145,041	28,617	114,815	531,621	146,238	79,241	128,617	114,815
TOTAL PROGRAMME COSTS	1,121,155	1,088,218	986,063	880,328	4,075,762	497,570	397,124	986,056	880,320

15

The Global MTCC Network (GMN) project is funded by the European Union and implemented by the IMO

IMO Work to Address Energy Efficiency and Greenhouse Gas Emissions

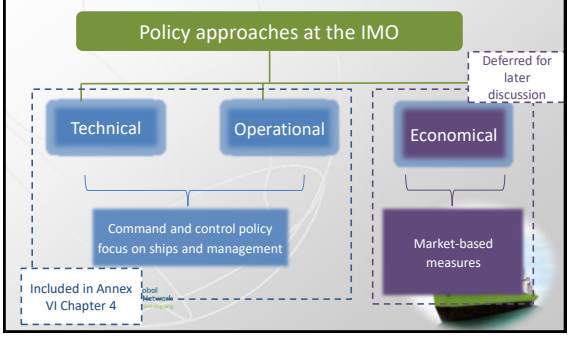
Dr. Leigh Mazany| GMN Consultant
MTCC-Pacific 1st Regional Workshop | 12-15 December 2017 | Suva, Fiji

The Global MTCC Network (GMN) project
is funded by the European Union and implemented by IMO
The views expressed in this presentation
can in no way be taken to reflect the views of the European Union

Content

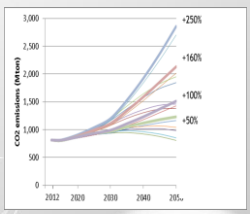
1 Marine Environment Protection Committee Work
2 Data Collection System
3 Roadmap for an IMO Strategy

2nd GHG Study (2009): Policy approaches to reduce GHG emissions



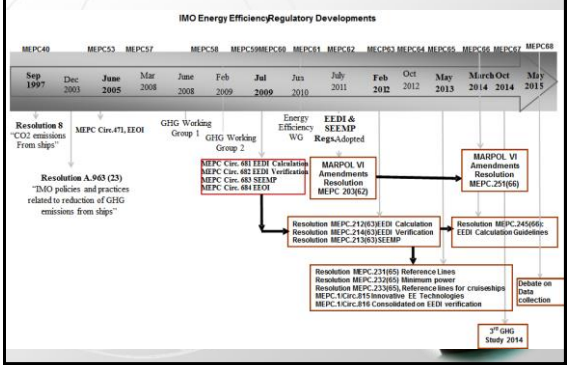
3rd IMO GHG Study (2014): CO_2 emissions projections

- Various scenarios modelled
- An estimated average increase of 50% to up to 250% in shipping emissions by 2050
- This study, plus a number of other studies, all point to an increase in future shipping GHG emissions
- Regulators aim to stabilise or reduce future emissions



Marine Environment Protection Committee Work

Work leading to Chapter 4 of MARPOL Annex VI



Data Collection System

IMO Data Collection System for Fuel Oil Consumption of Ships

- **Resolution MEPC.278(70)** adopted on 28 October 2016 amending MARPOL Annex VI
- New regulation 22A on «Collection and reporting of ship fuel oil consumption data»
  - SEEMP part II (Data Collection Plan)
  - Applies to ships 5,000 GT and above
  - Ship to report data to its Administration on an annual basis
  - Data to be transferred to the IMO database
  - Parties to MARPOL Annex VI shall have access to anonymized data for their analysis and consideration
  - Based on reported data, the Secretary-General shall produce an annual report to the MEPC

8

IMO Data Collection System for Fuel Oil Consumption of Ships in MARPOL Annex VI

9

IMO Data Collection System for Fuel Oil Consumption of Ships

- Information to be submitted to the IMO Ship Fuel Oil Consumption Database
  - IMO number of the ship
  - Start date - End date
  - Ship type
  - GT - NT
  - DWT (if applicable)
  - Power output of main and auxiliary engines over 130 kW
  - EEDI (if applicable)
  - Ice class (if applicable)
  - Fuel oil consumption, by fuel oil type, in metric tonnes
  - Methods used for collecting fuel oil consumption data
  - Distance travelled
  - Hours underway

10

IMO Data Collection System Guidelines and Database

- Guidelines on development of a ship energy efficiency management plan (SEEMP) – Resolution MEPC.282(70) adopted 28 October 2016
- Guidelines for Administration verification of ship fuel oil consumption data – Resolution MEPC.292(71) adopted 7 July 2017
- Guidelines on development and management of IMO ship fuel oil consumption data base – Resolution MEPC.293(71) adopted on 7 July 2017
- Development of the IMO Ship Fuel Oil Consumption Database
  - GISIS model user interface
  - Extensible Markup Language (XML) format
  - IMO Web Accounts system of permissioning

11

Submission of data from a State non-Party to MARPOL Annex VI

- **Circular MEPC.1/Circ.871** approved by MEPC 71 and issued on 4 September 2017
- Provides measures for voluntary submission of data from non-Party States
- Provisions for recognized organizations
- As of 8 November 2017, MARPOL Annex VI had 88 contracting States representing 96.16% of the world fleet

12

Roadmap for an IMO Strategy

Comprehensive IMO strategy on reduction of GHG emissions

MEPC 70 (October 2016)

- MEPC approved a Roadmap for developing a comprehensive IMO strategy on reduction of GHG emissions from ships, which foresees an initial GHG reduction strategy to be adopted in 2018.
- It contains a list of activities, including further IMO GHG studies and significant intersessional work, with relevant timelines and provides for alignment of those new activities with the ongoing work by the MEPC on the three-step approach to ship energy efficiency improvements.
  - This alignment provides a way forward to the adoption of a revised strategy in 2023 to include short-, mid-, and long-term further measures, as required, including implementation schedules.
- The Committee also agreed to hold intersessional working group meetings on reduction of GHG emissions from ships.

Development of an IMO Strategy on reduction of GHG emissions from ships

Outcome of MEPC 71 (July 2017) & ISWG-GHG 2 (October 2017)

- Draft outline of the structure of the initial IMO GHG Strategy (following the Roadmap for developing a comprehensive IMO Strategy agreed at MEPC 70)
  1. Preamble/introduction/context/objectives including emission scenarios
  2. Vision
  3. Levels of ambition
     Guiding principles
  4. List of candidate short-, mid- and long-term further measures with possible timelines and their impacts on States
  5. Barriers and supportive measures; capacity building and technical cooperation; R&D
  6. Follow-up actions towards the development of the revised Strategy
  7. Periodic review of the Strategy
- ISWG-GHG 3 planned for April 2018

Latest progress made in developing GHG Strategy for international shipping

Outcome of ISWG-GHG 2 (23-27 October 2017)

- ISWG-GHG 2 has made progress in starting to shape a draft initial IMO GHG Strategy including refining the **vision** for IMO, which will express IMO's further commitment to reducing GHG emissions from international shipping.
- While the structure of the Strategy has been largely agreed, the detailed text to be included is still under discussion.
- The group agreed that candidate short-term measures could be finalized and agreed by the Marine Environment Protection Committee (MEPC) between 2018 and 2023; candidate mid-term measures could be measures finalized and agreed by the MEPC between 2023 and 2030; and candidate long-term measures could be measures finalized and agreed by the MEPC beyond 2030.
- Dates of entry into force and when the measure can effectively start to reduce GHG emissions would be defined for each measure individually.
- The group supported the need for early action.

Thank you for your attention!
http://gmn.imo.org/

17

REGIONAL CONFERENCE AND OFFICIAL LAUNCH

GMN | The Global MTCC Network
A global network for energy-efficient shipping

GMN introduction and status Update

Anton Rhodes| Suva, Fiji, Dec 2017

The Global MTCC Network (GMN) project
is funded by the European Union and implemented by IMO
The views expressed in this presentation
can in no way be taken to reflect the views of the European Union

GMN | The Global MTCC Network
A global network for energy-efficient shipping

IMO's path to promote tech. transfer and capacity building

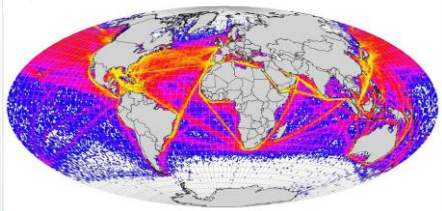
GMN Project

Res. 23 -MARPOL Annex VI (2013), MEPC Resolution on TT and CB,	ITCP: Awareness raising and capacity building tools	Major Projects: Capacity building & private sector partnerships	Global network to promote technology cooperation and transfer	Institutionalize capacity building and tech. transfer support at regional level for sustainable impact

IMO INTERNATIONAL MARITIME ORGANIZATION

Project funded by the EUROPEAN UNION

Initiation of a Network through GMN Project



- Focus on regions with large number of LDCs and SIDS (Africa, Asia, Caribbean, Latin America and Pacific)

GMN | The Global MTCC Network

5

GMN overview

- An initiative funded by EU and implemented by IMO
- Total Funding Euro 10,000,000
- Jan, 2016 – Dec, 2019 (4 Years)
- During the project, 5 MTCCs to become fully operational
- Beyond the project, the MTCCs will become financially self-sustained

Overall Objective:

- Build capacity – Improve energy efficiency – Reduce GHG emissions – Contribute to fight against climate change

6

GMN - Specific objectives

1. Establish 5 MTCCs
2. Improve human and instiutional capacities in maritime administrations and relevant stakeholders
3. Promote the uptake of low carbon technologies and operations
4. Establish pilot voluntary data collection and reporting systems
5. Communicate and disseminate information

GMN | The Global MTCC Network

Objective 1 - All MTCCs are established and operational

MTCC CARIBBEAN
Maritime Technology Cooperation Centre

University of Trinidad & Tobago

MTCC AFRICA
Maritime Technology Cooperation Centre

Jomo Kenyatta University of Agriculture & Technology, Kenya

MTCC ASIA
Maritime Technology Cooperation Centre

Shanghai Maritime university, China

MTCC PACIFIC
Maritime Technology Cooperation Centre

Pacific Community (SPC), Fiji

MTCC LATIN AMERICA
Maritime Technology Cooperation Centre

International Maritime University of Panama, Panama

GMN | The Global MTCC Network

11

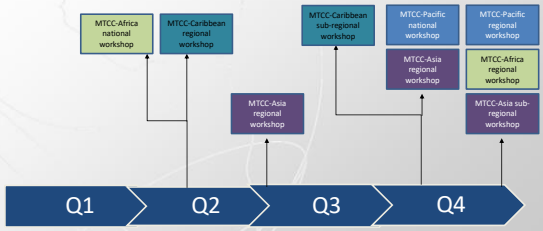
Launch of Global MTCCs Network

IMO HQ, London, 4 December

7

Objective 2 - Capacity building

MTCC capacity building activities in 2017



8

Objective 3 - Low Carbon technologies

- MTCC Africa - Establish on-shore power supply at Port of Mombasa (Cold Ironing)
- MTCC Asia - Update the ship trim optimization management system
- MTCC Caribbean - Capture the current baseline on the use of ship energy efficient technology within the region
- MTCC Pacific - Conduct energy audits in maritime ports
- MTCC Latin America - TBC

9

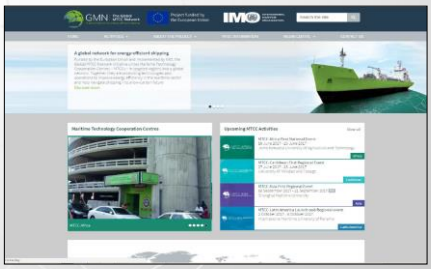
Objective 4 - Data collection

- **Caribbean** - collection of data
- **Pacific** – data templates developed
- **Africa** – Staff training on-board target vessels
- **Asia** – Data collection seminar – Role of key stakeholders confirmed

10

Objective 5 - Communication and dissemination

- Launch of GMN and MTCC websites



11

Objective 5 - Communication and dissemination

- GMN and MTCC newsletters, articles and social media

12

Objective 5 - Communication and dissemination

- International and regional events

13

Overview of GMN deliverables

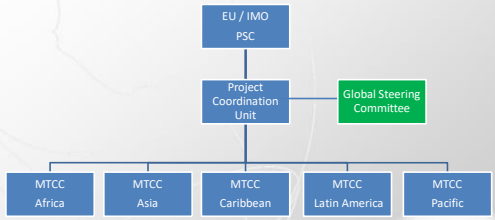
2016
- Established PCU
- Completed international tender process

2017
- MTCCs established and operational
- MTCC Launch Events and Regional Workshops
- Established MTCC network
- Established GSC

2018
- Develop pilot projects
- Deliver national and sub-regional workshops
- Develop sustainability plans

2019
- Complete Pilot Projects
- Deliver regional conferences
- Finalize sustainability plans

GMN MANAGEMENT STRUCTURE



GMN | The Global MTCC Network

15

Thank you for your attention!
http://gmn.imo.org/

GMN | The Global MTCC Network

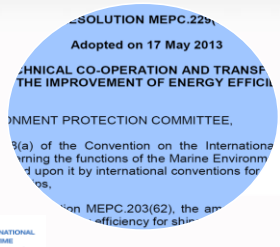
16

Roadmap for developing a comprehensive IMO Strategy on reduction Of GHG emissions from ships

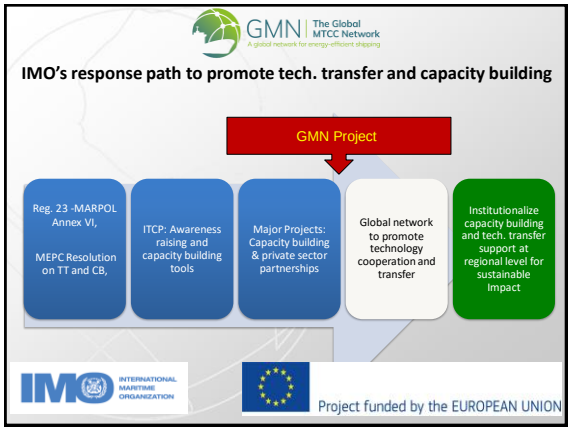
January 2019	- Start of Phase 1: Data collection (Ships to collect data)
Spring 2019 (MEPC 74)	- Discussion continues - Initiation of Fourth IMO GHG Study using data from 2012-2018
Summer 2020	- Data for 2019 to be reported to IMO
Autumn 2020 (MEPC 76)	- Start of Phase 2: data analysis (no later than autumn 2020) - Discussion continues - Publication of Fourth IMO GHG Study for consideration by MEPC 76[3]
Spring 2021 (MEPC 77)	- Initiation of work for adjustments on Initial IMO Strategy, based on DCS data - Secretariat report summarizing the 2019 data pursuant to regulation 22A.10
Summer 2021	- Data for 2020 to be reported to IMO
Spring 2022 (MEPC 78)	- Phase 3: Decision step - Discussion continues - Secretariat report summarizing the 2020 data pursuant to regulation 22A.10
Summer 2022	- Data for 2021 to be reported to IMO
Spring 2023 (MEPC 80)	- Adoption of Revised IMO Strategy, including short-, mid- and long-term further measure(s), as required, with implementation schedules - Secretariat report summarizing the 2021 data pursuant to regulation 22A.10

MARPOL Annex VI & Technical Cooperation

Regulation 23 of MARPOL Annex VI, Chapter 4: "Promotion of technical co-operation and transfer of technology relating to the improvement of energy efficiency of ships"



IMO INTERNATIONAL MARITIME ORGANIZATION


GMN
The Global MTCC Network
A global network for energy-efficient shipping
IMO's response path to promote tech. transfer and capacity building
GMN Project
Reg. 23 -MARPOL Annex VI,
MEPC Resolution on TT and CB,
ITCP: Awareness raising and capacity building tools
Major Projects: Capacity building & private sector partnerships
Global network to promote technology cooperation and transfer
Institutionalize capacity building and tech. transfer support at regional level for sustainable impact
IMO INTERNATIONAL MARITIME ORGANIZATION
Project funded by the EUROPEAN UNION

GloMEEP Project Components
Legal, Policy and Institutional Reforms (LPIR)
Capacity building and knowledge sharing
Public-private partnerships
IMO INTERNATIONAL MARITIME ORGANIZATION
gef
UNDP
GloMEEP

USP THE UNIVERSITY OF THE SOUTH PACIFIC

A joint collaboration between RMI Government and USP

MICRONESIAN CENTER FOR SUSTAINABLE TRANSPORTATION

Presentation to the MTCC-PACIFC REGIONAL CONFERENCE AND OFFICIAL LAUNCH

1. The History
2. MCST – A catalyst for transition
3. MCST Framework – priorities for action
4. 1.5 to stay alive – High level Policy Unit
5. GCF – a country driven regional strategy

MICRONESIAN CENTER FOR SUSTAINABLE TRANSPORTATION

Presentation to the MTCC-PACIFC REGIONAL CONFERENCE AND OFFICIAL LAUNCH

1. **The History**
2. MCST – A catalyst for transition
3. MCST Framework – priorities for action
4. 1.5 to stay alive – High level Policy Unit
5. GCF – a country driven regional strategy

2012

1st Sustainable Sea Transport Talanoa
- USP host the first international experts meeting in the region since 1984. USP commit to 6-year research program

2013

Majuro Declaration on Climate Leadership
- Pacific leaders commit to be climate leaders – 9 PICs commit to reducing fossil fuel electricity – only RMI sets transport target

Inaugural PIDF Leaders Summit
- Low carbon shipping named a key regional priority

2014

2nd Sustainable Sea Transport Talanoa
- 150 regional and international participants

RMI calls for a national program for low carbon transport
- RMI ask USP to design a whole of country transition program

UN SIDS – S.A.M.O.A. pathway
- Sustainable transport recognized as regional priority by UN

2015

MCST Framework Endorsed RMI
- MCST Framework national consultation

RMI submission to MEPC68
- Tony De Brum leads Pacific delegations to call for 1.5 shipping emissions target

MCST Framework Endorsed MPS
- Palau, FSM, RMI Presidents endorse MCST approach

Suva Declaration
- PA is silent on bunker emissions

Paris Agreement

2016

MCST Framework Endorsed USP
- USP Council includes 12 PIC Education Ministers

MEPC70 - IMO commits to emissions reduction roadmap 2018-2023
- Pacific and European states combine to form SHAC. PIDF provides facilitation and USP technical backstopping. IMO Roadmap agreed- 2 stage process to determine shipping emissions.

MCST Framework Endorsed CANCC
- Atoll nations endorse RMI request to lead through a whole of country example

2017

ISWG GHG 1/MEPC71
- SHAC continues to press for PA level reductions and high ambition

MCST Framework Endorsed SIS Leaders
- SIS leaders

MCST Framework Endorsed 3rd Regional Ministers Forum
- Regional Ministers agree to support the MCST approach and RMI concept note for $250m GCF application

ISWG GHG 2
- SHAC continues to grow in membership. Roadmap progress is limited. 1.5 is still alive

COP23
- MCST organize High Level Low Carbon Shipping Event. ICS call RMI irresponsible and unrealistic. Germany announces €9.5m project to support MCST.

Macron Summit -Tony de Brum Declaration
- France honors RMI at "Real World Summit" with TdB Declaration calling for IMO to adopt Paris Agreement targets with urgency. Declaration will be signed by multiple world Presidents and Ministers this week

MICRONESIAN CENTER FOR SUSTAINABLE TRANSPORTATION

Presentation to the MTCC-PACIFC REGIONAL CONFERENCE AND OFFICIAL LAUNCH

1. The History
2. **MCST – A catalyst for transition**
3. MCST Framework – priorities for action
4. 1.5 to stay alive – High level Policy Unit
5. GCF – a country driven regional strategy

RMI – 4 strand approach to low carbon shipping transition

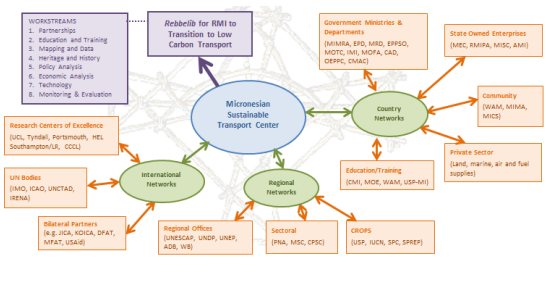
- Shipping High Ambition Coalition
  - All sectors must bear their 'fair share'
  - PSIDS issues must be accommodated and based on science
- Micronesia Center for Sustainable Transport
  - Whole of sector/whole of country low carbon transition
  - Catalyst for change cascading successes to the region
- Re-balance between transport/energy
  - Review NDCs to include transport and electricity emitting sectors
- Climate Financing for Pacific low carbon transition
  - RMI CN to 2nd GCF Council

MICRONESIAN CENTER FOR SUSTAINABLE TRANSPORTATION

Presentation to the MTCC-PACIFC REGIONAL CONFERENCE AND OFFICIAL LAUNCH

1. The History
2. MCST – A catalyst for transition
3. **MCST Framework – priorities for action**
4. 1.5 to stay alive – High level Policy Unit
5. GCF – a country driven regional strategy

Marshall Islands ***Rebbelib*** for Transition to Low Carbon Transport



1. Intra and Inter-Island Sailing Multi-Hulls

Historically, Marshallese developed fast, versatile multi-hull sailing vessels for local sea transport within and between its scattered atoll archipelagoes. Using the base built by WAM over the past two decades, this project will develop and trail designs for 2 classes of locally-built multi-hull sailing vessels for intra-lagoon and inter-atoll transport of passengers and cargo.

2. Wing in Ground 'Wingships'

Working with industry leaders from Korea, this project will undertake a full feasibility study and field trials of 'Wingships' capable of carrying 50 passengers or 10 ton of cargo on routes up to 500km as an alternative to small aircraft and high-speed craft. This evaluation phase with include full consideration of training, infrastructure, operations and maintenance requirements necessary to support uptake of this technology in a Micronesian operating environment.

3. MISC 15-Year Fleet Efficiency Strategy

The Marshall Islands Shipping Corporation operates a fleet of 5 vessels providing government services throughout the country. For most outer island communities this is their main source of connectivity. It is essential for Government service delivery, disaster response in particular. Vessels include conventional ships and landing craft aged from 3 to 30 years old. This project will plan how this fleet can be replaced over its operating life cycle with more efficient, low carbon based vessels and what operational and technology retrofits and changes can be effected within the current fleet to substantively reduce fuel use by the MISC fleet over a 15-year period.

4. Flettner Rotors

Flettner rotors offer strong potential for numerous applications including new builds and retrofits of existing vessels. Working with leading German innovators through Green Shipping Niedersachsen (GSN) at the HEL this project will identify and trial the use of Flettner rotors within a RMI context.

5. Wind Powered Interisland Freighter

Copra is an important economic commodity for Micronesian economies and was once traded using small sail-assisted shipping. This project will evaluate the best design option for a small wind powered island trader to support rejuvenation of the outer islands copra trade within RMI and with its neighboring States.

6. RE Options for Electric Vehicle Recharging

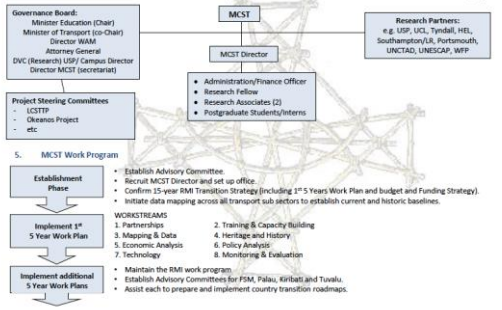
Hybrid and all-electric vehicles offer strong potential for cutting land vehicle fuel usage. However, for plug-in vehicles to be of use in atoll situations, the issue of sourcing the electricity has to resolved. This project will evaluate and trial leading candidates for powering recharging stations using renewables.

7. Marine Based/Sourced Biofuel

Biofuels have often been cited as a potential fossil fuel replacement for transport. Atolls pose the difficulty of a lack of land for source material. Coconut oil processing is one possibility. Recent research indicates that marine biota; especially seaweeds and algae may provide a solution. This

4. MCST Structure

MCST will be established as a small research team at USP-MI campus that coordinates and collaborates with all relevant stakeholders and projects.



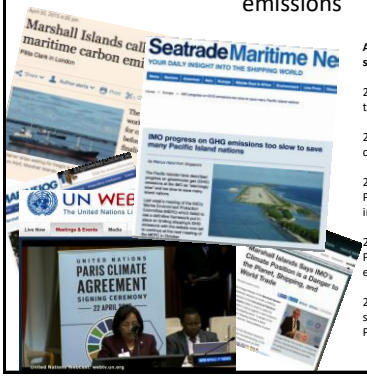
5. MCST Work Program

MICRONESIAN CENTER FOR SUSTAINABLE TRANSPORTATION

Presentation to the MTCC-PACIFC REGIONAL CONFERENCE AND OFFICIAL LAUNCH

1. The History
2. MCST – A catalyst for transition
3. MCST Framework – priorities for action
4. **1.5 to stay alive – High level Policy Unit**
5. GCF – a country driven regional strategy

RMI has been calling for shipping to reduce emissions



All sectors must contribute their fair share if Paris Agreement is to be met.

2013 - Majuro Declaration – RMI sets transport target

2015 - MEPC 68. RMI joins Pacific coalition to IMO

2015 - Suva Declaration – RMI joins Pacific leaders to ensure transport included

2016 - MEPC 69. RMI joins Pacific/European submission for emissions workplan

2017 – COP23 RMI states initial strategy must include commitment to PA targets

MICRONESIAN CENTER FOR SUSTAINABLE TRANSPORTATION

Presentation to the MTCC-PACIFC REGIONAL CONFERENCE AND OFFICIAL LAUNCH

1. The History
2. MCST – A catalyst for transition
3. MCST Framework – priorities for action
4. 1.5 to stay alive – High level Policy Unit
5. **GCF – a country driven regional strategy**

"Time is not our Friend"

Tony de Brum address to MEPC68

A shift to low carbon transport is a Now not a Future issue.
It needs a paradigm shift.

USP THE UNIVERSITY OF THE SOUTH PACIFIC

A joint collaboration between RMI Government and USP

MARITIME TECHNOLOGY COOPERATION CENTRE – PACIFIC (MTCC-PACIFIC)

CAPACITY BUILDING FOR CLIMATE MITIGATION IN THE MARITIME SHIPPING INDUSTRY
THE GLOBAL MTCC NETWORK (GMN) PROJECT

MTCC-PACIFC REGIONAL CONFERENCE AND OFFICIAL LAUNCH

12 – 15 December, 2017
Suva, Fiji

The Global MTCC Network (GMN) project is funded by the European Union and implemented by the IMO.

1. What are the drivers in your country for reducing greenhouse gas (GHG) from your country maritime transport including port and shipping?

- Our broader commitment as a party to the Paris Agreement via our NDCs. We are achieving this through major investment in renewable energy, with a target of 100% reliance on solar power by 2020. This has benefits for the maritime sector, through supplying power to our major ports which are linked to the national grid.
- The Cook Islands, through private sector initiatives, have trialled wind/diesel hybrid systems on board our inter-island domestic shipping vessels, primarily to lower costs of operation.

2. What are the needs to achieve the country objectives (if any) in reducing GHG emissions from maritime transport and/or to take action to address GHG emissions from maritime transport?

These could include a range of things including:

- promoting compensatory mechanisms for SIDS,
- subsidisation to offset economic impacts,
- provision of targeted technical assistance, and/or
- direct financial assistance with acquisition of new technology and/or new vessels to replace older less efficient technology.

3. What are the identified barriers that prevent from implementing country approach to reduce GHG emissions from maritime transport?

Cost. Efficacy. Potential negative impacts on vulnerable SIDS Economies:

- The Cook Islands obviously shares the same concerns as everyone else on the threats to our very existence as a nation by the slow onset impacts of human induced climate change, as well as by more severe short term impacts (i.e. cyclones) to our extremely vulnerable economy as a small island developing state.
- However since the contribution to global GHG from our region is extremely small (less than .03%), there is a comparatively limited amount we can do as a region to lower overall global emissions. This is largely beyond our control.
- We are however extremely sensitive to any measures which are taken which would significantly increase the costs of shipping in our region, which are already some of the highest in the world.
- A timely, reliable, efficient and sustainable shipping service is absolutely critical to the economic wellbeing of our country.
- Since our island economies rely heavily on imported goods and foodstuffs, there remains a very real risk that any measures to minimise GHG emissions will carry with them a cost that will be passed on down to the 'end consumer', and in this case that is the SIDS - who already face extremely high shipping costs – which may further inflate the ever increasing cost of living in our islands.
- In this case, we need to be carefully considering ways to minimise such impacts to zero – commensurate which any reductions of GHG. By example, we need to inform ourselves objectively as to what impacts moving to low carbon fuels will have for our region – including costs and availability of such fuels. The costs of modifications, including to engines, on ships which are largely older ships (retired from Europe, primarily) also need to be considered. Any proposed carbon offset schemes need to avoid adding to the costs of goods shipped from distant markets to our islands. Other proposed ideas such as slow-steaming will not work in our region either for the reasons already articulated elsewhere in this presentation.

4. What are the possible relevant actions to address GHG emissions from maritime transport?

- Since costs of shipping in our region are already very high and already impact on our ability to meet key SDGs related to the 2030 Development Agenda, we need to explore ways to bring these costs down, not increase them. This is why the Cook Islands often links these challenges with the GHG issue. Indeed, its important to acknowledge that in meeting the relevant SDGs, we also increase the ability for our countries to become more resilient to the impacts of human-induced climate change.
- So whatever actions we ultimately decide to take, these must always be considered within the broader context of meeting the development aspirations of our countries.

5. Conclusion

In summary, we are open to considering any and all options related to GHG reductions in shipping, but these must not penalise SIDS who are on the front line of climate change, they must recognise the vulnerability of our small economies and the high sensitivity to any increases in transport costs globally, and they must offer a better alternative than the situation we currently face.

GMN | The Global MTCC Network
MTCC PACIFIC
Meitaki Maata
Thank you

MARITIME TECHNOLOGY COOPERATION CENTRE – PACIFIC (MTCC-PACIFIC)

CAPACITY BUILDING FOR CLIMATE MITIGATION IN THE MARITIME SHIPPING INDUSTRY
THE GLOBAL MTCC NETWORK (GMN) PROJECT

MTCC-PACIFC REGIONAL CONFERENCE AND OFFICIAL LAUNCH

12 – 15 December, 2017
Suva, Fiji

The Global MTCC Network (GMN) project is funded by the European Union and implemented by the IMO.

What are the drivers in your country for reducing greenhouse gas (GHG) from your country maritime transport including port and shipping

- National Action Plan on Marine Pollution (Marine Pollution).
- Energy Management in Port
- Kiribati Nationally Determined Contributions (NDCs) on GHG reduction targets
- Strategies in reducing imported fossil fuel dependency

What are the needs to achieve the country objectives (if any) in reducing GHG emissions from maritime transport and/or to take action to address GHG emissions from maritime transport

- Strengthening GHG Policies, Strategies and Legislation;
- Delivering appropriate education, training and awareness programmes
- Improving knowledge and information generation, management and sharing
- Promoting the use of sustainable renewable sources of energy and energy efficiency
- Strengthening capacity to access finance, monitor expenditures and maintain strong partnerships;

What are the identified barriers that prevent from implementing country approach to reduce GHG emissions from maritime transport

- Lack of national GHG legislation in place in addressing GHG
- No Technical Expertise in GHG within the Country
- Lack of detailed information and understanding in anthropogenic CO2 emissions & its impacts by the Maritime, Port and Shipping Sectors
- No study undertaken yet in addressing the impacts of GHG by the maritime transport

What are the possible relevant actions to address GHG emissions from maritime transport

- Seeking support by Government - full enactment and implementation of GHG legislations, Policies and Strategies
- Assistance by foreign technical experts to assist in coordination of awareness programme on GHG to Port, Shipping Communities and relevant key stakeholders
- Entailing benefits to the shipping market, in terms of emission reductions (cost vs revenue)
- Provision of an energy-efficiency culture in both port and shipping sectors

And

thank you for your attention!

Kam rabwa

MARITIME TECHNOLOGY COOPERATION CENTRE – PACIFIC (MTCC-PACIFIC)

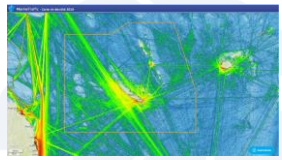
CAPACITY BUILDING FOR CLIMATE MITIGATION IN THE MARITIME SHIPPING INDUSTRY
THE GLOBAL MTCC NETWORK (GMN) PROJECT

MTCC-PACIFC REGIONAL CONFERENCE AND OFFICIAL LAUNCH

12 – 15 December, 2017
Suva, Fiji

The Global MTCC Network (GMN) project is funded by the European Union and implemented by the IMO.

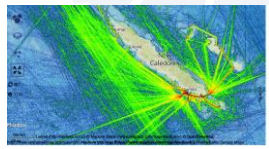
International and domestic maritime traffic



International routes around New-Caledonia

International routes concentrated on the west and south coast

Mining activities all around the main island



Maritime traffic in New-Caledonia

1. General international traffic (container carriers, tankers, car carriers…) in Noumea harbour and nickel factories

International bulk carriers in nickel mines

About 700 vessels /year

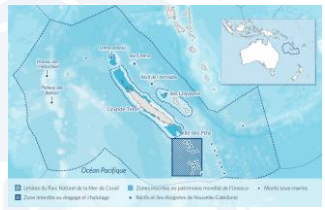
2. International Cruising liner

About 500 vessels / year

3. Domestic traffic

Using only gasoil

Division of competences



French state: French flag vessels on international voyages – Port state control – Safey of navigation on EEZ

New-Caledonia government: Environment on exclusive economic zone – natural park on coral sea – domestic vessels

New caledonia provinces: Environment on internal and territorial waters

Measures taken to limit GHG

1. French state: MARPOL VI, IEEC, MRV (Europe area), PSC

2. New-Caledonia government: energy transition plan for next 20 years

SECA to be implemented in the natural park on coral sea

3. Provinces: SECA to be implemented in harbour area (to limit emissions from cruising liner)

The main issues

1. To improve maritime traffic monitoring and to check the compliance to the regulation

2. To coordinate the different measures to reduce GHG

3. To support innovating technical projects for vessels propulsion

GMN
MTCC PACIFIC
Thank you for your attention

GMN | The Global MTCC Network
A global network for energy-efficient shipping

MTCC PACIFIC
Maritime Technology Cooperation Centre

MARITIME TECHNOLOGY COOPERATION CENTRE – PACIFIC (MTCC-PACIFIC)

CAPACITY BUILDING FOR CLIMATE MITIGATION IN THE MARITIME SHIPPING INDUSTRY
THE GLOBAL MTCC NETWORK (GMN) PROJECT

MTCC-PACIFC REGIONAL CONFERENCE AND OFFICIAL LAUNCH

12 – 15 December, 2017
Suva, Fiji

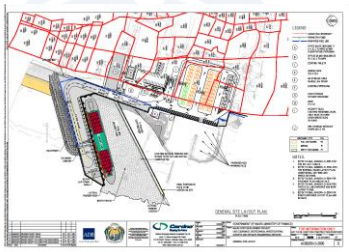
EUROPEAN UNION
IMO INTERNATIONAL MARITIME ORGANIZATION
Pacific Community Communauté du Pacifique
HOST INSTITUTIONS OF MTCC-PACIFIC
SPREP Secretariat of the Pacific Regional Environment Programme

The Global MTCC Network (GMN) project is funded by the European Union and implemented by the IMO.

Location

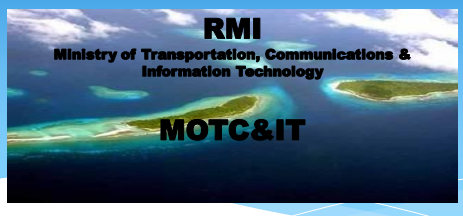
Open Sea Moorings

New Wharf



MTCC-PACIFC REGIONAL CONFERENCE AND OFFICIAL LAUNCH

12 – 15 December, 2017
Suva, Fiji



Geographical location

The Republic of the Marshall Islands (RMI) is located between latitude 4 degree to 14 degree north and longitude 160 degree to 173 degree east, consisting of 29 atolls and 5 independent islands widely scattered. RMI has a wide exclusive economic zone (EZZ) of 2 million square kilometer. The Marshall Islands is characterized by two chains the Ratak, or Sunrise, to the east, and the Ralik, or Sunset, to the west. The chains lie about 125 miles (200 kilometres) apart and extend some 800 miles northwest to southeast.

BACKGROUND OF MARITIME IN RMI

Maritime has always been a way of life for the Marshallese People. Connecting the whole Marshall Islands has been by means of sea transportation. Domestic Shipping services plays an indispensable role to keep the lifeline in their daily life of the Marshall Islands. The basic concept of shipping services is simply that the vessels load cargoes to the outer islands to be delivered to individual consignees. Furthermore, a vessel also carries a merchant onboard to sell the goods in the outer islands and on return trips, they would pick up the copra to bring back the main island.

Drivers in RMI for Reducing GHG

- Purchasing New Ships with fuel efficiency capacity, including wind/solar propelled vessels.
- Effective and efficient operations on Ports
- Aid to Navigation with light buoys in the other island for night operations
- RMI in the Heads of Transport Officials Meeting in November 2015 in Suva Fiji, RMI took the initiative in regard to low carbon transport transition and transport emissions reductions measures at global and local levels.

Cont.

- RMI submission to MEPC 68, May 2015 requesting IMO set a global sector target for shipping emission reduction and consequential actions;
- RMI submission of its INDC including specific targets of transport emissions reduction;
- RMI decision, endorsed by the 2015 Micronesian Presidents Summit, to establish the Micronesian Sustainable Transport Center (MSTC) as a sub-regional and regional catalyst for change.

Needs & Objectives

- Resources (equipment) to fasten both offloading and loading
- We need more sustainable sea transport like Vaka

Barriers:

- Second-handed vessels are usually purchased
- Underdeveloped infrastructure for docking and loading/offloading of passenger cargo in the outer islands.
- Limited Formal Training for RMI Maritime Officers and Seafarers.
- Minimal Maintenance of Government Vessels.
- Bridging the gap between the domestic and international maritime acts to bring about better coordination in which they can better compliment each other.

Actions

MARITIME TECHNOLOGY COOPERATION CENTRE – PACIFIC (MTCC-PACIFIC)

CAPACITY BUILDING FOR CLIMATE MITIGATION IN THE MARITIME SHIPPING INDUSTRY
THE GLOBAL MTCC NETWORK (GMN) PROJECT

MTCC-PACIFC REGIONAL CONFERENCE AND OFFICIAL LAUNCH

12 – 15 December, 2017
Suva, Fiji

The Global MTCC Network (GMN) project is funded by the European Union and implemented by the IMO.

Drivers for Reducing Greenhouse Gas (GHG) Emissions from Maritime Transportation in PNG

- International/Regional Conventions & Treaties/Agreements.
- Real Effects of Climate Change Experienced in the Country – Sea Level Risings & Islands Sinking.
- Promotion of Green Energy and Transportation Strategies.
- GHG & Emissions Control Regulations.

PNG objectives in addressing GHG emissions from maritime transport

- MARPOL Annex VI Ratification-Prevention of Air Pollution from Ships.
- Development and Enforcement of National Legislation.
- Capacity/ Resources Development
- Data Collections, Inspections & Monitoring Aspects

Implementation of Preventative Barriers of reducing GHG emissions from maritime transportation sector

- Lack of National Legislation for regulating both international and domestic shipping.
- Inspections & Monitoring capabilities.
- Emissions data collections, collations, analysis & interpretations.(Costs & Economical Benefits).
- Associated costs involved in shipping engine efficiencies improvements.

Reduction actions to address GHG emissions from maritime transport.

- GHG Emissions reduction strategy put in place.
- Ratification of MARPOL Annex VI-Prevention of Air Pollution.
- National Legislation for regulating both international and domestic shipping.
- Human Resources Capacity Development
- Data Collections, Inspections & Monitoring Aspects
- Possibility of declaring of Emissions Control Areas (ECA) within identified PNG Waters.
- LNG Powered Marine Engines
- GHG Emissions Control Curriculum –Maritime Training Institutions

THANK YOU

MARITIME TECHNOLOGY COOPERATION CENTRE – PACIFIC (MTCC-PACIFIC)

CAPACITY BUILDING FOR CLIMATE MITIGATION IN THE MARITIME SHIPPING INDUSTRY
THE GLOBAL MTCC NETWORK (GMN) PROJECT

MTCC-PACIFC REGIONAL CONFERENCE AND OFFICIAL LAUNCH

12 – 15 December, 2017
Suva, Fiji

The Global MTCC Network (GMN) project is funded by the European Union and implemented by the IMO.

Samoa's Report on GHG Emission

What are drivers for reducing greenhouse gas (GHG) from maritime Transport including ports and shipping in Samoa?

- International Conventions – complying to requirements that call for protection of marine environment from climate change
- National Framework - initiatives involving relevant maritime stakeholders that aligned with international maritime instruments to ensure enabling laws, action plans & procedures to reduce GHG are developed and implemented

What are the needs to achieve the country objective in reducing GHG emissions from maritime Transport and to take actions to address GHG emission?

- MARPOL – comply with applicable provisions of the Convention (eg. Annex VI)
- Implement Marine Pollution Prevention Act 2008
  (Regulations near completion)
- Enforce National Disaster Management Plan
- Implement Greenhouse gas Abatement Strategy 2008-2018.

What are the identified barriers that prevent from implementing country approach to reduce GHG emissions from maritime transport

- Scarcity of resources – finance; technology and lack of expertise (Human resource)
- Cost of New Technology on existing ships - to comply with Annex VI of MARPOL reduce Sox and Nox on vessels of > 500gt

What are the possible relevant actions to address GHG emissions from the maritime transport?

- Enforcement of Ports State & Flag State inspections.
- Public Awareness programmes
- Application of International Rules and Regulations.
- Developing of an Energy National Strategy plan to reduce the GHG emission in the maritime transport in Samoa

GMN | The Global MTCC Network
MTCC PACIFIC
Faafetai.

GMN | The Global MTCC Network
A global network for energy-efficient shipping

MTCC PACIFIC
Maritime Technology Cooperation Centre

MARITIME TECHNOLOGY COOPERATION CENTRE – PACIFIC (MTCC-PACIFIC)

CAPACITY BUILDING FOR CLIMATE MITIGATION IN THE MARITIME SHIPPING INDUSTRY
THE GLOBAL MTCC NETWORK (GMN) PROJECT

MTCC-PACIFIC REGIONAL CONFERENCE AND OFFICIAL LAUNCH

12 – 15 December, 2017
Suva, Fiji

EUROPEAN UNION

IMO INTERNATIONAL MARITIME ORGANIZATION

Pacific Community Communauté du Pacifique

HOST INSTITUTIONS OF MTCC-PACIFIC

SPREP Secretariat of the Pacific Regional Environment Programme

The Global MTCC Network (GMN) project is funded by the European Union and implemented by the IMO.

Solomon Islands

- Emissions from steaming;
  - Slower speed reduces emissions – dependant on demand for shipping services, freight rates and bunker prices.
  - Improved ship design (IMO's Energy Efficiency Design Index).
  - SI does not "design" ships; we "inherit" them second-hand
- Emissions in port;
  - Reduction cuts climate impact from CO_2; less NO_x; less CO_2
  - Cuts health risks; improves air quality in Honiara;
  - Depends on City Council's efforts to reach political climate goals;
  - Buses going up hills in Honiara are worse offenders compared to domestic shipping; some on the flat are very bad also.

Solomon Islands

- Measures to reduce GHG
  - Alternative fuels;
  - Ship design; and
  - Operation.
- Horses for Courses
  - Alternative fossil fuels are not realistic, solar and wind power are;
  - We "inherit" not "design", but retrofit is worth examining. Domestic vessels do not lend themselves to retrofit (essential cargo handling gear gets in the way; stability problems; etc.,)

Solomon Islands

- Improved operation holds most potential in domestic fleet;
  - "scientific" operations give best reductions;
  - Better utilisation of "liner" routes around the Solomon Islands – benefits of scale;
- International Shipping Port operations;
  - Potential for speeding up ship's turnaround times;
  - Port opening hours;
  - Stevedoring efficiency;
  - Berth availability;
  - Efficient cargo-handling equipment

Solomon Islands

- Needs:
  - Consciousness raising;
  - Better planning;
  - Better statistical information;
  - Understanding of "cost of ship's time";
  - Removal of socio-political barriers;
    - Provincial rivalry and exclusivity;
    - "Shipping grant ships" vs "private sector ships";
  - Removal of socio-economic barriers;
    - High cost of trading licences to more than one Province;
    - Rationalisation of shipping services.

SOLOMON ISLANDS HAS ONE OF THE CLEAREST GLOBAL ATMOSPHERES – LET'S KEEP IT THAT WAY

Solomon Islands

- Relevant actions to resolve issues:
  - Solomon Islands very active at IMO MEPC in the fight to reduce GHG internationally;
  - Solomon Islands Government to take more concern about all shipping matters; safety, environment and commercial;
  - Increase powers of SIMSA, the safety administration, to take on responsibility for marine environmental protection (as mandated, but not resourced) and also become involved in research into commercial shipping matters as an "Authority" (this is in the pipeline);
  - Rationalisation of port operations to increase productivity;
  - Higher levels of shipping and port education and training (this is also in the pipeline).

MARITIME TECHNOLOGY COOPERATION CENTRE – PACIFIC (MTCC-PACIFIC)

CAPACITY BUILDING FOR CLIMATE MITIGATION IN THE MARITIME SHIPPING INDUSTRY
THE GLOBAL MTCC NETWORK (GMN) PROJECT

MTCC-PACIFC REGIONAL CONFERENCE AND OFFICIAL LAUNCH

12 – 15 December, 2017
Suva, Fiji

The Global MTCC Network (GMN) project is funded by the European Union and implemented by the IMO.

CONTENTS

- Drivers reducing GHG emissions from Solomon Ports (Ports Perspective)
- What is needed to achieve reduction of GHG emissions (Ports Perspective)
- Barriers to implementation of country approach to GHG emission reduction
- Possible action from Ports to reduce GHG emissions
- Projects to reduce GHG emissions _ Solomon Ports

Drivers for the reduction of GHG emissions from Solomon Ports (Ports Perspective)

- Rising sea level – Climate change
- High electricity cost

What is needed to achieve the country objectives in reduction of GHG emissions (Ports Perspective)

- Government policy
- Ports initiative to become "GREEN"
- Commitment for environment

Barriers identified for the implementation of country approach to GHG emissions (Ports Perspective)

- Lack of awareness and importance of GHG reduction
- High cost of imported material to the country (solar panels + regulatory framework)
- High cost of power generation (Diesel power – as a result providing shore power for vessels will be very expensive)

Possible action from Ports to reduce GHG emissions

- Use of renewable energy for yard lighting, reefers a
- Use of hybrid vehicles
- Reduce un-necessary electricity usage
- Maintain vehicles for better performance

Electricity Consumption – Before and After the Energy Audit

August Electricity Bill

HONIARA ELECTRICITY						
INVOICE DATE	A/C NAME	A/C ALLOCATED	A/C #	AMPOUNT	CHQ #	DATE PAID
10/8/2017	ELECTRICITY	410	A/C 05097	$ 384,736.63	116560	15/9/2017
10/8/2017	ELECTRICITY	410	A/C 05098-07	$ 1,449.05	116560	15/9/2017
			TOTAL	$ 386,185.68		
INVOICE DATE		NORO ELECTRICITY				
15/8/2017	ELECTRICITY	410	A/C 81034	$ 1,176.25	116561	15/9/2017
15/8/2017	ELECTRICITY	410	A/C 81006	$ 25,026.73	116561	15/9/2017
15/8/2017	ELECTRICITY	410	A/C 81006-01	$ 3,156.33	116561	15/9/2017
15/8/2017	ELECTRICITY	410	A/C 81006.02	$ 3,760.74	116561	15/9/2017
			TOTAL	$ 33,120.05		
			Grand Total Bill	$ 419,305.73		

September Electricity Bill

HONIARA ELECTRICITY						
INVOICE DATE	A/C NAME	A/C ALLOCATED	A/C #	AMPOUNT	CHQ #	DATE PAID
10/9/2017	ELECTRICITY	410	A/C 05097	$ 343,065.29	116730	4/10/2017
10/9/2017	ELECTRICITY	410	A/C 05098-07	$ 1,402.92	116730	4/10/2017
			TOTAL	$ 344,468.21		
INVOICE DATE		NORO ELECTRICITY				DATE PAID
15/9/2017	ELECTRICITY	410	A/C 81034	$ 1,341.32	116728	3/10/2017
15/9/2017	ELECTRICITY	410	A/C 81006	$ 24,348.74	116728	3/10/2017
15/9/2017	ELECTRICITY	410	A/C 81006-01	$ 4,717.30	116728	3/10/2017
15/9/2017	ELECTRICITY	410	A/C 81006.02	$ 4,071.94	116728	3/10/2017
			TOTAL	$ 34,479.30		
			Grand Total Bill	$ 378,947.51		

Electricity Consumption – Before and After the Energy Audit

HONIARA ELECTRICITY						
INVOICE DATE	A/C NAME	A/C ALLOCATED	A/C #	AMPOUNT	CHQ #	DATE PAID
10/11/2017	ELECTRICITY	410	A/C 05097	$ 347,130.49	117106	8/11/2017
10/11/2017	ELECTRICITY	410	A/C 05098-07	$ 1,831.52	117106	8/11/2017
			TOTAL	$ 348,962.01		
INVOICE DATE		NORO ELECTRICITY				DATE PAID
15/10/2017	ELECTRICITY	410	A/C 81034	$ 785.88	117107	8/11/2017
15/10/2017	ELECTRICITY	410	A/C 81006	$ 18,377.73	117107	8/11/2017
15/10/2017	ELECTRICITY	410	A/C 81006-01	$ 4,018.35	117107	8/11/2017
15/10/2017	ELECTRICITY	410	A/C 81006.02	$ 2,667.76	117107	8/11/2017
			TOTAL	$ 25,849.72		
			Total	$ 374,811.73		

LED Yard Lights – Installation in Progress

LED Yard Lights – Installation in Progress

LED Yard Lights – Installation in Progress

LED Yard Lights – Installation in Progress

LED Yard Lights – Installation in Progress

THANK YOU TUMAS

GMN | The Global MTCC Network
A global network for energy-efficient shipping

MTCC PACIFIC
Maritime Technology Cooperation Centre

MARITIME TECHNOLOGY COOPERATION CENTRE – PACIFIC (MTCC-PACIFIC)

CAPACITY BUILDING FOR CLIMATE MITIGATION IN THE MARITIME SHIPPING INDUSTRY
THE GLOBAL MTCC NETWORK (GMN) PROJECT

MTCC-PACIFC REGIONAL CONFERENCE AND OFFICIAL LAUNCH

12 – 15 December, 2017
Suva, Fiji

TONGA- COUNTRY PRESENTATION (From the Port Perspective)

EUROPEAN UNION
IMO INTERNATIONAL MARITIME ORGANIZATION
Pacific Community Communauté du Pacifique
HOST INSTITUTIONS OF MTCC-PACIFIC
SPREP Secretariat of the Pacific Regional Environment Programme

The Global MTCC Network (GMN) project is funded by the European Union and implemented by the IMO.

Reducing GHG @ Port of Nuku'alofa

Key 3 Drivers to reduce GHG in the Port of Nuku'alofa…

1. Political Support
   - ✓ Strong support from the government/Cabinet (shareholder) will help PAT BoDs and Management successfully achieved the output objectives set in the PAT business Plan to reduce GHG in the Port of Nuku'alofa.
2. Board of Directors Support:
   - ➢ The BoDs must have the value of good governance with strong leadership in order to make concise and informative policy decisions in regards to reduce GHG in the port operations.
3. CEO and Management full commitments through:
   - PAT Internal Policies - *through our Business plan (Statement of Corporate Intent) set by the Board and Management- Strategic Objectives and KPIs.*
   - Procurement strategies - *through investing on low carbon emission handling equipment.*
   - Cargo handling process and wharf layout - *generally aiming at improving efficiency but at the same time reducing GHG emission through less movement by the forklifts.*

PAT needs to achieve its objectives on reducing GHG…

- Maritime Division as Regulatory Body to be more proactive in enforcing existing maritime environmental legislations and regulations.
- Maritime Division to set and monitor the level of GHG emission in the Maritime sector in Tonga.
- Collective effort from a national and regional level to seriously address this issue.

Barriers to achieve the objectives…

- Budget constraint- Funding
- poor enforcement from Regulatory body- Maritime Authority.
- Lack of clear legal framework to address GHG in the Maritime sector in Tonga.
- Lack of coordination between relevant stakeholders i.e Maritime Authority, PAT & Local boat owners.

PAT actions to address GHG emissions…

- Implement effective environmental policies and procedure to minimise the risks of environmental pollution.
- Work closely with Maritime Authority in enforcing maritime environmental legislations and regulations.
- More clear strategies and KPIs to address GHG.
- Report and analyse annual fuel consumption to determine GHG emission to assist setting KPIs.

GMN | The Global MTCC Network

MTCC PACIFIC Maritime Technology Cooperation Centre

Malo ‘Aupito & Thank you

GMN | The Global MTCC Network
A global network for energy-efficient shipping

MTCC PACIFIC
Maritime Technology Cooperation Centre

MARITIME TECHNOLOGY COOPERATION CENTRE – PACIFIC (MTCC-PACIFIC)

CAPACITY BUILDING FOR CLIMATE MITIGATION IN THE MARITIME SHIPPING INDUSTRY
THE GLOBAL MTCC NETWORK (GMN) PROJECT

MTCC-PACIFC REGIONAL CONFERENCE AND OFFICIAL LAUNCH

12 – 15 December, 2017
Suva, Fiji

EUROPEAN UNION
IMO INTERNATIONAL MARITIME ORGANIZATION
Pacific Community Communauté du Pacifique
HOST INSTITUTIONS OF MTCC-PACIFIC
SPREP Secretariat of the Pacific Regional Environment Programme

The Global MTCC Network (GMN) project is funded by the European Union and implemented by the IMO.

GMN | The Global MTCC Network
MTCC PACIFIC Maritime Technology Cooperation Centre

What are the drivers in your country for reducing greenhouse gas GHG from your country maritime transport including port and shipping

- **National Strategy for Sustainable Development 2016 -2020**
  - **STRATEGIC AREA 9 – Infrastructure and Support Services**
    - **THEMATIC AREA – 9.10** New infrastructure and better service support will, by definition, play central roles in combating the effects of climate change. GoT commits to embarking on aggressive climate change adaptation measures, as permitted by available funding.
- **Sustainable Development Goals**
  - **GOAL 13 – Take Urgent Action to Combat Climate Change and its Impacts**
  - **GOAL 14 - Conserve and sustainably use the oceans, seas and marine resources for sustainable development**

GMN | The Global MTCC Network
MTCC PACIFIC Maritime Technology Cooperation Centre

What are the needs to achieve the country objectives(if any) in reducing GHG emission from maritime transport and/or to take action to address GHG emissions from maritime transport

- ✓ Update National Legislation include requirements for reduction of GHG
- ✓ Expand workforce for small administrations
- ✓ All MS agree that shipping should reduce its GHG emissions in the long term
- ✓ Trading routes and level playing field should not suddenly change because of CO_2 reduction measures.

GMN | The Global MTCC Network
MTCC PACIFIC Maritime Technology Cooperation Centre

What are the identified barriers that prevent from implementing country approach to reduce GHG emissions from maritime transport

- No national legal framework in place
- Minimal public knowledge on GHG
- Stimulate/support cooperation between Governments – Research – Shipping sector (golden triangle)
- As in other sectors, development of appropriate technologies, fuels and infrastructure for shipping must be incentivized to reduce GHG emissions.
- Business relationships between charter and ship owner will need improvement.
- Measures of impacts on all States must be considered before implementation, especially impacts on SIDS and LDCs
- Negative impacts on vulnerable States, Ship registries, and the Shipping sector will be greater and more difficult to adapt later in the century when measures are implemented.
- ALL SIDs and LDC's must maintain solidarity in global debates to reduce GHG emissions.

GMN | The Global MTCC Network
MTCC PACIFIC Maritime Technology Cooperation Centre

What are the possible relevant actions to address GHG emission from maritime transport.

- ➢ All ships should target to have almost a zero Co_2 footprint in the near future by green (environmental friendly) ship, instead of black ships
- ➢ Energy efficiency measures alone will not achieve zero emission from the sector (2035,2050,2060,2100)
- ➢ A clear roadmap on CO_2 reduction to guide the shipping sector undertake proper planning for the future.

GMN | The Global MTCC Network
MTCC PACIFIC Maritime Technology Cooperation Centre

Fakafetai Lasi

GMN | The Global MTCC Network
A global network for energy-efficient shipping

MTCC PACIFIC
Maritime Technology Cooperation Centre

MARITIME TECHNOLOGY COOPERATION CENTRE – PACIFIC (MTCC-PACIFIC)

CAPACITY BUILDING FOR CLIMATE MITIGATION IN THE MARITIME SHIPPING INDUSTRY
THE GLOBAL MTCC NETWORK (GMN) PROJECT

MTCC-PACIFIC REGIONAL CONFERENCE AND OFFICIAL LAUNCH – VANUATU PRESENTATION

12 – 15 December, 2017
Suva, Fiji

EUROPEAN UNION
IMO INTERNATIONAL MARITIME ORGANIZATION
Pacific Community Communauté du Pacifique
HOST INSTITUTIONS OF MTCC-PACIFIC
SPREP Secretariat of the Pacific Regional Environment Programme

The Global MTCC Network (GMN) project is funded by the European Union and implemented by the IMO.

Country Profile - VANUATU

- KEY MARITIME INSTITUTIONS
  - Ports and Marine Department (Ports Act [Cap 26])
    - Responsible for the operation of Ports and Harbour of Port Vila/Santo
  - Office of the Maritime Regulator (Maritime Sector Regulatory Act 2016
    - Responsible for regulation of Maritime sector of Vanuatu including but not limited to the regulations of vessels and ports, vessel registration, licensing and safety, policy advice, certification of seafarers, trainings, implementation of conventions, etc.
  - Ministry of Infrastructure and Public Utilities
    - International shipping Registry
  - Vanuatu Maritime College
    - Maritime trainings – STCW trainings

Brief stats - Vanuatu

- 57 registered vessels (LC, tugs and passenger vessels;
- 3 International Ports (Longest 360 M);
- 1 privately Owned and 2 owned by Government;
- Stevedoring - operated by 2 different private companies in separate Ports under concessions;

Questions:

1. What are the drivers in your country for reducing greenhouse gas (GHG) from your country maritime transport including port and shipping?

- Vanuatu commitment towards implementing its National Energy Roadmap 2016-2030 in reducing GHG emission in the transport sector (land and sea);
- Vanuatu commitment towards the Global (GSDG) Sustainable Development Goals
- The NERM found that by 2030, Vanuatu consumption of imported petroleum products (petroleum, diesel, etc.) is expected to grow by 50%.
- All vessels operated in Vanuatu relies on diesel fuel
- No standards in place to assess GHG emissions from ships
- Demand for shipping in Vanuatu increases in the past 5 years
- Majority of the population relies on shipping for transportation
- More vessels imported into Vanuatu
- International port of entries upgraded and expanded with state of the art facilities-high demand of electricity

2. What are the needs to achieve the country objective (if any) in reducing GHG emissions from maritime transport and/or to take action to address GHG emissions from maritime transport?

- Need capacity building for maritime administrators and shipping stakeholders
- Need awareness on the importance of reducing GHG emissions for shipping
- Private sector engagement
- Proper policy and regulations needed to legalize the requirement of GHG emissions
- Access to green fund to support Vanuatu implement GHG programs

3. What are the identified barriers that prevent from implementing country approach to reduce GHG emissions from maritime transport?

- Lack of awareness on ship owners
- Lack of understanding and economic benefit for promoting energy efficiency in shipping and Ports
- Lack of partnership between stevedoring operators and Ports administrators (two different entities operate the Ports and stevedoring) in terms of reducing GHG emissions
- Lack of policies and regulations to regulate GHG emissions in Ports and vessels

4. What are the possible relevant actions to address GHG emissions from maritime transport?

- Established proper policies and regulations to address GHG emissions
- Engaged concessionaires in early discussions and negotiations to promote energy efficiencies
- Create a national WG to advocate and implement GHG emissions programs
- Provide capacity for maritime administrations and ship owners on the benefit of GHG emissions (economic and technical)
- Access green fund where possible
- Government must take ownership and support GHG programs (tax incentive, exemptions, subsidy, etc)

Thank You very much

An IMO Strategy on Reduction of GHG Emissions from International Shipping

Dr Leigh Mazany| GMN Consultant

MTCC-Pacific 1st Regional Workshop | 12-15 December 2017 | Suva, Fiji

The Global MTCC Network (GMN) project is funded by the European Union and implemented by IMO
The views expressed in this presentation can in no way be taken to reflect the views of the European Union

Content

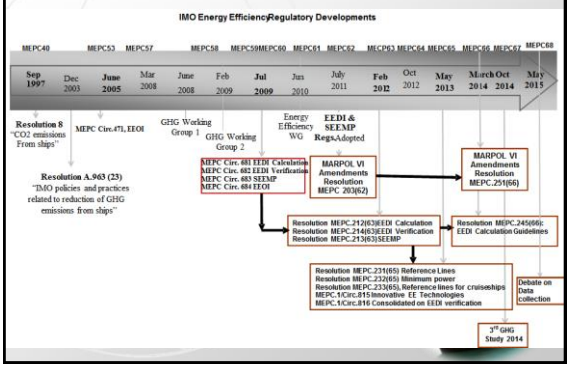
1 IMO initiatives
2 Data Collection System
3 An IMO Strategy on Reduction of GHG Emissions
4 Capacity Building and Technical Cooperation
5 Potential impacts, costs and benefits

IMO initiatives

IMO Initiatives to Reduce Greenhouse Gas (GHG) Emissions

- IMO's work on GHG emissions rooted in
  - Convention on the International Maritime Organization 1948
  - UNCLOS 1982
  - MARPOL 1973 and its 1997 Protocol + International Air Pollution Conference resolution 8
  - Assembly resolution A.963(23) on IMO policies and practices related to the reduction of GHG emissions from ships (2003)

Activities leading to Chapter 4 of MARPOL Annex VI



Energy efficiency regulations for ships

- In 2011, Chapter 4 (regulations 19 to 23) added to MARPOL Annex VI (Resolution MEPC.203(62))
- Entered into force on 1 January 2013
- First ever mandatory global energy efficiency standard for international shipping
  - Applies to internationally trading ships of ≥ 400 GT
  - Make mandatory:
    - **Energy Efficiency Design Index (EEDI)** for new ships
    - **Ship Energy Efficiency Management Plan (SEEMP)** for all ships
  - Address ship types responsible for 85% of GHG emissions from international shipping
  - Introduced new International Energy Efficiency Certificate (IEEC)

Data Collection System

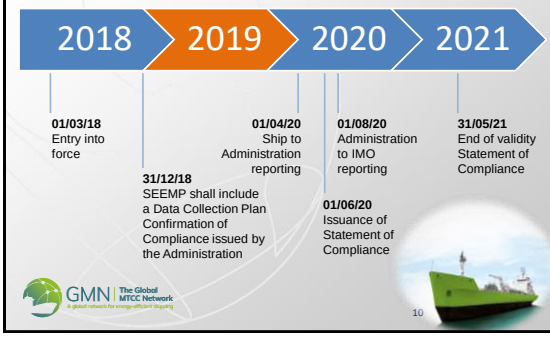
Data Collection System

- Work at MEPC 64 (Oct 2012) and MEPC 65 (May 2013) on a proposal by the US to enhance the energy efficiency of international shipping through a phased approach:
  - Data collection and analysis phase (Phase I)
  - Pilot phase (Phase II)
  - Full implementation phase (Phase III)
- MEPC agreed to add to its agenda work on "Further technical and operational measures for enhancing the energy efficiency of international shipping"

Data Collection System (cont'd)

- MEPC 68 (May 2015) agreed that development of a data collection system for ships should progress and follow a three-step approach:
  - data collection;
  - data analysis; and
  - followed by decision-making on what further measures, if any, are required
- Noted that a purpose of the data collection system was to analyse energy efficiency and that for this analysis to be effective some transport work data needs to be included
- MEPC 69 (April 2016) agreed amendments to MARPOL Annex VI (Regulation 22A) – Data collection system for fuel consumption
- Adopted at MEPC 70 (October 2016) and enters into force 1 March 2018, with reporting starting in 2019

IMO Data Collection System for Fuel Oil Consumption of Ships in MARPOL Annex VI



10

An IMO Strategy on Reduction of GHG Emissions

IMO strategy on the reduction of GHG emissions

MEPC 70 (October 2016)

- MEPC approved a Roadmap for developing a comprehensive IMO strategy on reduction of GHG emissions from ships, which foresees an initial GHG reduction strategy to be adopted in 2018.
- It contains a list of activities, including further IMO GHG studies and significant intersessional work, with relevant timelines and provides for alignment of those new activities with the ongoing work by the MEPC on the three-step approach to ship energy efficiency improvements.
  - This alignment provides a way forward to the adoption of a revised strategy in 2023 to include short-, mid-, and long-term further measures, as required, including implementation schedules.
- The Committee also agreed to hold intersessional working group meetings on reduction of GHG emissions from ships.

October 2016 (MEPC 70)	- Adoption of Data Collection System - Voluntary data collection and submission begins - Approval of Roadmap
Week before MEPC 71	- Intersessional meeting to start discussions on a comprehensive IMO strategy on reduction of GHG emissions from ships, taking into account inputs such as: (1) Third IMO GHG Study; (2) submissions on the elements below and on existing activities related to GHG emissions reductions by States and stakeholders; and (3) a technical paper by the Secretariat compiling a list of existing IMO activity related to reducing GHG emissions in the shipping sector. The discussions should include but not be limited to the elements below: • Levels of ambition and guiding principles for the strategy; • Emissions scenarios; • Assessment of the projected future demand for shipping; • Parameters/indicators on energy efficiency of ships (current status and long-term potential); • Emission reduction opportunities (near-, mid- and long-term actions), including alternative fuels; • Costs and benefits; • Capacity building and technical cooperation; • Barriers to emissions reductions and how to overcome them; • Priority areas for R&D, including in relation to technology; • Impact of EEDI; • Impacts on States, taking into account the HLAP (resolution A.1098(29)); and • Impacts of other regulations on GHG emissions
May 2017 (MEPC 71)	- Discussion continues[1]
September 2017	- Intersessional meeting
Week before MEPC 72	- Intersessional meeting
Spring 2018 (MEPC 72)	- Adoption of initial IMO Strategy[2], including, inter alia, a list of candidate short-, mid- and long term further measures with possible timelines, to be revised as appropriate as additional information becomes available

[1] Modality of further intersessional work after MEPC 71 to be considered based on written submissions.
[2] Initial IMO Strategy is subject to revision based on DCS data during 2019-2021 and does not prejudge any specific further measures that may be implemented in phase 3 of the 3-step approach.

Roadmap for developing a comprehensive IMO Strategy on reduction of GHG emissions from ships

Roadmap (cont'd)

January 2019	- Start of Phase 1: Data collection (Ships to collect data)
Spring 2019 (MEPC 74)	- Discussion continues - Initiation of Fourth IMO GHG Study using data from 2012-2018
Summer 2020	- Data for 2019 to be reported to IMO
Autumn 2020 (MEPC 76)	- Start of Phase 2: data analysis (no later than autumn 2020) - Discussion continues - Publication of Fourth IMO GHG Study for consideration by MEPC 76[3]
Spring 2021 (MEPC 77)	- Initiation of work for adjustments on Initial IMO Strategy, based on DCS data - Secretariat report summarizing the 2019 data pursuant to regulation 22A.10
Summer 2021	- Data for 2020 to be reported to IMO
Spring 2022 (MEPC 78)	- Phase 3: Decision step - Discussion continues - Secretariat report summarizing the 2020 data pursuant to regulation 22A.10
Summer 2022	- Data for 2021 to be reported to IMO
Spring 2023 (MEPC 80)	- Adoption of Revised IMO Strategy, including short-, mid- and long-term further measure(s), as required, with implementation schedules - Secretariat report summarizing the 2021 data pursuant to regulation 22A.10

Initial discussions on the IMO GHG Strategy

The initial IMO GHG Strategy is set to include:

1. Preamble/introduction/context/objectives including emission scenarios
2. Vision
3. Levels of ambition
   Guiding principles
4. List of candidate short-, mid- and long-term further measures with possible timelines and their impacts on States
5. Barriers and supportive measures; capacity building and technical cooperation; R&D
6. Follow-up actions towards the development of the revised Strategy
7. Periodic review of the Strategy

GMN | The Global MTCC Network

15

Latest progress made in developing GHG Strategy for international shipping

Outcome of ISWG-GHG 2 (23-27 October 2017)

- ISWG-GHG 2 has made progress in starting to shape a draft initial IMO GHG Strategy including refining the **vision** for IMO, which will express IMO's further commitment to reducing GHG emissions from international shipping.
- While the structure of the Strategy has been largely agreed, the detailed text to be included is still under discussion.
- The group agreed that candidate short-term measures could be finalized and agreed by the Marine Environment Protection Committee (MEPC) between 2018 and 2023; candidate mid-term measures could be measures finalized and agreed by the MEPC between 2023 and 2030; and candidate long-term measures could be measures finalized and agreed by the MEPC beyond 2030.
- Dates of entry into force and when the measure can effectively start to reduce GHG emissions would be defined for each measure individually.
- The group supported the need for early action.

GMN | The Global MTCC Network

Capacity Building and Technical Cooperation

GMN | The Global MTCC Network

Technical cooperation and transfer of technology relating to the improvement of energy efficiency of ships

Reg. 23, MARPOL Annex VI, MEPC Resolution, TT-EG

ITCP: Awareness raising and capacity building tools

Major Projects: Capacity building & private sector partnerships

Global network to promote technology cooperation and transfer

Catalyze institutions and financing for sustainable marine transport

GMN | The Global MTCC Network

Potential impacts, costs and benefits

Transportation Challenges in the Pacific

- Long routes – distance from trading partners
- Small economies, with limited economic diversification
- Heavy reliance on imported goods
  - Imbalance between goods coming in and goods going out
  - Limits ability to benefit from economies of scale
- Reliance on imported fossil fuels
- Financing barriers
- High infrastructure costs

All add up to high import costs and reduced export competitiveness

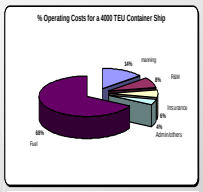
20

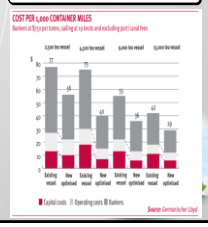
Estimating impacts of potential measures to reduce GHG emissions from international shipping

- Not easy to model economic costs and benefits of GHG mitigation measures on countries or regions
  - Complexity of the model needed to capture all costs and benefits
  - Lack of data on maritime shipping, especially for small island developing states
- Some work has been done to estimate impact on international shipping (UNCTAD, OECD, World Bank/IMF, academia)
  - Depends on impact of measure on freight rates and by how much such changes could be passed on to consumers and producers – this varies by country and product
  - Distance from major exporters is a key determinant, but not the only determinant
- Estimates of impact on GDP suggest less than a 2% effect, but varies by region or country
  - But even small percentage increases can have a large impact on consumers with low incomes

Cost of fuel to the economy

- Industrial activities are impacted by price of fuel
- Size of effect depends on energy intensity of the industrial activities:
  - The more energy intensive, the larger the percentage fuel costs are relative to other costs of production
- Transportation is generally an energy intensive industry
- Shipping fuel cost is a major part of total shipping costs
- Hence, reducing fuel costs can help lower shipping costs





Benefits of GHG emission mitigation

- Reduction of direct costs in the form of lower fuel (energy) costs
- Reduction of social cost to society in mitigating the negative impacts, including health costs, impacts on infrastructure, etc.
- Energy efficiency improvements and use of alternative fuel can have both economic and environmental benefits

23

Benefits of Regional Action

- Strong link between measures to address climate change mitigation and adaptation and sustainable development measures
- Measures to reduce GHG emissions usually imply reducing fuel consumption or switching to renewable energy sources
  - This results in cost savings, increasing competitiveness
  - Also reduces strain on foreign exchange earnings and public finance
- Increased inter-island shipping services and improved port infrastructure can help reduce some of the transport challenges
- Some debate whether cooperation could lead to improved procurement and supply chain management

24

Annex 1: Data Collection System for Fuel Consumption: Main elements

- Application: ships of 5,000 GT and above on international voyages
- Methodology for collecting the data to be outlined in the ship-specific SEEMP
- Data to be collected includes: IMO number, technical characteristics, fuel consumption (by fuel type, in metric tons), distance travelled from berth to berth, hours not at berth
- Data to be aggregated into an annual figure at the end of each calendar year and reported by ship to the Administration (flag State) within three months
- Upon receipt of data, Administration or organization duly authorized shall determine whether data has been reported in accordance with regulation 22A and if so issue a Statement of Compliance to the ship no later than 5 months from the beginning of the calendar year
- Administration to submit data to IMO for inclusion in a centralized database (no later than 1 month after issuing Statements of Compliance)
- Database shall be undertaken and managed by the IMO Secretary-General (pursuant to guidelines developed by IMO)
- Parties shall have access to the anonymized data strictly for their analysis and consideration

GMN | The Global MTCC Network

Thank you for your attention!
http://gmn.imo.org/

GMN | The Global MTCC Network

Low Carbon Sea Transport Project

Marshallese- German
Bilateral Technical Cooperation Project
2017 - 2022

Brief

- Funded by German Federal Ministry for the Environment, Nature Conservation, Building and Nuclear Safety (BMUB)
- Bilateral Technical Cooperation Project between Marshall Islands and Germany
- Bilateral Agreement signed by both Governments in May 2017
- 9.5M EUR (11M USD)

Implementation of Project

- Implemented by GIZ and RMI
- Responsible German Embassy: Manila
- Responsible in RMI: Ministry of Transport and Communication
- Focus: Fleet of Marshall Islands Shipping Corporation

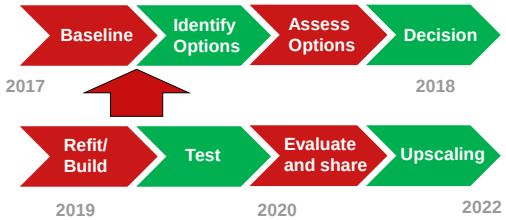
Implementation partners

- Cooperation planned with:
  - University of Emden (Technical Expertise/ Refitting of Cargo Vessels)
  - WAM (Building Intra Lagoon Vessel)
  - USP (Assessment of Social Aspects)
  - MCST (Trainings, Research)

Outcome

- Contribution to reduction of emissions from shipping
- Contribution to achieving RMI's NDC
- Lower shipping costs
- Saving on Fossil fuel imports
- Proposal for upscaling
- Inputs of RMI to UNFCCC, IMO & regional meetings

Project Steps

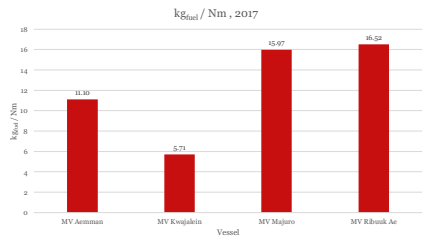


Preliminary baseline data

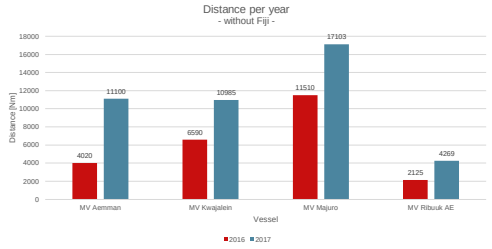
Data collected and analysed by

- Raffael Held, Marine Engineer (GIZ)
- Mark Oxley, Shipping Consultant (ADB)

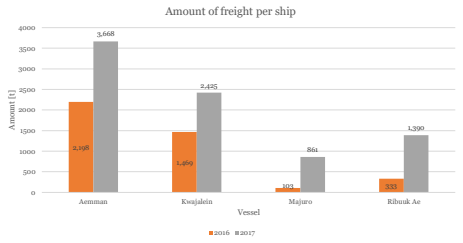
Fuel consumption per nautical mile



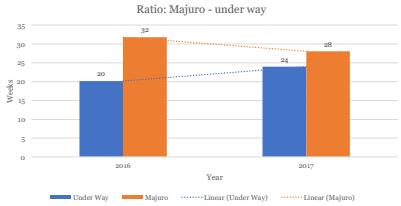
Distances per year



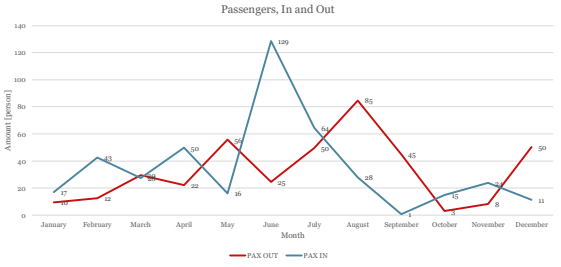
Amount of freight per ship



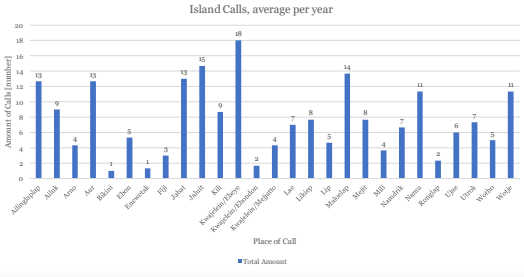
Ratio: In port Majuro – under way



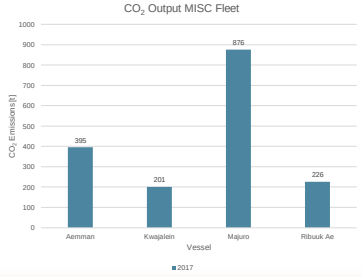
Passengers, In and out



Island/ Atoll calls, average per year



CO_2 Emissions of the MISC fleet



CO_2 Emissions of the MISC fleet

$CO2\ Emissions =$

$Amount\ of\ fuel\ x\ carbon\ emission\ factor =$

[ton fuel] x [ton CO_2 /ton fuel] =

1 t Marine Gas Oil x 3.207 ton CO_2

/ton fuel

= 3.207 tons CO_2

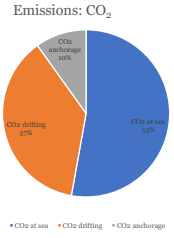
EEOI (Energy Efficiency Operational Indicator)

Calculation: gCO2/t.nm

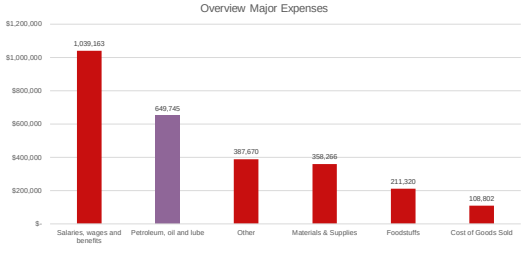
Passengers and cargo

Some data still missing

Ship‘s modes and CO_2 emissions

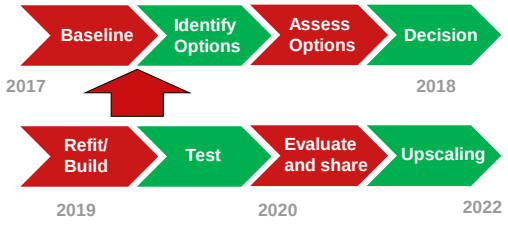


MISC expenses



Alexander Beetz, August 3, 2017

Project Steps



Alexander Beetz, December 6, 2017

Options

- Modification of existing vessels
- Building new vessels

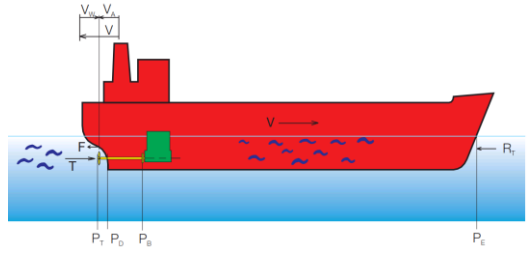
Approaches

- Institutional (Management)
- Operational
- Infrastructural
- Technical

Technology approaches

- Thrust efficiency
- Engine efficiency
- Hydrodynamics
- Auxiliary power
- Wind
- Operational

Thank you!



GMN | The Global MTCC Network
A global network for energy-efficient shipping

MTCC PACIFIC
Maritime Technology Cooperation Centre

MARITIME TECHNOLOGY COOPERATION CENTRE – PACIFIC (MTCC-PACIFIC)

CAPACITY BUILDING FOR CLIMATE MITIGATION IN THE MARITIME SHIPPING INDUSTRY

THE GLOBAL MTCC NETWORK (GMN) PROJECT

MTCC-PACIFC REGIONAL CONFERENCE AND OFFICIAL LAUNCH

12 – 15 December, 2017

Suva, Fiji

EUROPEAN UNION

IMO INTERNATIONAL MARITIME ORGANIZATION

Pacific Community
Communauté du Pacifique

HOST INSTITUTIONS OF MTCC-PACIFIC

SPREP
Secretariat of the Pacific Regional Environment Programme

The Global MTCC Network (GMN) project is funded by the European Union and implemented by the IMO.

« New-Caledonia Towards Sustainable Blue Growth »

Programme and Projects of the New-Caledonia Maritime Cluster (CMNC)

Lionel Loubersac, General Manager

Just a brief

- General Manager

- Oceanographer **(Engineer and PhD)**

  42 years of Oceanographic Applied Research

- Entrepreneur

  Maritime forward looking projects

  Environment, Oceanography and Marine Technology

Océan Avenir NC

creocean
Environnement & océanographie

lionel.loubersac@outlook.fr

http://annuaire.ifremer.fr/cv/16331/

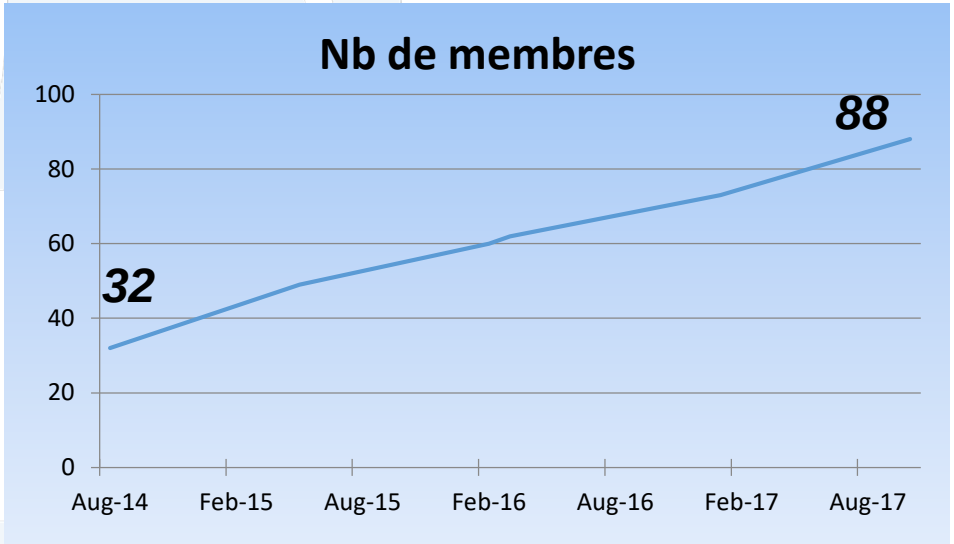
Key Data on CMNC

- ***Non lucrative organization***
- ***Established august 5th 2014***
- ***Represents today more than 3000 employments***
- ***Budget 160.000 euros (members fees and grants from local authorities)***
- ***Associated to Australian Pacific Islands Business Council (regional integration)***
- ***3 members types : active (private), associated (administration, research), observers (ONG, associations)***

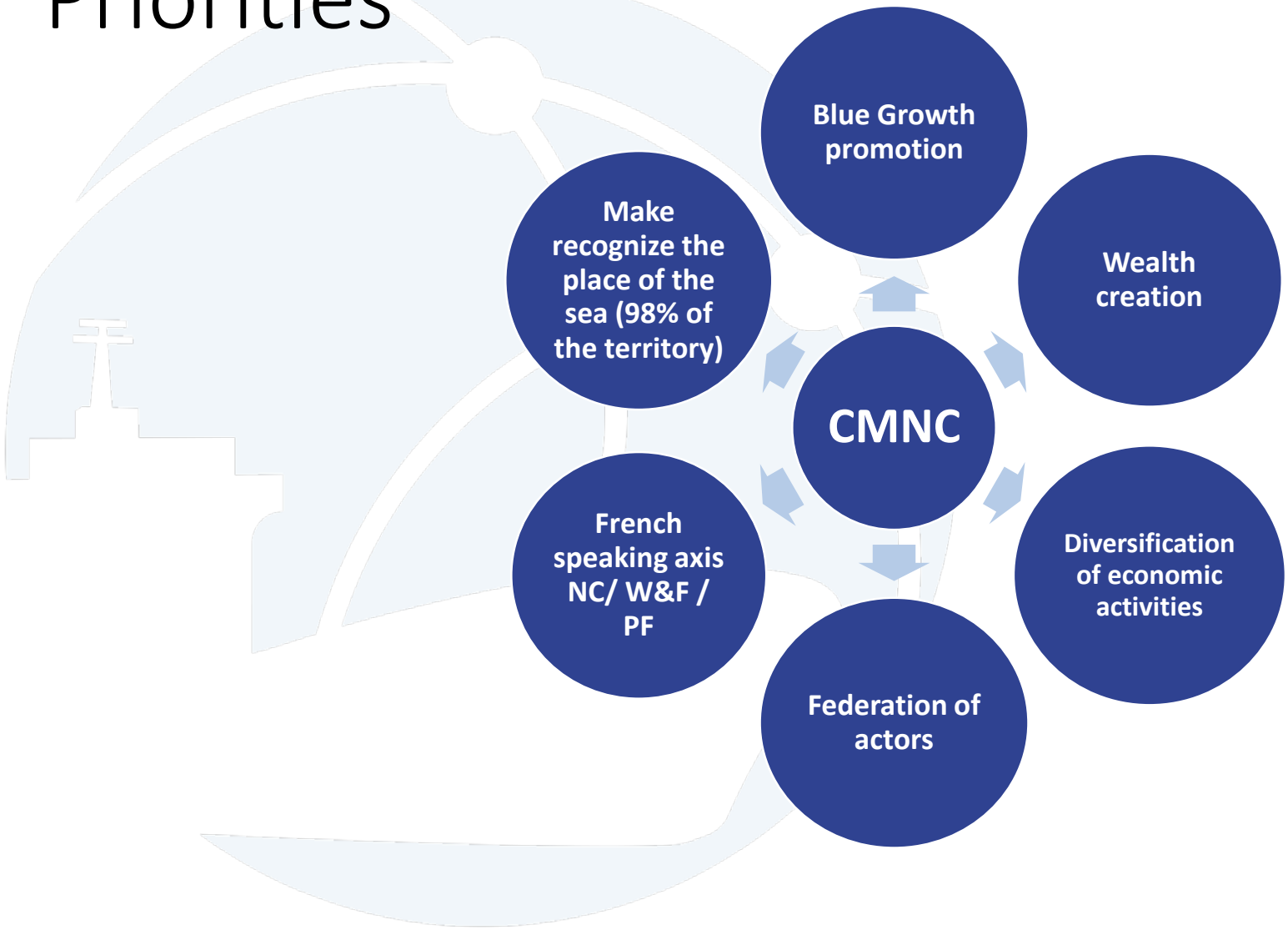
Collective intelligence

« None of us knows what we know together »

Euripides



Priorities


Blue Growth promotion
Wealth creation
Diversification of economic activities
Federation of actors
French speaking axis NC/ W&F / PF
Make recognize the place of the sea (98% of the territory)
CMNC

Key Reminders on NC EEZ

- **An EEZ on a submerged continent (Zelandia)**
- **An exceptional geodiversity associated to a world rank marine biodiversity « hot spot »**
- **As many marine species in a 20x20 km square in the New-Caledonia main lagoon than in the whole Mediterranea**
- **Below 300 meters depth : one species out of two, never described**
- **15.000 km^2 of reefs and lagoon registred on UNESCO World Heritage and 1.300.000 km^2 as Natural Marine Park of the Coral Sea**
- **26 maritime activity sectors, of wich 10 belonging to the 11 priorities identified by the European Union Blue Growth Strategy**

So major sustainabble development stakes and a unique marine « playground » for innovative projects, environmental responsability and Blue Growth implementation.

Flagships

- **Summary report on the maritime challenges and stakes in New-Caledonia**
- **Organization of the General States of the Sea (4-5 july 2016)**
- **Launch of the Maritime Economic Observatory (3500 structures identified)**
- **First Sea Day (12 july 2017)** *to be organized on an annual basis*

Working Groups - Synergies

➡ **16 active groups**

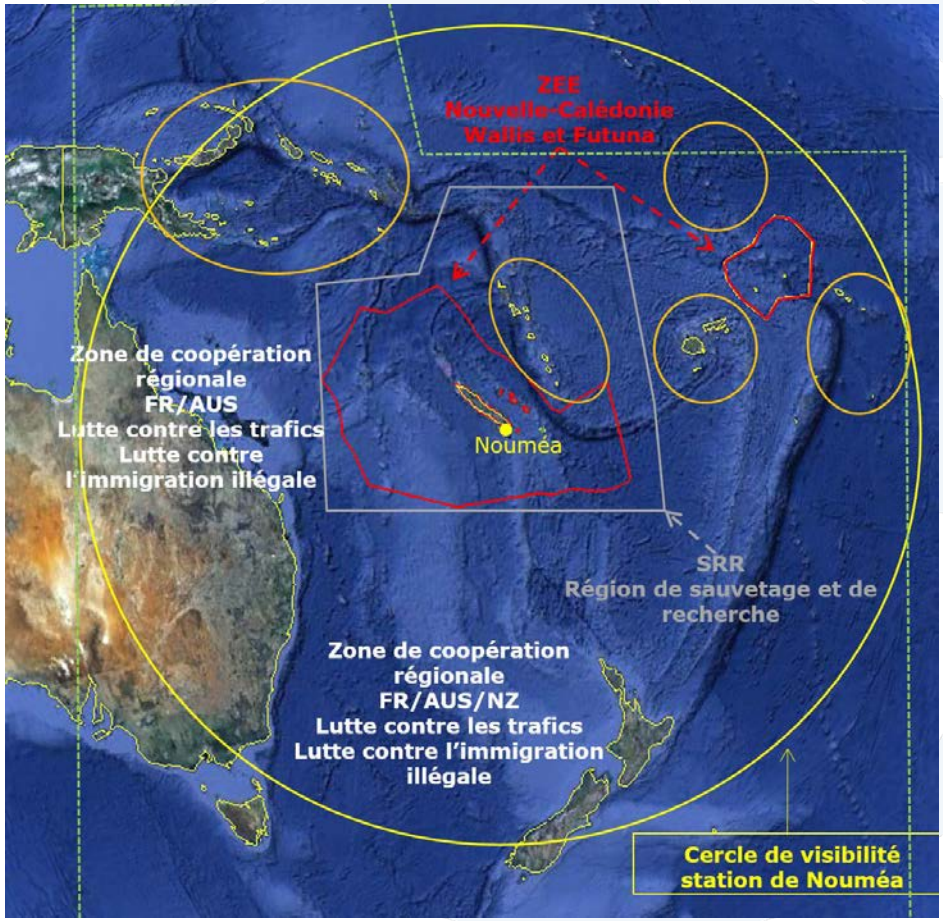
- ***Governance***
- ***Spatial Data & Marine Spatial Planning***
- ***Marine Technology Platform***
- ***Harbour Infrastructures***
- ***Pole of Excellence***
- ***Ship Repair***
- ***Cruise***
- ***Maritime Tourism***
- ***Yachting Sector***
- ***Security***
- ***End-of-life Vessels***
- ***Energy***
- ***Bio-Ressources & Biodiversity Valuation***
- ***Training***
- ***Communication & Heritage (natural and cultural)***
- ***Sustainable Blue Growth Laboratory***

✓ **Most significant projects at the interface between several groups**

✓ **Ecoresponsability as a cross cutting issue**

Project examples

Marine Technology platform



Very high resolution satellites receiving station (optic and radar)

Under study

- Maritime Surveillance
- Security at sea
- Natural disasters impacts
- Marine hydrography
- Environment (mangroves, coral reefs…)
- regional cooperation

Project examples

Marine Technology platform

Submarine Cables

- to promote « SMART » Cables on international roads
- to promote Multidisciplinary Seafloor and Water Column Observatories (EMSO Type)
- environment knowledge
- global change monitoring
- early warning (submarine earthquakes, tsunamis…)

Project examples

Marine Technology platform

« Clever boats »

to equip ships of opportunity with « ferry boxes »

- continuous and repetitive marine environment parameters monitoring at lower cost

- knowledge of the seasonnal, interannual and long term behaviour of the water masses (lagoons, territorial waters, open sea and our Marine Natural Park…)

Project examples

Marine Technology platform

Using AIS to estimate near real time currents fields

about 100.000 boats in operation

- An E_Odyn solution
- statistically significant
- flexible and low cost
- better routing and an opportunity for fuel economy (estimated 1 to 3% by CMA CGM)
- a demonstration on the NC EEZ
- other applications as drift of man at sea….

Project examples

Ship Repair

Floating Dock

- Economic feasibility under progress
- Response to local ships needs
- Response to transient boats and the fleets of neighbours countries
- Up to date and eco-friendly solutions

Project examples

Sustainable Sailing

Eco-sailing

End-of-life vessels

- An important stock of shipwrecks : recently the contener ship “Kea Trader” (180 m long)
- 6300 ships to destroy and recycle within the next 15 years

Project examples

Sustainable Cruise

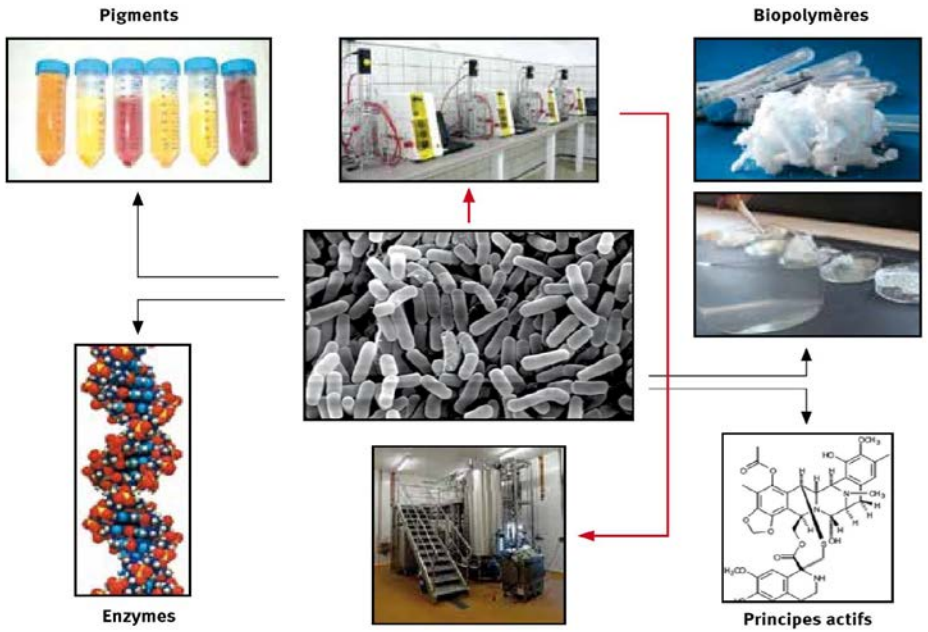
Cruise Club

- 500 cruise liners a year
- Strategy and choices on the cruise types to boost
- Diversification of destinations
- Social and environmental acceptability
- partnerships with neighbours

Project examples

Marine Biotechnologies and Bio-inspiration

(to adopt solutions offered by the Nature)



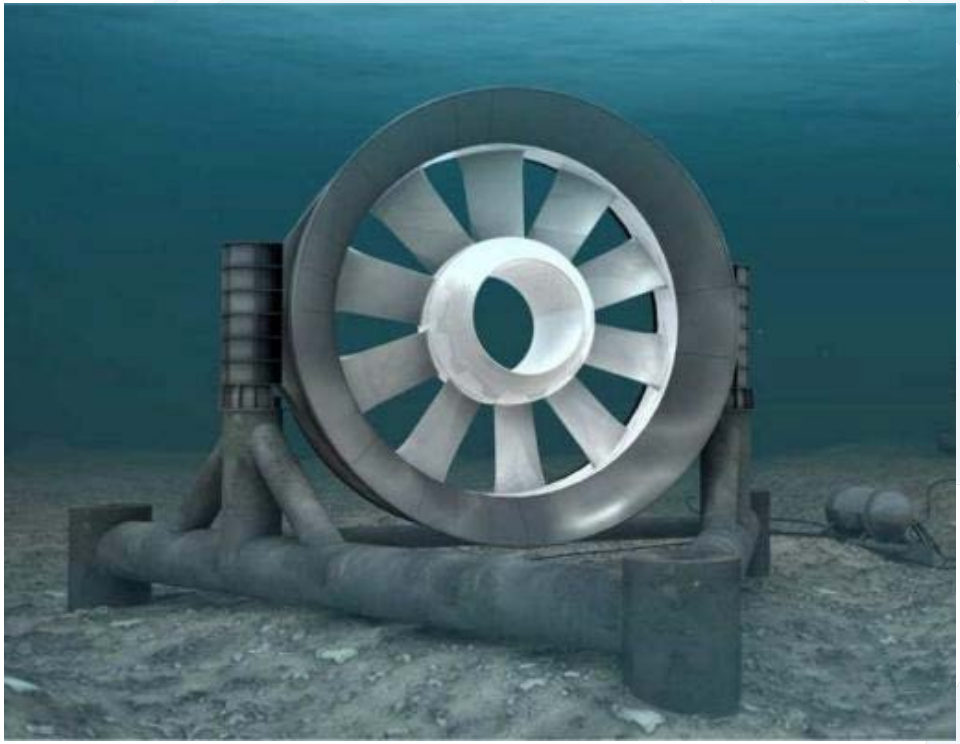
Marine biomolecules

The ultra-small : bacteria & micro-algae

- Health and Cosmetics
- Proteins, pigments,
- Bio-plastics
- Natural antifouling agents
- Soils and water quality decontamination
- Bioremediation and CO_2 sequestration
- Future bio-fuels?

Project examples

Energy



Marine Renewable Energies and Energy Transition

- ***Atlas of the potential of Marine Energies (offshore wind, currents, swell, SWAC, OTEC), finalized end of the year***

- ***Solutions for Energy Storage at sea?***

- ***Valuing the future SLN Doniambo nickel plant : power plant equipped with Liquefied Natural Gaz storage unit (decided) ► a project on alternative hybride engines for ships crossing the Coral Sea.***

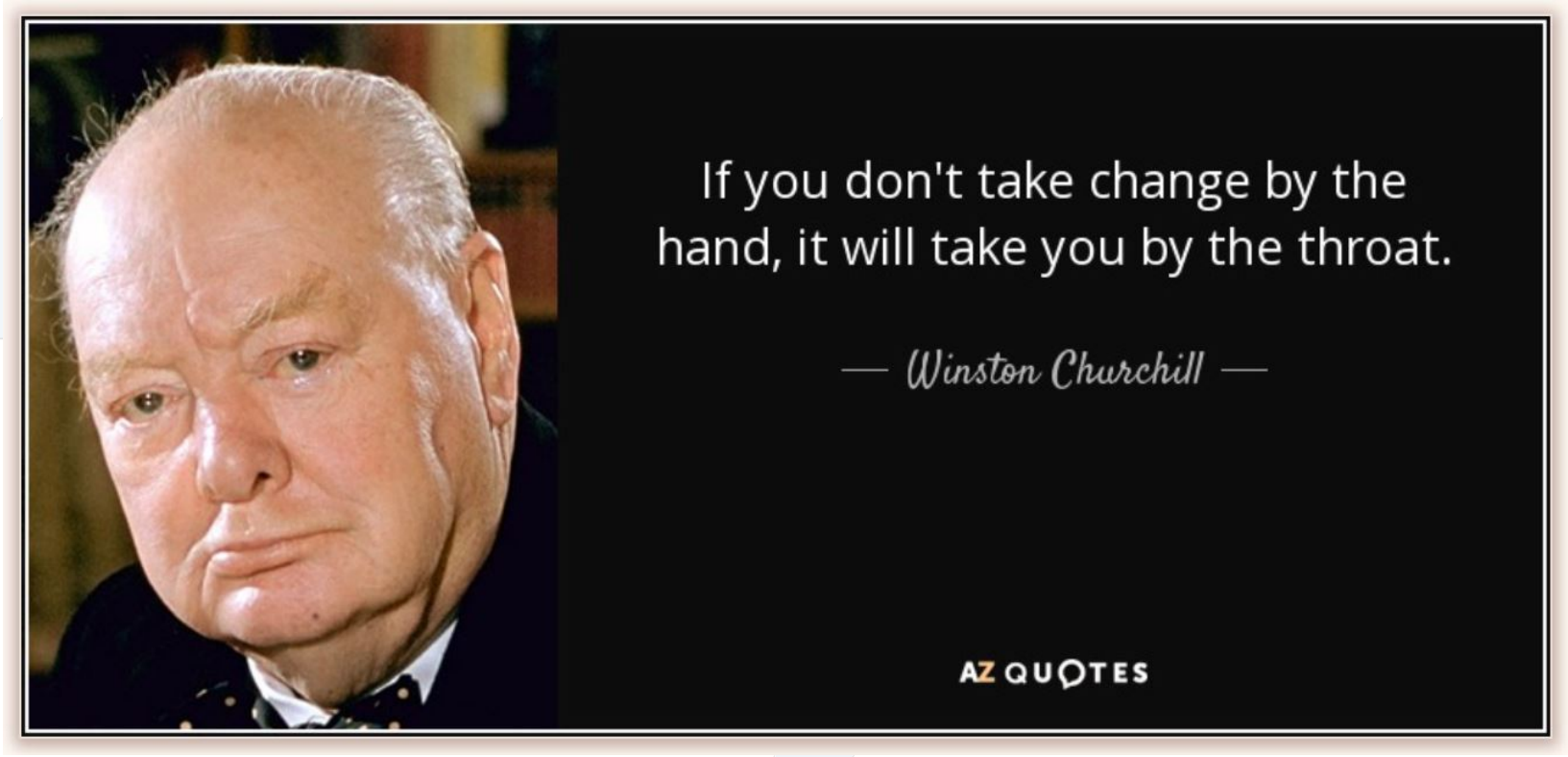
Suggestion

- **CMNC to contribute widering to the whole Pacific the Maritime Cluster governance concept as exposed : Business world (pilot) associated with Politics, Administrations, Research, ONG… ►PIDF**
- **A possible model : The Baltic Sea Forum**
- **Assistance to public decision making,**
- **Appraisal of projects,**
- **Setting up public/private partnerships**
- **Reaching sustainable Blue Growth excellence**

We are facing global change…

The ball is in our court!

Business as usual is nowadays out…



We do need to be innovative!!!

GMN | The Global MTCC Network
A global network for energy-efficient shipping

MTCC PACIFIC
Maritime Technology Cooperation Centre

Thank you!

The Local Shipowners Association

P.O. Box 14537 Suva, Fiji Islands, Phone: 3380615 Fax 3372035

Reducing GHG FROM Domestic Ships

Ship Owners perspective

Presentation by
Josateki Tagi
LSOA Member

Local Ship Owners Association

- BACKGROUND
- IMO Study - the maritime sector was responsible for nearly 3.3 % of the global greenhouse gas (GHG) emissions during 2007. Of these, 2.7 % of emissions was CO2 from International shipping
- In the absence of proper actions, emissions from the maritime sector may grow by 150–250 % by 2050, in comparison with the emissions in 2007
- Result of Growth in the maritime sector
- This warrants proactive action to achieve a greener shipping industry.

Local Ship Owners Association

PROACTIVE ACTION

- EU Funding, IMO Implementation
- 5 Centers MTCC
- Pacific (MTCC), SPC as Host , Suva, Fiji
- Mission : Reduction of GHG & Promote energy efficiency in Ships

Local Ship Owners Association

WHERE TO FROM HERE

Ship Owners IMO & MTCC Pacific Programme & COP 23

Local Ship Owners Associations (LSOA) – Fiji

- LSOA – Registered in the 1970's
- Fellowship *(Mutual – routes sharing)*
- Protection of Ship Operators
- Advisory Role to the Government - Marine Transportation
- Advocate fare trade
- Observe & Compliance *(IMO Conventions, MSAF)*

Local Ship Owners Association

Local Ship Owners Association

No.	Shipping Companies		Passenger Cargo (Ferries & Tourism Conventional)	Cargo	Fishing Vessel	Total Power Estimates	Fuel Consumption per hour	Fuel consumed per week	Fuel Consumed per year	Estimated carbon discharge
1	Goundar Shipping		**5**	**-**	**-**	**12000KW**				
2	Patterson Bros		**3**	**-**	**-**	**5000KW**				
3	Venu Shipping		**1**	**-**	**-**	**800KW**				
4	All Engineering		**-**	**5**	**-**	**2000KW**				
5	Victoria Marine		**2**	**2**	**-**	**2600KW**	**420L/hr**	**(40hrs) 16800L**	**873,600L**	
6	Cruz Holding		**--**	**7**	**-**	**4800KW**				
7	South Sea Shipping		**2**	**-**	**-**	**1000KW**				
8	Valesasa		**1**	**-**	**-**	**400KW**				
9	Tokalau Shipping		**1**	**-**	**-**	**50KW**				
10	Government Shipping		**4**	**8**	**-**	**10000KW**				
11	South Sea		**6**	**-**	**-**	**4000KW**				
12	Captain Cook		**5**	**-**	**-**	**3500KW**				
13	PBS		**-**	**7**	**-**	**5000KW**				
14	Fiji Fish		**-**	**-**	**7**	**3000KW**				
15	Solander		**-**	**-**	**6**	**2800KW**				
16	Sea Quest		**-**	**-**	**12**	**6000KW**				
17	Hangton		**-**	**-**	**7**	**5000kw**				

Local Ship Owners Association

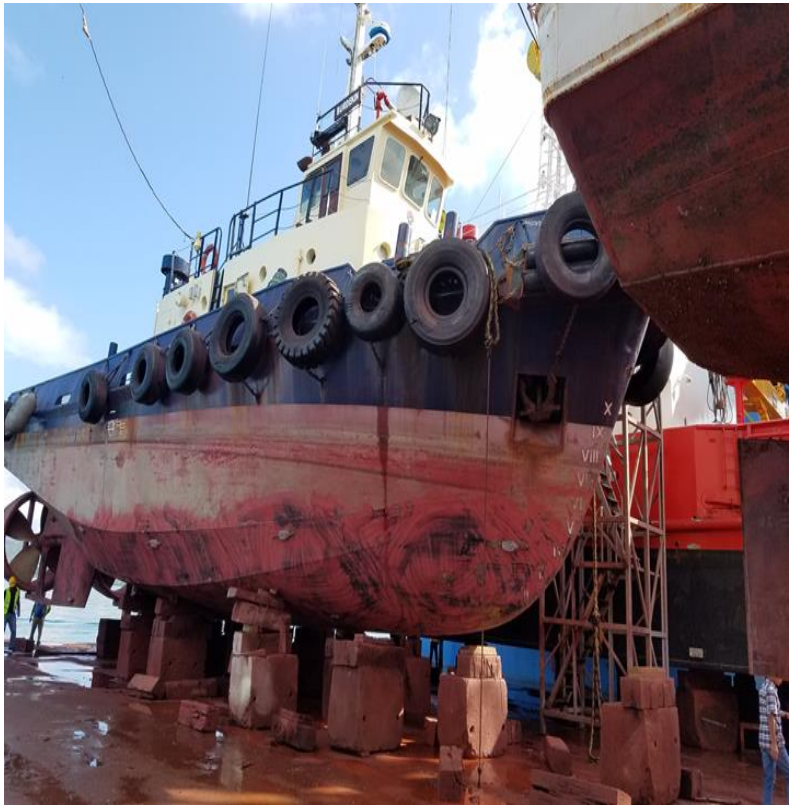
Local Ship Owners Association

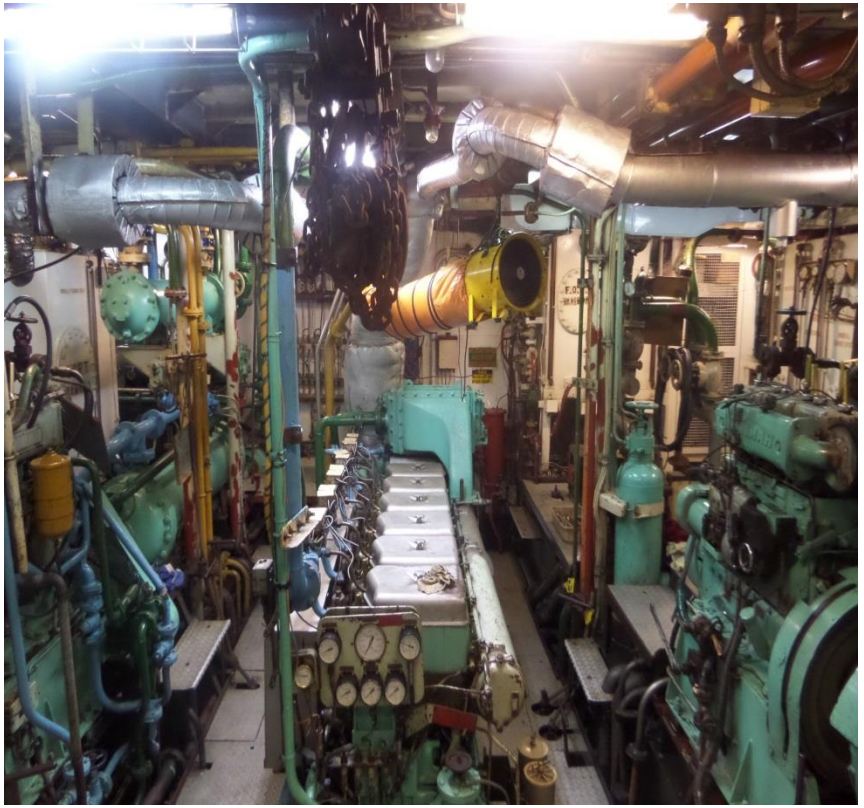
Local Ship Owners Association

WAY FORWARD

- Support MTCC Data Collection
  - Supply necessary information for analysis
  - Champion the initiative (Placard & MTCC Sticker on Board vessels)
  - Regular update of progress in the implementation
- Improve Ships Maintenance Program
  - Proper Ship Propulsion maintenances
  - Docking and Clean Hull
  - Shipboard Training and take ownership

Local Ship Owners Association





Local Ship Owners Association

WAY FORWARD

- Improve sailing Schedules
  - Voyage Plan (shortest Route)
  - Day and night Sailing (Nav Aid)

- New Designs (Sail Assistance) to improve efficiency
  - Make shift arrangement - Efficiency Improved by 30%
  - SOLAR Power assist Generators – *In Port*
  - Wind Generators/Crushing Alternators

- Outer Islands Infrastructure
  - Jetty, mooring, proper anchorage, access to outer islands
  - Dredge – proper depth

Local Ship Owners Association

Local Ship Owners Association

SUMMARY

- High Level Lead Role
  - Government - MOT
  - MTCC

- Sustainability and Profitability
  - Revise Freights & Charges
  - Uneconomical Routes – Subsidy x 2
  - Infrastructure Improvement – outer islands
  - Tax Incentives

Local Ship Owners Association

CONCLUSION

- Ideally – Improve/Modernise Domestic Fleet that comply with GHG emission requirement

MARITIME TECHNOLOGY COOPERATION CENTRE – PACIFIC (MTCC-PACIFIC)

CAPACITY BUILDING FOR CLIMATE MITIGATION IN THE MARITIME SHIPPING INDUSTRY
THE GLOBAL MTCC NETWORK (GMN) PROJECT

MTCC-PACIFC REGIONAL CONFERENCE AND OFFICIAL LAUNCH

12 – 15 December, 2017
Suva, Fiji

The Global MTCC Network (GMN) project is funded by the European Union and implemented by the IMO.

« New-Caledonia Towards Sustainable Blue Growth »

Programme and Projects of the New-Caledonia Maritime Cluster (CMNC)

Lionel Loubersac, General Manager

Just a brief

- General Manager
- Oceanographer **(Engineer and PhD)**
  - **42 years of Oceanographic Applied Research**
- Entrepreneur
  - **Maritime forward looking projects**
  - **Environment, Oceanography and Marine Technology**

lionel.loubersac@outlook.fr

http://annuaire.ifremer.fr/cv/16331/

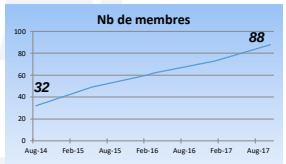
Key Data on CMNC

- ***Association (15 members in the board of directors)***
- ***Established august 5th 2014***
- ***Represents today more than 3000 employments***
- ***Budget 160.000 euros (members fees and grants from local authorities)***
- ***Associated to Australian Pacific Islands Business Council (regional integration)***
- ***3 members types (active, associated, observers)***

Collective intelligence

« None of us knows what we know together »

Euripides



Priorities

Key Reminders on NC EEZ

- **An EEZ on a submerged continent (Zelandia)**
- **An exceptional geodiversity associated to a world rank marine biodiversity « hot spot »**
- **As many marine species in a 20x20 km square in the New-Caledonia main lagoon than in the whole Mediterranea**
- **Below 300 meters depth : one species out of two, never described**
- **15.000 km² of reefs and lagoon registred on UNESCO World Heritage and 1.300.000 km² as Natural Marine Park of the Coral Sea**
- **26 maritime activity sectors, of wich 10 belonging to the 11 priorities identified by the European Union Blue Growth Strategy**

So major sustainabble development stakes and a unique marine « playground » for innovative projects, environmental responsability and Blue Growth implementation.

Flagships

- **Summary report on the maritime challenges and stakes in New-Caledonia**
- **Organization of the General States of the Sea (4-5 july 2016)**
- **Launch of the Maritime Economic Observatory (3500 structures identified)**
- **First Sea Day (12 july 2017)** *to be organized on an annual basis*

Working Groups - Synergies

➡ **16 active groups**

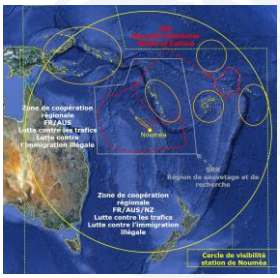
- ***Governance***
- ***Spatial Data & Marine Spatial Planning***
- ***Marine Technology Platform***
- ***Harbour Infrastructures***
- ***Pole of Excellence***
- ***Ship Repair***
- ***Cruise***
- ***Maritime Tourism***
- ***Yachting Sector***
- ***Security***
- ***End-of-life Vessels***
- ***Energy***
- ***Bio-Ressources & Biodiversity Valuation***
- ***Training***
- ***Communication & Heritage (natural and cultural)***
- ***Sustainable Development Laboratory and Showcase***

✓ **Most significant projects at the interface between several groups**

✓ **Ecoresponsability as a cross cutting issue**

Project examples

Marine Technology platform



Very high resolution satellites receiving station (optic and radar)

Under study

- Maritime Surveillance
- Security at sea
- Natural disasters impacts
- Marine hydrography
- Environment (mangroves, coral reefs…)
- regional cooperation

Project examples

Marine Technology platform

Submarine Cables

- to promote « SMART » Cables on international roads
- to promote Multidisciplinary Seafloor and Water Column Observatories (EMSO Type)
- environment knowledge
- global change monitoring
- early warning (submarine earthquakes, tsunamis…)

Project examples

Marine Technology platform

« Clever boats »

to equip ships of opportunity with « ferry boxes »

- continuous and repetitive marine environment parameters monitoring at lower cost
- knowledge of the seasonnal and interannual behaviour of the water masses (lagoons, territorial waters, open sea and Marine Natural Park…)

Project examples

Ship Repair

Floating Dock

- Economic feasibility under progress
- Response to local ships needs
- Response to transient boats and the fleets of neighbours countries
- Up to date and eco-friendly solutions

Project examples

Sustainable Sailing

Eco-sailing

End-of-life vessels

- An important stock of shipwrecks : recently the contener ship “Kea Trader” (180 m long)
- 6300 ships to destroy and recycle within the next 15 years

Project examples

Sustainable Cruise

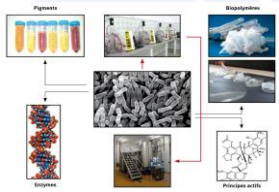
Cruise Club

- Strategy and choices on the cruise types to boost
- Diversification of destinations
- Social and environmental acceptability
- partnerships with neighbours

Project examples

Marine Biotechnologies and Bio-inspiration

(to adopt solutions offered by the Nature)



Marine biomolecules

The ultra-small : bacteria & micro-algae

- Health and Cosmetics
- Proteins, pigments,
- Bio-plastics
- Natural antifouling agents
- Soils and water quality decontamination
- Bioremediation

…

Project examples

Energy

Marine Renewable Energies and Energy Transition

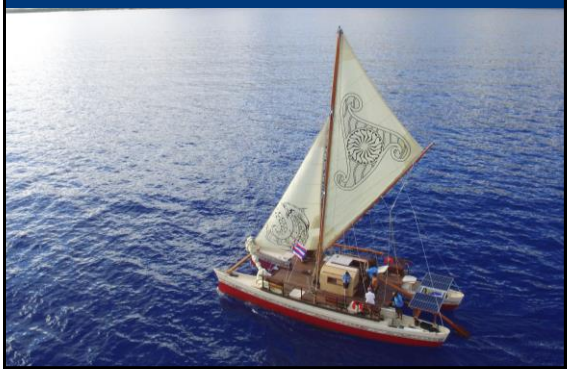
- ***Atlas of the potential of Marine Energies (offshore wind, currents, swell, SWAC, OTEC), finalized end of the year***
- ***Valuing the future Doniambo Liquefied Natural Gaz storage unit and projet appraisal on alternative hybride engines for ships crossing the Coral Sea***
- ***Solutions for Energy Storage at sea?***

Thank you!

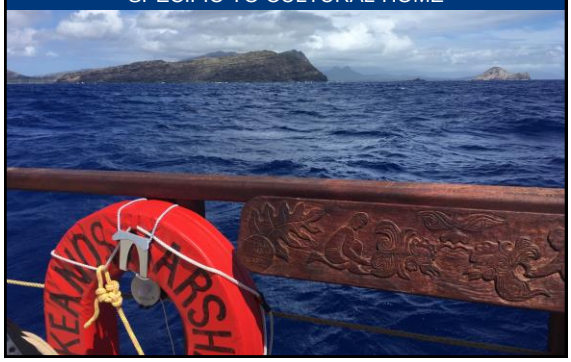
2011-2012 OKEANOS VAKA MOANA FLEET
TE MANA O TE MOANA (spirit of the ocean)

TE MANA O TE MOANA FLEET ASKED TO TRANSPORT GOODS, PEOPLE, MEDICINE BETWEEN THE ISLANDS

VAKA MOTU: BOAT FOR THE ISLANDS



EACH VAKA INDIVIDUALLY DESIGNED SPECIFIC TO CULTURAL HOME



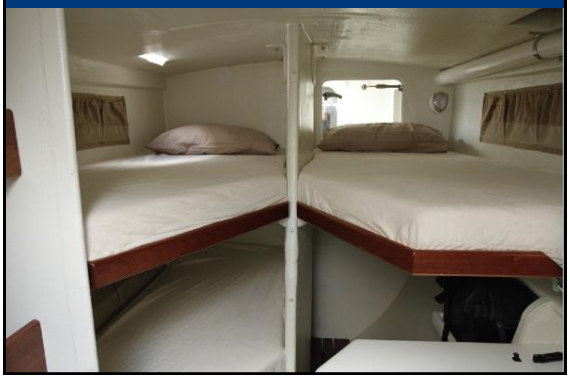
TRADITIONAL LASHINGS

TRADITIONAL SAILS

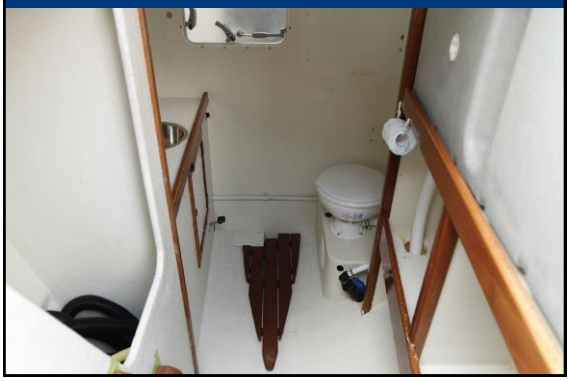
TRADITIONAL STEERING

DOCKS DIRECTLY ON THE BEACH

10 BUNKS



FLUSH TOILET



FULL GALLEY

DECK HOUSE WITH MODERN NAVIGATION

SOLAR ELECTRIC PROPULSION

COCONUT OIL BIO-FUEL

OKEANOS VANUATU

OKEANOS MARSHALL ISLANDS

OKEANOS MARIANAS

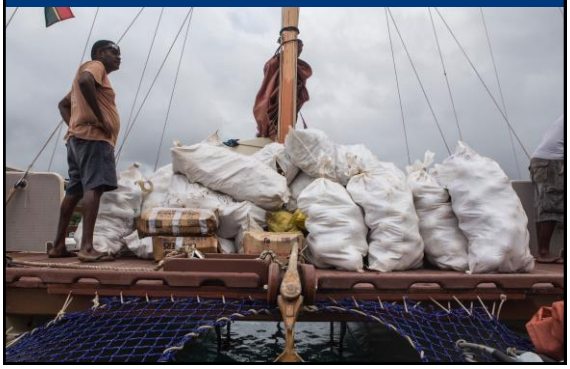
OKEANOS AOTEAROA IN TONGA

BOAT BUILDING

EFFICIENT PRODUCTION

WORK FORCE DEVELOPMENT

CARGO TRANSPORTATION

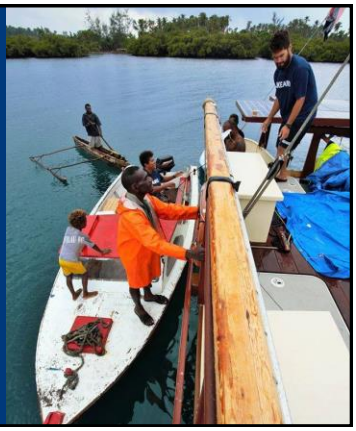


CARGO TRANSPORTATION

TRANSPORT PEOPLE & GOODS TO MARKET

COMMUNITY LED BUSINESS INITIATIVES

DISASTER RELIEF



SERVICING REMOTE COMMUNITIES

HOPE FOR THE FUTURE

VAKA MOTU OPERATIONS

Reviving sea transportation culture in Fiji

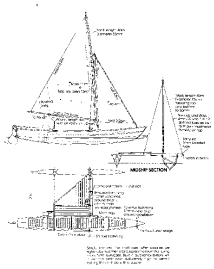
Kinship ties and trade routes…

Although the uptake was low, the design work is still relevant today

One of the critical lessons learned was that cultures with a living sailing tradition were much more likely to use sail effectively.

A whole sailing culture has to be learnt and used .

Sail options need to demonstrate economic viability for broad uptake



KIRIBATI – 7.1 Metre Canoe KIB-4

Transitioning to sustainable sea transport

Pacific Voyaging Revival

- In the past 30 yrs there has been a sustained revival and relearning of Oceanic sailing heritage.
- There are increasing numbers of traditional (and modern adaptations) of iconic Oceanic sailing craft and increasing interest by Pacific Islanders in their sailing heritage.
- The most recent fleet, including Uto ni Yalo has led to an unprecedented resurgence in interest in sailing and Drua culture in Fiji.

GMN | The Global MTCC Network
A global network for energy-efficient shipping

MTCC PACIFIC
Maritime Technology Cooperation Centre

MARITIME TECHNOLOGY COOPERATION CENTRE – PACIFIC (MTCC-PACIFIC)

CAPACITY BUILDING FOR CLIMATE MITIGATION IN THE MARITIME SHIPPING INDUSTRY

THE GLOBAL MTCC NETWORK (GMN) PROJECT

MTCC-Pacific Progress Report

Second MTCC-Pacific Steering Committee Meeting

Suva, Fiji, 12 December 2017

EUROPEAN UNION

IMO INTERNATIONAL MARITIME ORGANIZATION

Pacific Community
Communauté du Pacifique

HOST INSTITUTIONS OF MTCC-PACIFIC

SPREP
Secretariat of the Pacific Regional Environment Programme

The Global MTCC Network (GMN) project is funded by the European Union and implemented by the IMO.

Result 1: Establishment & Operations

IR1.1 MTCC established in the office of SPC

- SPC-IMO contract signed on 12 May 2017 - Project ending on 31 December 2019
- A closed office is allocated by SPC at the third floor of Lotus Building at its Regional Office in Suva, Fiji
- Website published at mtccpacific.spc.int

IR1.2 Appoint qualified staff to MTCC

- 2 October 2017: all MTCC-Pacific staff appointed
  - ✓ Head of MTCC-Pacific
  - ✓ Transport GHG Adviser
  - ✓ Administration and Information Assistant
  - ✓ Maritime Industry Energy Efficiency Officer
- SPC Project Kick-off Team de-established

Result 1: Establishment & Operations

IR1.4 Project workplan implementation

- 7 Project Monthly Meetings have been completed: 18 May, 28 June, 19 July, 24 august, 28 September, 12 October and 1 November
- First Quarterly Progress Report submitted and approved by IMO
- Second Quarterly Report submitted and approved by IMO
- Participated at COP23 GMN side-event
- Participated at the MTCCs Technical Workshop at IMO in December – MTCCs MoU signed and MTCCs Coordinating Committee established.

IR1.5 Partnerships implementation

- MTCC-Pacific Partners' Meeting on 27 June attended by EU delegation, FNU, FMA, PIDF, PIFS, USP
- First Steering Committee Meeting 16-17 August – ToR adopted
- Second Steering Committee Meeting 12 December

Result 2: capacity-building activities

Fiji National Workshop on Energy Efficient Operations of Ships

- Date: 24-27 October in Suva, Fiji
- Representatives from
  - Fiji Ministry of Infrastructure and Transport (MoIT)
  - Maritime Safety Authority of Fiji (MSAF)
  - Fiji Port Corporation Limited (FCPL)
  - Fiji Ships and Heavy Industry Limited (FSHIL)
  - Pacific Islands Development Forum (PIDF)
  - Patterson Shipping/Searoad Shipping
  - Seaquest Fiji Ltd/Sealand Processors
  - Inter Link Shipping Line Ltd
  - Government Shipping Services
  - Tokalau Shipping, All Barging and Marine (Fiji) Limited
  - Billett Wright & Associates (Fiji) Limited
  - Solander Pacific Limited.

- The objective of the National Workshop was to inform the participants on new energy efficient technologies, tools and methods
- This workshop elicited great interest in improved ship propulsion efficiency, especially in Energy Saving Devices (ESD).

Result 3: pilot-projects on energy efficiency

- Pilot-project on Energy Efficiency approved on 19 July by IMO
- Guide for Energy Management in Port and tools developed
- Port Energy audits completed:
  - In August in Samoa and Solomon Islands
  - In Fiji in September for trials, 2nd week of December and scheduled last visit in March
- Ship Energy Management visit in Kiribati in November

Result 4: Pilot-project on Data Collection

- The pilot-project on Energy Efficiency was approved by IMO on 19 July
- SPC visit in Vanuatu in July under the Pacific Islands Domestic Ship Safety (PIDSS) programme
- MTCC-Pacific visit in Kiribati in November in conjunction with PIDSS
- All shipowners provided with data collection templates – awaiting for on-board training

The Global MTCC Network (GMN) project is funded by the European Union and is implemented by the IMO

Result 5: communication and visibility

IR5.1 Communication and visibility plan designed and approved

- MTCC-Pacific C&V Plan approved by IMO on 26 June

IR5.2 Communication and Visibility materials

- All communication and visibility materials delivered on 31 July

IR5.3 Dedicated website for the MTCC-Pacific designed and launched

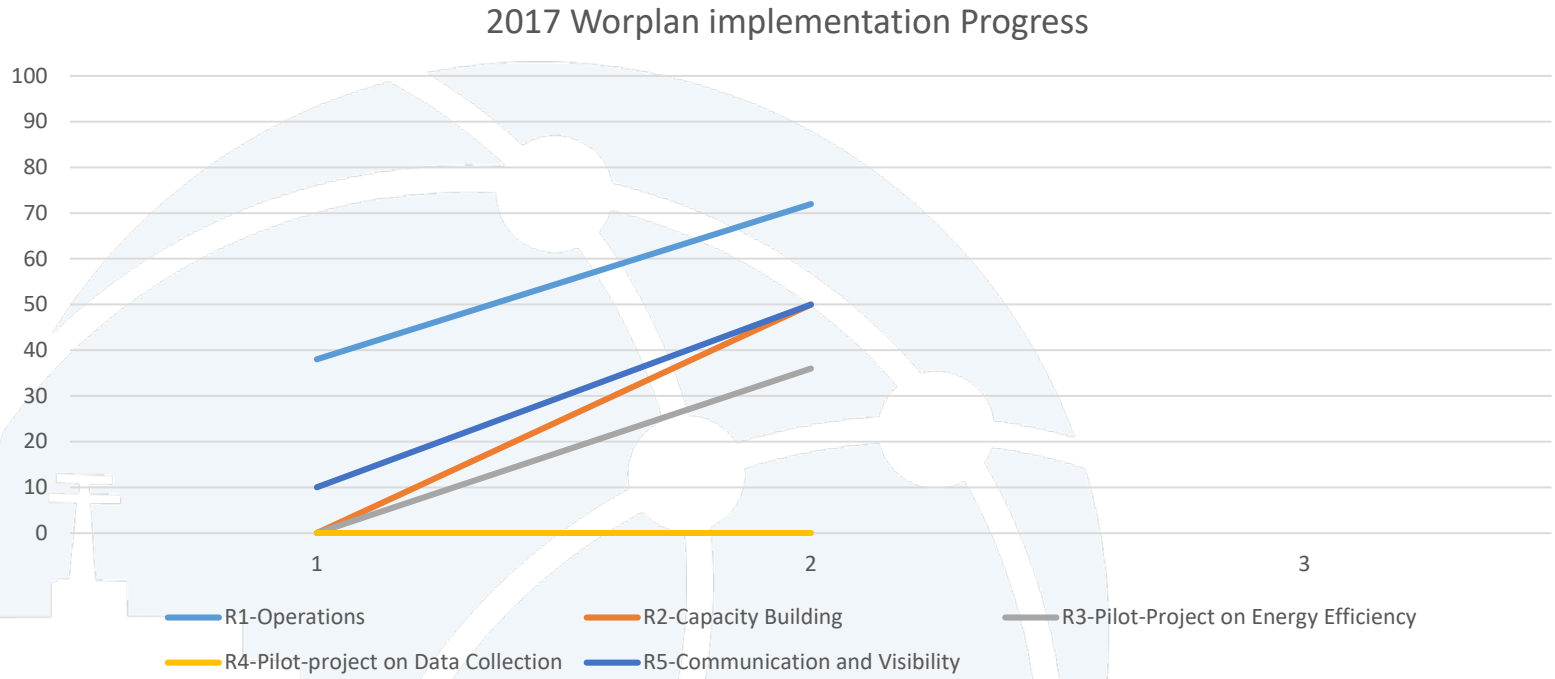
- Website published

IR5.5 Progress communicated

- GMN Project and MTCC-Pacific presented at SPC CRGA (with booth) in July and at September SPREP meeting

IR5.6 Relevant publications

- Article in August *Pacific Maritime Watch* and in next December issue



- As of 12 November, 72% of the Result 1 activities for 2017 completed - on track
- 50% of the Capacity-building activities, rest Regional Conference – on track
- 36 % of the pilot-project on Energy Efficiency activities have been completed – behind schedule for Fiji Port work and legal review that is postponed early 2018
- 0% Result 4 shows 0% completion as the indicators for 2017 activities are related to the dissemination of templates which will happen at the Regional Conference in December
- 50% of the communication and visibility activities – rest is website and dissemination of all template developed (completed this week) – on track

The Way Forward

- In 2018, MTCC-Pacific activities to plan:
  - ✓National workshop on Energy Efficient Operations of Ships
  - ✓Data Collection
  - ✓Development and implementation of Ship Energy Efficiency Management Plan (SEEMP)
  - ✓Energy Management in Port
- Targeted countries:
  - Fiji, Kiribati, Marshall Islands, Samoa, Solomon Islands, Tuvalu and Vanuatu

The Global MTCC Network (GMN) project is funded by the European Union and is implemented by the IMO

GMN | The Global MTCC Network
A global network for energy-efficient shipping

MTCC PACIFIC
Maritime Technology Cooperation Centre

The Global MTCC Network (GMN) project is funded by the European Union and is implemented by IMO

THANK YOU

EUROPEAN UNION

IMO INTERNATIONAL MARITIME ORGANIZATION

MARITIME TECHNOLOGY COOPERATION CENTRE – PACIFIC (MTCC-PACIFIC)

CAPACITY BUILDING FOR CLIMATE MITIGATION IN THE MARITIME SHIPPING INDUSTRY
THE GLOBAL MTCC NETWORK (GMN) PROJECT

Energy Management in Ports

Activities and Results

Suva, 14 December 2017

The Global MTCC Network (GMN) project is funded by the European Union and implemented by the IMO.

Activities

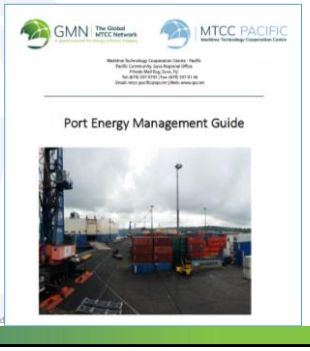
Activity overview

- Energy Management Workshop
- Energy Management Guide
- Energy Audits
- Trials

EM Workshop, Auckland

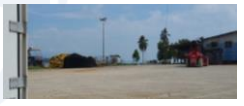
Development of the EM Guide

- Templates for EM Policy and Plan
- Spreadsheets for tracking energy use and quantifying savings opportunities
- EM Guide document



Energy Audits

- Solomon Islands: Ports of Honiara and Noro
- Samoa: Port of Apia



Trials – at port of Suva

- Eco-driving trial
- Tower lighting upgrade

Results

Results overview

- From the EM workshop
- From the audits
- From the trials

EM workshop

Drivers

- Cost of energy
- Capacity development
- National initiatives eg green growth, emissions targets
- Environmental concerns.
- Competitive advantage
- Keeping cost of EE down
- Reputation
- Safety
- International requirements (eg QA)
- Policy and procedures
- Corporate Social responsibility.

Barriers

- Cost of implementation
- Available budget
- Payback period
- Lack of government support
- Political interference
- Port profitability
- Lack of expertise
- Lack of legislation
- Lack of policies

Solutions to barriers

- Spread out cost over several years, having a long term approach.
- Communication policy / make it hard not to support (in line with environment and CSR drivers)
- Develop and implement policies and energy management plans

Cook Islands – post workshop intent

- Preparation of an Energy Policy, appointment of an Energy Manager, and development of an Energy Management Action Plan.

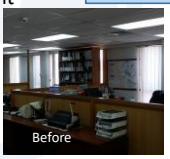
Initiatives already implemented

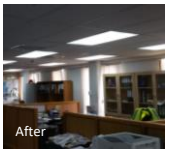
- Yard surface repair and maintenance
- Solar lighting
- Air compressor upgrade
- Reduced idling
- Minimisation of travel distances
- Electricity meters at each marina berth
- Changing of propeller on barge, better matched to vessel, engines

Fiji – post workshop intent

- Preparation of an Energy Policy, appointment of an Energy Manager, and development of an Energy Management Action Plan.
- Detailed audit focussed on eco-driving and yard lighting trial

Initiatives already implemented
- Energy audit
- Draft energy policy and draft EM action plan
- Energy Manager appointed
- Lighting upgrade of FPCL office





Kiribati – post workshop intent

- Preparation of an Energy Policy, appointment of an Energy Manager, and development of an Energy Management Action Plan

Initiatives already implemented
- Yard surface repair and maintenance
- Solar lighting
- Relocation of engineering workshop closer to the port
- Installation of port control tower
- Appointment of a project officer with Masters Degree in Sustainable Energy.

Nauru – post workshop intent

- Preparation of an Energy Policy, appointment of an Energy Manager, and development of an Energy Management Action Plan
- Include energy efficiency measures in specification and design of the proposed new port.

Initiatives already implemented
LED lighting is being installed to illuminate the port area.

Solomon Islands – post workshop intent

- Preparation of an Energy Policy, appointment of an Energy Manager, and development of an Energy Management Action Plan

Initiatives already implemented
- Upgrade of wharf lighting to LED at Honiara
- Energy audit
- Improved port lighting control following the energy audit

Tonga– post workshop intent

- Preparation of an Energy Policy, appointment of an Energy Manager, and development of an Energy Management Action Plan
- Integrate energy performance reporting with financial reporting

Initiatives already implemented
- Tracking of fuel and electricity usage
- Tracking productivity of each machine (eg number of lifts)
- Driver checklist
- Slow steaming
- Yard lighting control
- Hull cleaning every 3 years
- New reach stackers and trailers
- Strong focus on reducing op ex.
- 3% reduction in fuel use in 2016, whilst number of TEUs increased

Vanuatu – post workshop intent

- Preparation of an Energy Policy, and development of an Energy Management Action Plan
- Undertake an energy audit

Initiatives already implemented
- LED wharf lighting

Honiara EA

Very ROUGH Estimates!

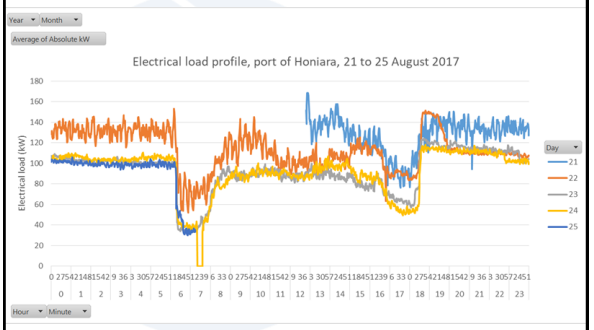
	ANNUAL ELECTRICITY SAVINGS (KWH)	TYPE OF FUEL SAVED	ANNUAL FUEL SAVINGS (LITRES)	ANNUAL ENERGY SAVINGS (GJ)	ANNUAL COST SAVINGS ($ SBD)	ANNUAL GHG SAVINGS (TONNES CO2-E) (TONNES CO2-E)	ESTIMATED IMPLEMENTATION COST ($ SBD)	EST SIMPLE PAYBACK (YEARS)
INSTALL A YARD LIGHTING CONTROL SYSTEM ON TERMINAL 2	57,000			200	$ 290,000	49	$ 430,000	1.5
UPGRADE TERMINAL 2 YARD LIGHTING TO LED	160,000			580	$ 830,000	140	$ 1,200,000	1.4
WHEN UPGRADING SPLIT SYSTEMS PURCHASE HIGH EFFICIENCY UNITS INSTEAD OF STANDARD EFFICIENCY	11,000			39	$ 56,000	9	$ 180,000	3.2
INSTALL A ROOFTOP SOLAR PV SYSTEM	79,000			290	$ 240,000	68	$ 1,100,000	4.7
UNDERTAKE AN ECO DRIVER TRAINING PROGRAM INCLUDING INSTALLATION OF FUEL FLOW METERS		Diesel	10,000	370	$ 71,000	28	$ 350,000	4.9
UPGRADE OFFICE LIGHTING TO LED	13,000			48	$ 69,000	11	$ 390,000	5.7
RATIONALIZE VEHICLE YARD MOVEMENTS WITH A TOS AND CHANGED MODES OF TRANSPORT		Diesel	55,000	2,000	$ 640,000	150	$ 3,700,000	5.8
INSTALL OCCUPANCY SENSORS ON INDOOR LIGHTING	7,100			25	$ 37,000	6	$ 240,000	6.5
TOTAL	330,000		66,000	3,500	$ 2,200,000	460	$ 7,600,000	3.4

Inc. $10,000 in maint. savings

Inc. $312,000 savings in maintenance 8 downtime

Yard lighting control implemented

- Implementation started immediately (straight after presentation of the energy audit report, on the same day)
- In month of September 2017, compared with August 2017:
  - saved 7,800 kWh of electricity
  - Equivalent to about 6.7 tonnes of GHG
  - Saving SBD $40,400



Very ROUGH Estimates!

Noro

	ANNUAL ELECTRICITY SAVINGS (KWH)	TYPE OF FUEL SAVED	ANNUAL FUEL SAVINGS (LITRES)	ANNUAL ENERGY SAVINGS (GJ)	ANNUAL COST SAVINGS ($ SBD)	ANNUAL GHG SAVINGS (TONNES CO2-E) (TONNES CO2-E)	ESTIMATED IMPLEMENTATION COST ($ SBD)	EST SIMPLE PAYBACK (YEARS)
RATIONALIZE VEHICLE YARD MOVEMENTS		Diesel	1,300	45	$ 15,000	3	$ 7,900	0.5
INSTALL A YARD LIGHTING CONTROL SYSTEM	4,700			17	$ 25,000	4	$ 64,000	2.6
UNDERTAKE AN ECO DRIVER TRAINING PROGRAM INCLUDING INSTALLATION OF FUEL FLOW METERS		Diesel	1,100	41	$ 7,000	3	$ 26,000	3.8
UPGRADE YARD LIGHTING TO LED	8,600			31	$ 45,000	7	$ 180,000	3.9
UPGRADE OFFICE LIGHTING TO LED	2,900			10	$ 15,000	3	$ 74,000	5.0
WHEN UPGRADING AIR CONDITIONERS PURCHASE HIGH EFFICIENCY UNITS INSTEAD OF STANDARD EFFICIENCY	2,500			9	$ 13,000	2	$ 71,000	5.6
TOTAL	**19,000**		**1,100**	**110**	**$ 100,000**	**19**	**$ 410,000**	**4.0**

Very ROUGH Estimates!

Apia

Recommendation	Annual Electricity Savings (kWh)	Type of fuel saved	Annual Fuel Savings (litres)	Annual energy savings (GJ)	Annual Cost Savings (WST)	Annual GHG savings (tonnes CO2-e)	Estimated Implementation Cost (WST)	Est simple payback (years)
Upgrade Inside lighting in offices , sheds, and control tower to LED	6,300	Elec	0	23	6,900	5	$19,000	2.7
Replace Main Office Old Split AC Units	810	Elec	0	3	680	1	$2,400	3.5
Yard Lighting Retrofit	24,000	Elec	0	86	23,000	19	56,000	2.4
TOTAL	31,000	Elec	0	110	31,000	25	$77,000	2.5

Port energy benchmarks

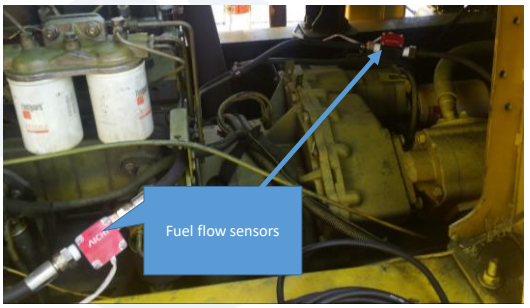
Port name	Organisation(s)	Port operations model	Annual TEUs	TEU estimate or actual	Boundary	Unusual inclusions	Exclusions
Suva	Fiji Ports Corporation Limited (Port Authority) and Fiji Ports Terminal Limited (Terminal Operator)	Landlord	90,000	Estimate	All energy use, including the use of shore cranes, most vehicles	Shore cranes	Terminal company vehicles operating outside the terminal.
Honiara	Solomon Islands Port Authority	Wholly operated by Ports Authority	31,000	Estimate	All energy use, including vehicles, pilot boat		
Noro	Solomon Islands Port Authority	Wholly operated by Ports Authority	2,400	Estimate	All energy use, including vehicles, pilot boat, staff boat	Power supply to the tenanted copra warehouse	
Apia	Samoa Ports Authority	Landlord, although SPA controls yard lighting.	35,876	Actual	Port authority electricity use (including yard lighting), diesel for generator and tug boats	Tug boat	Terminal operations, pilot boat, vehicles

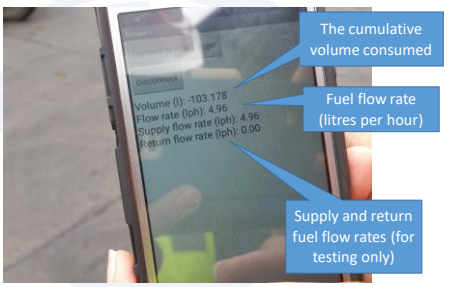
Port energy benchmarks

Port name	Number of reefer points	Annual Electricity (kWh)	Annual Diesel (l)	Annual Petrol (l)	Overall energy (kWh)	Overall GHG (tonnes CO2-e)	Overall Energy/TEU (kWh/TEU)	kg CO2-e/TEU
Suva	100	2,452,945	626,873	87,521	9,515,513	2,966	106	33
Honiara	20	797,286	249,000	12,200	3,388,246	1,391	109	45
Noro	20	58,195	24,000	10,400	395,555	140	165	58
Apia	72	1,130,480	46,700		1,594,678	681	44	19

Trials at Port of Suva

- Fuel trial: Baseline being developed
  - Fuel flow metering installed on an empty container handler
  - Record keeping – soon to start tracking number of movements between refuelling, to determine fuel use per movement
- After the baseline
  - Will install fuel flow meters/dashboard on each machine to provide feedback to drivers
  - Eco driver training
  - Monitoring/reporting of fuel productivity





Lighting trial

- Replacing metal halide lights on one lighting tower with LED
- Installing a dimming system (controlled manually)
- Meter to measure before/after energy use
- Expecting to reduce energy use by 70% to 80%.

Thank you

Bruce Rowse, 8020Green Pty Ltd
Port Energy Management Consultant to MTCC-Pacific
bruce@8020green.com

GMN | The Global MTCC Network
A global network for energy-efficient shipping

MTCC PACIFIC
Maritime Technology Cooperation Centre

MARITIME TECHNOLOGY COOPERATION CENTRE – PACIFIC (MTCC-PACIFIC)

CAPACITY BUILDING FOR CLIMATE MITIGATION IN THE MARITIME SHIPPING INDUSTRY

THE GLOBAL MTCC NETWORK (GMN) PROJECT

MTCC-PACIFC REGIONAL CONFERENCE AND OFFICIAL LAUNCH

12 – 15 December, 2017

Suva, Fiji

EUROPEAN UNION

IMO INTERNATIONAL MARITIME ORGANIZATION

Pacific Community
Communauté du Pacifique

HOST INSTITUTIONS OF MTCC-PACIFIC

SPREP
Secretariat of the Pacific Regional Environment Programme

The Global MTCC Network (GMN) project is funded by the European Union and implemented by the IMO.

CONTENTS

- Drivers reducing GHG emissions from Solomon Ports (Ports Perspective)
- What is needed to achieve reduction of GHG emissions (Ports Perspective)
- Barriers to implementation of country approach to GHG emission reduction
- Possible action from Ports to reduce GHG emissions
- Projects to reduce GHG emissions _ Solomon Ports

Drivers for the reduction of GHG emissions from Solomon Ports (Ports Perspective)

- Rising sea level – Climate change
- High electricity cost

What is needed to achieve the country objectives in reduction of GHG emissions (Ports Perspective)

- Government policy
- Ports initiative to become “GREEN”
- Commitment for environment

Barriers identified for the implementation of country approach to GHG emissions (Ports Perspective)

- Lack of awareness and importance of GHG reduction
- High cost of imported material to the country (solar panels + regulatory framework)
- High cost of power generation (Diesel power – as a result providing shore power for vessels will be very expensive)

Possible action from Ports to reduce GHG emissions

- Use of renewable energy for yard lighting, reefers a
- Use of hybrid vehicles
- Reduce un-necessary electricity usage
- Maintain vehicles for better performance

Electricity Consumption – Before and After the Energy Audit

August Electricity Bill

HONIARA ELECTRICITY						
INVOICE DATE	A/C NAME	A/C ALLOCATED	A/C #	AMPOUNT	CHQ #	DATE PAID
10/8/2017	ELECTRICITY	410	A/C 05097	$ 384,736.63	116560	15/9/2017
10/8/2017	ELECTRICITY	410	A/C 05098-07	$ 1,449.05	116560	15/9/2017
			TOTAL	**$ 386,185.68**		
INVOICE DATE		**NORO ELECTRICITY**				
15/8/2017	ELECTRICITY	410	A/C 81034	$ 1,176.25	116561	15/9/2017
15/8/2017	ELECTRICITY	410	A/C 81006	$ 25,026.73	116561	15/9/2017
15/8/2017	ELECTRICITY	410	A/C 81006-01	$ 3,156.33	116561	15/9/2017
15/8/2017	ELECTRICITY	410	A/C 81006.02	$ 3,760.74	116561	15/9/2017
			TOTAL	**$ 33,120.05**		
			Grand Total Bill	**$ 419,305.73**		

September Electricity Bill

HONIARA ELECTRICITY						
INVOICE DATE	A/C NAME	A/C ALLOCATED	A/C #	AMPOUNT	CHQ #	DATE PAID
10/9/2017	ELECTRICITY	410	A/C 05097	$ 343,065.29	116730	4/10/2017
10/9/2017	ELECTRICITY	410	A/C 05098-07	$ 1,402.92	116730	4/10/2017
			TOTAL	**$ 344,468.21**		
INVOICE DATE		**NORO ELECTRICITY**				**DATE PAID**
15/9/2017	ELECTRICITY	410	A/C 81034	$ 1,341.32	116728	3/10/2017
15/9/2017	ELECTRICITY	410	A/C 81006	$ 24,348.74	116728	3/10/2017
15/9/2017	ELECTRICITY	410	A/C 81006-01	$ 4,717.30	116728	3/10/2017
15/9/2017	ELECTRICITY	410	A/C 81006.02	$ 4,071.94	116728	3/10/2017
			TOTAL	**$ 34,479.30**		
			Grand Total Bill	**$ 378,947.51**		

Electricity Consumption – Before and After the Energy Audit

HONIARA ELECTRICITY						
INVOICE DATE	**A/C NAME**	**A/C ALLOCATED**	**A/C #**	**AMPOUNT**	**CHQ #**	**DATE PAID**
10/11/2017	ELECTRICITY	410	A/C 05097	$ 347,130.49	117106	8/11/2017
10/11/2017	ELECTRICITY	410	A/C 05098-07	$ 1,831.52	117106	8/11/2017
			TOTAL	**$ 348,962.01**		
INVOICE DATE		**NORO ELECTRICITY**				**DATE PAID**
15/10/2017	ELECTRICITY	410	A/C 81034	$ 785.88	117107	8/11/2017
15/10/2017	ELECTRICITY	410	A/C 81006	$ 18,377.73	117107	8/11/2017
15/10/2017	ELECTRICITY	410	A/C 81006-01	$ 4,018.35	117107	8/11/2017
15/10/2017	ELECTRICITY	410	A/C 81006.02	$ 2,667.76	117107	8/11/2017
			TOTAL	**$ 25,849.72**		
			Total	**$ 374,811.73**		

LED Yard Lights – Installation in Progress

LED Yard Lights – Installation in Progress

LED Yard Lights – Installation in Progress

LED Yard Lights – Installation in Progress

LED Yard Lights – Installation in Progress

GMN | The Global MTCC Network
A global network for energy-efficient shipping

MTCC PACIFIC
Maritime Technology Cooperation Centre

THANK YOU TUMAS

SIPA ports

GMN | The Global MTCC Network
A global network for energy-efficient shipping

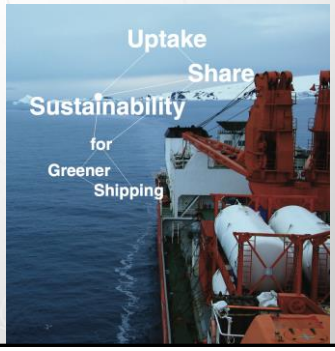
MTCC ASIA
Maritime Technology Cooperation Centre

MTCC-ASIA SUMMARY

XIE Jieying | Technical Officer of MTCC-Asia
Dec.2017| SUVA, Fiji

The Global MTCC Network (GMN) project
is funded by the European Union and implemented by IMO
The views expressed in this presentation
can in no way be taken to reflect the views of the European Union

Our Version



MTCC ASIA
Maritime Technology Cooperation Centre

General

- Venues and facilities: city office and campus office, plus all available SMU resources
- Human resources: full time, part time, consultants, etc,.
- Funding: IMO(EU)+SMG+SMU…;
- Stakeholders: Cambodia, Thailand, DNV, BSM(Germany), COSCO-Shipping, CCS, SIPG, and Shanghai Governmental Commissions, etc. (about 25 partners and stakeholders)
- Internal management system: independent but subject to the SMU's

MTCC ASIA
Maritime Technology Cooperation Centre

Main activities in 2017

No.	Activity Report	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1	Registration in government					5.15							
2	Office space supplement					5.15							
3	Staff recruitment and training					5.15							
4	Development and launching of official website					5.15							
5	Launching event					5.15							
6	Development and launching of Online Clean Shipping Air Forum									9.25			
7	Regional workshop in China									9.25			
8	Establishment of branch office in Cambodia									9.25			
9	Enrollment of postgraduate candidate on ship energy efficiency program								8.25				
10	Seminar on pilot project of energy efficiency (EE)						6.25						
11	Real ship data collection of EE pilot project												12.25
12	Seminar on pilot project of data collection (DC)						6.25						
13	E-System development of DC pilot project												12.25
14	National workshop in Cambodia											11.25	
15	Train the trainer course in Cambodia											11.25	
16	National technology needs, barriers and required relevant action assessment											11.25	

MTCC ASIA
Maritime Technology Cooperation Centre

Action plans in 2017

- Cambodia workshop and train the trainers in November
- Development and launching of Online Clean Shipping Air Forum
- Enrollment of postgraduate candidate on ship energy efficiency program
- Real ship data collection of EE pilot project
- E-System development of DC pilot project
- National technology needs,barriers and required relevant action assessment
- …

Something more:
•Establishment of focal points and network
•Exchange of technical staff and mutual visits
•Joint research programs
•Relevant courses and programs for the promotions of greener shipping industry
•…

MTCC ASIA
Maritime Technology Cooperation Centre

preparation activities

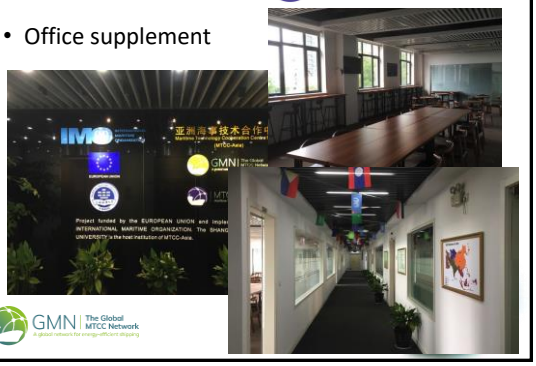
MTCC ASIA
Maritime Technology Cooperation Centre

- Office supplement

preparation activities

MTCC ASIA Maritime Technology Cooperation Centre

- Office supplement



GMN The Global MTCC Network

preparation activities

MTCC ASIA Maritime Technology Cooperation Centre

- Staff training

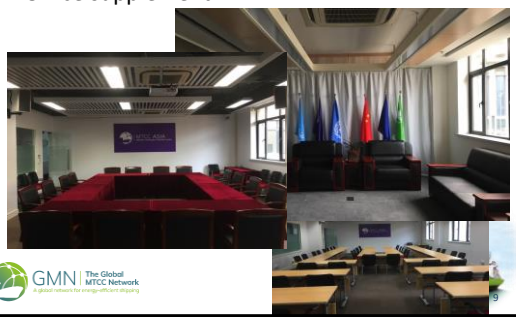
GMN The Global MTCC Network

8

preparation activities

MTCC ASIA Maritime Technology Cooperation Centre

- Office supplement



GMN The Global MTCC Network

9

Launch event

MTCC ASIA Maritime Technology Cooperation Centre

- 15 May, 2017

GMN The Global MTCC Network

10

Regional workshop

MTCC ASIA Maritime Technology Cooperation Centre

- 18th-22nd Sept.,2017

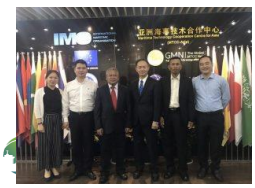
GMN The Global MTCC Network

11

National event

MTCC ASIA

- 18th-22nd Dec.,2017 in Cambod
- Opening of Branch office
- National workshop
- Train the trainer course



International Conference on GHG and EE

MTCC ASIA Maritime Technology Cooperation Centre

- In Oct.23- Nov.10

GMN The Global MTCC Network

13

Pilot project

MTCC ASIA Maritime Technology Cooperation Centre

- Seminars for two pilot projects in June

GMN The Global MTCC Network

14

Upcoming Events

MTCC ASIA GMN The Global MTCC Network

MTCC-Asia First Sub-Regional Workshop

18-22 December 2017
Phnom Penh, Cambodia

Welcome to MTCC-Asia

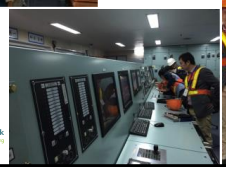
Working discussion with representatives from Ministry of Public Works and Transport, Cambodia

15

Pilot project

MTCC ASIA Maritime Technology Cooperation Centre

- Onboard visit in Oct.

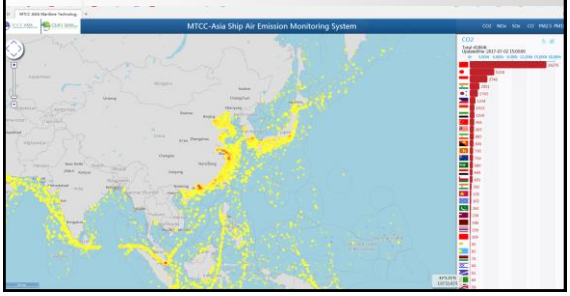


GMN The Global MTCC Network

Pilot project

MTCC ASIA Maritime Technology Cooperation Centre

- Ship Air Emission Monitoring System



Data Collection

Collect fuel consumption data as a ship company

Verify data as a administration

15 Target ships to be selected

18

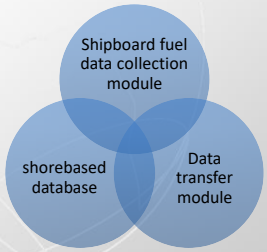
Data collection

Project participants and supporters

19

Data collecting

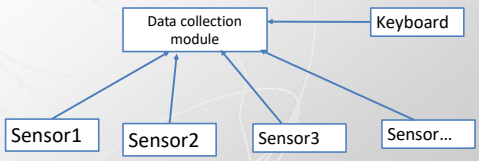
Modules of our fuel consumption data collection system



20

Data collection

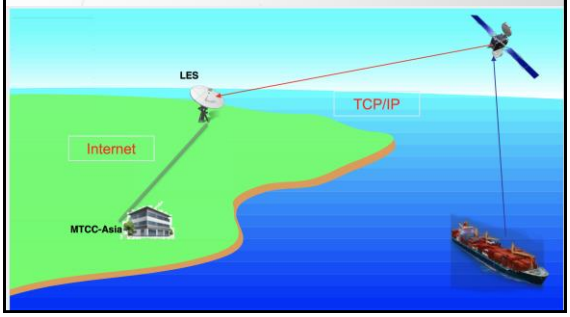
Shipboard fuel consumption data collection module



21

Data collection

Ship to shore communication module



Data collection

Shore based database:

- fuel data receiving,
- Storage and
- verification

23

Data collection

Working plan of the pilot project

•Organize seminars on fuel consumption data collection and reporting(2017.6)

•Develop fuel oil consumption data collection system (2017.6- 2017.12)

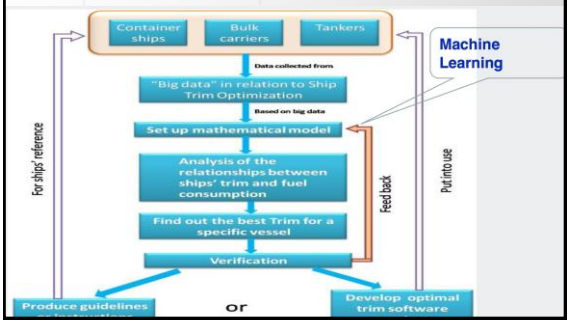
•Collect and analyse real-ship data(3 container ship, 3 bulk 3 tanker for 2017, other 6 ships by 2018)

•Publish and demonstrate ship fuel data collection and reporting process (2019)

24

Trim optimization

Methodology Big dada analytics and machine learni



Trim optimization

Data samples

Date and time	Voy. No	ME consumed (ton/day)	Shaft rpm	Relative wind direction	Wind wave direction	Primary wave period (s)	Wind wave period(s)	Trim (m)	Ture wind speed(m/s)	GPS speed(knots)	Log speed(knots)	Draft mean (m)	Trim(m)
2017/7/X X12:00	45N	45.95626786	48.38155	199.4953 84613611	98.18728 95424585	5.271576 52474108	5.271576 52474108	2.216944 44444444	4.054694 44444445	13.39757 89473684	13.67257 89473684	8.31	0.3
2017/7/X X 12:15	45N	34.61123 21428571	44.45045	199.4953 84613611	98.18728 95424585	5.271576 52474108	5.271576 52474108	2.011388 88888889	5.136944 44444444	11.58157 89473684	11.82657 89473684	8.31	0.9
2017/7/X X12:30	45N	32.06048 21428571	43.42065	199.4953 84613611	98.18728 95424585	5.271576 52474108	5.271576 52474108	1.956916 66666667	5.673444 44444444	11.43957 89473684	11.52757 89473684	8.31	1.1

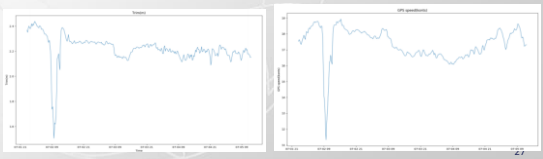
Voyage	Ship	Departure	From	To	EEDI	Duration	kg/nm	MT/day	Distance	Speed
045N	COSCO xxxx	May XX 2017	Hong Kong	Lianyungang, China	48.6	3d 23h 10m	93	26.6	1129	11.9

Over 20,000 data have been collected for a COSCO container ship

Trim Optimization

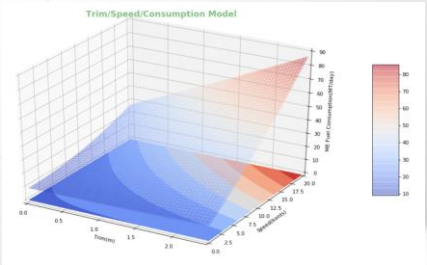
Data

- Relation between main engine fuel oil consumption rate and time
- Relation between trim and time
- Ship speed and time



Trim Optimization

MTCC ASIA Maritime Technology Cooperation Centre



GMN The Global MTCC Network

National technology needs, barriers and required relevant actions

MTCC ASIA Maritime Technology Cooperation Centre

- Based on information and data collected from the websites, database, research papers, etc,.
- Based on the questionnaire survey made during Regional workshops, international conferences, etc,.
- Based on individual meetings.

GMN The Global MTCC Network

Communication and visibility

MTCC ASIA Maritime Technology Cooperation Centre

- News media

CCTV.com 央视网 新闻

中国日报中文网 CHINADAILY.COM.CN

ShanghaiDaily.com

GMN The Global MTCC Network

Communication and visibility

MTCC ASIA Maritime Technology Cooperation Centre

- Wechat
- WhatApp

MTCC-ASIA FAMILY
+880 1672-252653

GMN The Global MTCC Network

Communication and visibility

MTCC ASIA Maritime Technology Cooperation Centre

- Website

http://www.mtccasia.org

Communication and visibility

MTCC ASIA Maritime Technology Cooperation Centre

- Brochure

Communication and visibility

MTCC ASIA Maritime Technology Cooperation Centre

- Video

GMN The Global MTCC Network

Other activities

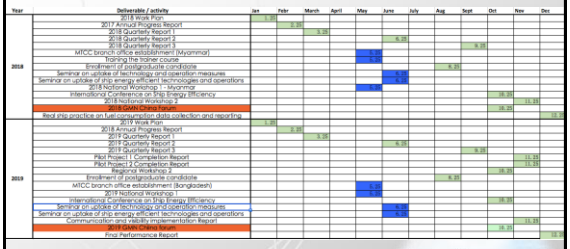
MTCC ASIA Maritime Technology Cooperation Centre

- Postgraduate enrollment
  - 3-year terms, full-English course, tailor-made for MTCC-Asia
  - lecture, field study, internship at mtcc-asia, etc.
  - master degree issued by SMU when completion
  - full scholarship provided by SMG (tuition, accommodation and living expenditures)
  - 5 students
  - …
- Various visitors (additionally)
  - Director-Generals of Ministry of Transport, China
  - Rector of WMU

GMN The Global MTCC Network

Plan for 2018

MTCC ASIA Maritime Technology Cooperation Centre



GMN The Global MTCC Network

MTCC ASIA
Maritime Technology Cooperation Centre

Thank you for your attention!
http://www.mtccasia.org/

GMN | The Global MTCC Network
A global network for energy-efficient shipping

37

GMN | The Global MTCC Network
A global network for energy-efficient shipping

Pilot Project & Data Collection

M. A. Zullah | Maritime Industry Energy Efficiency Officer
CAPACITY BUILDING WORKSHOP FOR MTCCs | 4-7 Dec. 2017 | London, UK

The Global MTCC Network (GMN) project
is funded by the European Union and implemented by IMO
The views expressed in this presentation
can in no way be taken to reflect the views of the European Union

MTCC-Pacific

- Capacity Building for Climate Mitigation in the Pacific Maritime Shipping Industry - Support Pacific SIDS and LDCs in limiting and reducing GHG emissions from their shipping sectors
- Consortium of hosting institutions: SPC & SPREP
- Fiji is the host country
- Targeted countries:
  - Fiji, Samoa, Marshall Islands, Kiribati, Solomon Islands, Tuvalu and Vanuatu
- MTCC-Pacific results:
  - Establishment and operation
  - Regional and national capacity building workshops
  - Pilot project on the "uptake of ship energy efficient technologies and operations
  - Pilot project on "fuel consumption data collection and reporting"

MTCC-Pacific Vision

- Pacific low-carbon maritime transport that supports the sustainable development goals of PICTs
- Support national approaches to uptake low-carbon technologies and operations within PICTs maritime sectors to reduce GHG emissions and reliance to fossil fuels
- Provide capacity-building activities to improve the capacity of PICTs to comply with international instruments and facilitate the implementation of energy efficient measures in the maritime industry
- Contribute to international and regional networks of centres of excellence to share information and experiences and promote the uptake of low carbon technologies and operations and energy efficient practises in the maritime industry

Development of a Fundraising Strategy 2017-2019

MTCC-Pacific pilot-projects

- Objective to lead by example and assist Pacific Islands Countries and Territories (PICTs) to reduce fuel oil consumption and GHG emissions from the maritime sector
- 2 pilot-projects have been approved for MTCC-Pacific:
  - ✓ Pilot-project on "uptake of ship energy efficient technologies and operations"
    - ➢ Focus mainly on domestic shipping and ports
    - ➢ Provide tools to develop/improve energy management
  - ✓ Pilot-project on "fuel oil consumption data collection and reporting"
    - ➢ Data collection on fuel oil consumption in domestic shipping
- Use programme/project activities already implemented by SPC and SPREP in the Pacific region
- SPC and SPREP are 2 regional organisations – 22 PICTs are members
  - ✓ SPC host the Regional Maritime Programme focusing on regional coordination, maritime policy and legal assistance, ship safety, security, trade facilitation
  - ✓ SPREP Waste Management and Pollution Division focus on marine pollution
  - ✓ SPC and SPREP closely collaborate in all areas and are the hosting institutions of MTCC-Pacific

Rationale

- 2013 Majuro Declaration, Pacific leader committed to act as climate leaders
- PICTs all declared NDCs under the Paris Agreement and committed to ambitious target however:
  - Data and information supporting the targets are not accurate, reliable
  - GHG emissions are not disaggregated by sector including transport and more specifically maritime transport
- In 2014 Transport Ministers reaffirmed the importance of access to timely, accurate and reliable data to provide the basis for informed decision-making and policy formulation
- In April 2017 Transport Ministers:
  - were informed about:
    - ✓ the issues related to technical and operational measures for enhancing energy efficiency of international shipping and its implications for the Pacific
    - ✓ the three-step approach towards data collection, analysis and decision-making on further measures
    - ✓ the approved Roadmap for developing a comprehensive IMO strategy on the reduction of GHG emissions from ships
  - Agreed to collect and share reliable, accurate and quality data for informed decision-making and adoption of relevant indicators on transport fuel use and GHG emissions
  - Requested regional organisations and development partners to assist PICs to review, when appropriate, their commitments made for transport emissions reductions in their NDCs well in advance of the 2018 UNFCCC facilitative dialogue.

Pilot-project implementation

- Implementation approach:
  - ✓ All activities related to dissemination of templates, guidelines and generic legislation – ALL PICTs
  - ✓ Activities focusing on implementing energy efficient measures and data collection – TARGETED PICTs (Fiji, Kiribati, Marshall Islands, Samoa, Solomon Islands, Tuvalu and Vanuatu)
- Use of PIDSS programme to target domestic ships:

PIDSS Progress Summary : 2010 - 2016	2010	2011	2012	2013	2014	2015	2016	Total as at end 2016
Countries participating*	2	2	2	4	5	6	8	**8**
Shipping Companies participating*	11	11	13	22	33	47	66	**66**
Ships considered under PIDSS*	14	14	17	30	50	72	114	**114**
Ships reported having SOPs approved*	4	7	8	8	6**	7	19	**20**
Ships reported having SOPs drafted*	0	4	6	6	5**	15	42	**42**
SOP Initial Audits conducted***	4	2	1	0	0	1	7	**15**
SOP Follow Up Audits conducted***	0	0	3	0	0	0	1	**4**
Number of personnel trained ***	39	0	32	27	19	27	92	**236**

- Kiribati and Tonga (2010); Marshall Islands and Vanuatu (Port Vila) (2013); Solomon Islands and Vanuatu (Luganville) (2014); Tuvalu (2015); and Federated States of Micronesia (Pohnpei) and Samoa (2016).

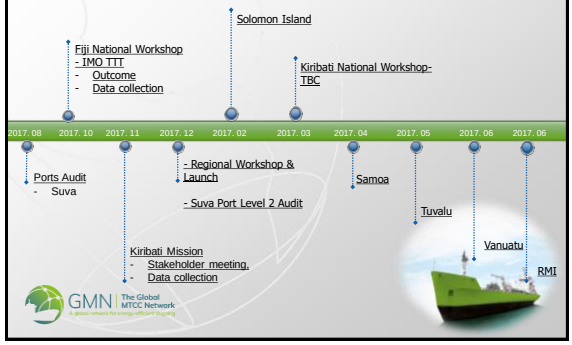
Pilot-project on EE & DC

Pilot-project on Energy Efficiency	Pilot-project on Data Collection
Develop generic laws for MARPOL Annex VI	Develop a data collection template
Extension of the Pacific Gender & Climate Change Toolkit	Develop a generic regulation for data collection
Develop templates for energy efficiency policy and adapted SEEMP and EEOI	Collect relevant data and information
Develop a guide for port energy management based on ISO 50000 standards	Analyse data and information and produce a report on GHG emissions
Implemented an adapted SEEMP onboard vessels	
Review the implementation of SEEMP measures	
Research, case study and expertise to uptake a low-carbon technology on board vessels	
Implement a low-carbon technology onboard vessel	
Conduct energy audit level 1 in pacific port	
GHG reduction project implemented in port	
Conduct energy audit level 2 or 3 in port	

GMN | MTCC Network

7

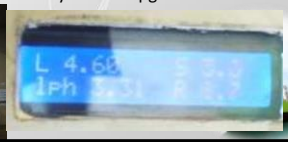
MTCC-Pacific Activities



Energy Saving of FPCL, Suva, Fiji

After level 1 audit, electricity consumption has dropped at FPCL HO by 21% compared with previous year

- Lighting was upgraded to LED
- Expected annual reduction is 75,000 kWh, saving $31,000 and 32 tonnes of GHG emissions.
- Cost of lights was under $23,000.
- LED lights have already paid for themselves (excluding labour costs). They have a 3 year warranty.
- Other indoor lighting has not yet been upgraded to LED



9

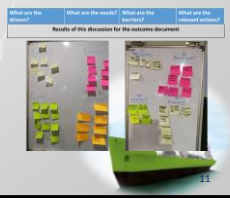
Energy Saving of SIPA, Honiara, Solomon

- Solomon Islands Port Authority (SIPA) – port of Honiara.
  - 800,000 kWh of electricity, 250,000 litres of diesel and 12,000 litres of petrol annually over 2016/17 – 1,400 tonnes of GHG (CO2-e) and cost SBD $5,900,000
  - Energy savings opportunities:
    - ✓ Installing a yard lighting control system and upgrade yard lighting on the new wharf to LED
    - ✓ Install a rooftop solar PV system and upgrade office lighting to LED
    - ✓ Undertake an eco-driver training program including installation of fuel flow meters
    - ✓ Rationalize vehicle yard movements with a TOS and changed modes of transport
    - ✓ Install occupancy sensors on indoor lighting
- SIPA had saved $40,358.20 on September Electricity bill compared to August with management of electricity power usage at the new berth – around 7,800 kWh corresponding to around 6.7 tonnes of GHG.

10

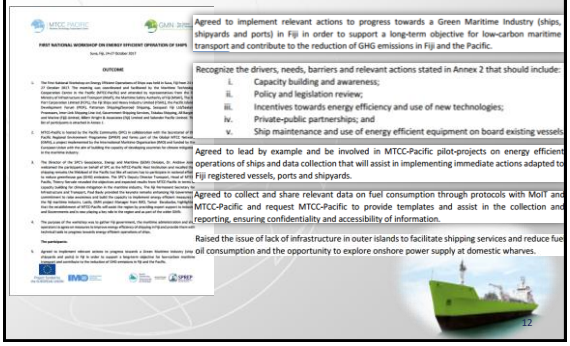
First National Workshop - Fiji

- 95% overall attendance as anticipated
- Group discussion on issues relating SEEO
- Discussion on CC, GHGE and P/SEM measures
- Interactive knowledge transfer on EEDI, EEOI, SEEMP, and PIDSS
- Interactive sessions on IMO GMN technology transfer and technical cooperation- SEET
  - PBCF, LED, WHRS, SG, and PV
- Discussion on the climate mitigation in the Fiji maritime industry: To reduce fuel oil consumption and GHG emission.
  - What are the **drivers**?
  - What are the **needs**?
  - What are the **barriers**?
  - What are the **relevant actions**?



11

Fiji National Outcome



Agreed to implement relevant actions to progress towards a Green Maritime Industry (ships, shipyards and ports) in Fiji in order to support a long-term objective for low-carbon maritime transport and contribute to the reduction of GHG emissions in Fiji and the Pacific.

Recognize the drivers, needs, barriers and relevant actions stated in Annex 2 that should include:

i. Capacity building and awareness;
ii. Policy and legislation review;
iii. Incentives towards energy efficiency and use of new technologies;
iv. Private-public partnerships; and
v. Ship maintenance and use of energy efficient equipment on board existing vessels.

Agreed to lead by example and be involved in MTCC-Pacific pilot-projects on energy efficient operations of ships and data collection that will assist in implementing immediate actions adapted to Fiji registered vessels, ports and shipyards.

Agreed to collect and share relevant data on fuel consumption through protocols with MoIT and MTCC-Pacific and request MTCC-Pacific to provide templates and assist in the collection and reporting, ensuring confidentiality and accessibility of information.

Raised the issue of lack of infrastructure in outer islands to facilitate shipping services and reduce fuel oil consumption and the opportunity to explore onshore power supply at domestic wharves.

12

Vessels in Fiji

2131 registered ships in Fiji by length as of 31/07/17

Length	<15m	15 – 20m	21 – 50m	>50m
No of Ships	1890	55	161	25

Lomaiviti Princess in the Suva harbour yesterday. Photo: Selita Bolanavanua

Details Of The Lomaiviti Princess V

- Previously known as Queen of Nanaimo.
- Built in 1964 for British Columbia Ferry Services Inc.
- Length 130 metres
- Speed: 16.5 knots (30.6 km/h; 19.0 mph)
- Capacity: 1,500 passengers and 192 cars
- Sold to Goundar Shipping Limited on September 2017

13

Kiribati Visit

- Met with Kiribati MCTTD and ships owners
- 4 vessel visit
- Conducted training for DC

With MCTTD Director, Consultant, Dep. Secretary & Secretary

14

Data Collection

- Fiji
  - Government Shipping Services
  - Patterson Shipping/Searoad Shipping
  - Inter Link Shipping Line Ltd
  - Solander Pacific Ltd
  - Billett Wright & Associates (Fiji) Ltd
- Kiribati
  - Kiribati Shipping Service Limited
  - Central Pacifc Producers Limited
  - Atinmarawa Company Limited
  - Coral Ace Shipping
  - Oceanic Shipping Services
  - Lu's Marine Shipping
  - Butaritari Shipping Services

GMN The Global MTCC Network

15

ESD's for the Pacific's "Leader-ship(s)"

- Photovoltaics Panel
  - ☞ most convenient tech. for the pacific
  - ☞ Easy to install and maintain
  - ☞ Tech. already exist in the region
  - ☞ Few Kiribati vessels have PV for lighting
- Light Emitting Diodes (LED) ***"LED the way"***
  - ☞ Last longer than conventional luminaire (6.5 yrs.)
  - ☞ Do not need to be replaced as often as conventional type (1.25yrs.)
  - ☞ More efficient than conventional luminaire

GMN The Global MTCC Network

16

ESD's for the Pacific's "Leader-ship(s)"

- Propeller Boss Cap Fins (PBCF) ***"The Boss Cap"***
  - ☞ It is suitable for slow speed vessel.
  - ☞ PBCF boost propulsive efficiency by about 5% and ship fuel efficiency by about 2~3%.
  - ☞ PBCF can be retrofitted easily to an existing ship.
- Optional data collection system
  - ☞ "If we cant measure it, we cant manage it"
  - ☞ Collects data through installed sensors (fuel meter, engine load, RPM sensors)
  - ☞ Eliminates the human error and non-consistency in data collection

GMN The Global MTCC Network

17

MTCC-Pacific Office and Staff

SPC, Lotus Building, Nabua, Fiji

Thierry Nervale – Head of MTCC

Mark H Davis CHGETA

M. S. Ullah / MHESO

Lora Cricket Admin & Info

18

MTCC-Pacific Website

GMN | The Global MTCC Network

19

Thank you for your attention!
http://gmn.imo.org/

GMN | The Global MTCC Network

20

ANNEX 5: Workshop Evaluation Analysis

EVALUATION QUESTIONNAIRE

MTCC-Pacific Regional Workshop and Official Launch

Suva, Fiji, December 2017

Arrangements prior to the activity

1	Was the invitation received in good time?	Yes ☐	No ☐	
2	Did you receive the information listed below about the event before your participation			
	• on its objective and scope	Yes ☐	No ☐	
	• subject areas and programme	Yes ☐	No ☐	
3	Were the instructions on the following clear and easy to understand?			
	• profile required of participant	Yes ☐	No ☐	
	• completion and submission of the nomination form	Yes ☐	No ☐	
4	Did you receive logistical information on			
	• venue	Yes ☐	No ☐	
	• travel arrangements	Yes ☐	No ☐	N/A ☐
	• DSA payments	Yes ☐	No ☐	N/A ☐
	• accommodation	Yes ☐	No ☐	N/A ☐
5	If you were given any pre-event assignment, was it useful?	Yes ☐	No ☐	N/A ☐

During the activity

6 To cover the topics fully, was the event (*please check the appropriate box*)

(1) too long ☐ (2) just right ☐ (3) too short ☐

7 How do you rate the event with regard to the following? (*tick one box in each case*)

	excellent	good	satisfactory	poor
Venue	☐	☐	☐	☐
Facilities	☐	☐	☐	☐
Equipment	☐	☐	☐	☐

8 How do you rate the following aspects of the materials? (*tick one box in each case*)

	excellent	good	satisfactory	poor
Presentation	☐	☐	☐	☐
Clarity	☐	☐	☐	☐
Technical content	☐	☐	☐	☐
Comprehensiveness	☐	☐	☐	☐
Quantity	☐	☐	☐	☐

9 How would you rate the following aspects of the presentations? *(tick one box in each case)*

	excellent	good	satisfactory	poor
Design and structure	☐	☐	☐	☐
Clarity	☐	☐	☐	☐
Technical contents	☐	☐	☐	☐
Comprehensiveness	☐	☐	☐	☐

10 How would you rate the use of the following? *(tick one box in each case)*

	excellent	good	satisfactory	poor	
Course materials	☐	☐	☐	☐	
IMO reference materials	☐	☐	☐	☐	
Other resource materials	☐	☐	☐	☐	
Group and practical activities	☐	☐	☐	☐	N/A☐
Field trips	☐	☐	☐	☐	N/A☐

At the end of the activity

11 Please rate the lecturers with regard to the following (*check one box in each case*)

.1	excellent	good	satisfactory	poor
content of lecture	☐	☐	☐	☐
delivery of presentation	☐	☐	☐	☐
ability to transfer knowledge	☐	☐	☐	☐
effectiveness in:				
• answering questions	☐	☐	☐	☐
• suggesting solutions to issues	☐	☐	☐	☐

12 What topics were of most interest and relevance to you?

_____

_____

13 Are there any topics which should be added? Yes ☐ No ☐
If yes, please list them:

_____

_____

14 Do you consider that the objective of the event was met? Yes ☐ No ☐

15 Are you likely to use the information you gained on the course when you return to work? Yes ☐ No ☐

16 Will you have the opportunity to transfer the knowledge gained to your colleagues at work? Yes ☐ No ☐

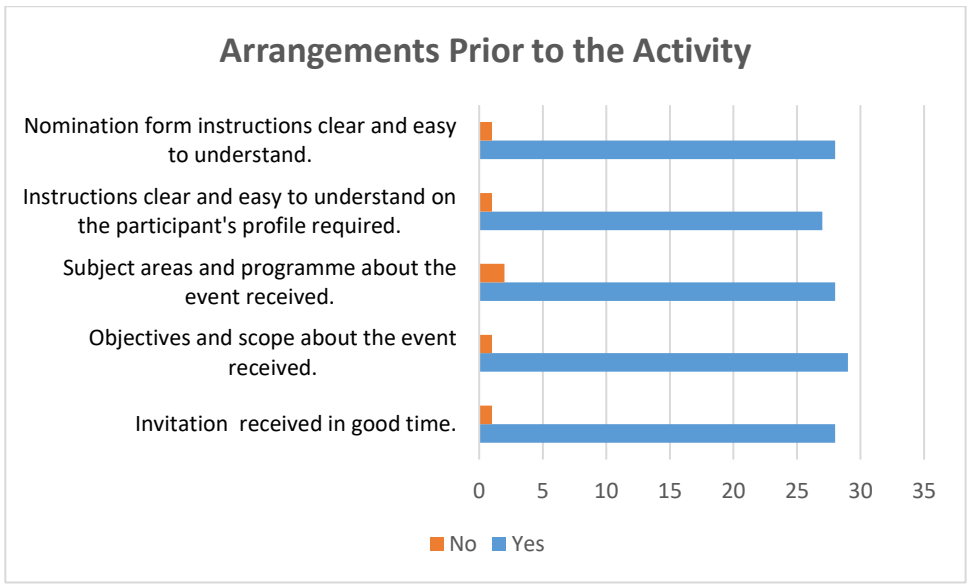
Comments:

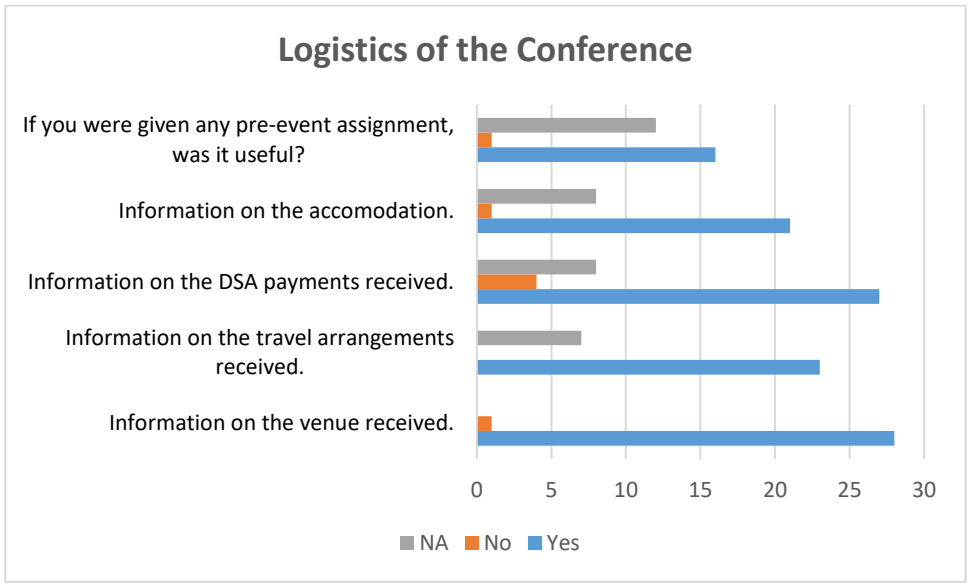
_____

_____

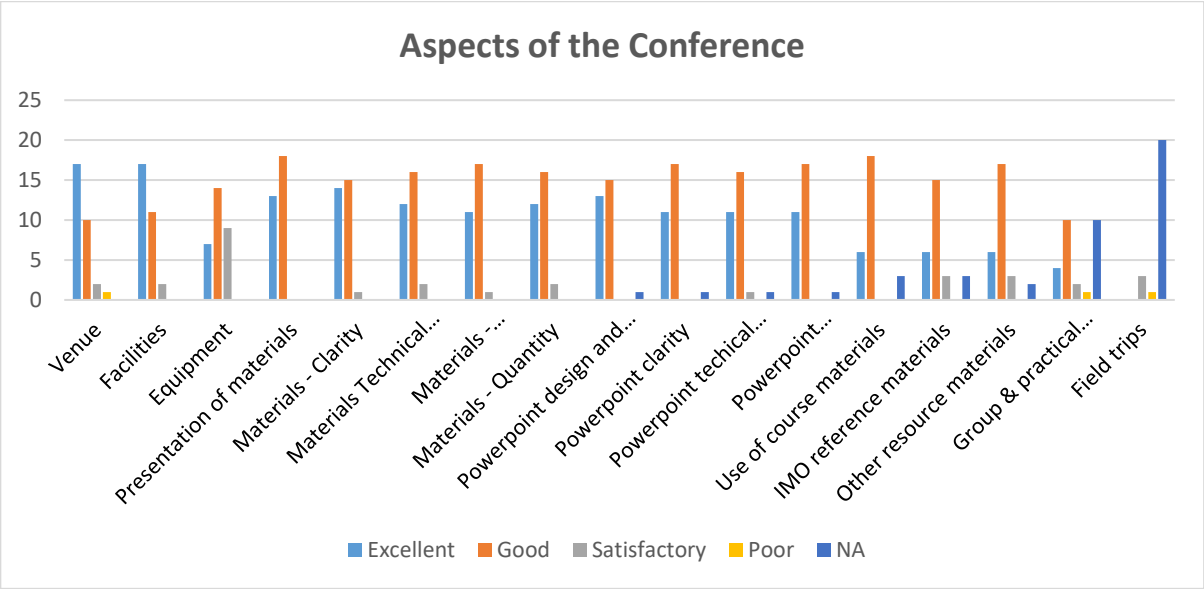
_____

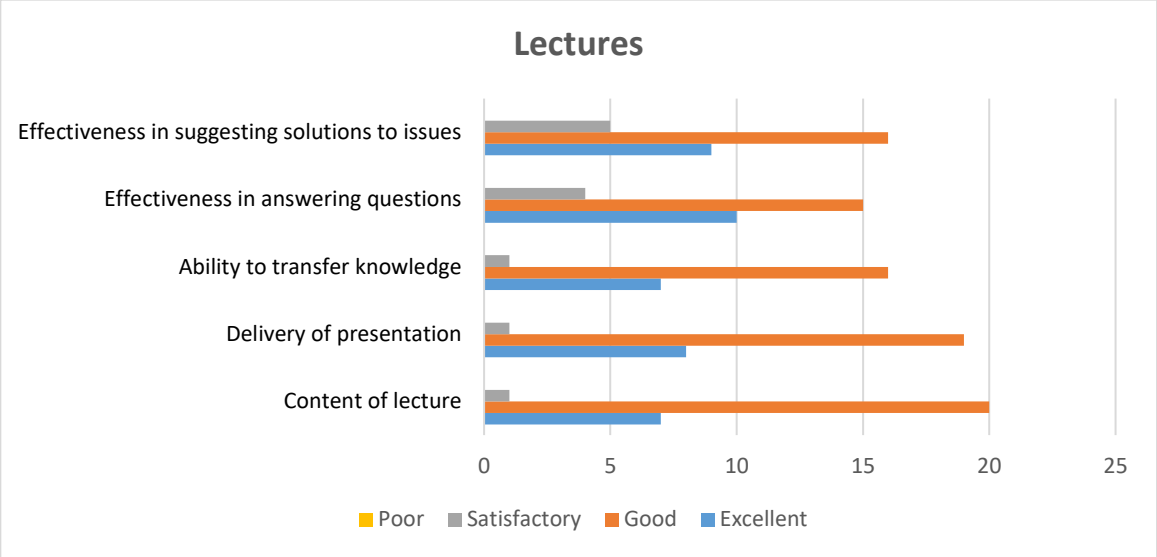
We greatly appreciate your time in completing this evaluation questionnaire. It contains important information that will assist IMO in determining the success and impact of the activity. Thank you.

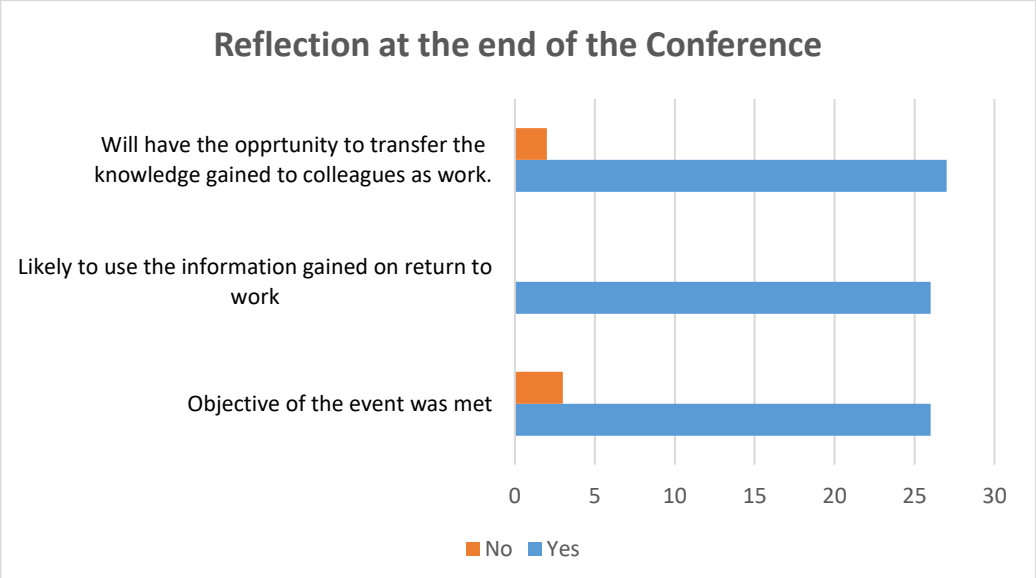

Arrangements Prior to the Activity
Nomination form instructions clear and easy to understand.
Instructions clear and easy to understand on the participant's profile required.
Subject areas and programme about the event received.
Objectives and scope about the event received.
Invitation received in good time.
0
5
10
15
20
25
30
35
No
Yes


Logistics of the Conference
If you were given any pre-event assignment, was it useful?
Information on the accomodation.
Information on the DSA payments received.
Information on the travel arrangements received.
Information on the venue received.
0
5
10
15
20
25
30
NA
No
Yes

Duration of the Conference
30
25
20
15
10
5
0
Event too long.
Event just right.
Event too short.


Aspects of the Conference
25
20
15
10
5
0
Venue
Facilities
Equipment
Presentation of materials
Materials - Clarity
Materials Technical...
Materials -...
Materials - Quantity
Powerpoint design and...
Powerpoint clarity
Powerpoint techical...
Powerpoint...
Use of course materials
IMO reference materials
Other resource materials
Group & practical...
Field trips
Excellent
Good
Satisfactory
Poor
NA


Lectures
Effectiveness in suggesting solutions to issues
Effectiveness in answering questions
Ability to transfer knowledge
Delivery of presentation
Content of lecture
0
5
10
15
20
25
Poor
Satisfactory
Good
Excellent


Reflection at the end of the Conference
Will have the opprtunity to transfer the knowledge gained to colleagues as work.
Likely to use the information gained on return to work
Objective of the event was met
0
5
10
15
20
25
30
No
Yes

Topics of most interest and relevance?	Are there any topics which should be added?	Comments
I would say that all topics were of interest to me as we got to learn from other island countries and what other developed countries have implemented in terms of maritime energy efficiency.		Great launch of MTCC-Pacific and good start to the project. All the best SPC.
Technology involved. Legal and other templates.		It's a great opportunity to attend this event and to know what happens in the Pacific region and their efforts toward GHG reduction and energy efficiency in the maritime sector.
Energy management. Ship owners' representatives.		From a ports perspectives this is a great training where we strive for green, clean, and efficient ports.
Session 1 - country presentations. Session 2 - partner perspectives.		Well done a getting a wide participation and having a diverse program. Keeping the program down to 2 and half to 3 days would have been nice.
Ships and port efficiency.		The arrangement with regards to DSA was poorly organised and the lack of feedback or constant update from the organisers made it a lot worse living the participants in the dark. Otherwise the host and other arrangements were good.
Methods of collecting data.		Grateful for attending the conference. Learnt a lot of things on climate change and its importance to reduce CO2 of its effect on the environment and humanity for the future.
Port energy management/policy. Global Maritime Network.		

ANNEX 6: Photos

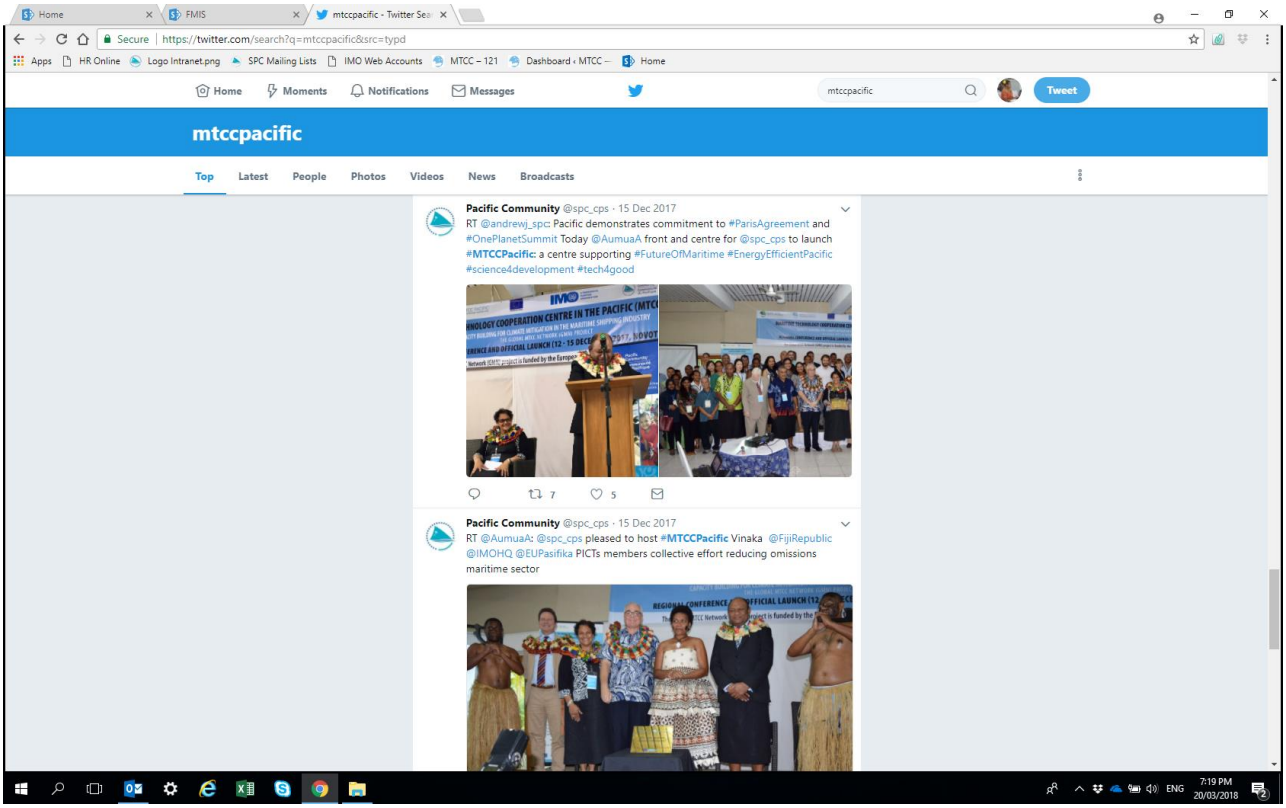
Workshop for Heads of Maritime, Fiji Dec 12 2017

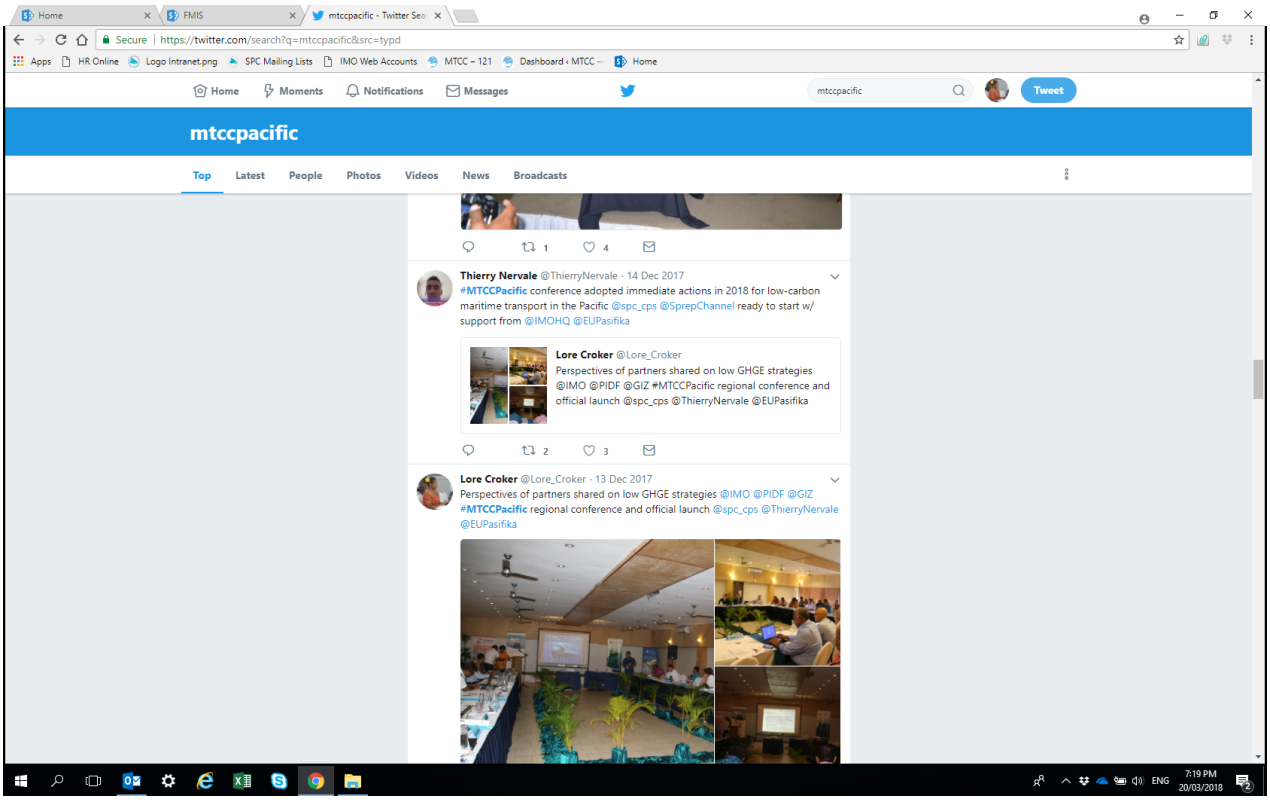
MTCC-Pacific Steering Committee Meeting, Fiji Dec 12 2017

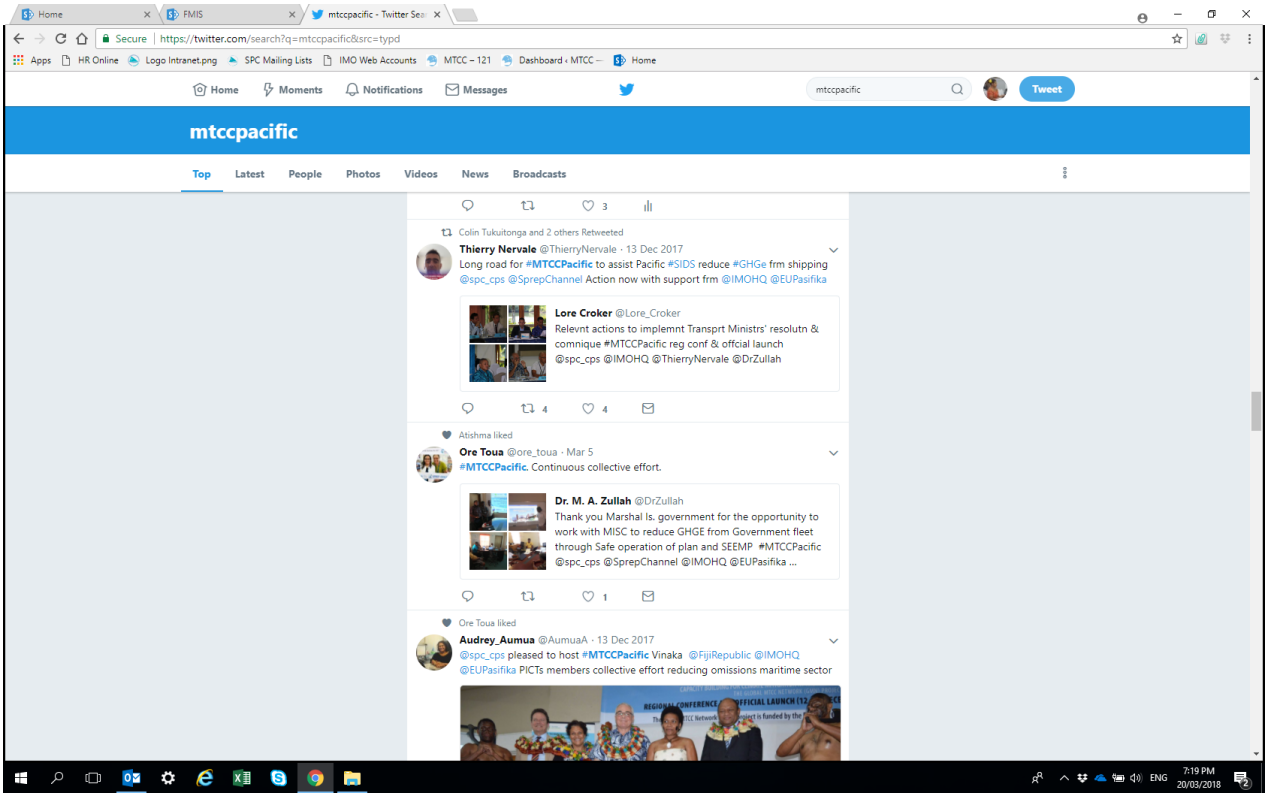
MTCC-Pacific Regional Conference and Official Launch, Fiji Dec 13-15 2017

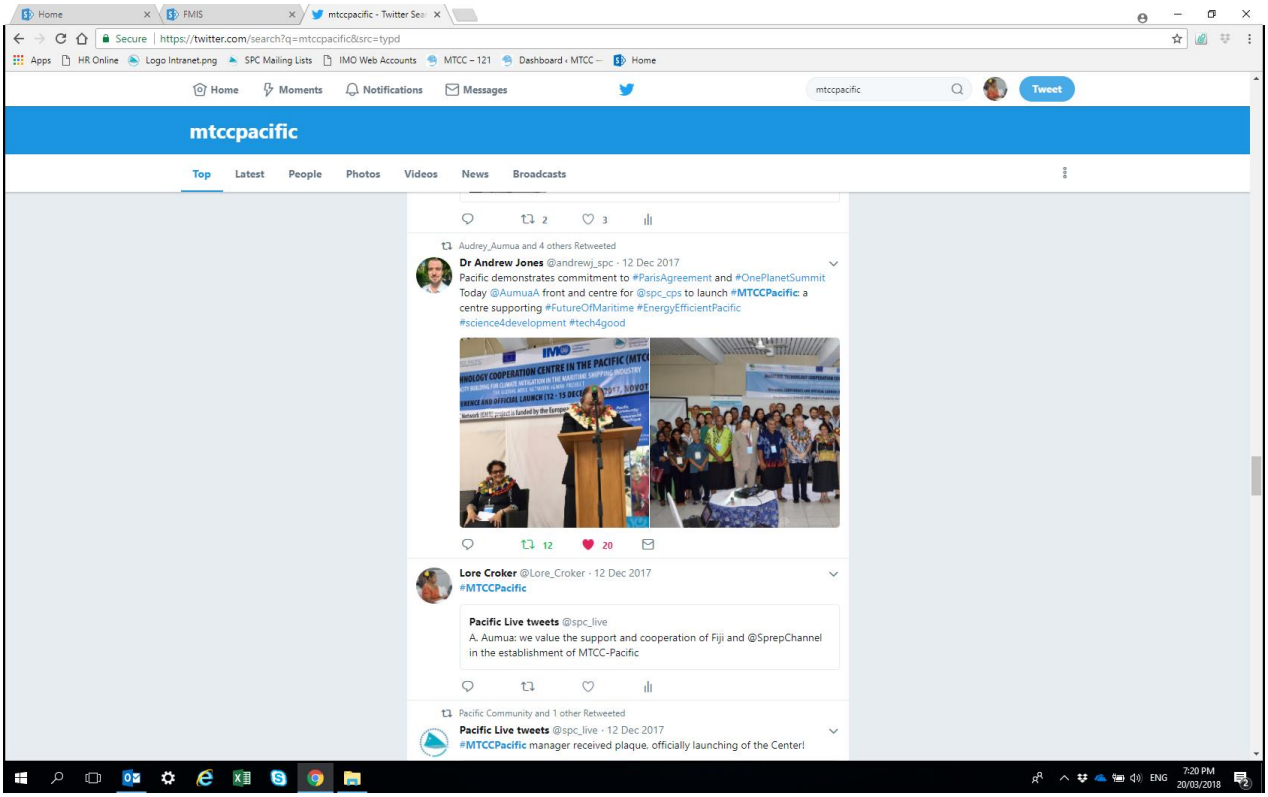
Solomon Islands	PROE SPREP MTCC-ASIA SUMMARY
PROE SPREP	

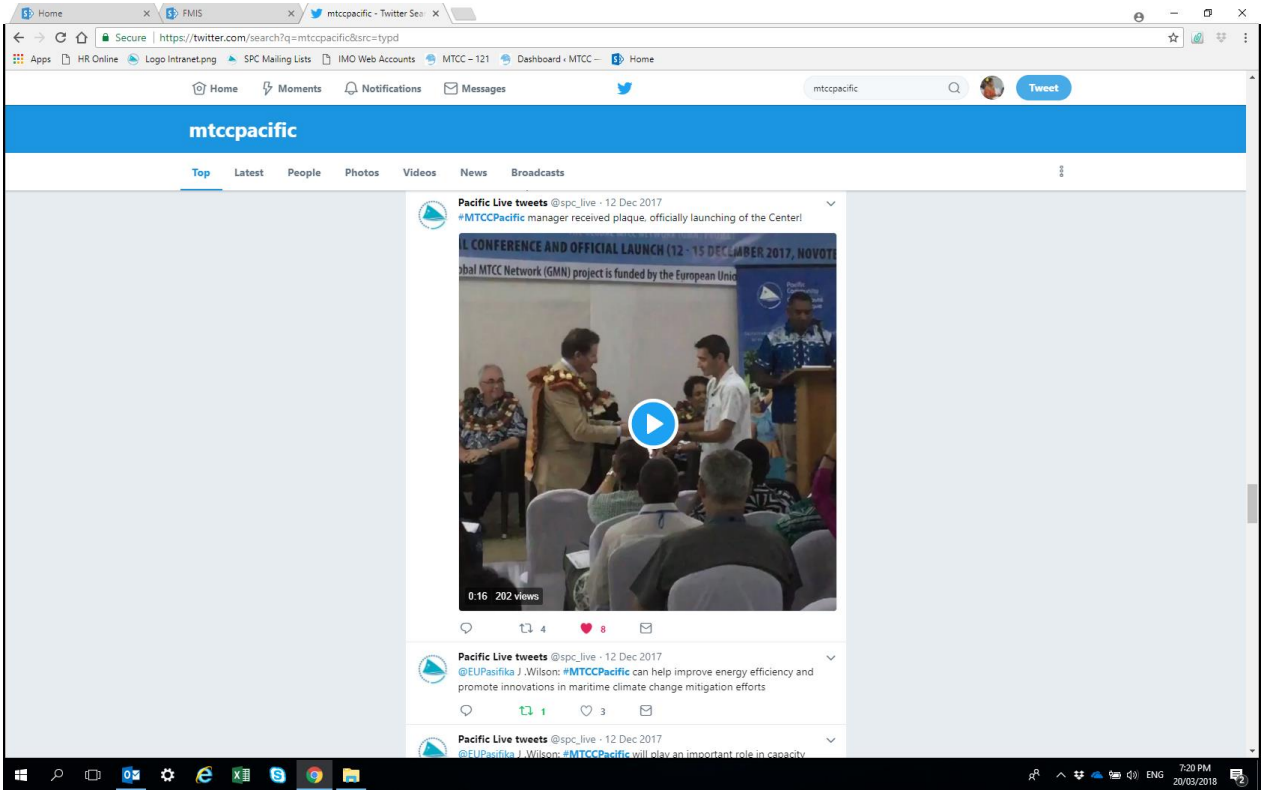
ANNEX 7: Communication and Visibility Activities

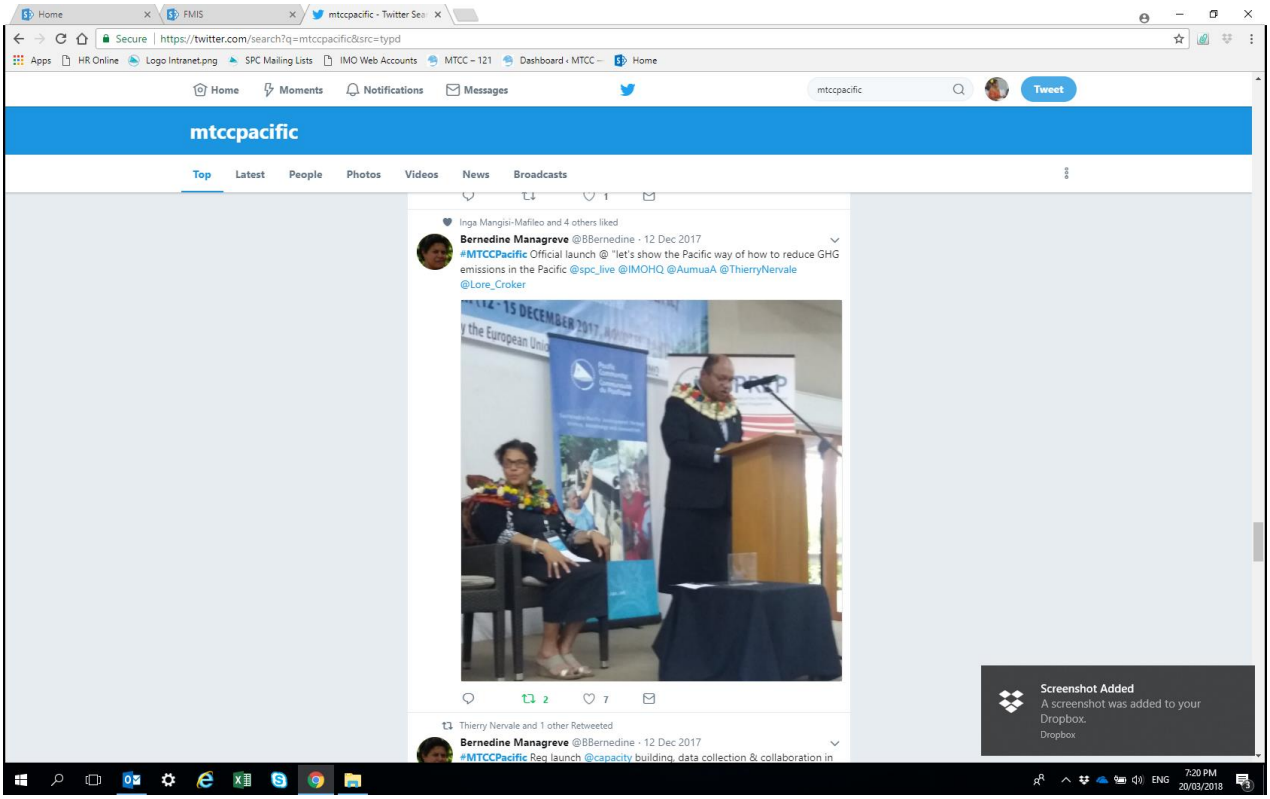

Home
FMIS
mtccpacific - Twitter Sea
Secure | https://twitter.com/search?q=mtccpacific&src=typd
Apps HR Online Logo Intranet.png SPC Mailing Lists IMO Web Accounts MTCC – 121 Dashboard ‹ MTCC – Home
Home Moments Notifications Messages
mtccpacific
Tweet
mtccpacific
Top Latest People Photos Videos News Broadcasts
Pacific Community @spc_cps · 15 Dec 2017
RT @andrewj_spc: Pacific demonstrates commitment to #ParisAgreement and #OnePlanetSummit Today @AumuaA front and centre for @spc_cps to launch #MTCCPacific: a centre supporting #FutureOfMaritime #EnergyEfficientPacific #science4development #tech4good
Pacific Community @spc_cps · 15 Dec 2017
RT @AumuaA: @spc_cps pleased to host #MTCCPacific Vinaka @FijiRepublic @IMOHQ @EUPasifika PICTs members collective effort reducing omissions maritime sector
ENG
7:19 PM
20/03/2018

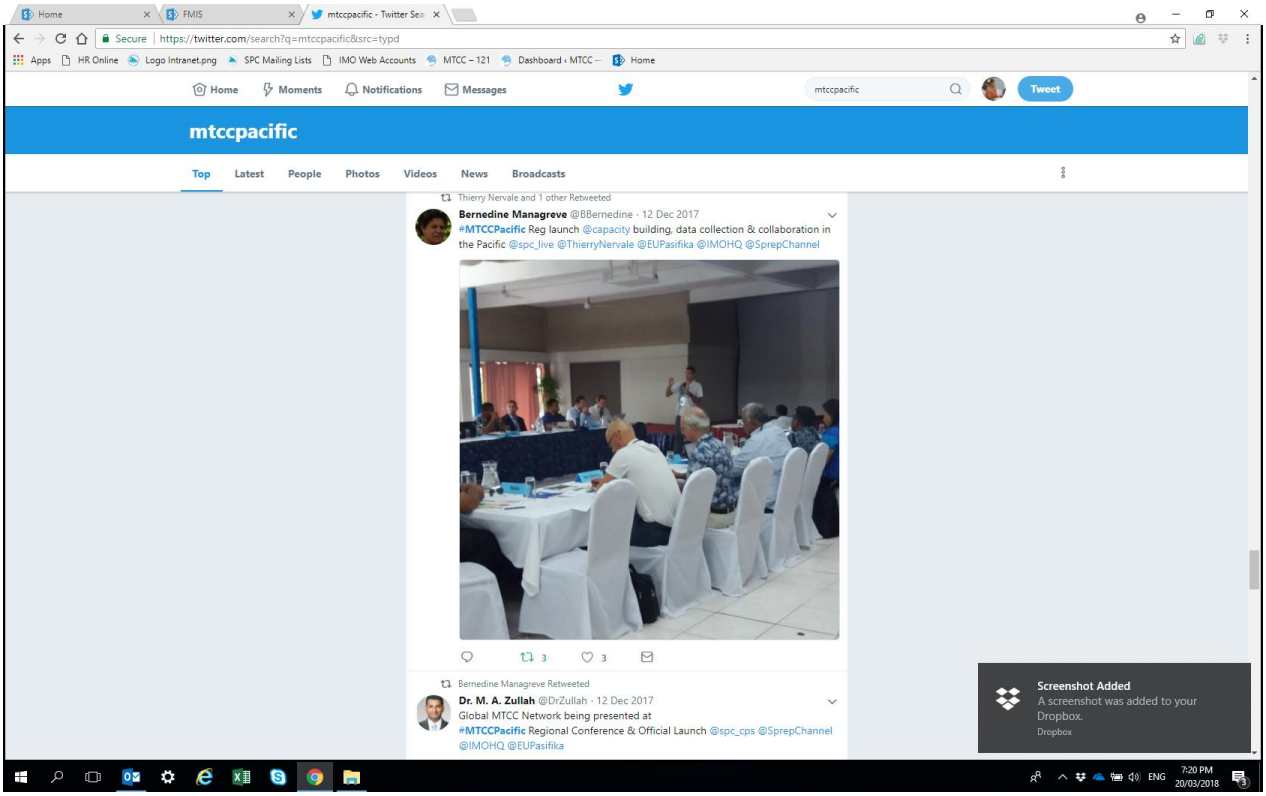

Home
FMIS
mtccpacific - Twitter Sea
Secure | https://twitter.com/search?q=mtccpacific&src=typd
Apps HR Online Logo Intranet.png SPC Mailing Lists IMO Web Accounts MTCC – 121 Dashboard ‹ MTCC – Home
Home Moments Notifications Messages
mtccpacific
Tweet
mtccpacific
Top Latest People Photos Videos News Broadcasts
Thierry Nervale @ThierryNervale · 14 Dec 2017
#MTCCPacific conference adopted immediate actions in 2018 for low-carbon maritime transport in the Pacific @spc_cps @SprepChannel ready to start w/ support from @IMOHQ @EUPasifika
Lore Croker @Lore_Croker
Perspectives of partners shared on low GHGE strategies @IMO @PIDF @GIZ #MTCCPacific regional conference and official launch @spc_cps @ThierryNervale @EUPasifika
Lore Croker @Lore_Croker · 13 Dec 2017
Perspectives of partners shared on low GHGE strategies @IMO @PIDF @GIZ #MTCCPacific regional conference and official launch @spc_cps @ThierryNervale @EUPasifika
ENG
7:19 PM
20/03/2018

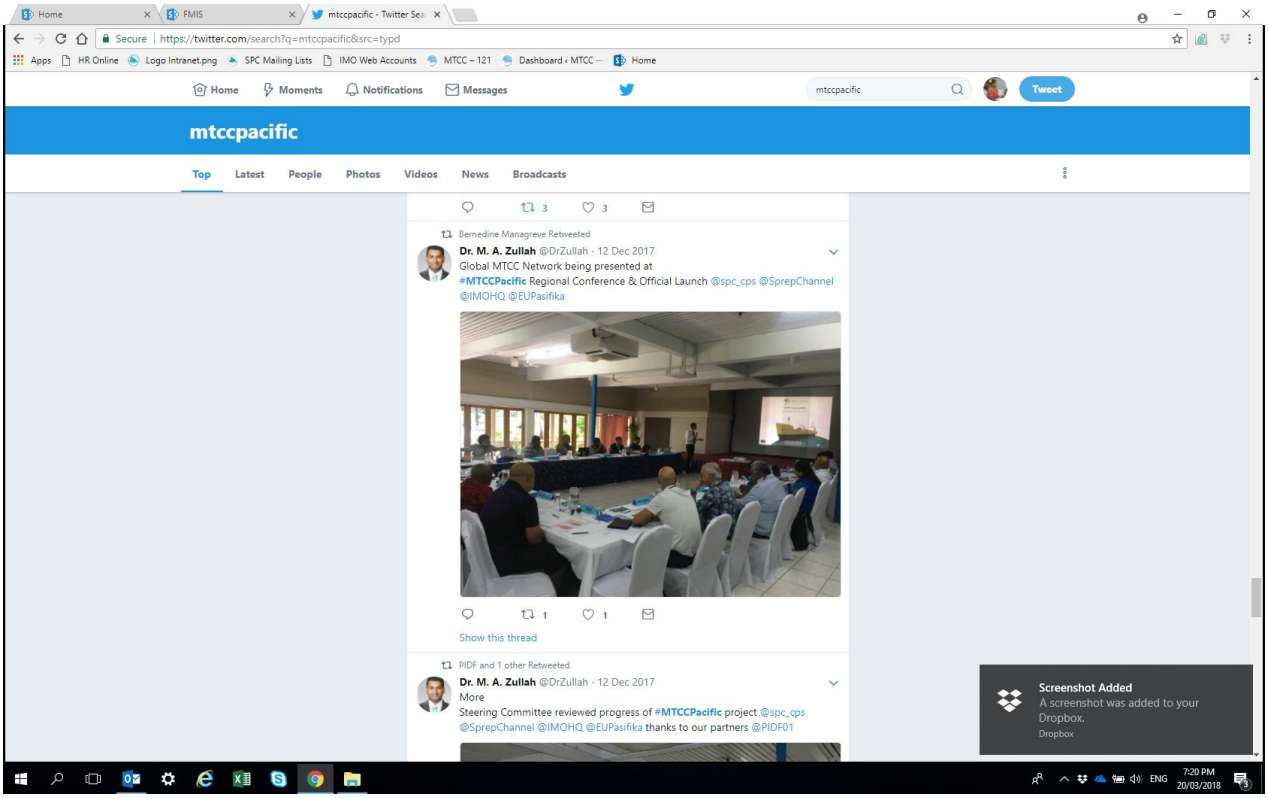

Home
FMIS
mtccpacific - Twitter Sea
Secure | https://twitter.com/search?q=mtccpacific&src=typd
Apps HR Online Logo Intranet.png SPC Mailing Lists IMO Web Accounts MTCC – 121 Dashboard ‹ MTCC – Home
Home Moments Notifications Messages
mtccpacific
Tweet
mtccpacific
Top Latest People Photos Videos News Broadcasts
Colin Tukuitonga and 2 others Retweeted
Thierry Nervale @ThierryNervale · 13 Dec 2017
Long road for #MTCCPacific to assist Pacific #SIDS reduce #GHGe frm shipping @spc_cps @SprepChannel Action now with support frm @IMOHQ @EUPasifika
Lore Croker @Lore_Croker
Relevnt actions to implemnt Transprt Ministrs' resolutn & comnique #MTCCPacific reg conf & offcial launch @spc_cps @IMOHQ @ThierryNervale @DrZullah
Atishma liked
Ore Toua @ore_toua · Mar 5
#MTCCPacific. Continuous collective effort.
Dr. M. A. Zullah @DrZullah
Thank you Marshal Is. government for the opportunity to work with MISC to reduce GHGE from Government fleet through Safe operation of plan and SEEMP #MTCCPacific @spc_cps @SprepChannel @IMOHQ @EUPasifika ...
Ore Toua liked
Audrey_Aumua @AumuaA · 13 Dec 2017
@spc_cps pleased to host #MTCCPacific Vinaka @FijiRepublic @IMOHQ @EUPasifika PICTs members collective effort reducing omissions maritime sector
ENG 7:19 PM 20/03/2018

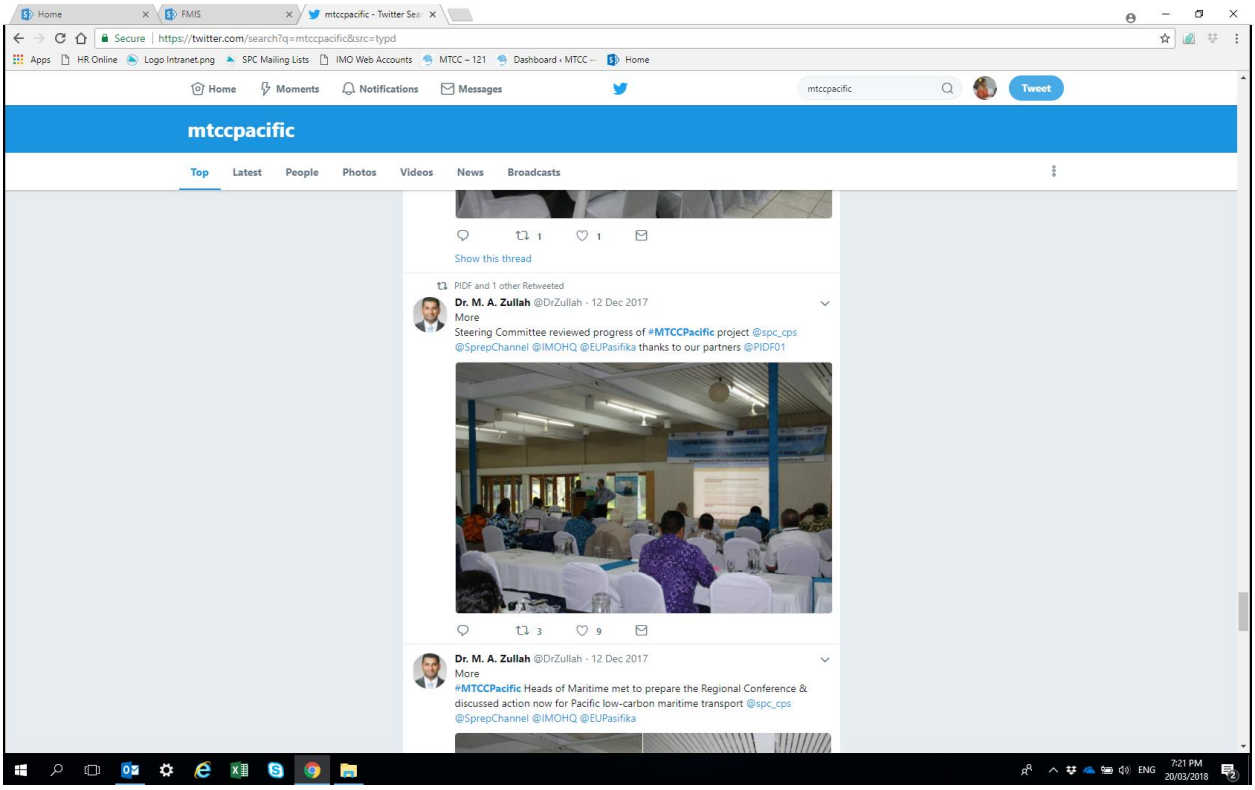

Home
FMIS
mtccpacific - Twitter Sea
Secure | https://twitter.com/search?q=mtccpacific&src=typd
Apps HR Online Logo Intranet.png SPC Mailing Lists IMO Web Accounts MTCC – 121 Dashboard ‹ MTCC – Home
Home Moments Notifications Messages
mtccpacific
Tweet
mtccpacific
Top Latest People Photos Videos News Broadcasts
Audrey_Aumua and 4 others Retweeted
Dr Andrew Jones @andrewj_spc · 12 Dec 2017
Pacific demonstrates commitment to #ParisAgreement and #OnePlanetSummit Today @AumuaA front and centre for @spc_cps to launch #MTCCPacific: a centre supporting #FutureOfMaritime #EnergyEfficientPacific #science4development #tech4good
Lore Croker @Lore_Croker · 12 Dec 2017
#MTCCPacific
Pacific Live tweets @spc_live
A. Aumua: we value the support and cooperation of Fiji and @SprepChannel in the establishment of MTCC-Pacific
Pacific Community and 1 other Retweeted
Pacific Live tweets @spc_live · 12 Dec 2017
#MTCCPacific manager received plaque. officially launching of the Center!
ENG 7:20 PM 20/03/2018

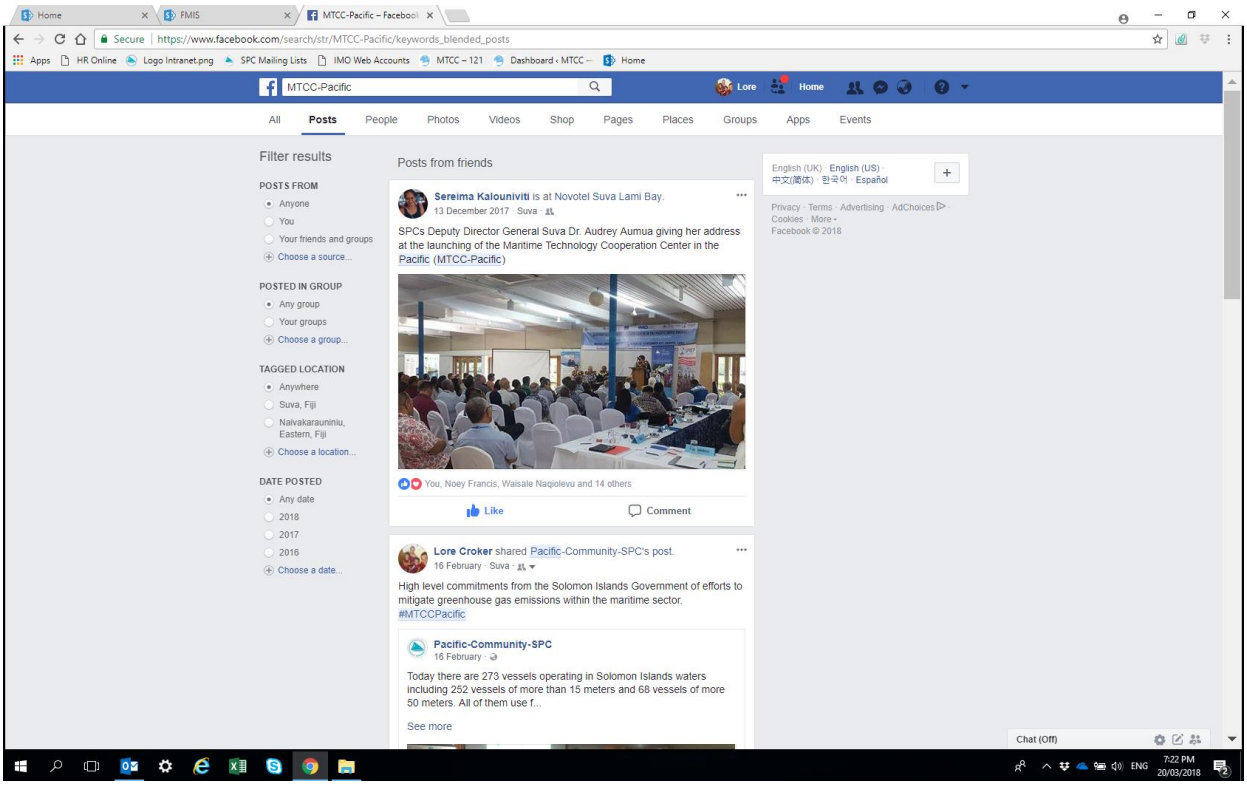

Home
FMIS
mtccpacific - Twitter Sea
Secure | https://twitter.com/search?q=mtccpacific&src=typd
Apps HR Online Logo Intranet.png SPC Mailing Lists IMO Web Accounts MTCC – 121 Dashboard ‹ MTCC – Home
Home Moments Notifications Messages
mtccpacific
Tweet
mtccpacific
Top Latest People Photos Videos News Broadcasts
Pacific Live tweets @spc_live · 12 Dec 2017
#MTCCPacific manager received plaque, officially launching of the Center!
CONFERENCE AND OFFICIAL LAUNCH (12 - 15 DECEMBER 2017, NOVOTEL
bal MTCC Network (GMN) project is funded by the European Union
0:16 202 views
4 8
Pacific Live tweets @spc_live · 12 Dec 2017
@EUPasifika J .Wilson: #MTCCPacific can help improve energy efficiency and promote innovations in maritime climate change mitigation efforts
1 3
Pacific Live tweets @spc_live · 12 Dec 2017
@EUPasifika J .Wilson: #MTCCPacific will play an important role in capacity
ENG 7:20 PM 20/03/2018

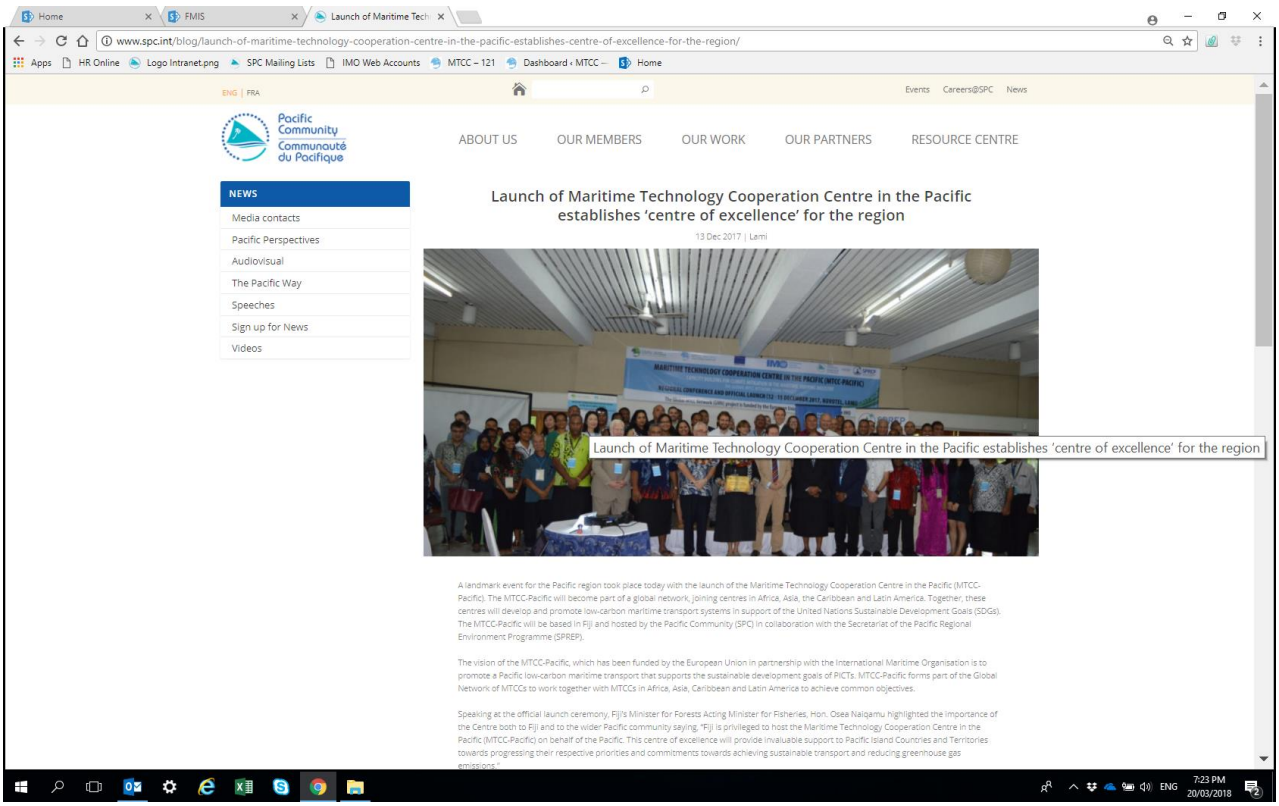

Home
FMIS
mtccpacific - Twitter Sea
Secure | https://twitter.com/search?q=mtccpacific&src=typd
Apps HR Online Logo Intranet.png SPC Mailing Lists IMO Web Accounts MTCC – 121 Dashboard ‹ MTCC – Home
Home Moments Notifications Messages
mtccpacific
Tweet
mtccpacific
Top Latest People Photos Videos News Broadcasts
Inga Mangisi-Mafileo and 4 others liked
Bernedine Managreve @BBernedine · 12 Dec 2017
#MTCCPacific Official launch @ "let's show the Pacific way of how to reduce GHG emissions in the Pacific @spc_live @IMOHQ @AumuaA @ThierryNervale @Lore_Croker
2 7
Thierry Nervale and 1 other Retweeted
Bernedine Managreve @BBernedine · 12 Dec 2017
#MTCCPacific Reg launch @capacity building, data collection & collaboration in
Screenshot Added
A screenshot was added to your Dropbox.
Dropbox
ENG 7:20 PM 20/03/2018

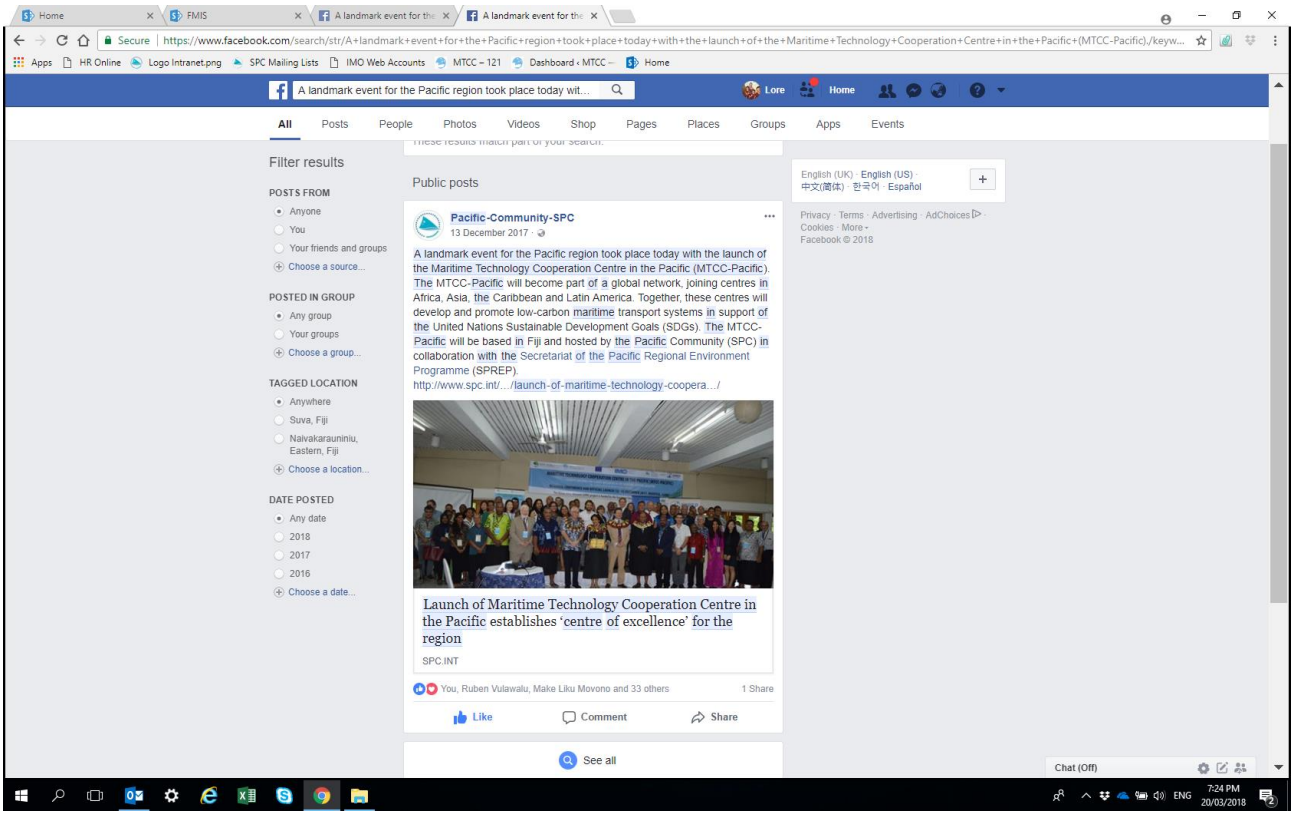

Home
FMIS
mtccpacific - Twitter Sea
Secure | https://twitter.com/search?q=mtccpacific&src=typd
Apps HR Online Logo Intranet.png SPC Mailing Lists IMO Web Accounts MTCC – 121 Dashboard ‹ MTCC – Home
Home Moments Notifications Messages
mtccpacific
Tweet
mtccpacific
Top Latest People Photos Videos News Broadcasts
Thierry Nervale and 1 other Retweeted
Bernedine Managreve @BBernedine · 12 Dec 2017
#MTCCPacific Reg launch @capacity building, data collection & collaboration in the Pacific @spc_live @ThierryNervale @EUPasifika @IMOHQ @SprepChannel
Bernedine Managreve Retweeted
Dr. M. A. Zullah @DrZullah · 12 Dec 2017
Global MTCC Network being presented at #MTCCPacific Regional Conference & Official Launch @spc_cps @SprepChannel @IMOHQ @EUPasifika
Screenshot Added
A screenshot was added to your Dropbox.
Dropbox
ENG
7:20 PM
20/03/2018


Home
FMIS
mtccpacific - Twitter Sea
Secure | https://twitter.com/search?q=mtccpacific&src=typd
Apps HR Online Logo Intranet.png SPC Mailing Lists IMO Web Accounts MTCC – 121 Dashboard ‹ MTCC – Home
Home Moments Notifications Messages
mtccpacific
Tweet
mtccpacific
Top Latest People Photos Videos News Broadcasts
Bernedine Managreve Retweeted
Dr. M. A. Zullah @DrZullah · 12 Dec 2017
Global MTCC Network being presented at #MTCCPacific Regional Conference & Official Launch @spc_cps @SprepChannel @IMOHQ @EUPasifika
Show this thread
PIDF and 1 other Retweeted
Dr. M. A. Zullah @DrZullah · 12 Dec 2017
More
Steering Committee reviewed progress of #MTCCPacific project @spc_cps @SprepChannel @IMOHQ @EUPasifika thanks to our partners @PIDF01
Screenshot Added
A screenshot was added to your Dropbox.
Dropbox
ENG
7:20 PM
20/03/2018


Home
FMIS
mtccpacific - Twitter Search
Secure | https://twitter.com/search?q=mtccpacific&src=typd
Apps HR Online Logo Intranet.png SPC Mailing Lists IMO Web Accounts MTCC – 121 Dashboard ‹ MTCC – Home
Home Moments Notifications Messages
mtccpacific
Tweet
mtccpacific
Top Latest People Photos Videos News Broadcasts
Show this thread
PIDF and 1 other Retweeted
Dr. M. A. Zullah @DrZullah · 12 Dec 2017
More
Steering Committee reviewed progress of #MTCCPacific project @spc_cps @SprepChannel @IMOHQ @EUPasifika thanks to our partners @PIDF01
Dr. M. A. Zullah @DrZullah · 12 Dec 2017
More
#MTCCPacific Heads of Maritime met to prepare the Regional Conference & discussed action now for Pacific low-carbon maritime transport @spc_cps @SprepChannel @IMOHQ @EUPasifika
ENG 7:21 PM 20/03/2018


Home
FMIS
MTCC-Pacific – Facebook
Secure | https://www.facebook.com/search/str/MTCC-Pacific/keywords_blended_posts
Apps HR Online Logo Intranet.png SPC Mailing Lists IMO Web Accounts MTCC – 121 Dashboard ‹ MTCC – Home
MTCC-Pacific
Lore Home
All Posts People Photos Videos Shop Pages Places Groups Apps Events
Filter results
POSTS FROM
Anyone
You
Your friends and groups
Choose a source...
POSTED IN GROUP
Any group
Your groups
Choose a group...
TAGGED LOCATION
Anywhere
Suva, Fiji
Naivakarauniniu, Eastern, Fiji
Choose a location...
DATE POSTED
Any date
2018
2017
2016
Choose a date...
Posts from friends
Sereima Kalouniviti is at Novotel Suva Lami Bay.
13 December 2017 · Suva
SPCs Deputy Director General Suva Dr. Audrey Aumua giving her address at the launching of the Maritime Technology Cooperation Center in the Pacific (MTCC-Pacific)
You, Noey Francis, Waisale Naqiolevu and 14 others
Like Comment
Lore Croker shared Pacific-Community-SPC's post.
16 February · Suva
High level commitments from the Solomon Islands Government of efforts to mitigate greenhouse gas emissions within the maritime sector.
#MTCCPacific
Pacific-Community-SPC
16 February
Today there are 273 vessels operating in Solomon Islands waters including 252 vessels of more than 15 meters and 68 vessels of more 50 meters. All of them use f...
See more
English (UK) · English (US)
中文(简体) · 한국어 · Español
Privacy · Terms · Advertising · AdChoices · Cookies · More ·
Facebook © 2018
Chat (Off)
ENG 7:22 PM 20/03/2018


Launch of Maritime Technology Cooperation Centre in the Pacific establishes 'centre of excellence' for the region
Launch of Maritime Technology Cooperation Centre in the Pacific establishes 'centre of excellence' for the region


Filter results
Public posts
Pacific-Community-SPC
13 December 2017
A landmark event for the Pacific region took place today with the launch of the Maritime Technology Cooperation Centre in the Pacific (MTCC-Pacific). The MTCC-Pacific will become part of a global network, joining centres in Africa, Asia, the Caribbean and Latin America. Together, these centres will develop and promote low-carbon maritime transport systems in support of the United Nations Sustainable Development Goals (SDGs). The MTCC-Pacific will be based in Fiji and hosted by the Pacific Community (SPC) in collaboration with the Secretariat of the Pacific Regional Environment Programme (SPREP).
http://www.spc.int/.../launch-of-maritime-technology-coopera.../
Launch of Maritime Technology Cooperation Centre in the Pacific establishes 'centre of excellence' for the region
SPC.INT

From: eddtm-infobulletin-bounces@lists.spc.int
To: EDDTM-infobulletin@lists.spc.int
Subject: [EDDTM-InfoBulletin] MEDIA ALERT: OFFICIAL LAUNCH OF THE MARITIME TECHNOLOGY COOPERATION CENTRE IN THE PACIFIC
Date: Friday, 8 December 2017 11:46:59 AM
Attachments: image001.png
ATT00001.txt

Official launch of the Maritime Technology Cooperation Centre in the Pacific

8 December 2017

Lami, Fiji – Over 100 high-level delegates and experts will convene in Lami next week (11–15 December) for a series of energy and maritime transport meetings to discuss the latest international and regional developments in energy and maritime transport sectors affecting the region.

Among the high-level speakers at the meeting will be the Minister for Fisheries and Forest, Hon Osea Naiqamu; Deputy Director General of the Pacific Community, Dr Audrey Aumua; Head of Delegation of the European Union for the Pacific, Ambassador Julian Wilson; and the Austrian Ambassador to Australia, Dr Bernhard Zimburg. High-level speakers from the International Maritime Organization and the Secretariat of the Pacific Regional Environment Programme will also be present.

The high-level meeting will cover the progress and plans from the 2017 Pacific Regional Energy and Transport Ministers' Meeting Communique' and the new institutional structure for the Georesources and Energy Programme as well as the Oceans and Maritime Programme.

As part of the week's events, the inauguration of the Fiji-based Maritime Technology Cooperation Centre in the Pacific (MTCC-Pacific) will also take place. MTCC-Pacific is hosted by SPC in collaboration with SPREP.

MTCC-Pacific forms part of the Global MTCCs Network (GMN), a project implemented by IMO and funded by the European Union with the overall objective of climate mitigation in the maritime industry and the reduction of greenhouse gas emissions from maritime transport.

Here is a list of all events happening next week:

1. **What:** 8th Meeting of the Pacific Energy Advisory Group
   When: Monday 11 December (0830 to 1800)
   Where: Novotel, Lami
2. **What:** 2nd Meeting of the Pacific Centre for Renewable Energy and Energy Efficiency Steering Committee
   When: Tuesday 12 December (0830 to 1500)
   Where: Novotel, Lami
3. **What:** GEF Pacific Alliance for Sustainability: Low Carbon Islands
   When: Tuesday 12 December (1530 to 1800)
   Where: Novotel, Lami
4. **What:** Workshop for the Heads of Maritime and 2nd MTCC-Pacific Steering Committee Meeting
   When: Tuesday 12 December (0830 to 1630)
   Where: Novotel, Lami
5. **What:** Side-event for Pacific Women in Maritime
   When: Tuesday 12 December (0830 to 1230)
   Where: Novotel, Lami
6. **What:** Official launch of Maritime Technology Cooperation Centre in the Pacific (MTCC-Pacific)
   When: Wednesday 13 December (1030 to 1230)
   Where: Novotel, Lami
7. **What:** Maritime Technology Cooperation Centre in the Pacific Conference (MTCC-Pacific)
   When: Wednesday 13 to Friday 15 December (0830 to 1630)
   Where: Novotel, Lami
8. **What:** Workshop on Accelerating Renewable Energy Deployment in Small Island Developing States
   When: Wednesday 13 to Friday 15 December (0830 to 1800)
   Where: Novotel, Lami

Media access:
Media representatives are welcome to attend. A Media interested in covering the Official launch of Maritime Technology Cooperation Centre in the Pacific (MTCC-Pacific) are invited to contact Atishma Lal, SPC Project Information Assistant, atishmal@spc.int or +679 337 9402 or +679 9338262.

About Us:
SPC is the principal scientific and technical organisation in the Pacific region, supporting sustainable development since 1947. It is an intergovernmental development organisation owned and governed by its 26 country and territory members.

Useful links:
To access agenda and meeting papers:

PEAG – http://prdrse4all.spc.int/node/4/content/8th-meeting-pacific-energy-advisory-group
PCREEE – http://prdrse4all.spc.int/node/4/content/second-meeting-pcreee-steering-committee-2017
Accelerating RE deployment in SIDS Workshop - http://prdrse4all.spc.int/node/4/content/accelerating-renewable-energy-deployment-small-island-developing-states-suva-fiji-11

Other useful links:
http://www.pcreee.org/
https://gmn.imo.org/mtcc/pacific/
https://gmn.imo.org/

From: eddtm-infobulletin-bounces@lists.spc.int
To: EDDTM-infobulletin@lists.spc.int
Subject: [EDDTM-InfoBulletin] Media Release: Launch of Maritime Technology Cooperation Centre in the Pacific establishes 'centre of excellence' for the region
Date: Wednesday, 13 December 2017 4:33:57 PM
Attachments: image003.png
ATT00001.txt

MEDIA RELEASE

Pacific Community
Communauté du Pacifique

Launch of Maritime Technology Cooperation Centre in the Pacific establishes 'centre of excellence' for the region

13 December 2017

Lami, Fiji – A landmark event for the Pacific region took place today with the launch of the Maritime Technology Cooperation Centre in the Pacific (MTCC-Pacific). The MTCC-Pacific will become part of a global network, joining centres in Africa, Asia, the Caribbean and Latin America. Together, these centres will develop and promote low-carbon maritime transport systems in support of the United Nations Sustainable Development Goals (SDGs). The MTCC-Pacific will be based in Fiji and hosted by the Pacific Community (SPC) in collaboration with the Secretariat of the Pacific Regional Environment Programme (SPREP).

The vision of the MTCC-Pacific, which has been funded by the European Union in partnership with the International Maritime Organisation is to promote a Pacific low-carbon maritime transport that supports the sustainable development goals of PICTs. MTCC-Pacific forms part of the Global Network of MTCCs to work together with MTCCs in Africa, Asia, Caribbean and Latin America to achieve common objectives.

Speaking at the official launch ceremony, Fiji's Minister for Forests Acting Minister for Fisheries, Hon. Osea Naiqamu highlighted the importance of the Centre both to Fiji and to the wider Pacific community saying, "Fiji is privileged to host the Maritime Technology Cooperation Centre in the Pacific (MTCC-Pacific) on behalf of the Pacific. This centre of excellence will provide invaluable support to Pacific Island Countries and Territories towards progressing their respective priorities and commitments towards achieving sustainable transport and reducing greenhouse gas emissions."

The European Union Ambassador to Fiji and the Pacific His Excellency Julian Wilson said: "The EU is proud of the partnership with IMO in supporting the establishment of the MTCC-Pacific in Fiji. This institution has an important role to play on climate mitigation in the maritime shipping sector, thus helping the global community reach the Paris Agreement goals,"

"The EU strongly hopes that this project will help the Pacific countries to accelerate the uptake of technical and operational possibilities to improve the energy efficiency of shipping and promote clean energy shipping. This will have an immediate impact on global GHG emissions." added Ambassador Wilson.

SPC's Deputy Director General, Dr Audrey Aumua, emphasised the historic nature of the MTCC launch and highlighted the importance of ensuring that the Pacific is well represented in the global network, given the central role that the maritime industry plays in the region. Dr. Anmua also stressed how this new 'maritime centre of excellence' will support the Framework for Resilient Development in the Pacific. "MTCC-Pacific compliments our regional efforts towards the FRDP and will provide us with an effective platform to accelerate progress towards the objectives of our regional climate change mitigation strategy."

MTCC-Pacific forms part of the Global MTCCs Network (GMN), a project implemented by IMO and funded by the European Union with the overall objective of promoting climate mitigation in the maritime industry and reducing greenhouse gas emissions from maritime transport.

International Maritime Organization (IMO) Deputy Director/Chief, Office of General Services, Mr. Aubrey Botsford welcomed the establishment of MTCC-Pacific noting that, "efforts to mitigate climate change and promote energy efficiency in the maritime sector, through actions taken by Pacific governments and maritime industry, show the dynamic approach taken by your region. IMO is strongly supportive of this approach and of the various actions taken in the Pacific."

Today's launch was part of a MTCC-Pacific Regional Conference, a week-long gathering of maritime industry leaders and experts from across the globe. The conference will conclude on Friday.

Media contacts
Atishma Lal, SPC Project Information Assistant |atishmal@spc.int or +679 337 9402 or +679 9338262.

Useful links:
https://gmn.imo.org/mtcc/pacific/
https://gmn.imo.org/
http://mtccpacific.spc.int/

Other useful links:
http://www.pcreee.org/
http://prdrse4all.spc.int/content/about

To subscribe to SPC's News mailing list, click the link below:
http://lists.spc.int/mailman/listinfo/press-releases